Topology and Geometry

Parts I and II

Lecture notes

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Part I

Topology and Manifolds

Lecture 1

Spaces and continuity

Objectives. *Continuity can be described without measuring distances. We keep the open sets of a metric space and use them to define a topological space. We then prove that metric continuity is equivalent to openness of inverse images. This gives a way to compare spaces and to put a topology on a subset or a product. Explicit maps identify an interval with a line and an open disc with a plane.*

By the end of this lecture you will be able to:

- Construct a topology from a metric or a basis.
- Test continuity by inverse images and verify a homeomorphism.
- Use the continuity tests for subspaces and finite products.
- Separate points in a Hausdorff space and prove uniqueness of sequence limits.

1.1 Open sets from a metric

Recall the distance function from *Linear Algebra for Physics*, Lecture 13. A *metric* on a set X is a function $d\colon X \times X \to [0,\infty)$ such that $d(x,y) = 0$ exactly when $x = y$, $d(x,y) = d(y,x)$, and $d(x,z) \le d(x,y) + d(y,z)$ for all $x, y, z \in X$. The last condition is the triangle inequality. The *open ball* with centre $x \in X$ and radius $r > 0$ is

$$B_r(x) = \{y \in X : d(x,y) < r\}.$$

A subset $U \subseteq X$ is *open* if every $x \in U$ has a ball $B_r(x) \subseteq U$ with $r > 0$. The radius may depend on the point.

For example, in $\mathbb{R}$ with $d(x,y) = |x - y|$, an interval (a,b) is open. At $x \in (a,b)$ take $r = \min(x - a, b - x) > 0$. Neither $[a,b]$ nor $[a,b)$ is open when $a < b$, because no ball about a stays in the interval. The empty set is open because it has no points to test.

Every point of an open ball has a smaller ball about it inside the original ball. The triangle inequality explains the inclusion in the figure.

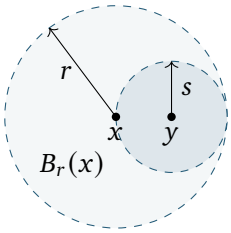


Figure 1.1: If $y \in B_r(x)$, the ball about y of radius $s = r - d(x,y)$ lies inside $B_r(x)$. The smaller boundary is also excluded.

Indeed, if $z \in B_s(y)$, then

$$d(x,z) \le d(x,y) + d(y,z) < d(x,y) + s = r.$$

Thus every open ball is open.

Proposition 1.1 (Open sets of a metric space). *The open subsets of a metric space include $\varnothing$ and X. Arbitrary unions and finite intersections of open sets are open.*

Proof. The whole set contains every ball. If x belongs to a union of open sets, it belongs to one member of the union. That member contains a ball about x, so the union does too. If $x \in U \cap V$ with U and V open, choose $B_r(x) \subseteq U$ and $B_s(x) \subseteq V$. Then $B_{\min(r,s)}(x) \subseteq U \cap V$. Repeating this argument gives any nonempty finite intersection. The intersection of no subsets is X. ∎

The restriction to finite intersections matters. In $\mathbb{R}$ the intersection

$$\bigcap_{n \ge 1} (-1/n, 1/n) = \{0\}$$

is not open.

1.2 Topological spaces and bases

Definition 1.2 (Topology). A *topology* on a set X is a collection $\mathcal{T}$ of subsets of X that contains $\varnothing$ and X and is closed under arbitrary unions and finite intersections. The pair $(X, \mathcal{T})$ is a *topological space*. Its members are called *open sets*. A subset is *closed* if its complement in X is open.

The preceding proposition constructs the *metric topology* of d. The standard topology on $\mathbb{R}^n$ is the topology of the Euclidean distance. We use this topology unless another one is specified. The same set may carry other topologies. The *discrete topology* contains every subset. The *indiscrete topology* contains only $\varnothing$ and X. Both satisfy the axioms directly.

A set can be both open and closed. The sets X and $\varnothing$ always are. A set can be neither open nor closed. The set $[0,1)$ in the standard real line is one example. Its complement is not open at 1.

For a subset $A \subseteq X$, its *interior* $\operatorname{int} A$ is the union of all open subsets of A. Its *closure* $\overline{A}$ is the intersection of all closed sets containing A. The interior is open by the union axiom. The closure is closed because its complement is a union of open sets. Every open subset of A is one of the sets being united, so lies in $\operatorname{int} A$. Every closed superset of A contains the defining intersection, so contains $\overline{A}$.

A *neighbourhood* of $x \in X$ is a set containing an open set that contains x. A neighbourhood itself need not be open.

In the standard real line, $\operatorname{int}[0,1) = (0,1)$ and $\overline{[0,1)} = [0,1]$. To check the second equality, note that every open interval about 1 meets $[0,1)$, so no closed superset can omit 1. The closed set $[0,1]$ itself contains $[0,1)$.

Listing all open sets is usually unnecessary. Open balls already determine the metric topology because every open set is a union of balls.

Definition 1.3 (Basis). A collection $\mathcal{B}$ of subsets of X is a *basis* if it covers X and has the following refinement property: whenever $x \in B_1 \cap B_2$ with $B_1, B_2 \in \mathcal{B}$, there is $B_3 \in \mathcal{B}$ such that

$$x \in B_3 \subseteq B_1 \cap B_2.$$

Proposition 1.4 (Topology from a basis). *The unions of members of a basis form a topology. A set U belongs to this topology exactly when every $x \in U$ lies in some $B \in \mathcal{B}$ with $B \subseteq U$.*

Proof. The empty union is $\varnothing$, and the covering property gives X. A union of unions is a union. For two unions U and V, a point $x \in U \cap V$ lies in basis members $B_1 \subseteq U$ and $B_2 \subseteq V$. Refine their intersection to B_3 as in the definition. The union of all these refinements is $U \cap V$. Finite intersections follow by repetition. The final assertion is precisely the condition that U be a union of basis members. ∎

The collection $\mathcal{B}$ is a *basis for a topology* $\mathcal{T}$ when $\mathcal{T}$ is exactly the topology it generates. Open balls form a basis for the metric topology. They cover the space, and the open intersection of two balls contains a ball about each of its points.

For example, open intervals form a basis for $\mathbb{R}$. Intervals with rational endpoints suffice: for $x \in (a, b)$ choose rational p and q with $a < p < x < q < b$.

The rational intervals are countable. A space is *second countable* if its topology has a countable basis.

In $\mathbb{R}^n$, boxes with rational endpoints give a countable basis as well. Every open ball contains such a box about each of its points, and every open box contains a ball about each of its points. Thus boxes and balls give the same topology. Second countability will be a condition in the definition of a manifold in Lecture 7.

1.3 Continuity and homeomorphisms

Definition 1.5 (Continuity). A map $f : X \to Y$ of topological spaces is *continuous* if $f^{-1}(V)$ is open in X for every open set $V \subseteq Y$. Here the *inverse image* is the set

$$f^{-1}(V) = \{x \in X : f(x) \in V\}.$$

An inverse image exists for every map. The notation does not assume that the map is invertible. It is enough to test basis members of Y, since inverse images preserve unions.

The inverse image can have several pieces even when the target set is an interval. For the map $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$, we obtain

$$f^{-1}((1, 4)) = (-2, -1) \cup (1, 2).$$

The figure shows the two pieces. Both are open, as required by continuity.

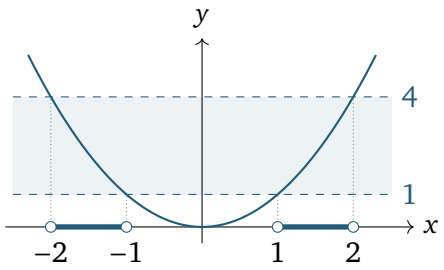


Figure 1.2: The horizontal band $1 < y < 4$ meets the graph $y = x^2$ above two open intervals on the x-axis. Endpoints are excluded.

Proposition 1.6 (Metric and topological continuity). *For metric spaces X and Y, the map $f : X \to Y$ is continuous exactly when, for every $x \in X$ and every $\varepsilon > 0$, there is $\delta > 0$ such that*

$$d_X(x, x') < \delta \quad \Longrightarrow \quad d_Y(f(x), f(x')) < \varepsilon \qquad (x' \in X).$$

Proof. Suppose inverse images of open sets are open. The inverse image of $B_\varepsilon(f(x))$ is an open set containing x. It contains a ball $B_\delta(x)$, which gives the implication.

Conversely, suppose the displayed condition holds. For an open set $V \subseteq Y$ and $x \in f^{-1}(V)$ choose $B_\varepsilon(f(x)) \subseteq V$. The condition supplies a ball $B_\delta(x) \subseteq f^{-1}(V)$. Thus $f^{-1}(V)$ is open. ∎

Continuity therefore depends on a metric only through its open sets. For example, two norms satisfying $c\,\|v\|_a \le \|v\|_b \le C\,\|v\|_a$ for every $v \in \mathbb{R}^n$, with $c, C > 0$, induce the same topology. Indeed, their balls satisfy

$$B^a_{r/C}(x) \subseteq B^b_r(x) \subseteq B^a_{r/c}(x).$$

Applying these inclusions at each point gives the same open sets. These are called *equivalent norms*. Equivalent norms give the same continuous maps, even though they may assign different lengths. Lecture 7 uses this when comparing charts.

Identity maps are continuous, and composites of continuous maps are continuous. The latter follows from $(g \circ f)^{-1}(W) = f^{-1}(g^{-1}(W))$ for maps $f\colon X \to Y$ and $g\colon Y \to Z$ and an open set $W \subseteq Z$.

Definition 1.7 (Homeomorphism)**.** A *homeomorphism* is a continuous bijection $f\colon X \to Y$ whose inverse $f^{-1}\colon Y \to X$ is continuous. Spaces joined by such a map are *homeomorphic*, written $X \cong Y$.

For example, $f\colon (0,1) \to \mathbb{R}$ given by $f(t) = \tan(\pi(t - \frac{1}{2}))$ is a homeomorphism. Its inverse is given by $f^{-1}(s) = \frac{1}{2} + \pi^{-1} \arctan s$. Both maps are continuous by ordinary one-variable calculus and the metric-continuity proposition.

1.4 Subspaces and products

The circle $S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$ is a subset of the plane. Its open sets must describe motion along the circle, although no plane ball lies inside the circle.

Definition 1.8 (Subspace and product topologies)**.** The *subspace topology* on $A \subseteq X$ consists of the sets $A \cap U$ with U open in X. For topological spaces X and Y, the *product topology* on $X \times Y$ has basis the rectangles $U \times V$ with $U \subseteq X$ and $V \subseteq Y$ open. The topology on a finite product has basis the products of open sets in its factors.

Intersections with A preserve unions and finite intersections, so the subspace sets form a topology. Rectangles cover the product, and the intersection of two rectangles is a rectangle. Hence the basis proposition constructs the product topology.

The restricted metric has balls $A \cap B_r(x)$ for $x \in A$. They are subspace-open. Conversely, if $x \in A \cap U$ with U open in X, a ball $B_r(x) \subseteq U$ gives a restricted ball inside $A \cap U$. Thus the two topologies agree.

For $\mathbb{R}^n$, every product of open sets is a union of open boxes, by choosing an interval about each coordinate. The box-and-ball comparison therefore identifies the finite product and Euclidean topologies.

The figure compares the two constructions. An open arc on the circle is open relative to the circle. An open rectangle in the plane is open in the product of two real lines.

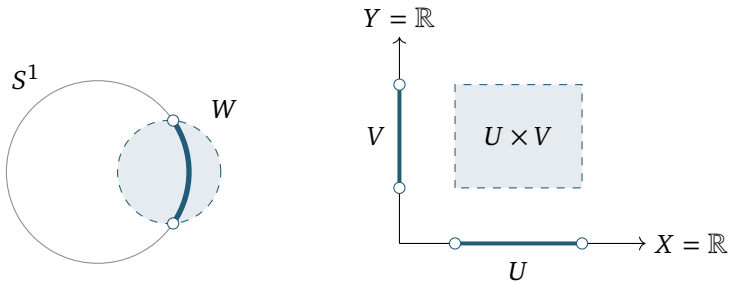


Figure 1.3: An open disc W cuts out an open arc on the circle. Open intervals U and V determine an open rectangle $U \times V$ in the plane.

Proposition 1.9 (Continuity tests for subspaces and products)**.** *For a map $f\colon Z \to A \subseteq X$, continuity into the subspace A is equivalent to continuity of $i \circ f\colon Z \to X$, where $i\colon A \to X$ is inclusion. A map $h\colon Z \to X \times Y$ is continuous exactly when its components $\pi_X \circ h$ and $\pi_Y \circ h$ are continuous. Here $\pi_X(x, y) = x$ and $\pi_Y(x, y) = y$.*

Proof. For an open set $U \subseteq X$, the inverse image under inclusion is $i^{-1}(U) = A \cap U$, so i is continuous. Continuity of f then gives continuity of the composite. Conversely, $f^{-1}(A \cap U) = (i \circ f)^{-1}(U)$, proving continuity into A.

The projections are continuous because $\pi_X^{-1}(U) = U \times Y$ and $\pi_Y^{-1}(V) = X \times V$ are open. Thus continuous h has continuous components. Conversely, for a basis rectangle the inverse image is

$$h^{-1}(U \times V) = (\pi_X \circ h)^{-1}(U) \cap (\pi_Y \circ h)^{-1}(V).$$

Continuous components make this intersection open. Testing basis members proves continuity of h. ∎

In particular, restricting a continuous map $f\colon X \to Y$ to $A \subseteq X$ preserves continuity, since the restriction is $f \circ i$.

For example, the map $t \mapsto (\cos t, \sin t)$ is continuous into $\mathbb{R}^2$ because both components are continuous. Its values lie in S^1, so the subspace test makes it continuous into the circle too. The same proof, with one projection and one inverse image per factor, gives the test for every finite product.

A continuous bijection need not have continuous inverse. Restrict the preceding map to $f\colon [0, 2\pi) \to S^1$. Each point of the circle has exactly one angle in this interval, so f is bijective. The set $[0, \pi)$ is open in $[0, 2\pi)$, but its image is not open in the circle. Every open neighbourhood of $(1, 0)$ in the circle contains points just below the horizontal axis. Thus the inverse map is not continuous. Lecture 2 supplies a useful sufficient condition for a continuous bijection to be a homeomorphism.

1.5 Hausdorff spaces and invariants

Definition 1.10 (Hausdorff space)**.** A space X is *Hausdorff* if any two distinct points have disjoint open neighbourhoods. A sequence $(x_n)_{n\geq 1}$ *converges* to $x \in X$ if every open neighbourhood of x contains all x_n for sufficiently large n.

Proposition 1.11 (Separation and limits)**.** *Metric spaces are Hausdorff. Subspaces and finite products of Hausdorff spaces are Hausdorff. A sequence in a Hausdorff space has at most one limit.*

Proof. For distinct points x and y of a metric space, put $r = d(x, y)/3$. The balls $B_r(x)$ and $B_r(y)$ are disjoint. A point in both would give $d(x, y) < 2r < d(x, y)$ by the triangle inequality. Intersecting disjoint open neighbourhoods with a subspace separates its points. Two distinct points in a product differ in some coordinate. Separate those coordinate values and take inverse images under the corresponding projection. The resulting product neighbourhoods are disjoint.

If a sequence had distinct limits x and y, disjoint open neighbourhoods U and V of those points would both contain every term after some common index. This would put a term in $U \cap V = \varnothing$. ∎

The Hausdorff condition can fail. The *cofinite topology* on $\mathbb{R}$ consists of $\varnothing$ and the subsets with finite complement. It is a topology. A nonempty union has complement contained in the finite complement of any one nonempty member. A finite intersection of nonempty open sets has complement a finite union of finite sets. Two nonempty open sets cannot be disjoint, since their intersection has finite complement in the infinite set $\mathbb{R}$. This space is not Hausdorff.

A *topological invariant* is a property shared by homeomorphic spaces. Hausdorffness is one example.

Proposition 1.12 (Transport of open sets)**.** *A homeomorphism $f : X \to Y$ gives a bijection $U \mapsto f(U)$ between their topologies, preserving unions, intersections and inclusion. Hausdorffness and second countability are topological invariants.*

Proof. If $f : X \to Y$ is a homeomorphism and $U \subseteq X$ is open, then $f(U) = (f^{-1})^{-1}(U)$ is open by continuity of the inverse. Continuity of f gives the converse correspondence. A bijection preserves the stated set operations. Images of disjoint open neighbourhoods separate the image points, proving Hausdorff invariance. If $\mathcal{B}$ is a countable basis for X, its images form a countable family of open sets. Bijectivity carries the covering and refinement properties across. Every open set of Y is the image of a union of members of $\mathcal{B}$, hence a union of these image sets. They form a basis for Y. ∎

The standard real line and the cofinite real line are therefore not homeomorphic. They have the same underlying set but different separation properties.

1.6 Explicit homeomorphisms

We compare the open interval, the line, a circle with a point removed, the open disc and the plane. All subsets have their subspace topologies. The interval and the line were identified above. To treat the punctured circle, remove $N = (0, 1)$ and define

$$p : S^1 \setminus \{N\} \longrightarrow \mathbb{R}, \qquad p(x, y) = \frac{x}{1 - y}.$$

The line through N and (x, y) meets the horizontal axis at $(p(x, y), 0)$. This construction is called *stereographic projection*. The figure shows this construction.

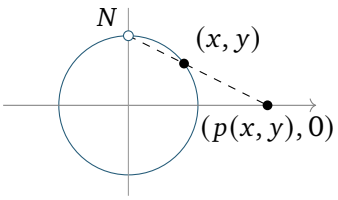


Figure 1.4: Stereographic projection from N onto the horizontal axis. The projection point N is omitted from the circle.

Solving the line and circle equations gives the inverse

$$p^{-1}\colon \mathbb{R} \longrightarrow S^1 \setminus \{N\}, \qquad p^{-1}(t) = \left(\frac{2t}{1+t^2}, \frac{t^2-1}{1+t^2}\right).$$

The squared coordinates sum to 1, and the second coordinate is less than 1. Substitution gives $p(p^{-1}(t)) = t$. Conversely, using $x^2 = 1 - y^2$ and $y \neq 1$ gives $p^{-1}(p(x,y)) = (x,y)$. Both maps have continuous coordinate functions with nonzero denominators on their domains. The subspace and product tests therefore make p a homeomorphism. By composition the punctured circle is homeomorphic to $(0,1)$ as well.

For the open disc $D = \{u \in \mathbb{R}^2 : \|u\| < 1\}$, move each point outward along its radius. Define

$$f_D\colon D \to \mathbb{R}^2, \qquad f_D(u) = \frac{u}{1-\|u\|}, \qquad f_D^{-1}\colon \mathbb{R}^2 \to D, \qquad f_D^{-1}(v) = \frac{v}{1+\|v\|}.$$

The norm here is Euclidean and continuous. The denominators are positive, so both formulas are continuous, including at 0. For $r = \|u\| < 1$, the norm of $f_D(u)$ is $r/(1-r)$, giving $f_D^{-1}(f_D(u)) = u$. For $s = \|v\|$, the inverse formula has norm $s/(1+s) < 1$, so lands in D. It gives $f_D(f_D^{-1}(v)) = v$. The subspace test proves continuity into D. Thus f_D is a homeomorphism.

The results proved so far form two families:

Spaces identified	Maps used
$(0,1) \cong \mathbb{R} \cong S^1 \setminus \{N\}$	tangent and stereographic projection
$D \cong \mathbb{R}^2$	radial map f_D

Nothing proved here distinguishes the two families. Lecture 2 proves that a line and a plane are not homeomorphic by examining what happens when a point is removed. It also distinguishes the circle from the line by compactness, and the circle from the closed interval by removing a point. Those distinctions are not assumed here.

Self-check. Here x^i and u^i label coordinates. A squared coordinate is written $(x^i)^2$. For the unit sphere $S^2 = \{(x^1, x^2, x^3) \in \mathbb{R}^3 : (x^1)^2 + (x^2)^2 + (x^3)^2 = 1\}$ and $N = (0,0,1)$, prove that the map

$$p_N\colon S^2 \setminus \{N\} \to \mathbb{R}^2, \qquad p_N(x) = \frac{(x^1, x^2)}{1 - x^3}$$

is a homeomorphism by finding its inverse.

Answer. For $u = (u^1, u^2) \in \mathbb{R}^2$, put $r^2 = (u^1)^2 + (u^2)^2$. The inverse is

$$p_N^{-1}(u) = \frac{(2u^1, 2u^2, r^2 - 1)}{1 + r^2}.$$

Its squared norm is $[4r^2 + (r^2 - 1)^2]/(1 + r^2)^2 = 1$ and its last coordinate is less than 1. Substitution gives $p_N(p_N^{-1}(u)) = u$. For $x \in S^2 \setminus \{N\}$, the sphere equation gives $\|p_N(x)\|^2 = (1 + x^3)/(1 - x^3)$, hence $p_N^{-1}(p_N(x)) = x$. The coordinate formulas and the subspace and product tests prove continuity of both maps.

Summary. A topology records which subsets are open. Continuity means that inverse images of open sets are open, and agrees with metric continuity when a metric is present. A homeomorphism requires continuity in both directions. Subspace and product topologies let us test continuity using inclusion and coordinate projections. Hausdorffness is preserved by homeomorphisms and ensures uniqueness of sequence limits. Lecture 2 introduces compactness and connectedness to distinguish more spaces.

Reading. Munkres, *Topology*, second edition, §§12–13 and 15–18: topologies, bases, products, subspaces, separation and continuity.

Lecture 2

Compactness, connectedness and gluing

Objectives. *Open sets let us define properties that distinguish spaces. Compactness expresses a finiteness condition on open covers. Connectedness rules out a separation into two open pieces. We prove how continuous maps preserve these properties, then use them to identify spaces formed by gluing. The main criterion turns a continuous bijection into a homeomorphism when its domain is compact and its target is Hausdorff.*

By the end of this lecture you will be able to:

- Prove the compact-to-Hausdorff criterion for a homeomorphism.
- Distinguish spaces by compactness, and by connectedness after removing a point.
- Test continuity out of a quotient and identify its points correctly.
- Describe the torus, Möbius band, projective plane and rotation group as quotients.

2.1 Compactness

Definition 2.1 (Compactness). An *open cover* of a space X is a family of open sets $(U_\alpha)_{\alpha\in I}$ whose union is X. A *subcover* is a subfamily that still covers X. The space is *compact* if every open cover has a finite subcover.

A subset $K \subseteq X$ is called compact when its subspace topology is compact. Equivalently, every cover of K by open sets of X has a finite subcover. This follows because the open sets of K are intersections $K \cap U$ with ambient open sets U (§ 1.4).

The figure illustrates the condition on an interval. A finite selection must cover every point, including both endpoints.

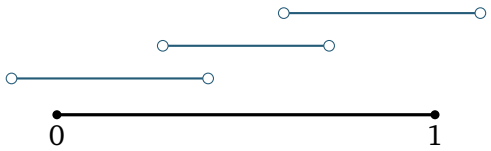


Figure 2.1: The three open intervals shown cover $[0, 1]$. For compactness, a finite subcover must exist for every open cover, not just this one.

Every finite space is compact. Choose one cover member containing each point. The resulting selection is finite. The empty space needs no members. In contrast, the real line is not compact. The open cover $\{(-n, n) : n \geq 1\}$ has no finite subcover, since a finite selection is contained in its member with the largest index.

Theorem 2.2 (Continuous images and closed subsets). *A continuous image of a compact space is compact. A closed subset of a compact space is compact.*

Proof. Let $f\colon X \to Y$ be continuous with X compact. Take an ambient open cover (V_α) of $f(X)$. Its inverse images cover X and are open. Choose finitely many of them covering X. The corresponding sets V_α cover $f(X)$.

Now let $C \subseteq X$ be closed, with X compact. Take an ambient open cover of C and add the open set $X \setminus C$. This covers X and therefore has a finite subcover. Discarding $X \setminus C$, if selected, leaves finitely many members covering C. ∎

We use two results about metric spaces with their proofs cited. *Heine–Borel* says that a subset of $\mathbb{R}^n$ is compact exactly when it is closed and bounded (Munkres, Theorem 27.3). Bounded means contained in some ball of finite radius. Thus closed intervals, closed balls and unit spheres are compact. The hypotheses refer to Euclidean space. Closed and bounded need not imply compact in a general metric space.

A metric space is *sequentially compact* if every sequence in it has a subsequence converging to a point of that space. This is the notion used in *Linear Algebra for Physics*, Lecture 13. It is equivalent to open-cover compactness for metric spaces (Munkres, Theorem 28.2). We use the open-cover definition on spaces without a specified metric.

The circle is compact by Heine–Borel, whereas the real line is not. They cannot be homeomorphic. A homeomorphism would take the compact circle onto the line. This settles one distinction promised in Lecture 1.

2.2 Compact spaces and Hausdorff targets

Hausdorffness means that distinct points have disjoint open neighbourhoods (§ 1.5). It supplies the other hypothesis in the homeomorphism criterion.

Proposition 2.3 (Compact subsets are closed). *A compact subset of a Hausdorff space is closed.*

Proof. Let $K \subseteq Y$ be compact and $y \in Y \setminus K$. For each $x \in K$, choose disjoint open neighbourhoods U_x of x and V_x of y. Finitely many sets $U_{x_1}, \dots, U_{x_m}$ cover K. The open neighbourhood $V_{x_1} \cap \cdots \cap V_{x_m}$ of y misses their union, so misses K. Thus $Y \setminus K$ is open. If K is empty, it is closed directly. ∎

Continuity can also be tested on closed sets. Inverse images commute with complements, so the open-set test is equivalent to the closed-set test.

Corollary 2.4 (Compact-to-Hausdorff criterion). *A continuous bijection from a compact space to a Hausdorff space is a homeomorphism.*

Proof. Let $f\colon X \to Y$ satisfy the hypotheses. For a closed subset $C \subseteq X$, compactness of X makes C compact. Its image $f(C)$ is compact, hence closed in Y. This image is the inverse image of C under $f^{-1}\colon Y \to X$. The closed-set test proves continuity of the inverse. ∎

Lecture 1's continuous bijection $[0, 2\pi) \to S^1$ does not meet the criterion. Its domain is not compact, since it is not closed in $\mathbb{R}$. We will instead use $[0, 1]$ and identify its two endpoints.

2.3 Connectedness

Definition 2.5 (Connectedness and paths). A *separation* of a space X is a pair of disjoint, nonempty open sets whose union is X. The space is *connected* if it has no separation. A *path* from x to y in X is a continuous map $\gamma\colon [0,1] \to X$ with $\gamma(0) = x$ and $\gamma(1) = y$. The space is *path-connected* if such a path exists for every pair $x, y \in X$.

A subset is connected or path-connected when its subspace topology has the corresponding property.

Theorem 2.6 (Intervals, images and paths). *The interval $[0,1]$ is connected. Continuous images preserve connectedness and path-connectedness. Every path-connected space is connected.*

Proof. Suppose $[0,1] = U \cup V$ is a separation. Choose $a \in U$ and $b \in V$, with $a < b$ after exchanging the names if necessary. Put $c = \sup(U \cap [a,b])$. If $c \in U$, then $c < b$, and openness of U gives a point of $U \cap [a,b]$ larger than c. If $c \in V$, then $c > a$, and openness of V forbids points of U immediately below c. This contradicts the defining property of the supremum. Both cases are impossible, so no separation exists.

For a continuous map $f\colon X \to Y$, a separation of $f(X)$ pulls back to a separation of X. Thus a connected domain has connected image. If X is path-connected, composing its paths with f gives paths between any two image points.

Finally, a path-connected space cannot have a separation $U \cup V$. A path joining a point of U to a point of V would have connected image, since $[0,1]$ is connected. Its intersections with U and V would separate that image. ∎

For example, a square and a ball are path-connected. The segment $\gamma(t) = (1-t)x + ty$ stays in the set and joins any two of its points.

A *connected component* is a connected subset maximal under inclusion. The component of $x \in X$ is the union of all connected subsets containing x. This union is connected. In any proposed separation, each of its connected subsets must lie wholly on the side containing x. It is maximal by its definition. Two components that meet therefore coincide, so components partition X.

Removing a point distinguishes a line from a plane, as the figure shows.

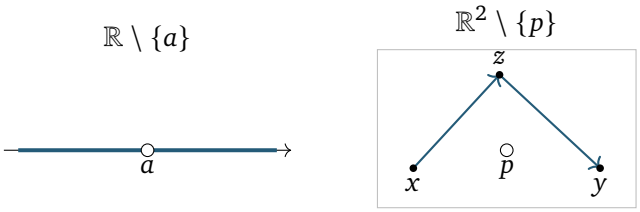


Figure 2.2: Removing a point separates a line into two open rays. In a plane, a broken line can join two points while avoiding the removed point.

The space $\mathbb{R} \setminus \{a\}$ has a separation into its two open rays. In $\mathbb{R}^n \setminus \{p\}$, with $n \geq 2$, take points x and y and choose z outside the two lines through p, x and through p, y. The segments from x to z and from z to y avoid p. They give the path

$$\gamma(t) = \begin{cases} (1-2t)x + 2tz, & 0 \leq t \leq \frac{1}{2}, \\ (2-2t)z + (2t-1)y, & \frac{1}{2} \leq t \leq 1. \end{cases}$$

The two formulas agree at $t = 1/2$. Each affine formula is continuous, so the preimage of a closed set is closed on its half-interval. It is also closed in $[0,1]$, since both half-intervals are closed there. Their finite union is closed, proving continuity of γ. Thus the punctured space is path-connected.

A homeomorphism restricts to a homeomorphism after deleting a point and its image. The punctured line and plane have different connectedness properties, so $\mathbb{R} \not\cong \mathbb{R}^2$. The radial map $x \mapsto x/\|x\|$ also shows that S^n is path-connected for $n \geq 1$, as an image of $\mathbb{R}^{n+1} \setminus \{0\}$.

The converse to path-connected implies connected is false. A cited example is the closure of $\{(x, \sin(1/x)) : 0 < x \leq 1\}$ in $\mathbb{R}^2$, the *topologist's sine curve* (Munkres, §24, Example 7). Its analysis is reserved for Sheet 2.

2.4 Quotients and orbit spaces

Definition 2.7 (Quotient topology)**.** For an equivalence relation $\sim$ on X, the *quotient set* $X/\!\sim$ consists of the equivalence classes. The projection $q\colon X \to X/\!\sim$ sends each point to its class. The *quotient topology* is defined by

$$V \subseteq X/\!\sim \text{ is open} \quad \Longleftrightarrow \quad q^{-1}(V) \text{ is open in } X.$$

Inverse images preserve unions and finite intersections, so this is a topology. The projection is continuous and surjective. Hence a quotient of a compact or path-connected space retains that property.

Proposition 2.8 (Continuity out of a quotient)**.** *A map $\bar{f}\colon X/\!\sim \to Y$ is continuous exactly when $\bar{f} \circ q$ is continuous. A map $f\colon X \to Y$ that is constant on equivalence classes induces a unique map $\bar{f}$ with $f = \bar{f} \circ q$.*

Proof. Composition with continuous q preserves continuity. Conversely, if $\bar{f} \circ q$ is continuous, then for open $W \subseteq Y$ the set $q^{-1}(\bar{f}^{-1}(W))$ is open. The quotient definition says that $\bar{f}^{-1}(W)$ is open. The induced map is given by $\bar{f}([x]) = f(x)$, which is well-defined precisely when f is constant on each class. Surjectivity of q makes it unique. ∎

Define the circle map $c\colon [0,1] \to S^1$ by

$$c(t) = (\cos(2\pi t), \sin(2\pi t)).$$

The figure depicts its endpoint identification.

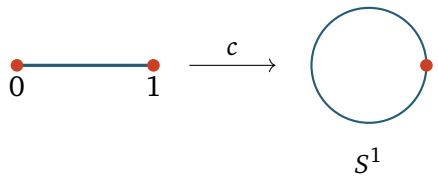


Figure 2.3: The circle map c sends the paired endpoints to the same point. Its induced map from the quotient will be a homeomorphism.

This map is continuous and surjective onto S^1. Its only repeated value on $[0,1]$ is the common endpoint value. The induced map $[0,1]/(0 \sim 1) \to S^1$ is therefore a continuous bijection. Its domain is compact and its target Hausdorff, so it is a homeomorphism by

Corollary 2.4. A *fibre* is the inverse image of one point. This gives a general method: construct a continuous map with exactly the desired classes as its fibres, then apply the criterion.

Recall that a group has an associative product, an identity e and inverses. An *action by homeomorphisms* of a group G on X assigns a homeomorphism $x \mapsto gx$ to each $g \in G$, with $ex = x$ and $(gh)x = g(hx)$. The *orbit* of x is $\{gx : g \in G\}$. The *orbit space* X/G has the quotient topology for these orbits.

One useful case lets us prove Hausdorffness explicitly. An *isometry* preserves a metric. Suppose $a\colon X \to X$ is an isometry with $a^2 = \mathrm{id}$. Its orbit classes are $[x] = \{x, ax\}$, with duplicates omitted. The orbit distance is

$$d_{\mathrm{orb}}([x],[y]) = \min\{d(x,y), d(x,ay)\}.$$

The isometry and $a^2 = \mathrm{id}$ give $d(ax,y) = d(a^2x,ay) = d(x,ay)$. Thus changing either representative interchanges the two distances. The minimum vanishes exactly when the orbits agree. Symmetry follows from the isometry. For the triangle inequality, choose maps $g, h \in \{\mathrm{id}, a\}$ realizing the two relevant minima. Then

$$d(x,ghz) \le d(x,gy) + d(gy,ghz) = d(x,gy) + d(y,hz).$$

This bounds the orbit distance from $[x]$ to $[z]$ by the sum of the other two. Thus the formula defines a metric. Its balls are $q(B_r(x))$, and their inverse images are $B_r(x) \cup B_r(ax)$. Thus orbit-metric balls are quotient-open. Conversely, the preimage of a quotient-open set containing $[x]$ contains some $B_r(x)$. Its image is an orbit-metric ball inside that set, so the set is metric-open. The topologies agree. The orbit space is therefore Hausdorff by Proposition 1.11.

For $n \ge 1$, the *real projective space* $\mathbb{R}\mathrm{P}^n$ is $S^n/(x \sim -x)$. Its points are the lines through the origin in $\mathbb{R}^{n+1}$. The antipodal map is an isometry, so $\mathbb{R}\mathrm{P}^n$ is Hausdorff. It is compact and path-connected as an image of S^n. Writing $D^n = \{v \in \mathbb{R}^n : \|v\| \le 1\}$, the map

$$D^n \longrightarrow \mathbb{R}\mathrm{P}^n, \qquad v \longmapsto [(v, \sqrt{1 - \|v\|^2})]$$

is continuous and surjective. The upper hemisphere consists of unit vectors whose last coordinate is nonnegative. Every line has a representative there. That representative is unique unless its last coordinate vanishes, when there are two antipodal representatives. Thus this realizes $\mathbb{R}\mathrm{P}^n$ as D^n with antipodal points of its boundary sphere identified, again by the compact-to-Hausdorff criterion.

The Hausdorff conclusion fails for general orbit spaces. Let the additive group $\mathbb{Q}$ act on $\mathbb{R}$ by translation. A nonempty open union U of orbits contains an interval. For any $x \in \mathbb{R}$, density of the rationals supplies $r \in \mathbb{Q}$ with $x - r$ in that interval. Invariance under translation by r gives $x \in U$. Hence the only open sets of $\mathbb{R}/\mathbb{Q}$ are the empty set and the whole space. There are distinct classes, for example those of 0 and $\sqrt{2}$, so this quotient is not Hausdorff. The Hausdorff condition will matter for the spaces constructed in Lectures 7 and 8.

2.5 Rotations as a glued ball

The rotation group is the matrix set

$$\mathrm{SO}(3) = \{R \in \mathbb{R}^{3\times 3} : R^T R = \mathbb{1},\ \det R = 1\},$$

with its Euclidean subspace topology. A vector v with $\|v\| \le \pi$ will specify an axis and an angle. The figure shows the identification that occurs at angle π, a half-turn. The vector n marks a unit axis.

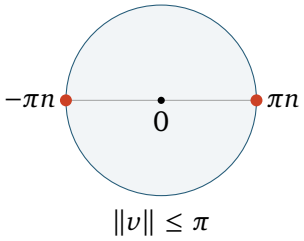


Figure 2.4: A cross-section of the closed rotation ball. Interior vectors are distinct rotations. The matching red boundary points πn and $-\pi n$ represent the same half-turn.

Proposition 2.9 (Axis–angle ball). *The closed ball* $\{v \in \mathbb{R}^3 : \|v\| \le \pi\}$ *with antipodal boundary points identified is homeomorphic to* $\mathrm{SO}(3)$. *Consequently,* $\mathrm{SO}(3) \cong \mathbb{R}\mathrm{P}^3$.

Proof. Define $A_v(w) = v \times w$ for $w \in \mathbb{R}^3$. Put $r = \|v\|$ and define

$$R(v) = \mathbb{1} + \frac{\sin r}{r} A_v + \frac{1 - \cos r}{r^2} A_v^2.$$

At $r = 0$, the coefficients have limiting values 1 and $1/2$. The map is therefore continuous there and $R(0) = \mathbb{1}$. For $v = rn$ with $\|n\| = 1$, the vector identity $n \times (n \times w) = n(n \cdot w) - w$ gives

$$R(v)w = (\cos r)w + (1 - \cos r)n(n \cdot w) + (\sin r)n \times w.$$

This fixes n and rotates its perpendicular plane through angle r. Thus it is orthogonal with determinant 1. At $r = \pi$ it equals $2nn^T - \mathbb{1}$, unchanged by $n \mapsto -n$.

Conversely, every $R \in \mathrm{SO}(3)$ has a unit fixed vector n. Indeed, orthogonality forces all eigenvalues to have modulus 1. Its nonreal eigenvalues occur in conjugate pairs of product 1. Its real eigenvalues are ± 1. After removing the conjugate pairs, an odd number of real eigenvalues remains. Their product is 1, so an even number are -1. At least one is therefore $+1$.

The perpendicular plane is invariant. Since $Rn = n$ and R is orthogonal, for $w \perp n$ we have $Rw \cdot n = Rw \cdot Rn = w \cdot n = 0$. The restriction has determinant 1, so is a plane rotation. Choosing its angle in $[-\pi, \pi]$ represents R as $R(\theta n)$.

There are no further identifications. The trace is $1 + 2\cos r$, so it determines $r \in [0, \pi]$. For $0 < r < \pi$, the difference $R(v) - R(v)^T = 2 \sin r \, A_n$ determines n. At $r = \pi$, the matrix $2nn^T - \mathbb{1}$ determines the line spanned by n, hence just the pair $\{n, -n\}$. At $r = 0$ there is only the vector 0.

We have a continuous bijection from the compact quotient ball to the Hausdorff matrix space $\mathrm{SO}(3)$. It is a homeomorphism. Rescaling the ball to D^3 and using the hemisphere construction above identifies it with $\mathbb{R}\mathrm{P}^3$. ∎

Lecture 4 will use this model to distinguish loops representing full rotations. No property of loops is needed for the identification here.

2.6 Three quotients of a square

We determine compactness, path-connectedness and Hausdorffness for the spaces obtained by the edge pairings in the figure. All start from $Q_0 = [0,1]^2$. Matching arrows indicate which points are identified. Apply also all identifications forced transitively by these pairings.

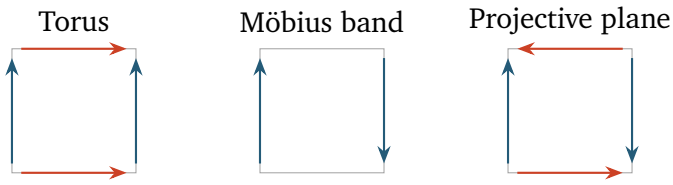


Figure 2.5: The torus pairs opposite edges without reversal. The Möbius band pairs only the vertical edges, with reversal. The projective plane pairs antipodal points of the square boundary.

Each quotient is compact and path-connected, since the square has both properties and the projection is continuous. Hausdorffness needs a separate argument. It need not survive a quotient, as $\mathbb{R}/\mathbb{Q}$ showed.

For the *torus*, the pairings are

$$(0,t) \sim (1,t) \quad \text{and} \quad (s,0) \sim (s,1).$$

Use the circle map c from § 2.4. The product $T^2 = S^1 \times S^1 \subseteq \mathbb{R}^4$ is the torus. The map $Q_0 \to T^2$ given by $(s,t) \mapsto (c(s), c(t))$ is continuous and onto.

Equality in the product is coordinatewise. In each coordinate, c identifies only the two endpoints. Thus the map's fibres are exactly the stated equivalence classes, including the single class of all four corners. The induced continuous bijection is a homeomorphism, because the quotient is compact and T^2 is Hausdorff.

For the *Möbius band*, the pairing is $(0,t) \sim (1, 1-t)$.

Identify S^1 with unit complex numbers. On the cylinder $S^1 \times [-1,1] \subseteq \mathbb{R}^3$, the map $(z,u) \mapsto (-z,-u)$ is an isometry whose square is the identity. Its orbit space is therefore Hausdorff by the orbit metric above. The map

$$Q_0 \longrightarrow (S^1 \times [-1,1])/\{\pm 1\}, \qquad (s,t) \longmapsto [(e^{i\pi s}, 2t-1)]$$

is continuous and onto. Every circle point or its negative lies on the upper semicircle.

For $0 < s < 1$ that representative is unique. At the two ends, changing 1 to -1 reverses the other coordinate, giving precisely the stated edge pairing. The induced bijection is a homeomorphism by the same compact-to-Hausdorff criterion.

For the *projective plane*, the pairings are

$$(0,t) \sim (1, 1-t) \quad \text{and} \quad (s,0) \sim (1-s, 1).$$

Put $w = (2s-1, 2t-1)$. The centred square maps homeomorphically to D^2 by radial rescaling:

$$w \longmapsto \frac{\|w\|_\infty}{\|w\|} w \quad (w \neq 0), \qquad 0 \longmapsto 0,$$

where $\|w\|_\infty$ is the maximum absolute coordinate and $\|w\|$ is Euclidean. The inverse sends nonzero u to $\|u\|\ u/\|u\|_\infty$. Their norms tend to zero at the origin, so both maps are continuous there as well. The square boundary maps to the unit circle and antipodes map to antipodes.

The hemisphere construction therefore identifies this square quotient with $\mathbb{R}P^2$, whose Hausdorffness was proved using its orbit metric.

The results, including the rotation ball, are as follows.

Quotient	Compact and path-connected	Hausdorff witness
Torus	yes	subspace of $\mathbb{R}^4$
Möbius band	yes	isometric orbit space
Projective plane	yes	antipodal orbit metric
Rotation ball	yes	$\mathrm{SO}(3) \subseteq \mathbb{R}^9$

Self-check. Prove that S^1 and $[0,1]$ are not homeomorphic by removing one point. Explain why deleting an endpoint of the interval is not a useful test.

Answer. Delete the midpoint of $[0,1]$. The two resulting half-intervals give a separation. Deleting any point from the circle leaves a space homeomorphic to $\mathbb{R}$. Rotate that point to the north pole, then use Lecture 1's stereographic projection. The line is path-connected by straight segments. A homeomorphism would restrict to a homeomorphism between these two punctured spaces, which is impossible. Deleting an endpoint leaves a path-connected half-open interval, so connectedness would not distinguish them.

Summary. Continuous images preserve compactness and connectedness. A continuous bijection from compact to Hausdorff is a homeomorphism. A quotient topology tests continuity after composition with its projection. Quotients preserve compactness and path-connectedness, but Hausdorffness needs its own proof. The rotation group is the closed three-ball with antipodal boundary points identified, hence $\mathbb{R}P^3$. Lecture 3 begins the study of loops.

Reading. Munkres, *Topology*, second edition, §§22–28; Heine–Borel is Theorem 27.3, metric sequential compactness is Theorem 28.2, and the sine curve is §24, Example 7.

Lecture 3

Homotopy and the fundamental group

Objectives. *We compare paths by continuous deformations that keep their endpoints fixed. A path returning to its starting point is a loop. The classes of loops at one fixed point form the fundamental group. For the circle this group is the integers: a continuously chosen angle records how many turns a loop makes. We prove this by lifting paths and homotopies, then use the integer to prove the fundamental theorem of algebra.*

By the end of this lecture you will be able to:

- Construct the fundamental group, induced homomorphisms and base-point isomorphisms.
- Prove that lifting computes the fundamental group of the circle as $\mathbb{Z}$.
- Compute the group of the punctured plane and prove the disc has no retraction onto its circle.
- Prove that a nonconstant complex polynomial has a root.

3.1 Homotopy and fixed endpoints

Write $I = [0, 1]$. Recall that a path in X is a continuous map $\gamma : I \to X$ (§ 2.3). A *loop* has equal endpoints. Their common value is its *base point*.

Definition 3.1 (Homotopy). A *homotopy* from $f : X \to Y$ to $g : X \to Y$ is a continuous map $h : X \times I \to Y$ with $h(x, 0) = f(x)$ and $h(x, 1) = g(x)$ for every $x \in X$. We write $f \simeq g$. For paths from x_0 to x_1, a homotopy *relative to the endpoints* also satisfies

$$h(0, s) = x_0, \qquad h(1, s) = x_1 \qquad (s \in I).$$

Write $h_s(x) = h(x, s)$ for a fixed-parameter map in the homotopy. Continuity is required jointly in the path parameter t and deformation parameter s. For example, the paths $\gamma(t) = (t, 0)$ and $\delta(t) = (t, t(1 - t))$ in $\mathbb{R}^2$ are homotopic relative to their endpoints by

$$h(t, s) = (t, s\, t(1 - t)).$$

This map is continuous because its coordinates are polynomials. At $t = 0$ it is $(0, 0)$ and at $t = 1$ it is $(1, 0)$, independently of s. The figure shows the intermediate paths.

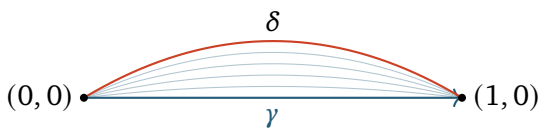


Figure 3.1: The homotopy keeps both endpoints fixed. Increasing s moves the straight path γ to the parabola δ.

We will glue formulas on finitely many closed pieces. If continuous maps on a finite closed cover agree on overlaps, their union is continuous. Indeed, the preimage of a closed set is a finite union of sets closed in those pieces, hence closed in the whole domain. This is the pasting argument of § 2.3, now also applied to squares.

Homotopy relative to endpoints is an equivalence relation. The constant family proves reflexivity, and $h(t, 1-s)$ reverses a homotopy. To join two homotopies, run the first on $0 \le s \le 1/2$ and the second on $1/2 \le s \le 1$, rescaling each parameter interval to I. They agree at the common edge, so the pasting argument proves transitivity.

3.2 The fundamental group

For paths γ from x to y and δ from y to z, define

$$(\gamma \cdot \delta)(t) = \begin{cases} \gamma(2t), & 0 \le t \le \frac{1}{2}, \\ \delta(2t-1), & \frac{1}{2} \le t \le 1. \end{cases}$$

The convention is γ **first, then** δ, as in Hatcher, §1.1. Define the reversed path by $\bar{\gamma}(t) = \gamma(1-t)$, and the constant path by $e_x(t) = x$. All these paths are continuous, by pasting at the matching endpoints. Write $[\gamma]$ for the endpoint-fixed homotopy class of γ.

Theorem 3.2 (Fundamental group). *The endpoint-fixed homotopy classes of loops at $x_0 \in X$ form a group, written $\pi_1(X, x_0)$, with*

$$[\gamma][\delta] = [\gamma \cdot \delta], \qquad 1 = [e_{x_0}], \qquad [\gamma]^{-1} = [\bar{\gamma}].$$

Proof. Products respect classes. Concatenate two endpoint-fixed homotopies in the t direction; their formulas agree along $t = 1/2$. Reversing the t parameter similarly respects classes.

A change of timing does not change a path class. For any continuous $\rho\colon I \to I$ with $\rho(0) = 0$ and $\rho(1) = 1$, the formula $h(t,s) = \gamma((1-s)\rho(t) + st)$ joins $\gamma \circ \rho$ to γ while fixing the endpoints. For associativity, the following timing changes $((\gamma \cdot \delta) \cdot \eta) \circ \rho$ into $\gamma \cdot (\delta \cdot \eta)$:

$$\rho(t) = \begin{cases} t/2, & 0 \le t \le \frac{1}{2}, \\ t - \frac{1}{4}, & \frac{1}{2} \le t \le \frac{3}{4}, \\ 2t - 1, & \frac{3}{4} \le t \le 1. \end{cases}$$

The formulas agree at the break times and fix 0 and 1. On the first interval the parameter in γ becomes $4\rho(t) = 2t$. On the middle interval it becomes $4\rho(t) - 1 = 4t - 2$ in δ. On the last it becomes $2\rho(t) - 1 = 4t - 3$ in η. These are precisely the timings of $\gamma \cdot (\delta \cdot \eta)$. The preceding homotopy therefore proves associativity of classes.

The paths $e_x \cdot \gamma$ and $\gamma \cdot e_y$ are γ with timings $\max(2t-1, 0)$ and $\min(2t, 1)$. Thus their classes equal $[\gamma]$. Finally, the homotopy

$$h(t,s) = \gamma\big((1-s)\min(2t, 2-2t)\big)$$

contracts $\gamma \cdot \bar{\gamma}$ to e_x, fixing both endpoints. Applying it to $\bar{\gamma}$ gives the other inverse identity. These homotopies also apply to joined paths with different endpoints. ∎

A *homomorphism* preserves group products. A bijective homomorphism is an *isomorphism*. A continuous map $f\colon X \to Y$ induces

$$f_*\colon \pi_1(X, x_0) \longrightarrow \pi_1(Y, f(x_0)), \qquad [\gamma] \longmapsto [f \circ \gamma].$$

Proposition 3.3 (Induced homomorphisms). *The induced map f_* is a homomorphism. Induced maps preserve composition and identity maps.*

Proof. Composing a path homotopy with f proves well-definedness. Composition also respects the two path pieces, so f_* preserves products. For continuous $g\colon Y \to Z$, this gives $(g \circ f)_* = g_* \circ f_*$. Identity maps give $(\mathrm{id}_X)_* = \mathrm{id}$. ∎

The base point requires care. For a path a from x_0 to x_1, the base-point change map is

$$c_a\colon \pi_1(X, x_0) \to \pi_1(X, x_1), \qquad c_a([\gamma]) = [\bar{a} \cdot \gamma \cdot a].$$

The figure gives the traversal order.

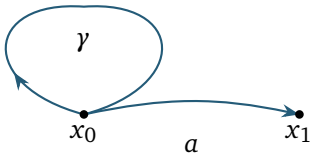


Figure 3.2: A loop at x_1 follows $\bar{a}$ to x_0, traverses γ, then follows a back.

Proposition 3.4 (Change of base point). *For every path a from x_0 to x_1, the map c_a is an isomorphism.*

Proof. Concatenating with fixed paths respects homotopy, so c_a is well-defined. Cancellation of $a \cdot \bar{a}$ proves that c_a preserves products. The inverse is $c_{\bar{a}}$, by the same cancellation. Thus it is an isomorphism. ∎

Elements u and v of a group are *conjugate* if $v = k^{-1}uk$ for some element k of that group. When $x_0 = x_1$, the map c_a is conjugation by $[a]$. If $[a][\gamma] \neq [\gamma][a]$, then $c_a([\gamma]) \neq [\gamma]$, whereas the constant choice of a gives the identity map. Thus the isomorphism can depend on the chosen path.

In a *free homotopy of loops*, each h_s is a loop, but its base point may move. The defining endpoint condition is

$$h(0, s) = h(1, s) \qquad (s \in I).$$

Let $\gamma_0 = h_0$ be based at x_0, and put $\gamma_1 = h_1$. The base point follows the path $a(s) = h(0, s)$. Put $a_s(t) = a(st)$. The loops $a_s \cdot h_s \cdot \bar{a}_s$ keep base point x_0. At $s = 0$, the inserted constant paths do not change the class, by the unit laws. At $s = 1$ this gives $[\gamma_0] = [a \cdot \gamma_1 \cdot \bar{a}]$. If γ_1 is also based at x_0, then a is a loop at x_0. The equality therefore makes $[\gamma_0]$ and $[\gamma_1]$ conjugate.

Conversely, let a be a loop at x_0. The family $\bar{a}_s \cdot \gamma \cdot a_s$ consists of loops with base point $a(s)$. Its end loops represent $[\gamma]$ and $[\bar{a} \cdot \gamma \cdot a]$. Both families are continuous by pasting their three path pieces. Free homotopy is an equivalence relation by the same pasting argument as before. Based homotopies are free homotopies, so conjugate classes are freely homotopic. Thus two loops based at x_0 are freely homotopic exactly when their classes are conjugate. In a path-connected space, move any base point to x_0 by a path. Free homotopy classes then correspond to conjugacy classes in $\pi_1(X, x_0)$, rather than to individual group elements.

3.3 Lifting phases on the circle

Identify S^1 with the unit complex numbers, and set

$$p\colon \mathbb{R} \to S^1, \qquad p(u) = e^{2\pi i u}.$$

We have $p(u) = p(v)$ exactly when $u - v \in \mathbb{Z}$. For a continuous map $f\colon Y \to S^1$, a *lift* through p is a continuous map $\tilde{f}\colon Y \to \mathbb{R}$ with $p \circ \tilde{f} = f$. For this projection, a lift is also called a *phase*.

For $r \in \mathbb{R}$ put

$$J_r = (r - 1/4, r + 1/4), \qquad U_r = \{z \in S^1 : \operatorname{Re}(p(-r)z) > 0\}.$$

These open semicircles cover the circle. The restriction $p\colon J_r \to U_r$ has the continuous inverse branch

$$b_0(z) = \frac{1}{2\pi} \arctan \frac{\operatorname{Im} z}{\operatorname{Re} z}, \qquad b_r(z) = r + b_0(p(-r)z).$$

For $z = e^{i\theta} \in U_0$ with $|\theta| < \pi/2$, the quotient $\operatorname{Im} z / \operatorname{Re} z$ is $\tan\theta$. The principal arctangent returns θ, so $b_0(z) = \theta/(2\pi) \in J_0$ and $p(b_0(z)) = z$. Conversely, for $u \in J_0$ the angle $2\pi u$ is in this range, so $b_0(p(u)) = u$. Rotating gives the inverse on U_r:

$$p(b_r(z)) = p(r)\, p(-r) z = z \qquad (z \in U_r).$$

The figure shows repeated real intervals for one semicircle.

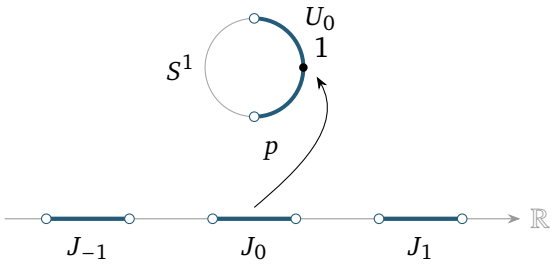


Figure 3.3: The intervals J_{-1}, J_0 and J_1 map to the same open semicircle U_0. Its boundary points are excluded.

We first justify the finite subdivisions used below. The *diameter* of a bounded nonempty set is the supremum of its pairwise distances.

> **Lemma 3.5** (Lebesgue number). *An open cover of a nonempty compact metric space K has a number $\delta > 0$ such that each nonempty subset of diameter less than δ lies in one cover member.*

Proof. For each $x \in K$ choose $\varepsilon_x > 0$ with $B_{2\varepsilon_x}(x)$ inside a cover member. Finitely many balls $B_{\varepsilon_x}(x)$ cover K; let δ be the minimum of their radii. For a set A of diameter less than δ, choose $a \in A$ and one selected ball $B_{\varepsilon_x}(x)$ containing a. For every $z \in A$,

$$d(z, x) \le d(z, a) + d(a, x) < \delta + \varepsilon_x \le 2\varepsilon_x.$$

Thus A lies in the associated cover member. ∎

> **Lemma 3.6** (Path and homotopy lifting for the circle). *A path $\gamma\colon I \to S^1$ has a unique lift after specifying $\tilde{\gamma}(0) \in p^{-1}(\{\gamma(0)\})$. A homotopy $h\colon I^2 \to S^1$ has a unique lift after a lift of its bottom edge $h(t, 0)$ has been specified.*

Proof. The interval and square are compact by Heine–Borel. Apply Lemma 3.5 to the cover of I by $\gamma^{-1}(U_r)$. For its Lebesgue number δ, choose N with $1/N < \delta$ and put $t_i = i/N$ for $i = 0, \dots, N$. Each subinterval maps into some U_r. On $[t_i, t_{i+1}]$, set $\tilde{\gamma} = b_r \circ \gamma + k$, with

$$k = \tilde{\gamma}(t_i) - b_r(\gamma(t_i)) \in \mathbb{Z}.$$

This matches the value already assigned at t_i. The pieces agree at their endpoints and paste continuously.

For a homotopy, use the cover of I^2 by $h^{-1}(U_r)$. Choose a square grid whose cells have diameter less than its Lebesgue number. Extend the given bottom lift across cells, row by row, from left to right. On a cell choose $b_r \circ h + k$ to match the assigned value at its lower-left corner. The only previously assigned points of a new cell are on its bottom and left edges. Along each already assigned bottom or left edge, the two lifts differ by a continuous integer-valued function. That function is constant by connectedness, and zero at the corner. Thus the new piece agrees on shared edges and pastes continuously.

Finally, two lifts on I or I^2 differ by a continuous integer-valued function. Such a function on a connected set is constant: a noninteger threshold would separate two different image values. Agreement at one point therefore proves uniqueness. ∎

The figure shows one extension step of the grid construction.

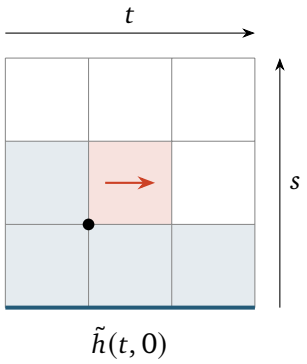


Figure 3.4: Previously assigned cells are shaded. A branch on the next cell is shifted by an integer to agree at the marked corner and hence on the shared edges.

For any loop γ in the circle, define its *winding number* by

$$\operatorname{wind} \gamma = \tilde{\gamma}(1) - \tilde{\gamma}(0) \in \mathbb{Z}.$$

Different initial lifts differ by an integer, so this number is independent of that choice. In a free homotopy of loops, lift the whole square. The endpoint difference is a continuous integer-valued function of s, so it is constant. Thus winding is invariant even when the base point moves.

Theorem 3.7 (Fundamental group of the circle). *Winding gives an isomorphism* $\operatorname{wind}\colon \pi_1(S^1, 1) \to \mathbb{Z}$. *The loop* $t \mapsto p(mt)$ *has winding* m *for every* $m \in \mathbb{Z}$.

Proof. Winding is invariant under free homotopy, hence under endpoint-fixed homotopy. It therefore defines a map on classes. Choose a lift starting at 0. If it ends at m, the homotopy

$$\tilde{h}(t, s) = (1 - s)\tilde{\gamma}(t) + smt$$

fixes the endpoints 0 and m. Its projection gives an endpoint-fixed homotopy from γ to $p(mt)$. Thus winding determines the class, and every integer occurs. For a product of loops with windings m and n, join the first lift to the second lift shifted by m. The joined lift ends at $m+n$, so winding preserves multiplication as addition. ∎

Counter-clockwise travel $p(t)$ has winding +1, fixing the sign convention.

3.4 Retractions and homotopy invariance

Definition 3.8 (Retraction and deformation retraction)**.** A *retraction* onto $A \subseteq X$ is a continuous map $r\colon X \to A$ with $r|_A = \mathrm{id}_A$. A *deformation retraction* also has a homotopy from id_X to $i \circ r$, where $i\colon A \to X$ is inclusion, that keeps every point of A fixed throughout.

Proposition 3.9 (Deformation retractions preserve the group)**.** *For a deformation retraction and $a_0 \in A$, the induced maps r_* and i_* are inverse isomorphisms of fundamental groups.*

Proof. We have $r \circ i = \mathrm{id}_A$. The other composite is homotopic to id_X while fixing a_0. Composing this homotopy with any loop at a_0 is an endpoint-fixed path homotopy. Hence $i_* r_* = \mathrm{id}$ as well as $r_* i_* = \mathrm{id}$. ∎

Identify $\mathbb{C}$ with $\mathbb{R}^2$. For example, on $\mathbb{C} \setminus \{0\}$ the formula

$$h(z,s) = \left(1 - s + \frac{s}{|z|}\right) z$$

deforms the identity to $r(z) = z/|z|$. Its coefficient is positive, so the image never reaches 0, and it fixes every point of S^1. Consequently $\pi_1(\mathbb{R}^2 \setminus \{0\}, 1) \cong \mathbb{Z}$. For a loop in the punctured plane, its winding means the winding of its normalization $\gamma/|\gamma|$.

A nonempty space is *simply connected* if it is path-connected and all based loops are endpoint-fixed homotopic to the constant loop.

A space is *contractible* if its identity is homotopic to a constant map with value x_* in the space.

Proposition 3.10 (Contractible spaces)**.** *Every contractible space is simply connected.*

Proof. The contraction gives paths from all points to x_*, so the space is path-connected. For a loop γ at x_0, it also gives a free homotopy to e_{x_*}. The moving-basepoint construction above converts this to a based homotopy from γ to $a \cdot e_{x_*} \cdot \bar{a}$, which cancels to e_{x_0}. ∎

Maps $f\colon X \to Y$ and $g\colon Y \to X$ give a *homotopy equivalence* if $g \circ f \simeq \mathrm{id}_X$ and $f \circ g \simeq \mathrm{id}_Y$. The general theorem that such f induces an isomorphism on fundamental groups at every base point is cited from Hatcher, Proposition 1.18. The deformation-retraction case used here was proved directly above.

3.5 No retraction of the disc onto its circle

Let $D^2 = \{z \in \mathbb{C} : |z| \le 1\}$ be the closed disc.

Corollary 3.11. *There is no retraction $r\colon D^2 \to S^1$.*

Proof. The homotopy $(z, s) \mapsto (1 - s)z$ contracts the disc. Thus $\pi_1(D^2, 1)$ is the trivial group, consisting only of its identity. If r were a retraction and $i\colon S^1 \to D^2$ the inclusion, then $r \circ i = \mathrm{id}_{S^1}$. Therefore $r_* i_*$ is the identity of $\pi_1(S^1, 1) \cong \mathbb{Z}$. But i_* takes values in the trivial group, so the composite sends every class to the identity element. It cannot be the identity map of this nontrivial group. ∎

The base point here is 1 on the circle. Contractibility gives the trivial group there even though the displayed contraction ends at the centre.

3.6 The fundamental theorem of algebra

Corollary 3.12 (Fundamental theorem of algebra)**.** *Every complex polynomial of degree at least one has a complex root.*

Proof. Write $f(z) = a_n z^n + \sum_{j=0}^{n-1} a_j z^j$ with $n \geq 1$ and $a_n \neq 0$. Choose

$$R > \max\left\{1, \frac{\sum_{j=0}^{n-1} |a_j|}{|a_n|}\right\}.$$

For $|z| = R$ the lower terms satisfy

$$\sum_{j=0}^{n-1} |a_j| R^j \leq R^{n-1} \sum_{j=0}^{n-1} |a_j| < |a_n| R^n.$$

Thus the loops

$$a_n R^n p(nt) + s \sum_{j=0}^{n-1} a_j R^j p(jt)$$

avoid zero for every $s \in I$. Normalizing gives a free homotopy of circle loops from the leading term to $f(Rp(t))$. Choose $b \in \mathbb{R}$ with $p(b) = a_n/|a_n|$. The leading loop has phase $b + nt$, so its winding is n.

The figure illustrates this comparison for a quadratic.

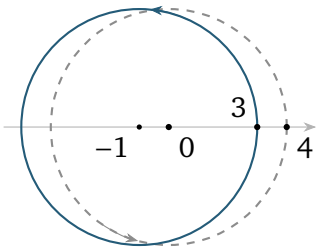


Figure 3.5: For $f(z) = z^2 - 1$ and $R = 2$, the comparison loops are $4p(2t) - s$, each traversed twice. Dashed: $s = 0$, with starting point 4; solid: $s = 1$, centred at -1, with starting point 3.

Now suppose f had no zero anywhere in $\mathbb{C}$. The map $h_f\colon I^2 \to \mathbb{C} \setminus \{0\}$ given by

$$h_f(t, s) = f((1 - s)Rp(t))$$

would be a homotopy of nonzero loops from $f(Rp(t))$ to the constant $f(0)$. These families may move the base point. After normalization, free-homotopy invariance would preserve winding and give its final value zero. This contradicts its value $n \geq 1$. Hence f has a root. ∎

Self-check. Compute $\pi_1(S^1 \times S^1, (1,1))$ using the coordinate projections. Prove the group identification, not just its surjectivity.

Answer. Send $[(\gamma, \delta)]$ to $(\operatorname{wind}\gamma, \operatorname{wind}\delta) \in \mathbb{Z}^2$. Projecting a product homotopy gives endpoint-fixed homotopies of the coordinates. Winding is invariant under these, so the map is well-defined. Projection respects concatenation, so the map preserves multiplication coordinatewise. Conversely, the component test of Lecture 1 makes a pair of loops a loop in the product, and a pair of endpoint-fixed homotopies a homotopy in the product. Every $(m,n) \in \mathbb{Z}^2$ is represented by $t \mapsto (p(mt), p(nt))$. If both windings vanish, each coordinate loop has an endpoint-fixed contraction; their pair contracts the product loop. If two classes have the same image, their product with the inverse of the second has both windings zero, so is trivial. The two classes therefore agree. The group is $\mathbb{Z}^2$.

Summary. Classes of loops at one fixed base point form the fundamental group. Changing the base point along a path gives an isomorphism that can depend on that path. In a path-connected space, free homotopy keeps only conjugacy classes. Lifting computes $\pi_1(S^1, 1) \cong \mathbb{Z}$; deformation retraction gives the same group for the punctured plane. This integer rules out a disc-to-circle retraction and proves the fundamental theorem of algebra. Lecture 4 extends the lifting construction to other maps.

Reading. Hatcher, *Algebraic Topology*, §1.1, pp. 25–38 (Theorem 1.8 and Proposition 1.18); Munkres, *Topology*, §§51–56, §58 and Lemma 27.5.

Lecture 4

Covering spaces and rotations

Objectives. *The circle projection of Lecture 3 has local inverse branches. A covering map has the same property over every sufficiently small open set. We lift paths using these branches and recover loop classes from their endpoints above a fixed point. The antipodal covering then computes the fundamental group of the rotation space: a full turn represents the nontrivial class, and two full turns represent the identity.*

By the end of this lecture you will be able to:

- Prove path and path-homotopy lifting for a covering map.
- Recover loop classes from a simply connected covering space.
- Compute $\pi_1(\mathbb{R}\mathrm{P}^n)$ for $n \geq 2$ and $\pi_1(\mathrm{SO}(3))$.
- Distinguish the lifted paths of a 2π and a 4π rotation.

4.1 Coverings and local inverse branches

Definition 4.1 (Covering map). A *covering map* is a continuous surjection $p\colon \widetilde{X} \to X$ such that each $x \in X$ has an open neighbourhood U with

$$p^{-1}(U) = \coprod_{\alpha} V_\alpha, \qquad p|_{V_\alpha} : V_\alpha \longrightarrow U \text{ a homeomorphism.}$$

The symbol $\coprod$ denotes a disjoint union. The sets V_α are open in $\widetilde{X}$. We call U *evenly covered* and each V_α a *sheet over* U.

The fibre $p^{-1}(\{x\})$ is the inverse image of the point, as in Lecture 2. It contains one point in each sheet over U. In particular, the fibre has the discrete subspace topology: every one of its points is open in the fibre. Its cardinality is constant as x varies in an evenly covered set. The figure shows the local structure.

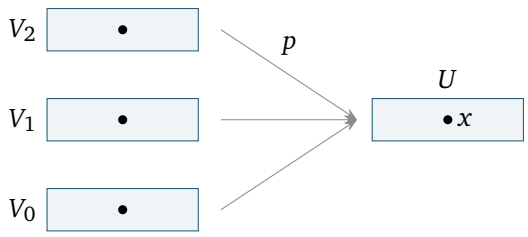


Figure 4.1: Each sheet maps homeomorphically onto the same open set U. The marked points belong to one fibre.

For example, the projection $u \mapsto e^{2\pi \mathrm{i} u}$ from $\mathbb{R}$ to S^1 is a covering. The semicircles U_r

of § 3.3 have inverse images $\coprod_{k\in\mathbb{Z}}(J_r + k)$. Each restriction has inverse $b_r + k$. The map $z \mapsto z^m$ on S^1, for an integer $m \geq 1$, is an m-sheeted covering. An arc of angle less than 2π has m disjoint inverse arcs, obtained by dividing its angles by m and adding $2\pi k/m$ for $k = 0, \ldots, m-1$.

A *local homeomorphism* restricts near each domain point to a homeomorphism onto an open subset of the target. Every covering has this property. The converse fails, even for a surjection. Consider

$$q\colon (0, 3\pi) \longrightarrow S^1, \qquad q(\theta) = e^{\mathrm{i}\theta}.$$

Small intervals give local inverses. The fibre over 1 is $\{2\pi\}$, but the fibre over $e^{\mathrm{i}\varepsilon}$ has two points whenever $0 < \varepsilon < \pi$. Every neighbourhood of 1 contains such points. Thus 1 has no evenly covered neighbourhood.



Figure 4.2: The truncated angle interval contains one preimage of 1, but two of $e^{\mathrm{i}\varepsilon}$. Open endpoints prevent a second branch at 1.

4.2 Lifting paths and their homotopies

Recall that a *lift* of $f\colon Y \to X$ is a continuous map $\tilde{f}\colon Y \to \widetilde{X}$ with $p \circ \tilde{f} = f$. Write $I = [0, 1]$.

> **Theorem 4.2** (Path and path-homotopy lifting). *For a covering p, a path $\gamma\colon I \to X$ has a unique lift after specifying $\tilde{\gamma}(0)$ above $\gamma(0)$. A homotopy $h\colon I^2 \to X$ has a unique lift after a lift of its bottom edge has been specified. If h fixes its path endpoints, so does its lift.*

Proof. First, uniqueness of a path lift follows locally. On a subinterval mapping into an evenly covered U, any lift stays in one sheet: the interval is connected and the sheets are disjoint and open. The sheet is determined by one lifted value. Its inverse branch determines the whole subinterval.

For existence, pull back evenly covered sets to an open cover of I. The Lebesgue-number lemma of Lemma 3.5 gives a finite subdivision such that each closed subinterval maps into one such set. On the first subinterval use the sheet containing the prescribed initial point. On each successive subinterval use the sheet containing the preceding endpoint. The formulas agree at the joins and paste continuously. Local uniqueness on successive subintervals gives uniqueness on I.

For a homotopy, pull the same open cover back to the compact square. Choose a grid with each cell mapping into an evenly covered set. Extend the bottom lift row by row, from left to right. Choose on each cell the inverse branch whose sheet contains the already assigned lower-left value. On its bottom and left edges, path-lift uniqueness forces agreement with every previously assigned value. There are no other previously assigned points on the cell. Finite pasting gives a continuous lift on the square. The same cell order proves uniqueness. This is the grid construction of Lemma 3.6, with sheets in place of integer translates.

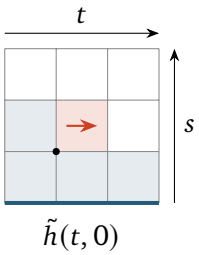


Figure 4.3: The new inverse branch is chosen at the marked corner. Uniqueness on shared edges makes it agree with the shaded region.

Finally, if an edge of h is constant, its lift is a path in a discrete fibre. A continuous image of the connected interval is connected, so this path is constant. Apply this to the two endpoint edges. ∎

In particular, endpoint-fixed homotopic paths have lifts with the same endpoint when their initial lifts agree. The induced homomorphism p_* is injective: lift a based contraction of a projected loop. The lift is a based contraction of the original loop. Also,

$$[\gamma] \in p_*\pi_1(\widetilde{X}, \tilde{x}_0) \quad \Longleftrightarrow \quad \text{the lift of } \gamma \text{ from } \tilde{x}_0 \text{ closes.}$$

For the reverse implication project the closed lift. For the forward implication compare γ with the projection of a representative loop upstairs and lift their based homotopy.

A space is *locally path-connected* if every neighbourhood of each point contains a path-connected open neighbourhood of that point. The following *lifting criterion* is stated with its proof sketched. For path-connected, locally path-connected Y, a continuous map $f\colon (Y, y_0) \to (X, x_0)$ lifts with value $\tilde{x}_0$ at y_0 exactly when

$$f_*\pi_1(Y, y_0) \subseteq p_*\pi_1(\widetilde{X}, \tilde{x}_0).$$

Necessity follows from $f_* = p_*\tilde{f}_*$. For sufficiency, choose a path from y_0 to y and lift its image under f. Define $\tilde{f}(y)$ as the lifted endpoint. The inclusion and the closed-lift test make this independent of the chosen path. On a small path-connected open set mapping into an evenly covered set, this rule is one inverse branch composed with f, so is continuous. This is Hatcher, Proposition 1.33.

For example, let $p(z) = z^3$ and $f(z) = z^6$ on S^1, with base point 1. The induced maps multiply winding by 3 and 6 respectively. The inclusion $6\mathbb{Z} \subseteq 3\mathbb{Z}$ permits a lift. It is $\tilde{f}(z) = z^2$, since $(z^2)^3 = z^6$ and $\tilde{f}(1) = 1$. In contrast, $z \mapsto z^5$ has no such lift because $5\mathbb{Z}$ is not contained in $3\mathbb{Z}$. Dividing the angle by three along the generating loop ends at $e^{2\pi i 5/3} \neq 1$ and therefore does not define a loop upstairs. The obstruction concerns a global lift, although every small arc has inverse branches.

4.3 Fibres, loop classes and deck transformations

Definition 4.3 (Deck transformation). A *deck transformation* of p is a homeomorphism $D\colon \widetilde{X} \to \widetilde{X}$ with $p \circ D = p$. Their group under composition is written $\mathrm{Deck}(p)$.

On a path-connected covering space, a deck transformation is determined by its value at one point. Indeed, for a path a from that point to any other point, $D \circ a$ is the unique lift of $p \circ a$ from the specified image point. In particular, a deck transformation fixing one point is

the identity. The deck group acts *transitively* on a fibre if for any two points of that fibre some deck transformation sends the first to the second.

Corollary 4.4 (Counting a simply connected covering). *Let $p\colon \widetilde{X} \to X$ be a covering with $\widetilde{X}$ simply connected. Fix $\tilde{x}_0$ above x_0. Sending a loop class to its lifted endpoint gives a bijection*

$$\pi_1(X, x_0) \longrightarrow p^{-1}(\{x_0\}).$$

If the deck group acts transitively on this fibre, it is isomorphic to $\pi_1(X, x_0)$.

Proof. The endpoint depends only on the based class by homotopy lifting. Every fibre point can be joined to $\tilde{x}_0$, since simple connectedness includes path-connectedness. Projecting this path proves surjectivity. If two lifted paths have the same endpoint, join the first to the reverse of the second. This loop contracts in $\widetilde{X}$. Its projection shows that the two original loop classes agree. This proves injectivity.

Under transitivity, let D_γ be the unique deck transformation taking $\tilde{x}_0$ to the endpoint of the lift $\tilde{\gamma}$. The preceding bijection and deck uniqueness give a bijection of groups. The lift of $\gamma \cdot \delta$ first follows $\tilde{\gamma}$, then $D_\gamma \circ \tilde{\delta}$. It therefore ends at

$$D_\gamma(D_\delta(\tilde{x}_0)), \qquad D_{\gamma\cdot\delta} = D_\gamma \circ D_\delta.$$

Thus the bijection preserves multiplication with the course's first-path-first convention. The isomorphism uses the chosen lifted base point. ∎

For example, the deck transformations of $\mathbb{R} \to S^1$ are exactly $u \mapsto u + k$ for $k \in \mathbb{Z}$. These are deck transformations, act transitively on the fibre, and exhaust the possibilities by uniqueness from the image of 0. They recover $\pi_1(S^1, 1) \cong \mathbb{Z}$.

4.4 The antipodal cover and rotations

Proposition 4.5 (Spheres of dimension at least two · *sketched*). *The sphere S^n is simply connected for $n \geq 2$.*

Proof sketch. Short spherical arcs join nonantipodal points. A path via a third point joins antipodes, so the sphere is path-connected.

First, cover the loop by open hemispheres. Compactness and Lemma 3.5 give a finite subdivision whose pieces each lie in one hemisphere. Replace each piece by the shortest spherical arc joining its endpoints in that hemisphere. Normalizing their straight interpolation gives the homotopy: its scalar product with the hemisphere centre stays positive.

Next, the resulting polygon lies in finitely many two-dimensional linear subspaces. For $n \geq 2$ these are proper subspaces of $\mathbb{R}^{n+1}$ and their union misses a point of the sphere. Finally, stereographic projection from that point identifies its complement with $\mathbb{R}^n$, where the polygon contracts with its base point fixed. The formulas of Lecture 1 apply in n coordinates. This sketches the sphere result of Hatcher, Proposition 1.14, using polygonal approximation. The original continuous loop need not miss a point.

Recall from Lecture 2 that $\mathbb{R}\mathrm{P}^n$ identifies antipodal points of S^n. The quotient map is a covering. To see this, for $x \in S^n$ take the cap $V = \{u \in S^n : u \cdot x > 1/2\}$. The inverse image of its quotient image is $V \coprod (-V)$. The two caps are disjoint. The quotient map is open, because the saturation of any open set W is the open set $W \cup (-W)$. Each cap therefore maps homeomorphically onto the same open quotient set. These sets cover $\mathbb{R}\mathrm{P}^n$.

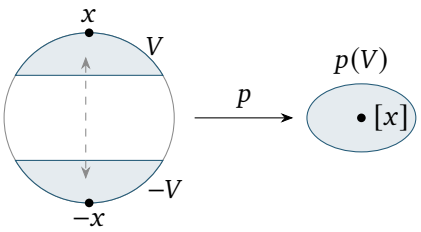


Figure 4.4: The two disjoint caps V and $-V$ lie above the same quotient neighbourhood. The antipodal map exchanges the two sheets.

Write $\mathbb{Z}_2$ for the integers modulo two.

Theorem 4.6 (Projective space and rotations · *proved given S^n simply connected*)**.** *For $n \geq 2$, the antipodal cover gives $\pi_1(\mathbb{R}\mathrm{P}^n) \cong \mathbb{Z}_2$. Consequently $\pi_1(\mathrm{SO}(3)) \cong \mathbb{Z}_2$.*

Proof. The sphere is simply connected by Proposition 4.5, at the sketched level just given. The identity and antipodal map act transitively on the two-point fibre. Deck uniqueness excludes any other transformation. The antipodal map has square the identity, so Corollary 4.4 gives $\mathbb{Z}_2$. The homeomorphism $\mathbb{R}\mathrm{P}^3 \cong \mathrm{SO}(3)$ was proved in Proposition 2.9 by the axis–angle construction. ∎

The circle case is different: $\mathrm{SO}(2) \cong S^1$ has fundamental group $\mathbb{Z}$. The general result $\pi_1(\mathrm{SO}(m)) \cong \mathbb{Z}_2$ for $m \geq 3$ is cited from Nakahara, Table 4.1; only $m = 3$ is proved in this lecture.

4.5 Universal covers and their classification

A simply connected covering space is called a *universal cover*. A space is *semilocally simply connected* if each $x \in X$ has an open neighbourhood U such that every loop in U based at x contracts in X. The neighbourhood itself need not be simply connected. A *covering isomorphism* is a homeomorphism between the total spaces commuting with their projections to X. In the based version it also preserves the chosen lifted point. Two subgroups are *conjugate* if they have the form H and $k^{-1}Hk$ for some group element k.

Theorem 4.7 (Existence and classification · *cited*)**.** *Let X be nonempty, connected, locally path-connected and semilocally simply connected. It has a universal cover. Isomorphism classes of based connected coverings correspond to subgroups of $\pi_1(X, x_0)$ by $p_*\pi_1(\widetilde{X}, \tilde{x}_0)$. Without chosen base points, they correspond to conjugacy classes of subgroups.*

This is Hatcher, Theorem 1.38. Local path-connectedness makes path components open, so connectedness implies path-connectedness here, both downstairs and in a connected cover.

4.6 A full turn and a double turn

We compute the class of rotation through angle $2\pi t$ about the third axis, for $t \in I$. Write a point of S^3 as (a, v), where $a \in \mathbb{R}$, $v \in \mathbb{R}^3$ and $a^2 + |v|^2 = 1$. The corresponding rotation is

$$w \longmapsto (a^2 - |v|^2)w + 2(v \cdot w)v + 2a(v \times w).$$

When $v = 0$, this is the identity. Otherwise, substituting $a = \cos\theta$ and $v = \sin\theta\, n$ gives Lecture 2's rotation $R(2\theta n)$ about the unit axis n. The formula is unchanged by $(a, v) \mapsto (-a, -v)$, so it induces a continuous map from $\mathbb{R}\mathrm{P}^3$ to $\mathrm{SO}(3)$.

To identify that map, choose representatives with $a \geq 0$ and $0 \leq \theta \leq \pi/2$. The corresponding axis–angle vector is

$$v \longmapsto \frac{2 \arcsin |v|}{|v|} v \quad (v \neq 0), \qquad 0 \longmapsto 0.$$

Its radius increases strictly and continuously from 0 to π, so it is a bijection from the hemisphere to the closed angle ball. Continuity at $v = 0$ follows because the image norm is $2 \arcsin |v| \to 0$. By Proposition 2.9, composing it with the angle-ball rotation identifies only antipodal boundary points. Thus the induced map $\mathbb{R}\mathrm{P}^3 \to \mathrm{SO}(3)$ is a continuous bijection and is a homeomorphism by Corollary 2.4. The displayed rotation formula is therefore the antipodal covering followed by this homeomorphism, and hence is itself a covering $S^3 \to \mathrm{SO}(3)$. Its radial coordinates differ from Lecture 2's linear hemisphere rescaling.

The *special unitary group* $\mathrm{SU}(2)$ consists of complex unitary 2×2 matrices of determinant one. Its matrices have the form below, identifying it with S^3:

$$U(a, v) = \begin{pmatrix} a - \mathrm{i}v_3 & -v_2 - \mathrm{i}v_1 \\ v_2 - \mathrm{i}v_1 & a + \mathrm{i}v_3 \end{pmatrix}, \qquad a^2 + |v|^2 = 1.$$

Indeed, orthonormal columns and determinant one give the form $\left(\begin{smallmatrix} z & w \\ -\bar{w} & \bar{z} \end{smallmatrix}\right)$ with $|z|^2 + |w|^2 = 1$. Taking real and imaginary parts gives the displayed coordinates and a continuous inverse. This recalls the spinor cover of *Symmetries in Physics*, Lecture 14, in explicit coordinates.

The lifted path from the identity for the chosen rotation is

$$(a(t), v(t)) = (\cos \pi t,\ \sin \pi t\, e_3), \qquad U(t) = \begin{pmatrix} e^{-\pi \mathrm{i} t} & 0 \\ 0 & e^{\pi \mathrm{i} t} \end{pmatrix}.$$

At $t = 1$ it ends at $(-1, 0)$, or $-\mathbb{1}$ in matrix form. Its projection is closed, but its lift is not. By Corollary 4.4, this is the nontrivial loop class in $\mathrm{SO}(3)$.

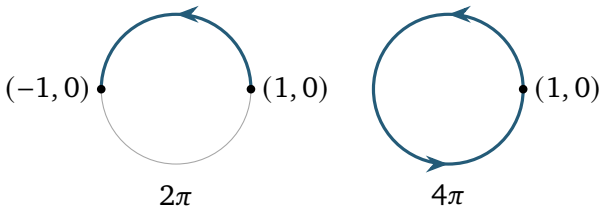


Figure 4.5: A full turn lifts from $(1, 0)$ to $(-1, 0)$ along a semicircle in S^3. A double turn completes the great circle and closes upstairs.

For a 4π rotation, replace t by $2t$ in the lifted formula. The lift ends at $(1, 0)$ and is a loop in the simply connected sphere. It contracts there, so its projection contracts in $\mathrm{SO}(3)$. In Dirac's belt experiment one end of a belt is fixed and the other is turned; a double full turn can be removed by moving the belt with its ends fixed. The covering calculation gives the corresponding loop-class statement for rotations. A spin-1/2 state in $\mathbb{C}^2$ transforms by $U(t)$: one full turn multiplies the state by -1, and two full turns multiply it by $+1$.

Self-check. Classify the connected coverings of the circle up to covering isomorphism.

Answer. The circle satisfies the classification theorem's hypotheses: small open arcs are intervals. Every subgroup of $\mathbb{Z}$ is $m\mathbb{Z}$ for exactly one integer $m \geq 0$. For a nonzero subgroup, its least positive element m and integer division prove this; the zero subgroup is $0\mathbb{Z}$. For $m \geq 1$, the map $z \mapsto z^m$ on S^1 induces multiplication by m on winding, so its image subgroup is $m\mathbb{Z}$. The cover $\mathbb{R} \to S^1$ has subgroup zero because $\mathbb{R}$ is contractible. The cited classification proves these exhaust the connected coverings. All subgroups are their own conjugacy classes, since $\mathbb{Z}$ is abelian.

Summary. Local inverse sheets lift paths and their homotopies uniquely. For a simply connected covering space, based loop classes correspond to fibre points. A transitive deck group gives the same group structure after choosing a lifted base point. The antipodal cover computes $\pi_1(\mathbb{R}\mathrm{P}^n) \cong \mathbb{Z}_2$ for $n \geq 2$, using the sketched sphere result. In $\mathrm{SO}(3)$, a full turn is nontrivial and a double turn is trivial. Lecture 5 computes fundamental groups by joining open pieces.

Reading. Hatcher, *Algebraic Topology*, Proposition 1.14, §1.3, especially Propositions 1.30–1.34 and Theorem 1.38; Munkres, *Topology*, §§53–54. Nakahara, *Geometry, Topology and Physics*, §4.6.5 and Table 4.1.

Lecture 5

Van Kampen, surfaces and braids

Objectives. *We compute loop classes in a space assembled from open pieces. Van Kampen's theorem records how loops in the overlap become relations. For a surface presented by a polygon, its boundary gives the relation. For identical particles, the space consists of unordered configurations of distinct points. Loops in this space describe exchanges. Planar exchanges retain more information than their endpoint permutations, allowing arbitrary scalar phases.*

By the end of this lecture you will be able to:

- Apply van Kampen with path-connected open sets and intersection.
- Read the fundamental-group presentation of a polygonal surface.
- Relate planar exchanges to braids and three-dimensional exchanges to permutations.
- Derive the allowed one-dimensional exchange phases in each case.

5.1 Free groups and relations

A *word* in letters $a_1, \dots, a_k$ and their formal inverses is a finite string of these symbols. Delete adjacent inverse pairs until none remain. The resulting *reduced word* is unique. One algorithm reads from left to right, keeping a stack: cancel the last stored letter when the incoming letter is its inverse, and append it otherwise. Inserting an adjacent inverse pair leaves the final stack unchanged, even when the first inserted letter cancels a stored letter. The second then restores it. Thus reduction is independent of the order of deletions. A reduced word passes through the stack unchanged. Hence every complete sequence of deletions ends at the same stack output.

The *free group* F_k consists of reduced words. Multiply by concatenating and reducing. The empty word is the identity. Reverse a word and invert each letter to get its inverse. Reduction is unchanged by inserting or deleting inverse pairs in any context, so reducing after either stage of a triple concatenation gives the same result. This proves associativity. For example,

$$ab^{-1}ba^{-1}b = b, \qquad aba^{-1}b^{-1} \neq 1 \quad \text{in } F_2.$$

The second word is already reduced. A choice of images for the generators in any group extends uniquely to a homomorphism, by substituting in words. Inverse cancellations do not change the substituted group element.

A subgroup N of G is *normal* if $gNg^{-1} = N$ for every $g \in G$. The quotient G/N identifies elements differing by an element of N. Its multiplication is inherited from G; normality makes this independent of representatives. The *normal closure* of specified words is the smallest normal subgroup containing them, equivalently the subgroup generated by all their conjugates and inverses. A *presentation* is written

$$\langle a_1, \dots, a_k \mid r_1, \dots, r_s \rangle = F_k / \langle\langle r_1, \dots, r_s \rangle\rangle.$$

It imposes $r_j = 1$ in every word context, not merely deletion of one printed copy of a relator. For example, $\langle a \mid a^m \rangle \cong \mathbb{Z}_m$ for $m \geq 1$, where $\mathbb{Z}_m$ is the group of integers modulo m. The *free product* $G * H$ joins two groups without additional relations: words alternate nonidentity elements of the two groups, combining adjacent elements from the same group. In particular, $\mathbb{Z} * \mathbb{Z} = F_2$.

5.2 Van Kampen's theorem

Let $X = U \cup V$, with $x_0 \in U \cap V$. The inclusions induce maps from the fundamental groups of these sets. Write i_{U*} and i_{V*} for the maps from the intersection to the two pieces.

Theorem 5.1 (Van Kampen · *cited; surjectivity sketched*)**.** *Suppose U and V are open and path-connected, and $U \cap V$ is path-connected. The natural homomorphism*

$$\pi_1(U, x_0) * \pi_1(V, x_0) \longrightarrow \pi_1(X, x_0)$$

is surjective. Its kernel is the normal closure of $i_{U*}(w) i_{V*}(w)^{-1}$ for $w \in \pi_1(U \cap V, x_0)$.

This is Hatcher, Theorem 1.20. It identifies the two images of each overlap loop. We sketch the surjectivity construction. Subdivide a loop into pieces γ_j in U or V, using the Lebesgue-number lemma. Merge successive pieces in the same set. At a change of set the joining point is in $U \cap V$. Choose a path a_j in the intersection from x_0 to each joining point, with constant paths at the start and end. The based loops $a_{j-1} \cdot \gamma_j \cdot \bar{a}_j$ each lie in one piece. Their product represents the original loop, since the inserted reverse paths cancel. The kernel statement is cited, not proved by this construction.

The connected-intersection hypothesis matters. On S^1, take the north point $N = (0, 1)$ and south point $S = (0, -1)$. Cover the circle by their complements. Both sets are open intervals and have trivial fundamental group. Their intersection has two components. The circle group is $\mathbb{Z}$, so the conclusion of the theorem cannot hold without the hypothesis.

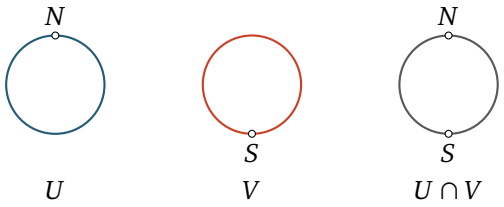


Figure 5.1: Removing N or S gives an open arc. Removing both leaves two components, so this circle cover does not satisfy van Kampen's hypotheses.

The *wedge* $S^1 \vee S^1$ identifies one point of each of two circles to a common point x_0. Remove a non-base point y_b of the second circle to get U, and a non-base point y_a of the first to get V. These are open in the wedge. The first retracts onto the first circle, the second onto the second circle. Collapse the remaining open arcs linearly to x_0; these homotopies fix their retained circles. The formulas agree at x_0, so finite closed pasting proves continuity. The intersection is four open arms joined at x_0 and contracts there. Hence the cited theorem gives

$$\pi_1(S^1 \vee S^1, x_0) \cong \mathbb{Z} * \mathbb{Z} = F_2.$$

The loops a and b run round the two circles. Inductively the same open-neighbourhood construction gives F_r for a finite wedge of r circles.

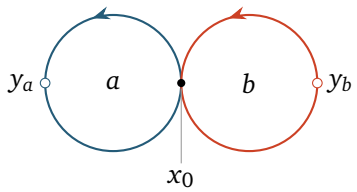


Figure 5.2: Removing the marked point from the opposite circle leaves an open neighbourhood that retracts onto the chosen circle. Both circles share x_0.

5.3 Polygonal surfaces

A *surface* is a Hausdorff space with a countable basis in which every point has a neighbourhood homeomorphic to an open disc in $\mathbb{R}^2$. A compact surface with these local models is called *closed*. For $g \geq 1$, let Σ_g be the standard closed surface obtained from a $4g$-gon by pairing equally labelled edges in the boundary word

$$a_1 b_1 a_1^{-1} b_1^{-1} \cdots a_g b_g a_g^{-1} b_g^{-1}.$$

The integer g is its *genus*. The paired arrows specify the same direction along each identified edge. Edge interiors join two half discs, and the cyclic corner identifications join to a disc. All vertices become one vertex, and the edges become a wedge of $2g$ circles. This is the polygonal model used in Hatcher, §1.2.

Proposition 5.2 (Surface groups · *proved given van Kampen*). *For $g \geq 1$, the fundamental group is*

$$\pi_1(\Sigma_g) \cong \left\langle a_1, b_1, \ldots, a_g, b_g \;\middle|\; \prod_{j=1}^{g} [a_j, b_j] \right\rangle, \qquad [a, b] = aba^{-1}b^{-1}.$$

Proof. Choose a point o in the face interior. Let $U = \Sigma_g \setminus \{o\}$ and let V be a small open disc about o inside that face. The punctured polygon retracts radially to its boundary, fixing it. This deformation respects the edge identifications and retracts U onto the wedge of $2g$ circles. The set V is contractible. The overlap is a punctured disc, with fundamental group $\mathbb{Z}$. Its positive generator follows the face boundary once after radial retraction. Choose a path from its base point to the polygon vertex to read the word in the stated generators. The result is $[a_1, b_1] \cdots [a_g, b_g]$. Van Kampen kills exactly its normal closure, giving the presentation. A different connecting path conjugates the relator and leaves its normal closure unchanged. ∎

For $g = 1$ the surface is the torus of Lecture 2; we compute its presentation explicitly in the final example.

5.4 Configuration spaces and endpoint permutations

Fix integers $n \geq 2$ and $d \geq 2$. The *ordered configuration space* and *unordered configuration space* are

$$\mathrm{Conf}_n^{\mathrm{ord}}(\mathbb{R}^d) = \{(x_1, \ldots, x_n) \in (\mathbb{R}^d)^n : x_i \neq x_j \text{ for } i \neq j\}, \qquad \mathrm{Conf}_n(\mathbb{R}^d) = \mathrm{Conf}_n^{\mathrm{ord}}(\mathbb{R}^d)/S_n.$$

We keep the dimension in the notation to distinguish the plane from space. In the planar case this is the space $C_n = \mathrm{Conf}_n(\mathbb{R}^2)$ of the endpoint-permutation convention. Here S_n is the group of permutations of $\{1, \dots, n\}$, with right-to-left composition. Its action is $(s \cdot x)_i = x_{s^{-1}(i)}$, which forgets labels in the quotient. The projection is a covering: take disjoint small balls around the n points. Their product and its $n!$ permuted copies are disjoint sheets over the same quotient neighbourhood. The quotient map is open because the saturation of an open set is a finite union of its permuted copies.

A loop downstairs lifts to n labelled particle trajectories. Label their initial positions $1, \dots, n$. The *outgoing permutation* $p_\gamma(i)$ is the final position of the particle starting at i. With the first-path-first convention,

$$p_{\gamma\cdot\delta} = p_\delta \circ p_\gamma, \qquad \rho([\gamma]) = p_\gamma^{-1}, \qquad \rho([\gamma][\delta]) = \rho([\gamma])\rho([\delta]).$$

Thus the inverse outgoing permutation is the homomorphism. Homotopy lifting makes it well-defined on loop classes. For the noncommuting test $p_\gamma = (12)$ and $p_\delta = (13)$, it gives

$$p_{\gamma\cdot\delta} = (123), \qquad \rho([\gamma][\delta]) = (132) = (12)(13).$$

In the plane, fix an unordered base configuration x_0. A *geometric braid* is n collision-free trajectories $(z_j(t), t)$ with that fixed set of positions at both $t = 0$ and $t = 1$. Time increases along every strand. Deformations keep the endpoint positions fixed and preserve the collision-free condition. Write B_n for their classes, with multiplication by running the first braid and then the second.

Proposition 5.3 (Exchange classes · *sketched; proved in* $\mathbb{R}^3$)**.** *In three dimensions,* $\pi_1(\mathrm{Conf}_n(\mathbb{R}^3)) \cong S_n$. *In the plane, the fundamental group is the braid group* B_n.

For $\mathbb{R}^3$, we prove that the ordered space is simply connected. Its collision sets are linear subspaces $L_{ik} = \{x_i = x_k\}$, for $1 \le i < k \le n$, of $\mathbb{R}^{3n}$, each of dimension $3n - 3$. A finite union of proper linear subspaces misses a point. Indeed, for each subspace choose a nonzero linear functional vanishing on it. On the curve $(1, t, \dots, t^{3n-1})$ each functional becomes a nonzero polynomial, with finitely many roots. Choose t outside those finite root sets.

First, a loop γ has positive distance ε from the closed union of collision sets, by compactness. Uniform continuity gives a subdivision $0 = t_0 < \cdots < t_m = 1$ with each image piece of diameter less than $\varepsilon/4$. Join successive vertices $v_j = \gamma(t_j)$ by straight segments to form η. At matching parameters, the interpolation between η and γ stays within $\varepsilon/2$ of $\gamma(t)$. This is a based homotopy avoiding collisions.

Next, choose an apex z_* outside all subspaces $L_{ik} + \mathrm{span}\{v_j, v_{j+1}\}$. Each has dimension at most $3n - 1$, so the finite-union argument applies. With two spatial coordinates, that sum can fill the ambient space, so the apex argument does not apply. The triangle with vertices v_j, v_{j+1}, z_* misses L_{ik}. Otherwise a collision point with nonzero apex coefficient would express z_* as an element of that forbidden subspace. A collision with zero apex coefficient would lie on the already collision-free polygon edge. Thus the map $h\colon [0,1]^2 \to \mathrm{Conf}_n^{\mathrm{ord}}(\mathbb{R}^3)$ given by

$$h(t, s) = (1 - s)\eta(t) + sz_*$$

contracts the loop freely. Lecture 3's moving-base-point construction turns this into a based contraction. Apply the same apex choice to two configurations instead of polygon vertices to join them by two straight segments. Hence the ordered space is path-connected and simply connected.

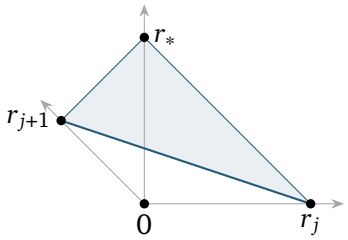


Figure 5.3: For two particles, project to relative coordinates $r = x_1 - x_2$. The projected vertices and apex span a triangle missing the collision point 0.

Coordinate permutations are homeomorphisms commuting with the projection, so they are deck transformations. They act transitively on a fibre and exhaust the deck group by uniqueness from one point. Now Corollary 4.4 gives S_n. The permutation reaching the lifted endpoint is $\rho([\gamma])$, by our chosen action, so this is the isomorphism stated above. Simon, §3.3.2, describes the exchange-space conclusion.

For the plane we sketch the identification. Lifting a configuration loop labels its strands. Conversely, their unordered positions give a configuration loop. The same correspondence applies to homotopies. Thus the two groups are identified. This is Simon, §3.3.1, in the course's path convention.

For two planar particles, write the positions as $c+u$ and $c-u$ in $\mathbb{C}$, where $u \neq 0$. Forgetting labels identifies u with $-u$. The centre c, radius $|u|$ and squared direction $(u/|u|)^2$ give

$$\mathrm{Conf}_2(\mathbb{R}^2) \cong \mathbb{R}^2 \times (0,\infty) \times S^1.$$

The inverse chooses either square root of the direction; the two choices give the same unordered pair. Locally a square-root branch is continuous, so the inverse is continuous. Contracting the first two factors gives $B_2 \cong \mathbb{Z}$. The counter-clockwise exchange $u(t) = e^{\pi \mathrm{i} t}$ makes the squared direction wind once. Two exchanges have winding two, even though their endpoint permutation is the identity.

5.5 Braid relations and exchange phases

Arrange the base positions on a real line. Let σ_i exchange positions i and $i+1$ counter-clockwise, fixing the other positions. Its inverse exchanges them clockwise. We cite Artin's presentation:

$$B_n = \left\langle \sigma_1, \dots, \sigma_{n-1} \;\middle|\; \begin{array}{ll} \sigma_i\sigma_j = \sigma_j\sigma_i, & |i-j| \geq 2, \\ \sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1}, & 1 \leq i \leq n-2 \end{array} \right\rangle.$$

The first relation exchanges separated pairs in either order. The second changes the order of three adjacent exchanges. The completeness of these relations is cited from Simon, §3.3.1 and Exercise 3.1. His braid words are read rightmost first and his positive exchange is clockwise. Our words are read leftmost first and positive exchanges are counter-clockwise. His positive generator is therefore our σ_i^{-1}.

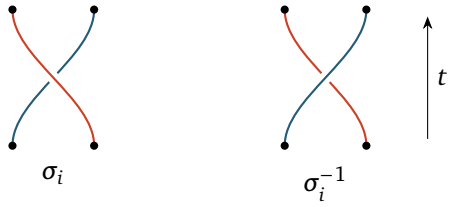


Figure 5.4: Time runs upward and the real axis to the right. At a crossing, the strand with larger imaginary coordinate is drawn on top.

A one-dimensional unitary representation is a homomorphism $\chi\colon G \to U(1)$, where $U(1) = \{z \in \mathbb{C} : |z| = 1\}$. Its value on an exchange is the *exchange phase*.

> **Corollary 5.4** (Scalar exchange phases · *proved given the presentations*)**.** *For $n \geq 2$, the one-dimensional unitary representations of S_n give exchange phases $+1$ or -1. For B_n, any phase $e^{i\theta}$, with $\theta \in \mathbb{R}$, is possible, with the same phase on every σ_i.*

Proof. Adjacent transpositions generate S_n and are conjugate to one another. Scalar images of conjugate elements agree. Each transposition squares to the identity, so its common image q satisfies $q^2 = 1$. Both possibilities occur: the constant character and the sign character, the latter given by the determinant of the permutation matrix.

For B_n, write $q_i = \chi(\sigma_i)$. All q_i are nonzero scalars. The adjacent braid relation gives

$$q_i^2 q_{i+1} = q_i q_{i+1}^2, \qquad q_i = q_{i+1}.$$

For $n \geq 3$ this makes all images equal. For $n = 2$ there is only one generator. Conversely, assigning the same $q \in U(1)$ to every generator satisfies both families of defining relations. Thus every $q = e^{i\theta}$ defines a character. ∎

Thus a character with value $e^{i\theta}$ on our positive exchange has value $e^{-i\theta}$ on Simon's positive exchange. The choices $q = 1$ and $q = -1$ describe scalar bosonic and fermionic statistics. The other planar choices describe *abelian anyons*. Higher-dimensional unitary representations of S_n give matrix-valued exchange operators and the possibility of parastatistics. The scalar conclusion does not classify those representations.

5.6 The torus square

We compute the group of the square with opposite sides identified in the same direction. The quotient is the torus T^2 of Lecture 2. All four corners give one base point, and the two edge loops are a and b.

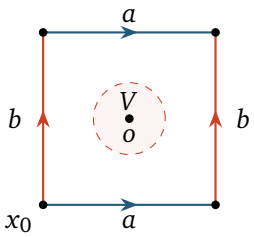


Figure 5.5: The paired arrows give $aba^{-1}b^{-1}$ on the counter-clockwise boundary. The dashed disc V and puncture o give the open cover imposing that relator.

Removing the centre retracts the square quotient onto the two edge loops, so its group is F_2. Adding the centre disc imposes the boundary relator by the open cover used in Proposition 5.2. Therefore

$$\pi_1(T^2) \cong \langle a, b \mid aba^{-1}b^{-1} \rangle \cong \mathbb{Z}^2.$$

For the last isomorphism, the relation makes a and b commute, so every word becomes $a^m b^n$. The exponent-sum map to $(m, n) \in \mathbb{Z}^2$ kills the relator and is inverse to $(m, n) \mapsto a^m b^n$. This agrees with the coordinate-winding calculation of Lecture 3.

Self-check. Prove that the group of $S^1 \vee S^1$ is nonabelian by finding a surjective homomorphism onto S_3.

Answer. Send a to (12) and b to (13). The universal property of F_2 gives a homomorphism. Its image contains $(23) = (12)(13)(12)$ and the two three-cycles, as well as the identity. Thus it contains all six permutations. Since $(12)(13) \neq (13)(12)$, the image is nonabelian, and F_2 is nonabelian. In particular, its commutator $aba^{-1}b^{-1}$ cannot be removed before a face imposes that relation.

Summary. Van Kampen combines path-connected open pieces with path-connected intersection by imposing the overlap relations. A polygonal surface has one relation, the product of its handle commutators. Planar configuration loops are braids; in three dimensions the ordered-space contraction reduces loop classes to permutations. The inverse outgoing permutation is the homomorphism in our path convention. Scalar exchange phases are arbitrary for braids and restricted to ± 1 for permutations. Lecture 6 counts holes of every dimension by linear algebra.

Reading. Hatcher, *Algebraic Topology*, §1.2; Nakahara, *Geometry, Topology and Physics*, §4.4; Simon, *Topological Quantum*, §§3.3–3.5 and Exercise 3.1.

Lecture 6

Homology and the Euler characteristic

Objectives. *Homology distinguishes closed chains from chains that bound higher-dimensional pieces. A decomposition into triangles makes this distinction a calculation with boundary matrices. We compute these matrices for the torus and the projective plane. Their alternating dimensions give the Euler characteristic. On a periodic lattice, the same quotient describes classes of closed strings in the toric code.*

By the end of this lecture you will be able to:

- Construct boundary matrices from ordered simplices.
- Compute cycles, boundaries and homology with specified coefficients.
- Prove the Euler characteristic formula for a finite complex.
- Identify the two independent binary string classes on a torus.

6.1 Simplices and their identifications

A *k-simplex* is the convex hull of $k+1$ affinely independent points, with a chosen ordering of these vertices. The standard simplex is

$$\Delta^k = \{(t_0, \dots, t_k) \in \mathbb{R}^{k+1} : t_i \geq 0,\ t_0 + \cdots + t_k = 1\}.$$

Deleting one vertex gives a *face*, with its inherited order. A point, an interval and a triangle are simplices of dimensions zero, one and two. The order of the vertices orients an edge from the first vertex to the second. It also specifies the signs of higher-dimensional faces.

A Δ-*complex structure* on X is a collection of continuous maps $\sigma\colon \Delta^k \to X$, called its simplices, with the following properties. Each map is injective on the interior of its domain. These interiors have disjoint images and together cover X. The restriction to each face is a designated lower-dimensional simplex, using the affine identification preserving the vertex order. The topology is specified by these maps: a subset $A \subseteq X$ is closed exactly when $\sigma^{-1}(A)$ is closed for every simplex σ. Thus the space is obtained by attaching the faces in the specified way.

Different vertices of one domain simplex may have the same image in X. An edge can therefore become a loop. By contrast, in a *simplicial complex* each simplex has distinct vertices in the space and any two simplices intersect in a common face or not at all. The torus model below has one vertex and two triangles. It is a Δ-complex, not a simplicial complex. Vertex names such as $[v_0, v_1, v_2]$ refer to ordered domain vertices even when some images coincide.

6.2 Chains and the boundary map

Fix a *coefficient ring* R. Our default is $R = \mathbb{Z}$. The alternatives used here are $\mathbb{Z}_2$, the field with arithmetic modulo two, and the real field $\mathbb{R}$. Integers form a ring, not a field. A *k-chain* is a finite formal sum of the k-simplices with coefficients in R. These sums form the *chain group* $C_k(X; R)$. Chains are added and multiplied by coefficients componentwise. Every chain has a unique expression in the simplex generators. When R is a field, they form a vector-space basis. Write $C_{-1} = 0$.

The *boundary map* $\partial_k \colon C_k(X; R) \to C_{k-1}(X; R)$ is the linear map defined on a simplex by alternating deletion:

$$\partial_k[v_0, \ldots, v_k] = \sum_{i=0}^{k} (-1)^i [v_0, \ldots, \widehat{v_i}, \ldots, v_k] \quad (k \geq 1), \qquad \partial_0 = 0.$$

The hat means that the vertex is omitted. Each face is read as the designated simplex of the complex. This ∂ acts on chains; it is not a coordinate derivative.

For an ordered triangle, the formula is

$$\partial[v_0, v_1, v_2] = [v_1, v_2] - [v_0, v_2] + [v_0, v_1].$$

The negative sign reverses the contribution of the middle edge. Over $\mathbb{Z}_2$ the negative sign equals a positive sign. Repeated occurrences of an edge still have to be added in the chosen ring.

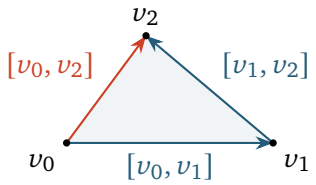


Figure 6.1: The ordered edges point from lower to higher vertex index. The boundary traverses the middle edge with a negative coefficient.

Theorem 6.1 (The boundary of a boundary)**.** *For every $k \geq 1$, the composition $\partial_{k-1}\partial_k$ is zero.*

Proof. For $k = 1$ the target boundary is $\partial_0 = 0$. For $k \geq 2$, fix indices $i < j$ in an ordered k-simplex. Deleting v_i first moves v_j to position $j - 1$. The resulting face has coefficient $(-1)^i(-1)^{j-1}$. Deleting v_j first leaves v_i in position i and gives coefficient $(-1)^j(-1)^i$. These are opposite signs. Every face with two deleted vertices occurs in exactly this pair. The pairs cancel before the face identifications, so they also cancel after those identifications. Linearity proves the assertion for every chain. ∎

For the triangle, applying the boundary once more gives

$$(v_2 - v_1) - (v_2 - v_0) + (v_1 - v_0) = 0.$$

Here a vertex name denotes its generator in C_0. A sequence of groups linked by homomorphisms whose consecutive compositions vanish is called a *chain complex*.

6.3 Cycles, boundaries and coefficients

A *cycle* is a chain with zero boundary. A *boundary* is a chain that is the boundary of a chain one dimension higher. Their groups and the resulting *homology group* are

$$Z_k(X;R) = \ker \partial_k, \qquad B_k(X;R) = \operatorname{im} \partial_{k+1}, \qquad H_k(X;R) = Z_k(X;R)/B_k(X;R).$$

The boundary-of-boundary theorem gives $B_k \subseteq Z_k$, so the quotient is defined. Two cycles are *homologous* when their difference is a boundary. Homology records which cycles remain after this identification. The symbols Z_k and B_k denote groups, distinct from the coefficient ring $\mathbb{Z}$ and the braid group B_n of Lecture 5.

For example, take the boundary of a triangle, with no filled face. The sum of its cyclically directed edges is a cycle. A general cycle has equal coefficients on those three directed edges, since its boundary vanishes at each vertex. There are no two-dimensional chains, so this cycle generates $H_1 \cong R$. Filling the triangle makes that same sum a boundary and gives $H_1 = 0$.

For a finite complex, order the simplices of each dimension. The boundary matrix D_k has a column for each source k-simplex and a row for each target $(k-1)$-simplex. Its entries are the signed face coefficients. Write c_k for the number of k-simplices and r_k for the rank of D_k over a specified field. Rank–nullity and the dimension formula for a quotient, recalled from *Linear Algebra for Physics*, give

$$\dim H_k(X;R) = (c_k - r_k) - r_{k+1} \qquad \text{when } R \text{ is a field.}$$

Over $\mathbb{Z}$, integer relations determine an abelian-group quotient instead. Real row reduction alone does not determine that group.

An element is a *torsion element* if it is nonzero and some positive integer multiple is zero. For an algebraic example, take the chain complex with $C_2 = C_1 = \mathbb{Z}$, boundary ∂_2 given by multiplication by two, and $\partial_1 = 0$. Its first homology is $\mathbb{Z}/2\mathbb{Z}$. With real coefficients, multiplication by two is onto, so the first homology is zero. With $\mathbb{Z}_2$ coefficients it is the zero map, so the first homology is $\mathbb{Z}_2$. This illustrates why coefficients are part of the answer. For a finite complex, the *kth Betti number* means $b_k = \dim_{\mathbb{R}} H_k(X;\mathbb{R})$. Real coefficients discard torsion information.

6.4 The Euler characteristic

For a finite Δ-complex of dimension d, its *Euler characteristic* is the alternating simplex count

$$\chi(X) = \sum_{k=0}^{d} (-1)^k c_k.$$

For a triangulated surface this is vertices minus edges plus triangles. The symbol χ denotes a topological count here, not the character of Lecture 5.

Theorem 6.2 (Euler characteristic and homology). *For a finite Δ-complex and any field R, $\chi(X) = \sum_{k=0}^{d} (-1)^k \dim_R H_k(X;R)$. In particular, $\chi(X) = \sum_k (-1)^k b_k$.*

Proof. Set $r_0 = r_{d+1} = 0$. The dimension formula for homology gives $c_k = \dim H_k + r_k + r_{k+1}$. On summing with alternating signs, the rank terms cancel:

$$\sum_{k=0}^{d} (-1)^k c_k = \sum_{k=0}^{d} (-1)^k \dim H_k + \sum_{k=0}^{d} (-1)^k r_k + \sum_{k=0}^{d} (-1)^k r_{k+1} = \sum_{k=0}^{d} (-1)^k \dim H_k.$$

Indeed, the coefficient of each r_j with $1 \le j \le d$ is $(-1)^j + (-1)^{j-1} = 0$. The endpoint ranks vanish. ∎

For a single filled triangle, the counts are 3, 3 and 1, giving $\chi = 1$. Its homology has one degree-zero generator and no higher generators, which gives the same alternating sum.

6.5 Invariance and closed surfaces

The calculation uses a decomposition. Its independence of that choice requires a theorem, rather than a comparison of a few examples.

Theorem 6.3 (Homology invariance · *cited*). *The homology of a Δ-complex agrees with its singular homology. Homotopy equivalences induce isomorphisms of these groups, with the coefficient ring fixed. Consequently the Euler characteristic of a finite Δ-complex is invariant under homotopy equivalence.*

The comparison is Hatcher, Theorem 2.27. Homotopy invariance is Corollary 2.11; coefficients are treated in §2.2, pp. 153–155. Singular homology uses all continuous maps $\Delta^k \to X$ as generators, without the interior-injectivity requirement. That construction applies without choosing a decomposition. The Euler assertion follows from the theorem above, for example with real coefficients.

A *triangulation* is a presentation of a surface by triangles glued along edges. A triangulated surface is *orientable* if its triangles can be oriented so that their induced directions on every shared edge are opposite. We use as part of the cited surface classification below that every closed surface admits such a presentation and that this condition is independent of the choice. The torus has this property; the projective plane does not.

Theorem 6.4 (Closed surfaces · *cited*). *Closed connected surfaces are classified, up to homeomorphism, by their orientability and Euler characteristic. For the orientable surface Σ_g of genus $g \ge 0$, the Euler characteristic is $2 - 2g$, where $\Sigma_0 = S^2$.*

This is the classification theorem of Kosniowski, Theorem 11.3, together with the Euler counts in Hatcher, §2.2, pp. 146–147. For $g \ge 1$, the polygon model of Lecture 5 has one vertex, $2g$ edges and one face. Using such polygonal cells instead of triangles gives the same Euler characteristic by Hatcher, Theorem 2.44. Thus its count is $1 - 2g + 1$.

There is also a direct connection to Lecture 5. For any path-connected space, $H_1(X;\mathbb{Z})$ is the *abelianization* of $\pi_1(X)$: the quotient by the normal subgroup generated by all commutators $aba^{-1}b^{-1}$. This result is cited from Hatcher, Theorem 2A.1. Homology therefore forgets the order information in noncommuting loop classes. It need not recover the fundamental group.

6.6 Boundary matrices for two surfaces

We compute the torus over $\mathbb{Z}$ and then $\mathbb{Z}_2$. Take a square with corners P, Q, R', S counterclockwise from the bottom left. Pair the horizontal edges to the right and the vertical edges upward. All corners become one vertex v. Call the horizontal edge a, the vertical edge b and the diagonal $P \to R'$ edge c. The ordered triangles are $T_0 = [P, Q, R']$ and $T_1 = [P, S, R']$.

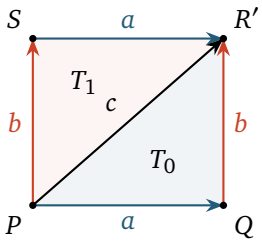


Figure 6.2: The torus has one vertex, three edges and two triangles. The upper triangle's vertex order has the opposite planar orientation to the lower one's.

Example 6.5 (The torus)**.** Both triangle boundaries are $a+b-c$. Each edge has zero boundary, since both endpoints become v. In the bases (a, b, c) and (T_0, T_1), the boundary matrices over $\mathbb{Z}$ are

$$D_1 = \begin{pmatrix} 0 & 0 & 0 \end{pmatrix}.$$

$$D_2 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ -1 & -1 \end{pmatrix}.$$

Every one-chain is a cycle. The boundaries are integer multiples of $a + b - c$. The list $(a, b, a + b - c)$ is an integer basis of C_1: its change-of-basis determinant is -1, so its inverse has integer entries. Quotienting by its last generator leaves a and b free. The kernel of D_2 consists of pairs $(m, -m)$, and there are no three-chains. The zero-dimensional quotient retains the single vertex. We have therefore obtained

$$H_0(T^2; \mathbb{Z}) = \mathbb{Z}, \qquad H_1(T^2; \mathbb{Z}) = \mathbb{Z}^2, \qquad H_2(T^2; \mathbb{Z}) = \mathbb{Z}.$$

The top generator is $T_0 - T_1$. Its signs give both triangles the same planar orientation, and their edge contributions cancel.

With $\mathbb{Z}_2$ coefficients, every entry -1 becomes 1. The ranks of D_1 and D_2 are zero and one. The homology dimensions are therefore 1, $3 - 0 - 1 = 2$ and $2 - 1 = 1$. The classes of a and b still generate H_1, and the top cycle is $T_0 + T_1$. Both coefficient choices give $\chi(T^2) = 1 - 3 + 2 = 0$.

For the projective plane, pair opposite square edges with opposite arrows. The corners P and R' become v, while Q and S become w. The edge a runs from P to S and from R' to Q. The edge b runs from P to Q and from R' to S. The diagonal c runs from P to R'.

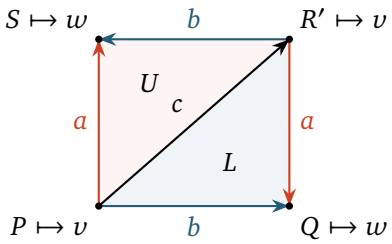


Figure 6.3: The projective plane has two vertices after identification. The diagonal c is a loop at v; both boundary-edge generators join v to w.

Example 6.6 (The projective plane over $\mathbb{Z}_2$)**.** Use the vertex basis (v, w), edge basis (a, b, c) and ordered triangles $U = [P, R', S]$ and $L = [P, R', Q]$. Work over $\mathbb{Z}_2$ throughout. Both edges a and b have boundary $v+w$, and c has zero boundary. Both triangles have boundary $a+b+c$. Thus

$$D_1 = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}.$$

$$D_2 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Both matrices have rank one. The kernel of D_1 is generated by $a + b$ and c. Its boundary subspace is generated by $a + b + c$, so the quotient is one-dimensional, represented by c. The kernel of D_2 is generated by $U + L$. The image of D_1 identifies the two vertex classes in H_0. Consequently $H_k(\mathbb{R}\mathrm{P}^2; \mathbb{Z}_2) \cong \mathbb{Z}_2$ for $k = 0, 1, 2$, and $\chi(\mathbb{R}\mathrm{P}^2) = 2 - 3 + 2 = 1 = 1 - 1 + 1$. These are binary homology groups; they do not assert the same answer with integer or real coefficients.

6.7 Binary cycles and the toric code

On a square lattice with opposite sides identified, put one qubit on each edge. A qubit has basis states $|0\rangle$ and $|1\rangle$. Its phase-flip operator Z fixes $|0\rangle$ and multiplies $|1\rangle$ by -1. This operator symbol is distinct from the cycle group Z_k. A *Z-string* applies this operator on a chosen set of edges. Represent that set by a one-chain with $\mathbb{Z}_2$ coefficients: an edge has coefficient one exactly when it is selected. Adding chains means toggling the edges, since applying Z twice is the identity.

The relevant local operations are read from vertices and square faces, called *plaquettes*. A vertex condition on a string requires an even number of selected edges at that vertex. Algebraically this is $\partial c = 0$. A plaquette boundary is also a cycle. Adding it replaces a part of the string by the other edges of that face. The drawing shows this change beside two independent closed strings.

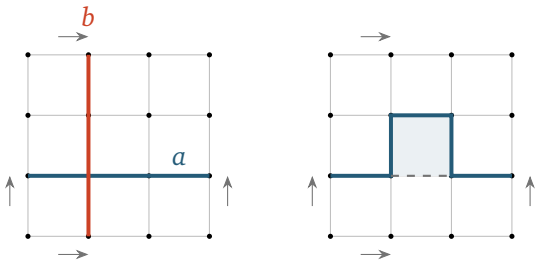


Figure 6.4: Opposite sides of each square are identified. The first panel shows the two periodic cycles. The second adds one plaquette boundary; the dashed edge is removed.

Draw every square's diagonal from lower left to upper right and order its triangles as in the torus example. A triangle boundary replaces each added diagonal in a cycle by its other two edges. Conversely, each diagonal occurs in exactly the two triangles of its square. A binary sum of triangle boundaries with no diagonal term therefore uses the two triangles of each square with equal coefficients. The two triangle boundaries add to the square boundary, so the sum is a sum of square boundaries. This proves that square-lattice

cycles modulo square boundaries agree with the homology of this triangulation. By the cited Theorem 6.3, its homology agrees with that of the two-triangle torus in Example 6.5.

Example 6.7 (Toric-code string classes · *cited; two-triangle homology proved*)**.** The binary string classes form $H_1(T^2;\mathbb{Z}_2) \cong \mathbb{Z}_2^2$. Representatives wind once in the horizontal and vertical directions. Every class is a sum of these two, so there are four classes. The identification of these lattice representatives is part of Kitaev's cited construction.

The *code subspace* is a chosen subspace of the joint qubit state space used to store quantum information. A *logical operator* preserves that subspace and acts on the states stored there. In the toric code, closed Z-strings preserve the code subspace. Plaquette boundaries act as the identity on that subspace, so homologous strings act in the same way. The two independent classes give the two logical Z operators. The full code construction supplies corresponding complementary operators and a four-dimensional code subspace: two logical qubits. This physical interpretation is cited from Kitaev (2003), §1, equations (1)–(2) and the following logical-operator argument. The homology calculation identifies the string classes; the Hilbert-space and operator assertions use that cited construction.

Self-check. Compute the Euler characteristic of the sphere from the boundary of a tetrahedron and from the boundary of an octahedron.

Answer. The tetrahedron has four vertices. Every pair is an edge, giving $\binom{4}{2} = 6$ edges, and every triple is a face, giving four triangles. The count is $4 - 6 + 4 = 2$. The octahedron has vertices at the positive and negative coordinate-axis unit vectors. Every vertex joins the four vertices other than itself and its opposite, giving $6 \cdot 4/2 = 12$ edges. A face chooses one endpoint from each axis, giving $2^3 = 8$ triangles. Its count is $6 - 12 + 8 = 2$, in agreement with the tetrahedron.

Summary. The boundary map has square zero. Homology is the quotient of cycles by boundaries, with the coefficient ring specified. Over a field, boundary-matrix ranks give its dimensions and their alternating sum is the Euler characteristic. Invariance and surface classification are cited. The torus has two independent binary one-cycle classes, which describe the two logical Z-string classes in the toric code. Lecture 7 adds smooth coordinates to spaces so that calculus can be defined on them.

Reading. Hatcher, *Algebraic Topology*, §2.1, pp. 102–107, Theorems 2.27 and 2A.1, and §2.2, pp. 146–147; Nakahara, *Geometry, Topology and Physics*, §3.3; Kosniowski, *A First Course in Algebraic Topology*, Theorem 11.3 and Chapter 29; Kitaev, *Fault-tolerant quantum computation by anyons* (2003), §1.

Lecture 7

Smooth manifolds

Objectives. *A smooth manifold has local coordinates whose changes are smooth. This allows derivatives to be compared between coordinate systems. We construct coordinates on spheres and projective spaces, then prove that regular constraints define manifolds using the implicit function theorem. The space of pure qubit states gives a concrete example: a sphere described by two complex coordinate charts.*

By the end of this lecture you will be able to:

- Check chart transitions and construct a smooth atlas.
- Test smooth maps in coordinates and apply the chain rule.
- Prove that a nonempty regular level set is an embedded submanifold.
- Relate the two projective qubit coordinates by $w = 1/z$.

7.1 Charts and smooth structures

A *topological n-manifold* is a Hausdorff, second-countable space M locally homeomorphic to open subsets of $\mathbb{R}^n$. Here second-countable means that its topology has a countable basis. A *chart* is a pair (U, φ), with $U \subseteq M$ open and $\varphi\colon U \to \varphi(U) \subseteq \mathbb{R}^n$ a homeomorphism onto an open subset. We consider manifolds without boundary. The component functions of the chart are its *coordinates*, written $x^1, \dots, x^n$. The superscripts are coordinate labels, not powers.

Two charts are *smoothly compatible* when both coordinate changes on their overlap are smooth. For charts (U, φ) and (V, ψ), one such change is

$$\psi \circ \varphi^{-1}\colon \varphi(U \cap V) \longrightarrow \psi(U \cap V).$$

Smooth means C^∞: all orders of partial derivatives exist and are continuous. An empty overlap imposes no condition. A *smooth atlas* is a collection of pairwise compatible charts covering M. A *smooth structure* is a maximal smooth atlas, containing every chart compatible with it. A *smooth manifold* is a topological manifold with a smooth structure.

Any smooth atlas determines a unique smooth structure. To prove this, adjoin every chart compatible with the given atlas. For two added charts ψ and θ and a point in their overlap, choose an original chart φ containing that point. Locally the transition is

$$\theta \circ \psi^{-1} = (\theta \circ \varphi^{-1}) \circ (\varphi \circ \psi^{-1}).$$

Both factors are smooth. Reversing the roles gives a smooth inverse. Thus the enlarged collection is an atlas. Any chart compatible with the enlarged collection was compatible with the original one and has already been added. This proves maximality. Every chart of

any maximal atlas containing the original atlas belongs to this enlarged collection. That maximal atlas must therefore equal the enlarged collection, proving uniqueness.

The smooth structure is extra data. On the topological line, the global charts $x(t) = t$ and $y(t) = t^3$ individually define smooth structures. Their transition $y \circ x^{-1}(s) = s^3$ is smooth, but its inverse is not differentiable at zero. Along $s = 1/k^3$, its difference quotient is $(1/k)/(1/k^3) = k^2$, which is unbounded as $k \to \infty$. Hence the two charts do not belong to one smooth atlas. The resulting manifolds will nevertheless be diffeomorphic, by a map different from the identity.

We also fix the index convention used in later calculus. A repeated upper and lower index is summed from 1 to n, unless another range is stated. For a row of coefficients a_i and a column of components X^i, write $a_i X^i = \sum_{i=1}^{n} a_i X^i$. The index position distinguishes transformation rules. It does not mean that a metric has been used. For example, a linear coordinate change and the corresponding rules are

$$J = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}, \qquad x' = Jx, \qquad X' = JX, \qquad a' = aJ^{-1}, \qquad a'X' = aX.$$

The last equality follows by cancelling $J^{-1}J$. Lower coordinate labels x_i in some prerequisite notes become x^i here. This relabelling does not raise the index of a linear functional.

7.2 Spheres, projective spaces and products

Write a point of the unit sphere $S^n \subset \mathbb{R}^{n+1}$ as (y, t), where $y \in \mathbb{R}^n$ and $t \in \mathbb{R}$. Let $N = (0, 1)$ and $S = (0, -1)$ be its north and south poles. Stereographic projection gives charts

$$\varphi_N(y, t) = \frac{y}{1 - t}, \qquad \varphi_S(y, t) = \frac{y}{1 + t}$$

on $S^n \setminus \{N\}$ and $S^n \setminus \{S\}$ respectively. Both have image $\mathbb{R}^n$. The projection is shown in a two-dimensional slice.

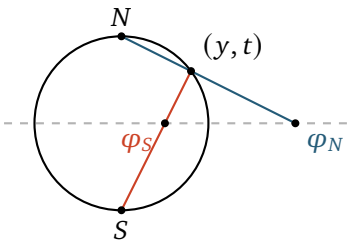


Figure 7.1: The ray from the omitted pole meets the equatorial coordinate plane. The two projections describe the same sphere away from their respective poles.

Proposition 7.1 (Standard atlases). *For $n \geq 1$, the sphere S^n and real projective space $\mathbb{R}\mathrm{P}^n$ are smooth n-manifolds. Products of smooth manifolds have smooth product atlases. In particular, $T^2 = S^1 \times S^1$ is a smooth surface.*

Proof. First, the inverse north chart is

$$\varphi_N^{-1}(u) = \left(\frac{2u}{1 + |u|^2}, \frac{|u|^2 - 1}{1 + |u|^2} \right).$$

Its squared norm is one. Substitution in $y/(1 - t)$ gives u. Conversely, for $(y, t) \neq N$, the relation $|y|^2 = 1 - t^2$ gives $|y/(1 - t)|^2 = (1 + t)/(1 - t)$. Substituting this into the displayed

formula returns (y, t). Thus it is the continuous inverse. The inverse south chart changes the sign of the last component. On the overlap, substitution gives $\varphi_S \circ \varphi_N^{-1}(u) = u/|u|^2$ for $u \neq 0$. This map is smooth and is its own inverse. The sphere is Hausdorff and second-countable as a subspace of Euclidean space. These charts therefore define its smooth structure.

Next, use Lecture 2's quotient $q\colon S^n \to \mathbb{R}\mathrm{P}^n$ by antipodes. The map q is open: the inverse image of the image of an open set W is $W \cup (-W)$. Distinct two-point orbits have disjoint small antipodally invariant neighbourhoods, proving Hausdorffness of the quotient. Images of a countable sphere basis give a countable quotient basis.

Write a class of unit vectors as $[X^0 : \cdots : X^n]$. A nonzero vector represents the class of its normalisation. Multiplying it by a nonzero scalar leaves the same antipodal class, so the notation also describes its line through zero. The open patch U_i consists of classes with $X^i \neq 0$. Its coordinates and inverse are

$$\varphi_i([X]) = (X^0/X^i, \ldots, \widehat{X^i/X^i}, \ldots, X^n/X^i), \qquad \varphi_i^{-1}(u) = [u^0 : \cdots : 1 : \cdots : u^n],$$

with the 1 in position i. The remaining coordinate labels omit i. The ratios are invariant under antipodes and continuous upstairs. Since q is open, its restriction over U_i is a quotient map, so the ratios are continuous on U_i. For the inverse, normalise the displayed representative vector and apply q. These continuous maps are inverse: normalising does not change ratios, and the ratios recover the original vector up to a nonzero scalar. On $U_i \cap U_j$, divide every component of the representative with ith component 1 by its nonzero jth component, then omit component j. All new coordinates are smooth rational functions with nonzero denominator. The same description with i and j exchanged gives the inverse transition.

The $n + 1$ patches cover, completing the projective-space assertion.

Finally, take products of charts. Their coordinate changes are products of smooth maps, so they are smooth. Finite products preserve Hausdorffness and second countability: products of the respective basis elements give a countable basis. This proves the product assertion and the torus case. ∎

7.3 Smooth maps and the chain rule

Let M and N be smooth manifolds of dimensions n and m. A continuous map $f\colon M \to N$ is *smooth* if, near every $p \in M$, its coordinate expression is smooth. For charts φ near p and ψ near $f(p)$, this expression is

$$\widehat{f} = \psi \circ f \circ \varphi^{-1}\colon \varphi(U \cap f^{-1}(V)) \longrightarrow \psi(V).$$

The domain is open because f is continuous. A *diffeomorphism* is a bijective smooth map with a smooth inverse.

> **Proposition 7.2** (Independence of coordinates)**.** *Smoothness is independent of the chosen charts. Compositions of smooth maps are smooth. Their coordinate Jacobians obey the chain rule.*

Proof. With different charts $\widetilde{\varphi}$ and $\widetilde{\psi}$, the new expression, on its common local domain, is

$$\widetilde{\psi} \circ f \circ \widetilde{\varphi}^{-1} = (\widetilde{\psi} \circ \psi^{-1}) \circ \widehat{f} \circ (\varphi \circ \widetilde{\varphi}^{-1}).$$

The outer maps are smooth transitions. This proves independence at each point. For a further smooth map $g\colon N \to P$, use the same middle chart on N. The coordinate expression of $g \circ f$ is $\widehat{g} \circ \widehat{f}$, and the Euclidean chain rule gives

$$D(\widehat{g} \circ \widehat{f})_x = D\widehat{g}_{\widehat{f}(x)}\, D\widehat{f}_x.$$

Both smoothness of the composition and the Jacobian formula follow. ∎

Here $D\widehat{f}_x$ denotes the $m \times n$ matrix of first partial derivatives. In component notation, the same formula is

$$\frac{\partial (g \circ f)^a}{\partial x^i} = \frac{\partial g^a}{\partial y^b}\frac{\partial f^b}{\partial x^i},$$

with the repeated index b summed. These are coordinate expressions; Lecture 8 defines the differential without choosing coordinates.

For the two smooth structures on the line above, the map $f(t) = \sqrt[3]{t}$ from the x-structure to the y-structure has coordinate expression $y \circ f \circ x^{-1}(s) = s$. Its inverse has the identity coordinate expression too. Thus it is a diffeomorphism, although the identity map between those structures is not a diffeomorphism.

7.4 Regular level sets and embeddings

A subset S of a smooth N-manifold M is an *embedded d-submanifold* if, with the subspace topology, it satisfies the following condition. At every $p \in S$ there is an ambient smooth chart (U, φ) containing p such that

$$\varphi(S \cap U) = \varphi(U) \cap (\mathbb{R}^d \times \{0\}).$$

The restricted coordinates give its smooth structure.

For a smooth map $f\colon W \to \mathbb{R}^k$, where $W \subseteq \mathbb{R}^N$ is open, a point $p \in W$ is *regular* when Df_p is onto. A *regular value* is a point $c \in \mathbb{R}^k$ such that every point of $f^{-1}(c)$ is regular. A value outside the image is regular vacuously.

Theorem 7.3 (Regular level sets · *proved given the implicit function theorem*)**.** *If c is a regular value of f, its nonempty level set $f^{-1}(c)$ is an embedded submanifold of W of dimension $N - k$.*

Proof. Fix $p \in f^{-1}(c)$. Surjectivity gives a nonsingular $k \times k$ minor of Df_p. Permute the input coordinates and write them as $(s, t) \in \mathbb{R}^{N-k} \times \mathbb{R}^k$, with this minor in the t coordinates. The smooth implicit function theorem, cited in the form of Tu, Theorem B.2, gives open neighbourhoods in which

$$f(s,t) = c \quad \Longleftrightarrow \quad t = h(s)$$

for a smooth function h. In this neighbourhood the ambient coordinate change $\Phi(s,t) = (s, t - h(s))$ has smooth inverse $\Phi^{-1}(s,u) = (s, u + h(s))$. It sends the level set to $u = 0$. These are the required ambient charts. On their overlaps, the restricted transitions are smooth, because they are restrictions of smooth ambient transitions followed by projection onto the first $N - k$ components. The subspace is Hausdorff and second-countable, inherited from W. Thus these charts define the claimed embedded manifold. ∎

This is the Euclidean case of Tu, Theorem 9.9, pp. 105–106. For the sphere, use $f(x) = |x|^2$. At $p \in S^n$, its derivative is $Df_p(v) = 2p \cdot v$, which takes the value 1 on $v = p/2$. Hence 1 is regular and the sphere has dimension $(n+1) - 1 = n$. The resulting structure agrees with the stereographic one, since the stereographic maps and their inverses are smooth ambient formulas on their respective domains.

For the orthogonal group, the constraint takes values in the vector space Sym_n of real symmetric matrices:

$$f\colon \mathbb{R}^{n\times n} \to \mathrm{Sym}_n, \qquad f(A) = A^T A.$$

The target has dimension $n(n+1)/2$, since its diagonal and entries above the diagonal determine it. Expanding $f(A+tH)$ to first order in the real parameter t gives

$$Df_A(H) = A^T H + H^T A.$$

At $A \in \mathrm{O}(n)$, any symmetric matrix S is the image of $H = AS/2$, since $A^T A = I$. Thus I is a regular value and $\dim \mathrm{O}(n) = n^2 - n(n+1)/2 = n(n-1)/2$. Taking all n^2 matrix entries as independent target constraints would incorrectly count the repeated off-diagonal equations.

A smooth map is an *immersion* if its coordinate Jacobian is injective at every point. An immersion is an *embedding* if it is also a homeomorphism onto its image with the subspace topology. Chart changes multiply the Jacobian by invertible matrices, so the immersion condition is independent of charts. An injective immersion need not be an embedding. Fix an irrational number $\alpha > 0$ and consider

$$\gamma : \mathbb{R} \longrightarrow T^2, \qquad \gamma(t) = (e^{\mathrm{i}t}, e^{\mathrm{i}\alpha t}).$$

Equal images require a difference $2\pi m$ with $m \in \mathbb{Z}$ and $\alpha m \in \mathbb{Z}$. Irrationality forces $m = 0$, proving injectivity. The first circle component has nonzero derivative, proving immersion. The square model shows the repeated returns of its image.

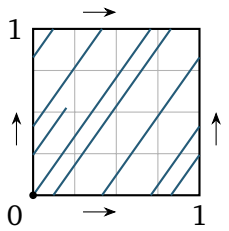


Figure 7.2: A finite segment of the line of slope $\sqrt{2}$ in the square quotient $\mathbb{R}^2/\mathbb{Z}^2$. Each exit continues at the identified opposite edge.

There are unbounded positive integers q_j with distance from αq_j to an integer tending to zero. To see this, divide $[0,1)$ into N equal intervals and place the $N+1$ fractional parts of $0, \alpha, \ldots, N\alpha$ into them. Two occupy the same interval. Their difference gives $1 \le q \le N$ with distance less than $1/N$. Such choices must be unbounded along a subsequence, since each of finitely many nonzero irrational multiples has positive distance from the integers. Consequently $\gamma(2\pi q_j) \to \gamma(0)$, although $2\pi q_j$ does not tend to zero. The inverse from the image is not continuous. Thus γ is not an embedding. Lecture 9 describes motion along this curve by prescribing its velocity at every point.

7.5 The qubit's projective state space

A pure qubit state is a nonzero column $(a, b) \in \mathbb{C}^2$, with two columns representing the same state when they differ by a nonzero complex scalar. Normalising the column leaves precisely the usual overall phase freedom. The quotient is the *complex projective line* $\mathbb{C}\mathrm{P}^1$, with points written $[a : b]$. Give it the quotient topology from $\mathbb{C}^2 \setminus \{0\}$. The patches with $a \neq 0$ and $b \neq 0$ have coordinates

$$z = b/a, \qquad w = a/b.$$

Their inverses are $z \mapsto [1 : z]$ and $w \mapsto [w : 1]$. The quotient projection is open because saturating an open set takes the union of its images under nonzero scalar multiplications.

Thus the patches are open and the ratios define continuous charts. On the overlap $z \neq 0$, the transition is $w = 1/z$. For $z = u + \mathrm{i}v$, this is the real smooth map

$$(u, v) \longmapsto \left(\frac{u}{u^2 + v^2}, \frac{-v}{u^2 + v^2} \right).$$

It is its own inverse. The missing point of either chart is the origin of the other chart, as the sphere picture indicates. Lecture 20 differentiates this transition to describe vectors attached to the sphere.

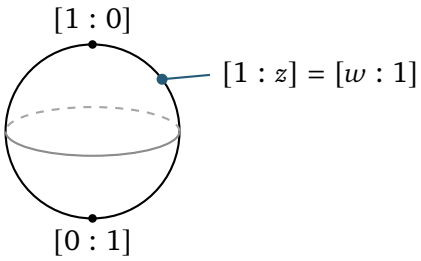


Figure 7.3: The north state $[1 : 0]$ has $z = 0$ and the south state $[0 : 1]$ has $w = 0$. Both coordinates are available away from the poles, with $w = 1/z$.

Example 7.4 (Projective states and the Bloch sphere)**.** The map $B\colon \mathbb{C}\mathrm{P}^1 \to S^2$ to the unit sphere is given by

$$B([a : b]) = \frac{1}{|a|^2 + |b|^2} \big(2\,\mathrm{Re}(\bar{a}b),\ 2\,\mathrm{Im}(\bar{a}b),\ |a|^2 - |b|^2\big).$$

Multiplying a and b by a common nonzero scalar multiplies numerator and denominator by the same positive number, so the map is well-defined. The formula on nonzero columns is continuous with nonvanishing denominator. Lecture 2's quotient criterion therefore makes B continuous. Its squared norm is one because $4|a|^2|b|^2 + (|a|^2 - |b|^2)^2 = (|a|^2 + |b|^2)^2$. In the z chart it becomes

$$B([1 : z]) = \frac{(2\,\mathrm{Re}\,z,\ 2\,\mathrm{Im}\,z,\ 1 - |z|^2)}{1 + |z|^2}.$$

For a sphere point (X, Y, Z) away from the south pole, the inverse coordinate is $z = (X + \mathrm{i}Y)/(1 + Z)$. Away from the north pole it is $w = (X - \mathrm{i}Y)/(1 - Z)$. These formulas are inverse on their patches and agree by $w = 1/z$ on the overlap. Indeed, the displayed z formula for B is exactly the inverse south stereographic chart, with real coordinates $(\mathrm{Re}\,z, \mathrm{Im}\,z)$. On the w patch it is the inverse north chart with coordinates $(\mathrm{Re}\,w, -\mathrm{Im}\,w)$. The already proved stereographic identities establish both inverse compositions. Hence B is a homeomorphism. In particular, $\mathbb{C}\mathrm{P}^1$ is Hausdorff and second-countable, inherited through this homeomorphism from the sphere. Its compatible charts therefore make it a smooth surface. The same formulas are smooth in both directions, so B is a diffeomorphism.

Take angles $0 \le \theta \le \pi$ and $0 \le \phi < 2\pi$. For the normalised representative $(a, b) = (\cos(\theta/2), e^{\mathrm{i}\phi}\sin(\theta/2))$, the z coordinate, when $\theta < \pi$, is $z = \tan(\theta/2)e^{\mathrm{i}\phi}$. At $\theta = \pi$ the state is $[0 : 1]$, with $w = 0$. The sphere image for the full angle range is $(\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$. Thus the two-component state and the sphere point describe the same pure state, with no overall phase coordinate.

Self-check. Use the regular level set theorem to show that $\mathrm{U}(n)$ is a real manifold of dimension n^2. Specify the real target of the constraint.

Answer. Use $f(A) = A^\dagger A$ on $\mathbb{C}^{n\times n} \cong \mathbb{R}^{2n^2}$ with target the real vector space of Hermitian matrices. That target has n real diagonal entries and one complex entry for each pair above the diagonal, so its real dimension is $n + 2\binom{n}{2} = n^2$. Differentiating with respect to a real parameter gives $Df_A(H) = A^\dagger H + H^\dagger A$. At a unitary A, every Hermitian S is reached by $H = AS/2$. Thus I is a regular value and its level set has dimension $2n^2 - n^2 = n^2$. Treating this as a complex-linear constraint would be incorrect: conjugation is part of the formula.

Summary. A smooth structure is specified by charts with smooth transitions. Smooth maps are tested in these charts, and the Euclidean chain rule makes the test independent of coordinates. The implicit function theorem turns independent smooth constraints into embedded submanifolds. The qubit state space has the two projective charts z and $w = 1/z$, and is diffeomorphic to the Bloch sphere. Lecture 8 uses coordinate changes to define tangent vectors and differentials intrinsically.

Reading. Tu, *An Introduction to Manifolds*, §§5–7, Theorem 9.9 and §11; the irrational-line example is Example 15.9. Nakahara, *Geometry, Topology and Physics*, §5.1.

Lecture 8

Tangent spaces, Lie groups and homogeneous spaces

Objectives. *A tangent vector records the initial velocity of a curve without choosing an ambient Euclidean space. Coordinate changes relate its components by their Jacobian matrices. We use these vectors to differentiate maps and describe the infinitesimal directions of matrix groups. A transitive smooth action of a Lie group presents a manifold as a quotient by the subgroup fixing a point. The sphere and the qubit state space give two concrete quotient descriptions.*

By the end of this lecture you will be able to:

- Define tangent vectors by curves and transform their components.
- Compute differentials and tangent spaces to regular level sets.
- Identify the tangent space of $\mathrm{SO}(3)$ at its identity.
- Determine stabilisers and justify the sphere quotient descriptions.

8.1 Tangent vectors as velocities

Let p be a point of a smooth n-manifold M. A *smooth curve through* p is a smooth map $\gamma\colon (-\varepsilon,\varepsilon) \to M$ with $\gamma(0) = p$, where $\varepsilon > 0$. In a chart $x = (x^1, \dots, x^n)$ near p, its coordinate velocity is $(x \circ \gamma)'(0) \in \mathbb{R}^n$. Two such curves are equivalent if these velocities agree. A *tangent vector* at p is an equivalence class of curves. The set of these classes is the *tangent space*, written T_pM. The drawing shows curves with different second-order behaviour but the same initial velocity in one chart.

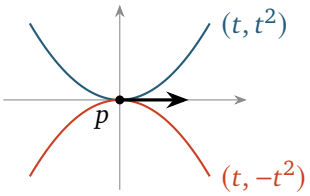


Figure 8.1: The coordinate curves (t, t^2) and $(t, -t^2)$ have the same velocity $(1, 0)$ at the marked point. Their common tangent vector does not record their bending.

Theorem 8.1 (Velocity classes and components)**.** *Curve equivalence is independent of the chart. A chart identifies T_pM with $\mathbb{R}^n$, giving it an n-dimensional vector-space structure independent of that chart. Under a coordinate change $y = y(x)$, components transform by*

$$v^a_y = \frac{\partial y^a}{\partial x^i}(p)\, v^i_x.$$

Proof. The chain rule gives $(y \circ \gamma)'(0) = D(y \circ x^{-1})_{x(p)}(x \circ \gamma)'(0)$. Thus equal velocities in x give equal velocities in y. The reverse coordinate change proves the converse. This also gives the displayed component formula.

The velocity map $T_pM \to \mathbb{R}^n$ is injective by the definition of equivalence. For any $v \in \mathbb{R}^n$, the curve $\gamma(t) = x^{-1}(x(p) + tv)$ is defined for sufficiently small $|t|$ because the chart image is open. It has velocity v, proving surjectivity. Transport addition and scalar multiplication through this bijection. A change of chart acts by an invertible linear Jacobian, so it preserves these operations. This proves that the vector-space structure is independent of the chart. ∎

The basis vector with coordinate components equal to the ith unit column is written $\partial_i|_p = \partial/\partial x^i|_p$. Thus $v = v^i \partial_i|_p$, with the upper/lower index convention of Lecture 7. The symbols $\partial_i|_p$ denote tangent vectors, not points of M.

A tangent vector differentiates smooth functions near p by

$$v(f) = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} f(\gamma(t)) = v^i \frac{\partial (f \circ x^{-1})}{\partial x^i}(x(p)).$$

The right side shows independence of the representative curve. The ordinary product rule gives $v(fg) = v(f)g(p) + f(p)v(g)$. Such a real linear rule on functions is called a *point derivation*. Only the function near p matters. Formally a *germ* at p is an equivalence class of functions defined near p, with two functions equivalent when they agree on some smaller neighbourhood. The product rule makes sense on germs. Conversely, all point derivations on smooth germs have the displayed coordinate form. This comparison is cited from Tu, §8, Proposition 8.9; its derivation formulation is developed on Sheet 8.

8.2 Differentials and regular constraints

For a smooth map $f\colon M \to N$, its *differential* at p is

$$\mathrm{d}f_p\colon T_pM \longrightarrow T_{f(p)}N, \qquad \mathrm{d}f_p[\gamma] = [f \circ \gamma].$$

In coordinates the image velocity is $D\widehat{f}_{x(p)}v_x$, by the chain rule. This depends only on the velocity class and is linear. It therefore defines the stated linear map. For $g\colon N \to P$, composition on representatives gives the intrinsic chain rule

$$\mathrm{d}(g \circ f)_p = \mathrm{d}g_{f(p)} \circ \mathrm{d}f_p. \tag{8.1}$$

The identity map has identity differential. Applying this formula to a diffeomorphism and its inverse shows that their differentials are inverse linear maps.

For example, if $f\colon \mathbb{R}^2 \to \mathbb{R}^3$ is given by $f(s,t) = (s, t, s^2 + st)$, then

$$\mathrm{d}f_{(s,t)}(u,v) = (u, v, (2s+t)u + sv).$$

Its first two components recover (u,v), so the differential is injective. Recall from Lecture 7 that an injective differential defines an *immersion*. An onto differential defines a *submersion*. An immersion that is a homeomorphism onto its image is an *embedding*.

Proposition 8.2 (Tangent space of a level set · *proved given the implicit function theorem*)**.** *Let $S = f^{-1}(c) \subset W \subseteq \mathbb{R}^N$ be a nonempty regular level set of a smooth map $f\colon W \to \mathbb{R}^k$, with W open. The inclusion identifies T_pS with $\ker Df_p$ for every $p \in S$.*

Proof. For a curve in S, differentiating $f(\gamma(t)) = c$ gives $Df_p\gamma'(0) = 0$. In the graph chart of Theorem 7.3, the inclusion has the form $s \mapsto (s, h(s))$. Its differential sends u to $(u, Dh\,u)$, so it is injective. Its image has dimension $N - k$. The kernel of the onto map Df_p has that same dimension by rank–nullity. The inclusion already proved is therefore equality. ∎

For S^2, the constraint $|p|^2 = 1$ gives $T_pS^2 = \{v \in \mathbb{R}^3 : p \cdot v = 0\}$, since the derivative is $v \mapsto 2p \cdot v$. Thus the intrinsic velocity classes agree with the familiar tangent plane in this embedded example.

8.3 The tangent bundle

The *tangent bundle* is the disjoint union of all tangent spaces:

$$TM = \{(p, v) : p \in M,\ v \in T_pM\}.$$

The projection $\pi_{TM} : TM \to M$ is given by $\pi_{TM}(p, v) = p$. Its *fibre* over p is the inverse image $\pi_{TM}^{-1}(p)$, identified with T_pM. Vectors at different base points belong to different fibres. They cannot be added without a further identification.

Proposition 8.3 (Manifold structure on the tangent bundle · *sketched*)**.** *If M is a smooth n-manifold, then TM has a natural smooth structure of dimension $2n$. The projection is a smooth submersion.*

We sketch its chart construction. A chart x on $U \subseteq M$ gives coordinates $(x(p), v_x)$ on $\pi_{TM}^{-1}(U)$ with image $x(U) \times \mathbb{R}^n$. If y is another chart, the overlap map is

$$(x, v_x) \longmapsto \big(y(x), Dy_x v_x\big).$$

It is smooth and has the inverse obtained by the inverse coordinate change. The base and velocity coordinates give $2n$ independent real coordinates. They also specify the topology. Distinct base points can be separated downstairs. Distinct vectors over the same point can be separated in a common bundle chart. A countable atlas downstairs gives a countable basis in these product coordinates. This gives Hausdorffness and second countability. In each chart the projection is $(x, v) \mapsto x$, whose derivative is onto. The detailed chart verification is in Tu, §12, and Sheet 8.

For the circle, write $p = (\cos\theta, \sin\theta)$. Every tangent vector has the unique form $v = r(-\sin\theta, \cos\theta)$ with $r \in \mathbb{R}$. These formulas identify TS^1 with $S^1 \times \mathbb{R}$: the angle is periodic but the velocity coefficient is unrestricted. The figure displays this cylinder and its projection.

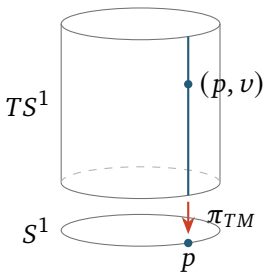


Figure 8.2: A fibre of TS^1 is a line of velocity coefficients. Projection forgets the coefficient and retains the point of the circle.

8.4 Lie groups and tangent matrices

A *Lie group* is a smooth manifold G that is also a group, with smooth multiplication $G \times G \to G$ and smooth inversion $G \to G$. Write e for its identity. For $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, the *general linear group* is

$$\mathrm{GL}(n, \mathbb{K}) = \{A \in \mathbb{K}^{n \times n} : \det A \neq 0\}.$$

The *special orthogonal group* is

$$\mathrm{SO}(n) = \{A \in \mathrm{O}(n) : \det A = 1\}.$$

A *matrix subgroup* is a subgroup of a general linear group. For complex matrices, the ambient space is regarded as a real manifold.

Proposition 8.4 (Regular matrix groups · *proved given the regular level set theorem*)**.** *A matrix subgroup that is a regular level set has the induced structure of a Lie group. The tangent space $T_I\mathrm{SO}(3)$ is the vector space $\mathfrak{so}(3) = \{A \in \mathbb{R}^{3\times 3} : A^T = -A\}$.*

Proof. The regular level set theorem gives an embedded manifold. Matrix multiplication is polynomial in real entries. Matrix inversion is smooth on invertible matrices, by the adjugate formula divided by the nonzero determinant. These maps preserve the subgroup. In adapted submanifold charts, their restrictions retain the first manifold-coordinate components of their ambient expressions, so the restrictions are smooth. They are the group's multiplication and inversion, as required.

The orthogonal constraint and its surjective derivative were computed in Lecture 7. The determinant-one part is open in $\mathrm{O}(3)$, since its only determinants are $+1$ and -1. It therefore has the same tangent space at I. By Proposition 8.2, that space is the kernel of $A \mapsto A^T + A$. This is precisely the antisymmetric matrices. ∎

Such a matrix has the unique form

$$A(a, b, c) = \begin{pmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{pmatrix}, \qquad (a, b, c) \in \mathbb{R}^3.$$

Thus $\mathfrak{so}(3)$ has basis $A(1, 0, 0)$, $A(0, 1, 0)$ and $A(0, 0, 1)$. These are infinitesimal rotations in the fixed Cartesian basis.

A *Lie algebra* is a real vector space with an antisymmetric bilinear bracket satisfying the Jacobi identity $[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = 0$. For a Lie group embedded as a matrix subgroup, the tangent space $\mathfrak{g} = T_I G$ carries the matrix commutator $[A, B] = AB - BA$. To show closure, choose curves $a(s)$ and $b(t)$ in the group with initial value I and velocities A and B. Differentiating $a(s)b(t)a(s)^{-1}$ first in t gives $a(s)Ba(s)^{-1} \in \mathfrak{g}$. Differentiating in s at zero gives $AB - BA$, since $(a^{-1})'(0) = -A$, obtained by differentiating $a(s)a(s)^{-1} = I$. This derivative stays in the finite-dimensional vector subspace $\mathfrak{g}$. Bilinearity and antisymmetry follow from the formula. One nested commutator expands as

$$[A, [B, C]] = ABC - ACB - BCA + CBA.$$

In the sum with its two cyclic versions, each product occurs once with each sign. This proves Jacobi. For an abstract Lie group, Lecture 9 constructs the bracket on $T_e G$ using vector fields.

8.5 Actions and homogeneous spaces

A *smooth left action* of a Lie group on M is a smooth map $G \times M \to M$, written $(g, p) \mapsto g \cdot p$, satisfying $e \cdot p = p$ and $(gh) \cdot p = g \cdot (h \cdot p)$. The *orbit* of p is $\{g \cdot p : g \in G\}$. The *orbit map* at p is the map $G \to M$ given by $g \mapsto g \cdot p$. The action is *transitive* when this orbit is all of M. A manifold with a transitive smooth action is a *homogeneous space*. The *stabiliser* (or isotropy subgroup) of p is

$$H = \{g \in G : g \cdot p = p\}.$$

It is a subgroup by the action laws. It is closed, being the inverse image of the closed singleton $\{p\}$ under the continuous orbit map.

For a subgroup H, the *left coset* gH is $\{gh : h \in H\}$. The set of left cosets, with the quotient topology, is written G/H. Two elements have the same coset exactly when $g_1^{-1} g_2 \in H$. An *embedded Lie subgroup* is a subgroup with an embedded smooth manifold structure and the restricted smooth group operations.

> **Theorem 8.5** (Closed subgroups and homogeneous spaces · *cited*)**.** *A closed subgroup H of a Lie group is an embedded Lie subgroup. The quotient G/H has a smooth manifold structure of dimension $\dim G - \dim H$ for which the quotient map is a smooth submersion. For a transitive smooth action on M with stabiliser H at p, the map $gH \mapsto g \cdot p$ is a diffeomorphism $G/H \to M$.*

These are Lee, second edition, Theorems 20.12, 21.17 and 21.18. The action identifies cosets with points: if $g_1 \cdot p = g_2 \cdot p$, then $g_1^{-1} g_2 \in H$, and the converse follows from the action law. Transitivity gives surjectivity, completing the bijection. The smooth-manifold and diffeomorphism assertions require the cited theorems.

Closedness cannot be omitted. The irrational line of Lecture 7 is a subgroup H of T^2, since adding its parameters multiplies its circle components. It is dense. Indeed, arbitrarily small nonzero fractional parts of integer multiples of α exist by the pigeonhole argument there, allowing either sign of the integer. If such a fractional part is $0 < \delta < \varepsilon$, the multiples $0, \delta, 2\delta, \ldots$ up to the last below 1 approach every point of $[0, 1)$ within ε. Thus integer multiples of α are dense modulo one. Fixing the first circle coordinate of the line leaves all these choices for the second, proving density in the torus. The subgroup is proper: over first coordinate 1, only the countably many points with second coordinate $e^{2\pi i \alpha m}$, $m \in \mathbb{Z}$, occur. Hence H is not closed. If T^2/H were Hausdorff, its singleton identity coset would be closed. Its inverse image H would then be closed, a contradiction. This quotient is not a manifold of our definition.

8.6 The sphere and the Hopf map

The standard action of $\mathrm{SO}(n+1)$ on S^n is transitive. Given a unit vector, extend it to an oriented orthonormal basis by Gram–Schmidt, recalled from *Linear Algebra for Physics*. Use it as the last column of an orthogonal matrix, completing the other columns with positive orientation. That matrix sends the last coordinate unit vector to the chosen point. The stabiliser consists of matrices with a block in $\mathrm{SO}(n)$ on its orthogonal complement and a last diagonal entry 1. The group $\mathrm{O}(n+1)$ is a regular level set by Lecture 7, so it is a Lie group by Proposition 8.4. Its determinant-one part is open and hence is also a Lie group. The sphere action is the restriction of the polynomial map $(g, p) \mapsto gp$, so it is smooth. Applying

the cited theorem gives

$$S^n \cong \mathrm{SO}(n+1)/\mathrm{SO}(n) \qquad (n \geq 1).$$

The identification is a diffeomorphism. In particular, the orbit map $g \mapsto ge_3$ exhibits $S^2 = \mathrm{SO}(3)/\mathrm{SO}(2)$. The subgroup $\mathrm{SO}(2)$ rotates the other two columns while leaving the last one fixed. These quotient descriptions concern smooth manifolds; the stabiliser is not a normal subgroup in general, so the quotient need not be a group.

For the qubit, use the left action $g \cdot [a : b] = [g(a,b)^T]$ of $\mathrm{SU}(2)$ on $\mathbb{C}\mathrm{P}^1$. The notation means the line spanned by the image column. If $|a|^2 + |b|^2 = 1$, the matrix with that first column is

$$U(a,b) = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}.$$

Its columns are orthonormal and its determinant is one, so $U(a,b) \in \mathrm{SU}(2)$. Conversely, a unitary matrix with first column $(a,b)^T$ has second column $\lambda(-\bar{b},\bar{a})^T$ for some $|\lambda| = 1$. Its determinant is λ, so determinant one forces $\lambda = 1$. Thus every matrix in $\mathrm{SU}(2)$ has this form. Transporting the smooth structure of S^3 through U makes $\mathrm{SU}(2)$ a smooth manifold. Its product and inverse are smooth: the first columns of the corresponding matrix formulas are polynomial in the real and imaginary coordinates. This makes $\mathrm{SU}(2)$ a Lie group. Every ray has a unit representative, proving transitivity. Fixing $[1 : 0]$ requires the first column to be $(e^{it}, 0)^T$. Unitarity and determinant one then force the other diagonal entry. Thus the stabiliser is the subgroup

$$H = \left\{ \begin{pmatrix} e^{it} & 0 \\ 0 & e^{-it} \end{pmatrix} : t \in \mathbb{R} \right\} \cong \mathrm{U}(1).$$

The action is smooth in projective charts: each output ratio is a ratio of complex linear expressions, on the open set where its denominator is nonzero. Consequently the cited theorem applies and gives $\mathbb{C}\mathrm{P}^1 \cong \mathrm{SU}(2)/\mathrm{U}(1)$ as smooth manifolds.

The figure places the two orbit maps and their stabilisers side by side.

$$\begin{array}{ccc} \mathrm{SO}(2) & \longrightarrow & \mathrm{SO}(3) \\ & & \downarrow{\scriptstyle g \mapsto ge_3} \\ & & S^2 \end{array} \qquad\qquad \begin{array}{ccc} \mathrm{U}(1) & \longrightarrow & \mathrm{SU}(2) \\ & & \downarrow{\scriptstyle g \mapsto [ge_1]} \\ & & \mathbb{C}\mathrm{P}^1 \end{array}$$

Figure 8.3: Each orbit map retains one datum: the last unit column for a rotation, or the line of the first unit column for $\mathrm{SU}(2)$. Multiplication on the right by its stabiliser leaves that datum fixed.

Example 8.6 (The Hopf map · *proved given the homogeneous-space theorem*)**.** The first-column map identifies $\mathrm{SU}(2)$ with $S^3 = \{(a,b) \in \mathbb{C}^2 : |a|^2 + |b|^2 = 1\}$. The inverse is $U(a,b)$, so both maps are smooth. The projection to the first-column line is the *Hopf map*

$$S^3 \longrightarrow \mathbb{C}\mathrm{P}^1, \qquad (a,b) \longmapsto [a : b].$$

Its fibre consists of the unit representatives $(e^{it}a, e^{it}b)$ of the same ray. Indeed, two unit columns span the same line exactly when the scalar relating them has modulus one. The matrix identity $U(e^{it}a, e^{it}b) = U(a,b)\,\mathrm{diag}(e^{it}, e^{-it})$ shows that these are precisely the left cosets $U(a,b)H$, the fibres of the orbit map $g \mapsto [ge_1]$. Composing with the Bloch diffeomorphism of Example 7.4 gives $S^3 \to S^2$. It forgets the common phase of a normalised qubit column. Lecture 20 develops its structure as a circle bundle.

Self-check. Find the dimension of $\mathrm{SO}(3)/\mathrm{SO}(2)$ using its stabiliser, and compare it with the tangent-plane calculation for the sphere.

Answer. The matrices in $\mathfrak{so}(3)$ have three independent parameters. The group $\mathrm{SO}(2)$ is a circle, so it has dimension one. The quotient dimension is $3 - 1 = 2$. The plane $p \cdot v = 0$ in $\mathbb{R}^3$ also has dimension two, since $p \neq 0$. The two calculations agree.

Summary. Tangent vectors are velocity classes. A chart identifies them with coordinate columns, and its Jacobian gives their transformation law. Differentials send curve velocities to image velocities and satisfy the chain rule. The tangent space of a regular level set is the kernel of the constraint's derivative. Closed stabilisers give smooth quotient manifolds by cited theorems. The rotation and qubit actions identify the sphere with $\mathrm{SO}(3)/\mathrm{SO}(2)$ and $\mathrm{SU}(2)/\mathrm{U}(1)$ respectively. Lecture 9 studies curves whose velocities are prescribed at every point by a vector field.

Reading. Tu, *An Introduction to Manifolds*, §§8, 12, 15–16; Lee, *Introduction to Smooth Manifolds*, second edition, Theorems 20.12, 21.17 and 21.18.

Lecture 9

Vector fields and flows

Objectives. *A smooth vector field prescribes a velocity at every point of a manifold. Its integral curves describe the resulting motion. Comparing a second field along that motion gives the Lie bracket. We prove the connection between a zero bracket and commuting complete flows, then compute the rotation fields and their signs.*

By the end of this lecture you will be able to:

- Find integral curves and identify their maximal time domains.
- Compute a Lie bracket and derive it from backward transport by a flow.
- Prove the commuting-flow criterion for complete fields.
- Reconcile rotation-field and angular-momentum commutator signs.

9.1 Vector fields and related fields

Let M be a smooth n-manifold. A *smooth vector field* on M is a smooth map $X\colon M \to TM$ with $X(p) \in T_pM$. Such a map is called a *section* of the tangent bundle: composing it with the bundle projection gives the identity on M. In local coordinates it has the form $X = X^i\partial_i$, where the coefficient functions X^i are smooth. Indeed, its bundle-chart expression is $x \mapsto (x, X^1(x), \dots, X^n(x))$, so these two smoothness conditions are equivalent.

The field acts on smooth real functions by

$$Xf = X^i\partial_i f.$$

This is a smooth function. The product rule gives $X(fg) = (Xf)g + f(Xg)$, so X acts as a derivation of functions. The value $(Xf)(p)$ is the point derivation $X_p(f)$ of Lecture 8. The same definition applies to smooth functions on any open subset of M.

For a diffeomorphism $f\colon M \to N$, the *push-forward field* f_*X on N is given by

$$(f_*X)_q = \mathrm{d}f_{f^{-1}(q)}X_{f^{-1}(q)}.$$

The inverse selects exactly one starting point for each q. For a general smooth map f, fields X on M and Y on N are *f-related* when $\mathrm{d}f_pX_p = Y_{f(p)}$ for every $p \in M$. A field need not descend under a general map. For $f\colon \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$ and $X = \partial_x$, the image vector is $2x\partial_y$ at $y = x^2$. The starting points 1 and -1 both map to 1 but give opposite nonzero vectors there. Thus no f-related field Y exists on the target. Lecture 10 compares this restriction with the pullback of forms.

9.2 Integral curves and maximal flows

An *integral curve* of X is a smooth curve $\gamma\colon I \to M$ on an open time interval satisfying $\gamma'(t) = X_{\gamma(t)}$. For initial point p, impose $0 \in I$ and $\gamma(0) = p$. In coordinates this is the

ordinary differential equation (ODE)

$$\dot{x}^i(t) = X^i(x(t)).$$

A curve is *maximal* if it has no extension to a larger open time interval as a solution with that initial condition. The resulting solution map is the *maximal flow*, written $\varphi\colon \mathcal{D} \to M$, with $\varphi_t(p) = \gamma_p(t)$. Its time domain at p is the interval $I_p = \{t : (t,p) \in \mathcal{D}\}$. A field is *complete* if $I_p = \mathbb{R}$ for every $p \in M$.

The *support* of a field is $\operatorname{supp} X = \overline{\{p : X_p \neq 0\}}$. The field has *compact support* when this closed set is compact.

Theorem 9.1 (Flow properties · *cited; laws and completeness proved from the cited flow properties*)**.** *A smooth vector field has a unique smooth maximal flow on an open set $\mathcal{D} \subseteq \mathbb{R} \times M$ containing $\{0\} \times M$. The time domains I_p are intervals. Its time maps satisfy*

$$\varphi_0 = \mathrm{id}, \qquad \varphi_{s+t}(p) = \varphi_s(\varphi_t(p))$$

whenever both sides are defined. Each time map is a diffeomorphism between its open source and image, with inverse φ_{-t}. Every smooth field with compact support is complete.

Existence and uniqueness of maximal curves, openness of $\mathcal{D}$ and joint smoothness in (t,p) are cited from Lee, second edition, Theorem 9.12. Its local input is the ODE theorem in charts. We prove the laws and completeness from these properties.

Proof. First fix $t \in I_p$ and put $q = \varphi_t(p)$. The curve $u \mapsto \varphi_{t+u}(p)$ is an integral curve starting at q. Uniqueness identifies it with $u \mapsto \varphi_u(q)$ on their common interval. Maximality gives $I_q = I_p - t$: either larger interval would, by shifting back and using uniqueness, extend the other solution. This proves the group law and ensures that $\varphi_{-t}(q) = p$. Define $M_t = \{p \in M : t \in I_p\}$, open because $\mathcal{D}$ is open. Applying the inverse identity with $-t$ shows that $\varphi_t\colon M_t \to M_{-t}$ is onto and both inverse compositions are identities. Joint smoothness gives smooth time maps and inverses.

Next let $K = \operatorname{supp} X$ be compact. If K is empty, the field is zero and every solution is constant. Otherwise, openness of $\mathcal{D}$ gives, near each $p \in K$, an open set U_p and a time $\varepsilon_p > 0$ for which the flow is defined on $(-\varepsilon_p, \varepsilon_p) \times U_p$. Choose a finite subcover of K and let $\delta > 0$ be the minimum of these finitely many times. Solutions starting in K exist for at least time δ in both directions. Outside K, the constant curve is a solution and uniqueness makes it the maximal solution. Thus every starting point has a solution for $|t| < \delta$. If a maximal interval had a finite right endpoint b, restart the solution at a time $t < b$ with $b - t < \delta$. The new solution extends past b, contradicting maximality and uniqueness. The same argument at a finite left endpoint proves completeness. ∎

On $\mathbb{R}$, the field $X = x^2\partial_x$ has maximal flow

$$\varphi_t(x) = \frac{x}{1 - tx}, \qquad \mathcal{D} = \{(t,x) : 1 - tx > 0\}.$$

Differentiating gives $\partial_t \varphi_t = \varphi_t^2$, and the initial value is x. At $x = 0$ the solution is constant for all time. For $x > 0$ it diverges at $t = 1/x$. For $x < 0$ the divergence is at that negative time. These intervals are maximal because a continuous real-valued extension cannot pass through a divergence. Requiring only a nonzero denominator would include a separate branch beyond the pole. This field has noncompact support. Lecture 21 contrasts this example with smooth linear transport equations, whose solutions extend across each finite path interval. The graph shows the failure at $x = 1$.

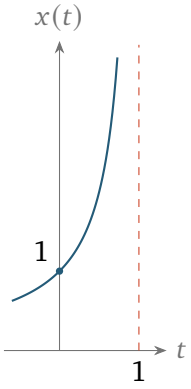


Figure 9.1: The solution starting at $x = 1$ diverges as t approaches 1 from below. The rational expression beyond the pole is not part of this integral curve.

9.3 The Lie bracket and matrix signs

For fields X and Y, define their *Lie bracket* by its action on smooth functions:

$$[X, Y]f = X(Yf) - Y(Xf).$$

The first composition expands as

$$X(Yf) = X^i(\partial_i Y^j)\partial_j f + X^i Y^j \partial_i \partial_j f.$$

In the second composition, swap the indices i and j in its second-derivative term. Equality of mixed partial derivatives makes those terms cancel. The result is

$$[X, Y] = \left(X^i \partial_i Y^j - Y^i \partial_i X^j\right)\partial_j.$$

The coefficients are smooth, so this expression is a smooth vector field. Its action on functions is intrinsic, making it independent of coordinates. Bilinearity, antisymmetry and Jacobi follow from the commutator calculation for operators, exactly as in Lecture 8.

The bracket preserves related fields under any smooth map f. Suppose X_j is f-related to Y_j for $j = 1, 2$. For every local smooth function h on the target, the chain rule gives $X_j(h \circ f) = (Y_j h) \circ f$. Applying it twice yields

$$[X_1, X_2](h \circ f) = ([Y_1, Y_2]h) \circ f.$$

Testing coordinate functions proves that the brackets are f-related. For a diffeomorphism this says $[f_* X, f_* Y] = f_*[X, Y]$. For a Lie group, write $\ell_g(h) = gh$ for left translation. A field is *left-invariant* when $(\ell_g)_* X = X$ for all g. Here a *matrix Lie group* means an embedded matrix subgroup with the smooth structure of Lecture 8. For tangent matrices, the formula $A^L(g) = gA$ follows by differentiating left multiplication.

Proposition 9.2 (Linear fields and left-invariant fields). *For $A, B \in \mathbb{R}^{n\times n}$, the fields $X_A(x) = Ax$ on $\mathbb{R}^n$ satisfy*

$$[X_A, X_B] = X_{-[A,B]}, \qquad [A, B] = AB - BA.$$

For tangent matrices A and B at the identity of a matrix Lie group, the left-invariant fields $A^L(g) = gA$ instead satisfy $[A^L, B^L] = [A, B]^L$.

Proof. For linear fields, $DX_A = A$ and $DX_B = B$. The coordinate formula gives $[X_A, X_B](x) = BAx - ABx$, proving the minus sign. The group fields are related by the inclusion i to the ambient fields $g \mapsto gA$ and $g \mapsto gB$. Their ambient derivatives in direction K are KA and KB. The related-fields result identifies their ambient bracket with the inclusion of the group bracket. Thus

$$\mathrm{d}i_g([A^L, B^L]_g) = (gA)B - (gB)A = g[A, B].$$

The inclusion has injective differential, so this identifies the group bracket. ∎

The operations act on different sides of their matrix arguments. They therefore give different signs when the coordinate bracket is computed.

For an abstract Lie group, each $A \in T_eG$ determines a left-invariant field by $A^L(g) = \mathrm{d}(\ell_g)_e A$. Smoothness follows by differentiating the smooth multiplication map in its second argument. Associativity and the chain rule give

$$(\ell_g)_* A^L(h) = \mathrm{d}(\ell_g)_h \mathrm{d}(\ell_h)_e A = \mathrm{d}(\ell_{gh})_e A = A^L(gh).$$

This proves invariance, and evaluation at the identity recovers A. Conversely invariance forces this formula, so the correspondence is a linear bijection. Naturality of the bracket makes $[A^L, B^L]$ left-invariant. Define $[A, B] = [A^L, B^L](e)$. This gives T_eG its Lie algebra structure. For matrix groups it agrees with the commutator just computed.

For example, the group of maps $r \mapsto ar+b$, where $a > 0$, has law $(a, b)(a', b') = (aa', b+ab')$. Set $H = \partial_a$ and $N = \partial_b$ at the identity $(1, 0)$. Differentiating left translation gives $H^L = a\partial_a$ and $N^L = a\partial_b$. The coordinate bracket gives $[H^L, N^L] = a\partial_b = N^L$.

The integral curve of A^L starting at the identity is a *one-parameter subgroup*: a smooth map $c\colon \mathbb{R} \to G$ with $c(s + t) = c(s)c(t)$ and $c(0) = e$. To prove this, take a local solution on $(-\delta, \delta)$ starting at the identity. Left translation gives a solution on that same interval starting at every group element. The uniform-time continuation argument above proves completeness. Left invariance and uniqueness identify the two curves $t \mapsto c(s + t)$ and $t \mapsto c(s)c(t)$, proving the group law. Conversely, differentiating a one-parameter subgroup's law makes it an integral curve of A^L, with $A = c'(0)$. For matrices the solution is $c(t) = e^{tA}$, where

$$e^{tA} = \sum_{k=0}^{\infty} \frac{t^k A^k}{k!}.$$

If each $|A_{ij}| \le m$, multiplication of matrices gives $|(A^k)_{ij}| \le n^{k-1} m^k$ for $k \ge 1$, by induction. The resulting scalar exponential bounds the series and its differentiated series uniformly on bounded time intervals. Termwise differentiation gives $c'(t) = c(t)A$ and $c(0) = I$. The group-valued curve, viewed through the inclusion, solves this ambient equation. Ambient uniqueness identifies it with the series. The full field flow is $g \mapsto ge^{tA}$.

9.4 The Lie derivative of a field

Vectors at different points cannot be subtracted intrinsically. For $t \in I_p$, the map $\varphi_{-t}\colon M_{-t} \to M_t$ is a diffeomorphism and $\varphi_t(p) \in M_{-t}$. Thus it pushes $Y_{\varphi_t(p)}$ back to p by $\mathrm{d}\varphi_{-t}|_{\varphi_t(p)}$. The *Lie derivative* is the resulting derivative in T_pM:

$$(L_X Y)_p = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_0 ((\varphi_{-t})_* Y)_p.$$

For functions, define $L_X f = Xf$, the derivative of $f(\varphi_t(p))$ at zero.

Theorem 9.3 (Flow derivative and bracket · *proved given local flow existence and smooth dependence*)**.** *The bracket $[X,Y]$ is a smooth vector field. The Lie derivative exists and satisfies $L_X Y = [X,Y]$.*

Proof. The bracket's coordinate formula above proves its smoothness. In a coordinate neighbourhood, the ODE gives $\varphi_t(p) = p + tX(p) + O(t^2)$. Differentiating the ODE $\partial_t \varphi_t = X(\varphi_t)$ in p, as joint smoothness permits, gives $\partial_t D\varphi_t = DX(\varphi_t) D\varphi_t$. Hence $D\varphi_t(p) = I + tDX_p + O(t^2)$. Inverting gives the inverse Jacobian $I - tDX_p + O(t^2)$ at $\varphi_t(p)$. Thus

$$\begin{aligned}((\varphi_{-t})_* Y)_p &= \big(I - tDX_p + O(t^2)\big)\big(Y_p + tDY_p X_p + O(t^2)\big) \\ &= Y_p + t\big(DY_p X_p - DX_p Y_p\big) + O(t^2).\end{aligned}$$

The coefficient of t is the bracket's coordinate formula. The definition by intrinsic push-forward makes this computation independent of the chosen neighbourhood. ∎

This is Tu, Theorem 20.4, with the same backward-push convention. Forward push would give the opposite sign.

9.5 Commuting complete flows

Suppose X and Y have flows φ_t and ψ_s. The fields *commute* when $[X,Y] = 0$. Their complete flows commute when $\varphi_t \circ \psi_s = \psi_s \circ \varphi_t$ for every $s, t \in \mathbb{R}$.

Theorem 9.4 (Commuting flows · *proved given flow existence, uniqueness and smooth dependence*)**.** *For two complete smooth fields, their bracket vanishes if and only if their flows commute for all real times.*

Proof. The flow group law, the push-forward chain rule $(f \circ g)_* = f_* g_*$ and Theorem 9.3 give

$$\frac{\mathrm{d}}{\mathrm{d}t}(\varphi_{-t})_* Y = (\varphi_{-t})_* [X,Y].$$

Indeed, write the value at $t + h$ as $(\varphi_{-t})_*((\varphi_{-h})_* Y)$ and differentiate in h at zero. If the bracket is zero, this family is constant and equals Y. Hence $(\varphi_t)_* Y = Y$ for every t. For fixed t, both curves $s \mapsto \varphi_t(\psi_s(p))$ and $s \mapsto \psi_s(\varphi_t(p))$ solve the Y equation with the same initial point. Uniqueness makes them equal. Completeness gives the equality for all $s, t \in \mathbb{R}$.

Conversely, differentiate commuting flows in s at zero. The result is $\mathrm{d}\varphi_t|_p Y_p = Y_{\varphi_t(p)}$, so $(\varphi_{-t})_* Y = Y$. Differentiating in t at zero gives $[X,Y] = 0$. ∎

The coordinate fields ∂_x and ∂_y translate in independent directions, giving the rectangle in the figure. By contrast ∂_x and $x\partial_y$ have bracket ∂_y. Running their flows for times $u, v, -u, -v$ in that order ends at $(x, y + uv)$, rather than (x, y).

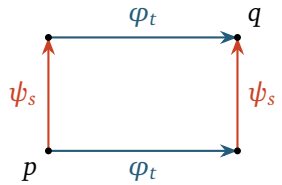


Figure 9.2: Translations in the two coordinate directions commute. The two orders reach the same opposite corner of the rectangle.

9.6 Distributions and Frobenius's theorem

A *smooth rank-k distribution* assigns a k-dimensional subspace $D_p \subseteq T_pM$ to every point, locally spanned by k independent smooth fields. It is *involutive* when the bracket of any two local fields taking values in D also takes values in D. An *integral manifold* is an immersion $i\colon S \to M$ satisfying

$$\mathrm{d}i_s(T_sS) = D_{i(s)} \qquad (s \in S).$$

Theorem 9.5 (Frobenius · *cited*)**.** *Every smooth involutive distribution of constant rank k has, near each point, coordinates in which it is spanned by $\partial_1, \dots, \partial_k$. Fixing the other coordinates gives local integral manifolds.*

This is Lee, second edition, Theorem 19.12. For example, the planes spanned by ∂_x and ∂_y in $\mathbb{R}^3$ have integral surfaces $z = c$. Involutivity checks consistency of prescribed tangent directions. The constant-rank hypothesis is part of the theorem.

9.7 Rotation fields and angular momentum

In Cartesian coordinates on $\mathbb{R}^3$, the rotation fields are

$$L_x = y\partial_z - z\partial_y, \qquad L_y = z\partial_x - x\partial_z, \qquad L_z = x\partial_y - y\partial_x.$$

For example, $L_x(x^2 + y^2 + z^2) = 2yz - 2zy = 0$. Cyclic permutation gives the same result for the other fields. Thus each is tangent to every sphere centred at zero. Its integral curves retain that radius. Here L_x names a field, distinct from the operator L_X acting on fields.

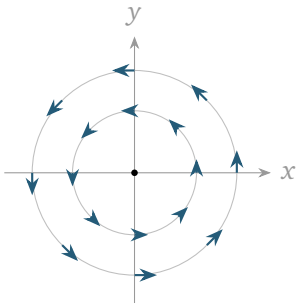


Figure 9.3: The field $L_z = (-y, x, 0)$ has circular trajectories in each horizontal plane. The arrows are its tangent velocities; the axis consists of fixed points.

Example 9.6 (Rotation flows and brackets)**.** The three flows are defined for all real t:

$$\begin{aligned}
\varphi^x_t(x, y, z) &= (x,\ y\cos t - z\sin t,\ y\sin t + z\cos t),\\
\varphi^y_t(x, y, z) &= (x\cos t + z\sin t,\ y,\ -x\sin t + z\cos t),\\
\varphi^z_t(x, y, z) &= (x\cos t - y\sin t,\ x\sin t + y\cos t,\ z).
\end{aligned}$$

For the first flow, differentiating its last two components gives $(-y\sin t - z\cos t,\ y\cos t - z\sin t)$, which is $(-z(t), y(t))$. At $t = 0$ the components are the initial coordinates. The same differentiation in the other coordinate planes verifies the other flows. Thus all three restrict to complete flows on S^2.

Their linear matrices are the basis matrices $A(1,0,0)$, $A(0,1,0)$ and $A(0,0,1)$ of Lecture 8. One bracket can also be read directly on coordinates: $[L_x, L_y]x = L_x(z) = y$, $[L_x, L_y]y =$

$-L_y(-z) = -x$ and $[L_x, L_y]z = 0$. Therefore

$$[L_x, L_y] = -L_z, \qquad [L_y, L_z] = -L_x, \qquad [L_z, L_x] = -L_y.$$

The matrices have the positive cyclic commutator, so this is precisely the linear-field minus sign of Proposition 9.2.

On complex smooth functions, define the angular-momentum differential operators $\widehat{L_a} = -\mathrm{i}\hbar L_a$ for $a \in \{x, y, z\}$, where $\hbar$ is the reduced Planck constant. Their operator commutator is

$$[\widehat{L_x}, \widehat{L_y}] = (-\mathrm{i}\hbar)^2(-L_z) = \hbar^2 L_z = \mathrm{i}\hbar\widehat{L_z}.$$

Thus the rotation-field sign agrees with the positive quantum angular-momentum relation. This is the differential-operator identification; no quantum representation theorem is assumed.

Self-check. Find the flow of $x\partial_y - y\partial_x$ on $\mathbb{R}^2$ and its value at time $\pi/2$ from $(1, 0)$. Include the trajectory from the origin.

Answer. The solution is $(x\cos t - y\sin t, x\sin t + y\cos t)$ for every $t \in \mathbb{R}$. From $(1, 0)$ it reaches $(0, 1)$ at time $\pi/2$, so positive time gives counter-clockwise rotation. The origin is a constant solution.

Summary. A vector field prescribes an ordinary differential equation in every chart. Uniqueness gives the flow group law, and compact support gives completeness. The Lie derivative of a field uses backward push-forward and equals its bracket with the generating field. Complete fields have zero bracket exactly when their flows commute. Linear fields Ax have the negative matrix commutator, whereas left-invariant matrix fields gA have the positive one. Lecture 10 introduces forms and their pullback under arbitrary smooth maps.

Reading. Tu, *An Introduction to Manifolds*, §14 and §20.2; Lee, *Introduction to Smooth Manifolds*, second edition, Theorems 9.12 and 19.12.

Lecture 10

Differential forms

Objectives. *A differential form assigns an alternating multilinear measurement to tangent vectors at each point. Forms pull back under every smooth map. Their exterior derivative extends the differential of a function and combines the Cartesian gradient, curl and divergence. We construct this derivative in coordinates, prove that it is independent of coordinates, and compute the sphere's area form.*

By the end of this lecture you will be able to:

- Evaluate forms with the determinant wedge convention.
- Compute exterior derivatives and pullbacks under smooth maps.
- Detect nonexactness of a closed 1-form by a nonzero loop integral.
- Derive both angular and stereographic expressions for sphere area.

10.1 Covectors and the cotangent bundle

Let M be a smooth n-manifold. The *cotangent space* at $p \in M$ is the dual vector space $T_p^*M = \mathrm{Hom}(T_pM, \mathbb{R})$. Its elements are *covectors*. For a smooth function $f\colon M \to \mathbb{R}$, its *differential* is given by

$$(\mathrm{d}f)_p(v) = v(f) \qquad (v \in T_pM).$$

This is the differential of Lecture 8 with $T_{f(p)}\mathbb{R}$ identified with $\mathbb{R}$ by its standard coordinate. In coordinates $x^1, \dots, x^n$, the covectors $\mathrm{d}x^i$ are dual to ∂_j, since $\mathrm{d}x^i(\partial_j) = \delta^i_j$. Thus a covector has a unique expression $\alpha = \alpha_i \mathrm{d}x^i$, and $\alpha(v) = \alpha_i v^i$. In particular,

$$\mathrm{d}f = \frac{\partial f}{\partial x^i}\mathrm{d}x^i.$$

The *cotangent bundle* is the disjoint union $T^*M = \coprod_{p\in M} T_p^*M$, with projection to p. Its local coordinates are $(x^1, \dots, x^n, \alpha_1, \dots, \alpha_n)$. If $y = y(x)$ has Jacobian $J = Dy$, invariance of $\alpha(v)$ requires the coefficient row to change by $\alpha' = \alpha J^{-1}$. These are smooth invertible bundle transitions. They give the cotangent manifold by the same construction as sketched for the tangent bundle in Lecture 8. The Hausdorff and countable-basis arguments there apply with these transitions.

A *smooth 1-form* is a smooth section of T^*M. Equivalently, its coefficients α_i are smooth in every chart. The form $\mathrm{d}f$ is smooth because the partial derivatives of f are smooth. A covector measures a tangent vector, as the figure illustrates.

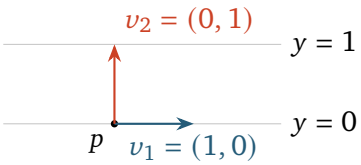


Figure 10.1: The horizontal lines are levels of the coordinate y. The covector $\mathrm{d}y$ gives 0 on the horizontal vector and 1 on the vertical vector.

For example, on $\mathbb{R}^2$ let $\alpha = 2x\mathrm{d}x + \mathrm{d}y$. At $(1, 0)$ it gives $\alpha(3\partial_x - \partial_y) = 6 - 1 = 5$. This form is the differential of $x^2 + y$.

10.2 Alternating tensors and the wedge product

A *tensor field of type* (r, s) assigns at each point a multilinear real-valued map with r cotangent arguments and s tangent arguments. Its coordinate coefficients must be smooth. A tensor with $r = 0$ is *covariant*. A *k-form* is a covariant k-tensor field that is alternating: interchanging two arguments reverses its sign. We write $\Omega^k(M)$ for the vector space of smooth k-forms. A 0-form is a smooth function. Alternation implies that every k-form is zero if $k > \dim M$.

Recall the exterior powers from *Linear Algebra*, Lecture 12, the wedge-and-determinant example.

Proposition 10.1 (Exterior powers · *imported definition; evaluation proved*)**.** *For a finite-dimensional real vector space V, the space $\Lambda^k V^*$ consists of alternating k-linear forms on V. For covectors $\alpha_1, \dots, \alpha_k$, the determinant convention gives*

$$(\alpha_1 \wedge \cdots \wedge \alpha_k)(v_1, \dots, v_k) = \det\big(\alpha_i(v_j)\big)_{i,j=1}^{k}.$$

Proof of the evaluation formula. LA12 identifies covariant tensors with multilinear maps. Its antisymmetric tensors are therefore alternating maps. Evaluating its unnormalized antisymmetrization gives the sum $\sum_\pi \operatorname{sgn}(\pi) \prod_j \alpha_{\pi(j)}(v_j)$. The Leibniz determinant formula identifies this with the stated determinant. ∎

For an increasing index tuple $I = (i_1 < \cdots < i_k)$, write $\mathrm{d}x^I = \mathrm{d}x^{i_1} \wedge \cdots \wedge \mathrm{d}x^{i_k}$. Every k-form has a unique expansion

$$\omega = \sum_I a_I \mathrm{d}x^I, \qquad a_I = \omega(\partial_{i_1}, \dots, \partial_{i_k}).$$

To prove this, the determinant gives $\mathrm{d}x^I(\partial_{j_1}, \dots, \partial_{j_k}) = \delta^I_J$ for increasing tuples I and J. This forces the displayed coefficients. Conversely, subtracting the displayed sum from ω gives an alternating form vanishing on every increasing tuple of coordinate vectors. Alternation makes it zero on every ordered tuple, including repeated indices. Multilinearity then makes it zero on arbitrary vectors. This proves existence and uniqueness of the expansion. The sum runs over increasing tuples only, so it has no factorial. If instead every index ranges independently from 1 to n, set $\omega_{i_1 \cdots i_k} = \omega(\partial_{i_1}, \dots, \partial_{i_k})$. Then

$$\omega = \frac{1}{k!} \sum_{i_1, \dots, i_k} \omega_{i_1 \cdots i_k} \mathrm{d}x^{i_1} \wedge \cdots \wedge \mathrm{d}x^{i_k}.$$

Each increasing tuple occurs $k!$ times with compensating signs in the component and basis wedge, which explains the factorial.

The *wedge product* of forms of degrees k and ℓ is the bilinear extension of concatenation of these basis wedges. A repeated index gives zero; otherwise sorting the concatenated indices gives its permutation sign. This definition agrees with the determinant convention. Intrinsically it is

$$(\alpha \wedge \beta)(v_1, \dots, v_{k+\ell}) = \frac{1}{k!\ell!} \sum_{\pi \in S_{k+\ell}} \operatorname{sgn}(\pi) \alpha(v_{\pi(1)}, \dots, v_{\pi(k)}) \beta(v_{\pi(k+1)}, \dots, v_{\pi(k+\ell)}).$$

Indeed, grouping the permutations by which k arguments enter α leaves $k!\ell!$ identical contributions within each group. By bilinearity take $\alpha = \mathrm{d}x^I$ and $\beta = \mathrm{d}x^J$. Both the permutation expression and the sorted basis wedge are alternating in their arguments, so the preceding basis test reduces their comparison to increasing tuples of coordinate vectors. Only shuffles placing the first index block in its matching slots contribute. They give its sorting sign; a repeated index gives zero. Thus the formula equals the sorting rule. The intrinsic formula proves that wedge is independent of coordinates and has smooth coefficients. Concatenating three index lists proves associativity: both parenthesizations have the same sorting sign, or both vanish because an index repeats. Moving a block of k indices past a block of ℓ indices makes $k\ell$ swaps. Bilinearity therefore gives graded commutativity:

$$\alpha \wedge \beta = (-1)^{k\ell} \beta \wedge \alpha.$$

For example, $(\mathrm{d}x + 2\mathrm{d}y) \wedge (3\mathrm{d}x - \mathrm{d}y) = -7\mathrm{d}x \wedge \mathrm{d}y$. Evaluating this on $(2, 1)$ and $(1, -1)$ gives 21. For variable coefficients the same rule gives

$$(x\mathrm{d}x + y\mathrm{d}y) \wedge (z\mathrm{d}x + x\mathrm{d}y) = (x^2 - yz)\mathrm{d}x \wedge \mathrm{d}y.$$

For two orthonormal one-particle states, *QM-I*, Lecture 14, writes the normalized antisymmetric tensor as $(u \otimes v - v \otimes u)/\sqrt{2}$. That state convention includes a norm factor absent from the determinant wedge on covectors.[1] As a tensor, our two-factor wedge is $\alpha \otimes \beta - \beta \otimes \alpha$. For k factors the normalized state uses $1/\sqrt{k!}$ times LA12's unnormalized antisymmetrization. This states the tensor-factor translation without identifying states with covectors. Here $(\mathrm{d}x \wedge \mathrm{d}y)(\partial_x, \partial_y) = 1$. This value also fixes the normalization used by the Hodge operator in Lecture 15.

10.3 The exterior derivative

The *exterior derivative* raises form degree by one. On a coordinate chart define it by

$$\mathrm{d}\omega = \sum_I \sum_j (\partial_j a_I) \mathrm{d}x^j \wedge \mathrm{d}x^I \quad \text{when} \quad \omega = \sum_I a_I \mathrm{d}x^I.$$

On functions this agrees with the differential already defined.

Theorem 10.2 (Exterior derivative · *sketched; existence and identities proved*)**.** *The coordinate formula defines a coordinate-independent real-linear map* $\mathrm{d}\colon \Omega^k(M) \to \Omega^{k+1}(M)$*. It satisfies* $\mathrm{d}^2 = 0$ *and, for* $\alpha \in \Omega^k(M)$*,*

$$\mathrm{d}(\alpha \wedge \beta) = \mathrm{d}\alpha \wedge \beta + (-1)^k \alpha \wedge \mathrm{d}\beta.$$

These properties and its value on functions determine it uniquely.

[1] The source is QM-I, Lecture 14, Many fermions: Slater determinants. Its tensor normalization does not change the present wedge.

Proof of existence and identities. Work first in one chart. The formula $\mathrm{d}(b\mathrm{d}x^J) = \mathrm{d}b \wedge \mathrm{d}x^J$ also holds for a tuple J that is not increasing: reordering changes both sides by the same sign, and a repeated index makes both sides zero. Applying d twice gives terms $(\partial_r \partial_j a_I)\mathrm{d}x^r \wedge \mathrm{d}x^j \wedge \mathrm{d}x^I$. Terms with $r = j$ vanish. Each remaining term cancels its partner with j and r exchanged, because mixed derivatives agree and wedge is antisymmetric. Thus $\mathrm{d}^2 = 0$.

For the product rule, expand into terms $\alpha = a\mathrm{d}x^I$ and $\beta = b\mathrm{d}x^J$, where $|I| = k$. The ordinary product rule differentiates ab into $b\mathrm{d}a + a\mathrm{d}b$. The first contribution is $\mathrm{d}\alpha \wedge \beta$. In the second, moving $\mathrm{d}b$ past the k factors of $\mathrm{d}x^I$ gives $(-1)^k \alpha \wedge \mathrm{d}\beta$. Summing proves the identity.

Now let $y = y(x)$ be another chart. The chain rule gives $\mathrm{d}y^a = (\partial_i y^a)\mathrm{d}x^i$. The operator defined in x coordinates annihilates $\mathrm{d}y^a$ by the same mixed-derivative cancellation. For a coefficient function b, the chain rule also identifies its x and y differentials. Apply the product rule just proved to a monomial $b\mathrm{d}y^{i_1} \wedge \cdots \wedge \mathrm{d}y^{i_k}$. Only the differentiated coefficient survives, so the result is precisely the y coordinate formula. Such monomials span every form. Thus the definitions agree on chart overlaps and give a global smooth form. ∎

Sketch of uniqueness. Let D have the stated properties. A smooth *cutoff function* equals 1 near a chosen point and 0 outside a chosen chart neighbourhood. Such functions are constructed in a smaller coordinate ball using $e^{-1/t}$ for $t > 0$, extended by zero for $t \le 0$. If α vanishes near p, choose a cutoff χ supported where $\alpha = 0$. Since $\chi\alpha = 0$, the product rule at p gives $(D\alpha)_p = 0$. Thus D depends only on the form near p. Use a cutoff to extend local coefficient and coordinate functions smoothly to M, without changing them near p. The identity $D\mathrm{d}f = D^2 f = 0$ for these extensions and the product rule force $D(\sum_I a_I \mathrm{d}x^I) = \sum_I \mathrm{d}a_I \wedge \mathrm{d}x^I$ at p. This proves uniqueness, with the cutoff construction left at sketch level. See Tu, §19.4. ∎

For example, $\mathrm{d}(x\mathrm{d}y) = \mathrm{d}x \wedge \mathrm{d}y$, while $\mathrm{d}(x\mathrm{d}x) = \mathrm{d}x \wedge \mathrm{d}x = 0$.

10.4 Pullback under a smooth map

Let $f\colon M \to N$ be any smooth map. For $\omega \in \Omega^k(N)$ its *pullback* is given by

$$(f^*\omega)_p(v_1, \ldots, v_k) = \omega_{f(p)}(\mathrm{d}f_p v_1, \ldots, \mathrm{d}f_p v_k).$$

For functions, set $f^*h = h \circ f$. The differential sends each argument to the required target tangent space. No inverse of f or of its differential is needed.

Theorem 10.3 (Pullback calculus). *Pullback takes smooth forms to smooth forms and satisfies*

$$f^*(\alpha \wedge \beta) = f^*\alpha \wedge f^*\beta, \qquad \mathrm{d}(f^*\alpha) = f^*(\mathrm{d}\alpha).$$

For smooth $g\colon N \to P$ it also satisfies $(g \circ f)^* = f^*g^*$ and $\mathrm{id}^* = \mathrm{id}$.

Proof. Substituting $\mathrm{d}f_p v_j$ in the permutation formula proves wedge compatibility. The chain rule gives $f^*\mathrm{d}h = \mathrm{d}(h \circ f)$. In local target coordinates y, write $f^a = y^a \circ f$. The pullback of $\alpha = \sum_I a_I \mathrm{d}y^I$ is

$$f^*\alpha = \sum_I (a_I \circ f)\mathrm{d}f^{i_1} \wedge \cdots \wedge \mathrm{d}f^{i_k}.$$

Its coefficients are smooth. Differentiating this expression leaves only $\mathrm{d}(a_I \circ f)$, since $\mathrm{d}^2 f^a = 0$. Pulling back $\mathrm{d}\alpha$ gives the same expression by the chain rule on coefficient functions. This proves commutation with d without using the uniqueness sketch. Finally, substituting $\mathrm{d}(g \circ f) = \mathrm{d}g \circ \mathrm{d}f$ into the definition proves the composition rule, and identity differentials prove the identity rule. ∎

For $f\colon \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$, pullback gives $f^*(\mathrm{d}y) = 2x\mathrm{d}x$, including at $x = 0$. Lecture 9 showed that ∂_x does not descend under this same map. Forms pull back even when a map loses rank or identifies distinct points.

10.5 Gradient, curl and divergence

Fix Cartesian coordinates on Euclidean $\mathbb{R}^3$. Identify a vector (A_1, A_2, A_3) with the 1-form $\alpha = A_1\mathrm{d}x + A_2\mathrm{d}y + A_3\mathrm{d}z$. Identify (B_1, B_2, B_3) with the 2-form $\beta = B_1\mathrm{d}y\wedge\mathrm{d}z + B_2\mathrm{d}z\wedge\mathrm{d}x + B_3\mathrm{d}x\wedge\mathrm{d}y$. Identify a scalar g with the 3-form $g\mathrm{d}x\wedge\mathrm{d}y\wedge\mathrm{d}z$. These are specified coordinate identifications.

Proposition 10.4 (Vector calculus from forms). *Under these identifications, exterior differentiation gives gradient, curl and divergence in successive degrees.*

Proof. The defining formula gives the following coefficients:

$$\begin{aligned}
\mathrm{d}f &= (\partial_x f)\mathrm{d}x + (\partial_y f)\mathrm{d}y + (\partial_z f)\mathrm{d}z,\\
\mathrm{d}\alpha &= (\partial_y A_3 - \partial_z A_2)\mathrm{d}y\wedge\mathrm{d}z + (\partial_z A_1 - \partial_x A_3)\mathrm{d}z\wedge\mathrm{d}x\\
&\quad + (\partial_x A_2 - \partial_y A_1)\mathrm{d}x\wedge\mathrm{d}y,\\
\mathrm{d}\beta &= (\partial_x B_1 + \partial_y B_2 + \partial_z B_3)\mathrm{d}x\wedge\mathrm{d}y\wedge\mathrm{d}z.
\end{aligned}$$

For example, the $\mathrm{d}y\wedge\mathrm{d}z$ coefficient receives $\partial_y A_3$ from $\mathrm{d}(A_3\mathrm{d}z)$ and $-\partial_z A_2$ from $\mathrm{d}(A_2\mathrm{d}y)$. Cyclic permutation gives the other two coefficients. For $\mathrm{d}\beta$, repeated factors vanish and each surviving cyclic order has positive sign. The displayed coefficients are the Cartesian definitions of gradient, curl and divergence. ∎

Applying $\mathrm{d}^2 = 0$ in degrees zero and one respectively gives

$$\operatorname{curl}\operatorname{grad} f = 0, \qquad \operatorname{div}\operatorname{curl} A = 0.$$

10.6 Closed and exact forms

A form ω is *closed* if $\mathrm{d}\omega = 0$. A k-form with $k \geq 1$ is *exact* if it equals $\mathrm{d}\eta$ for some $(k-1)$-form η. Every exact form is closed by $\mathrm{d}^2 = 0$. For a smooth 1-form α on M and a smooth path $\gamma\colon [a,b] \to M$, define the line integral by

$$\int_\gamma \alpha = \int_a^b \alpha_{\gamma(t)}(\gamma'(t))\,\mathrm{d}t.$$

Here a smooth path on $[a,b]$ means the restriction of a smooth curve on an open interval containing $[a,b]$. The pullback definition gives $\gamma^*\alpha = \alpha_{\gamma(t)}(\gamma'(t))\mathrm{d}t$, so this is also the integral of the pullback's coefficient. If $\alpha = \mathrm{d}f$, the chain rule and the fundamental theorem of calculus give $\int_\gamma \mathrm{d}f = f(\gamma(b)) - f(\gamma(a))$.

Proposition 10.5 (A closed form that is not exact). *On $\mathbb{R}^2 \setminus \{0\}$ the angular form*

$$\omega_{\mathrm{ang}} = \frac{x\mathrm{d}y - y\mathrm{d}x}{x^2 + y^2}$$

is closed but not exact.

Proof. Its exterior derivative has $dx \wedge dy$ coefficient

$$\partial_x \frac{x}{x^2+y^2} + \partial_y \frac{y}{x^2+y^2} = \frac{2}{x^2+y^2} - \frac{2(x^2+y^2)}{(x^2+y^2)^2} = 0.$$

For $\gamma(t) = (\cos t, \sin t)$, $0 \le t \le 2\pi$, substitution gives $\gamma^* \omega_{\mathrm{ang}} = dt$. Its integral is 2π. If it were exact, the endpoint formula would give zero on this closed path. This contradiction proves nonexactness. ∎

For a smooth loop $\gamma(t) = r(t)(\cos\phi(t), \sin\phi(t))$, where $r(t) > 0$, differentiation gives $x\dot{y} - y\dot{x} = r^2\dot{\phi}$. Thus $\gamma^*\omega_{\mathrm{ang}} = \dot{\phi}dt$. Lecture 3 lifts $\gamma/|\gamma|$ in turns; multiplying that lift by 2π gives ϕ. The angle ϕ is smooth because it locally differs from a smooth angle branch along γ by a constant multiple of 2π. The integral divided by 2π is therefore the winding number. The same computation with variable r and ϕ gives $\omega_{\mathrm{ang}} = d\phi$ on an angle chart. There is no global smooth angle function on the punctured plane.

10.7 The area form of the sphere

The inclusion $i\colon S^2 \to \mathbb{R}^3$ pulls the ambient form $\beta = x\,dy \wedge dz + y\,dz \wedge dx + z\,dx \wedge dy$ back to $\omega_{S^2} = i^*\beta$. At $p \in S^2$, for $v, w \in T_pS^2$, it evaluates as

$$(\omega_{S^2})_p(v, w) = \det[p\ v\ w] = p \cdot (v \times w).$$

This is signed area measured with the outward unit normal p. The next figure shows the sign test at the south pole for the north-pole stereographic chart of Lecture 7.

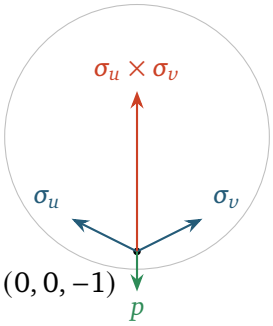


Figure 10.2: Viewed from below the equator, the north chart's tangents at the south pole have cross product $+4e_3$. The outward normal is $-e_3$.

Example 10.6 (Angular and stereographic area)**.** For angular parameters put $s(\theta, \phi) = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$, with $0 < \theta < \pi$ and ϕ in an open interval of length less than 2π. Its coordinate tangents are

$$s_\theta = (\cos\theta\cos\phi, \cos\theta\sin\phi, -\sin\theta), \qquad s_\phi = (-\sin\theta\sin\phi, \sin\theta\cos\phi, 0).$$

Their cross product is $s_\theta \times s_\phi = \sin\theta\, s$, so

$$s^*\omega_{S^2} = \sin\theta\, d\theta \wedge d\phi.$$

For the inverse stereographic map from the north pole, set $D = 1 + u^2 + v^2$ and $\sigma(u, v) = (2u, 2v, u^2 + v^2 - 1)/D$. Its derivative matrix is

$$(\sigma_u\ \sigma_v) = \frac{1}{D^2}\begin{pmatrix} 2(D - 2u^2) & -4uv \\ -4uv & 2(D - 2v^2) \\ 4u & 4v \end{pmatrix}.$$

The cross product of these columns is

$$\sigma_u \times \sigma_v = \frac{(-8uD, -8vD, 4D(1-u^2-v^2))}{D^4} = -\frac{4\sigma}{D^2}.$$

Since $\sigma \cdot \sigma = 1$, the pullback is

$$\sigma^* \omega_{S^2} = -\frac{4}{(1+u^2+v^2)^2} \mathrm{d}u \wedge \mathrm{d}v.$$

At $(u, v) = (0, 0)$ the point is $(0, 0, -1)$, while the tangent columns are $2e_1$ and $2e_2$. Their cross product points opposite to the outward normal. This verifies the negative sign. Lecture 12 formalizes this comparison as orientation reversal.

Self-check. Compute $\mathrm{d}(x\mathrm{d}y \wedge \mathrm{d}z)$ on $\mathbb{R}^3$ and its pullback to S^2.

Answer. The derivative is $\mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z$. Its pullback is zero: an alternating trilinear form vanishes on the two-dimensional tangent space of the sphere.

Summary. Forms alternate in their tangent arguments and use the determinant wedge. The exterior derivative is independent of coordinates, squares to zero and satisfies the graded product rule. Pullback exists under every smooth map and commutes with this derivative. A closed form need not be exact, as the angular form's nonzero loop integral proves. The sphere's outward area has negative coefficient in the north-pole stereographic coordinates. Lecture 11 differentiates forms along flows.

Reading. Tu, *An Introduction to Manifolds*, §§17–19; *Linear Algebra*, Lecture 12, exterior powers and the determinant wedge.

Lecture 11

Cartan's calculus

Objectives. *Pullback compares a form at a moving point with a form at the starting point. Differentiating this comparison gives its Lie derivative. Cartan's formula expresses that derivative using exterior differentiation and insertion of a vector field. We prove the formula, including its time-dependent version, and apply it to volume preservation and Hamiltonian motion.*

By the end of this lecture you will be able to:

- Compute contraction with its first-slot sign convention.
- Prove Cartan's formula and differentiate a time-dependent pullback.
- Determine whether a local flow preserves a chosen volume form.
- Derive Liouville's theorem with the fixed Hamiltonian signs.

11.1 Contraction with a vector field

Let X be a smooth vector field on a smooth manifold M. For a k-form α with $k \geq 1$, its *interior product*, or *contraction*, with X fills the first argument:

$$(\iota_X \alpha)_p(v_1, \ldots, v_{k-1}) = \alpha_p(X_p, v_1, \ldots, v_{k-1}).$$

For a 1-form this is the function $\alpha(X)$. For a function f set $\iota_X f = 0$. The resulting form is smooth because its coordinate coefficients are sums of products of smooth coefficients of X and α.

Theorem 11.1 (Contraction rules). *Contraction is real-linear, lowers degree by one, and satisfies*

$$\iota_X(\alpha \wedge \beta) = (\iota_X \alpha) \wedge \beta + (-1)^p \alpha \wedge \iota_X \beta, \qquad \iota_X^2 = 0,$$

where α has degree p. It is also linear over smooth functions in both X and the form.

Proof. For 1-forms $\alpha_1, \ldots, \alpha_k$, expand the determinant defining their wedge along its first column. This gives

$$\iota_X(\alpha_1 \wedge \cdots \wedge \alpha_k) = \sum_{j=1}^{k} (-1)^{j-1} \alpha_j(X) \alpha_1 \wedge \cdots \wedge \widehat{\alpha_j} \wedge \cdots \wedge \alpha_k.$$

The hat means that the factor is omitted. For the wedge of a block of p factors and a second block, split this sum according to the block containing the omitted factor. Omissions in the second block acquire $(-1)^p$. This is the stated product rule. Basis wedges span all forms, so bilinearity extends the identity to arbitrary forms. Applying contraction twice inserts X twice. Alternation makes this zero; on degrees zero and one it is zero by definition. Pointwise multilinearity proves both function-linear assertions and real-linearity. ∎

For example, $\iota_{\partial_x}(dx \wedge dy) = dy$. For the radial field $X = x\partial_x + y\partial_y$,

$$\iota_X(dx \wedge dy) = x\,dy - y\,dx.$$

Filling the last argument instead would reverse this sign.

11.2 Lie derivatives of forms and tensors

Let φ be the maximal local flow of X, supplied by Theorem 9.1. Near each $p \in M$, its maps are defined for all sufficiently small times. Pullback gives a form in the fixed exterior space at p, as in the figure.

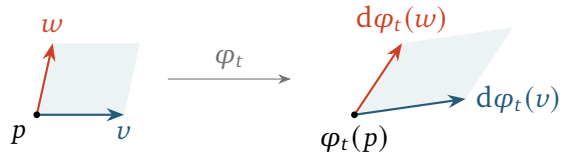


Figure 11.1: Pullback evaluates the form at $\varphi_t(p)$ on the transported vectors $d\varphi_t(v)$ and $d\varphi_t(w)$. The pullback at p gives this same evaluation.

The *Lie derivative of a form* is

$$(L_X\alpha)_p = \left.\frac{d}{dt}\right|_0 (\varphi_t^*\alpha)_p.$$

In coordinates the pullback has smooth coefficients jointly in time and position. Their time derivatives therefore define a smooth form. For a function, differentiating $f(\varphi_t(p))$ gives $L_X f = Xf$, agreeing with Lecture 9.

Theorem 11.2 (Lie derivative rules · *proved given smooth local flows*)**.** *The Lie derivative is real-linear and preserves degree. It satisfies*

$$L_X(\alpha \wedge \beta) = L_X\alpha \wedge \beta + \alpha \wedge L_X\beta, \qquad L_X d\alpha = d L_X\alpha.$$

Proof. Pullback preserves wedge by Theorem 10.3. Differentiating this identity at zero and using the ordinary product rule proves the first identity. No graded sign occurs because the derivative preserves degree. For the second, differentiate $\varphi_t^* d\alpha = d(\varphi_t^*\alpha)$. In a fixed chart, exterior differentiation differentiates coefficient functions in space. Time and space derivatives commute by joint smoothness. Thus the time derivative passes through d, proving the assertion. ∎

For example, the field $X = x\partial_x$ has flow $\varphi_t(x) = e^t x$. It gives $\varphi_t^* dx = e^t dx$, so $L_X dx = dx$.

Lie derivatives extend to tensors using the same comparison. For a diffeomorphism $f\colon M \to N$ and a tensor of type (r,s) on N, define f^*T by sending each tangent argument forward by df_p and each cotangent argument forward by the dual of $(df_p)^{-1}$. These arguments all lie at $f(p)$, where T is evaluated. Define $L_X T$ by differentiating $\varphi_t^* T$ at zero. This exists by the same smooth-coordinate argument. For vectors this pullback is $(\varphi_{-t})_* Y$, so it recovers $L_X Y = [X, Y]$ from Lecture 9. General tensor pullback here requires a diffeomorphism. Lecture 10's pullback formula for forms applies verbatim to covariant tensors along any smooth map.

11.3 Cartan's formula and time dependence

Theorem 11.3 (Cartan's formula · *proved given smooth local flows*)**.** *On differential forms,*

$$L_X = \mathrm{d}\iota_X + \iota_X \mathrm{d}.$$

Proof. Write $C_X = \mathrm{d}\iota_X + \iota_X\mathrm{d}$. The two graded product rules imply that C_X is an ordinary derivation. To see the cancellation explicitly, let $\deg\alpha = p$. In $C_X(\alpha\wedge\beta)$ the terms $(\iota_X\alpha)\wedge\mathrm{d}\beta$ have total coefficient $(-1)^{p-1} + (-1)^p = 0$. The terms $\mathrm{d}\alpha\wedge\iota_X\beta$ have coefficient $(-1)^p + (-1)^{p+1} = 0$. The remaining terms are $(C_X\alpha)\wedge\beta + \alpha\wedge C_X\beta$. The calculation also holds when $p = 0$, since contraction of a function is zero.

On functions, $C_X f = \iota_X \mathrm{d}f = Xf = L_X f$. On their differentials, $C_X\mathrm{d}f = \mathrm{d}(Xf) = L_X\mathrm{d}f$, using $\mathrm{d}^2 = 0$ and Theorem 11.2. Both sides depend only on the form near the point in question. For small time the trajectory stays in any chosen chart neighbourhood. Uniqueness makes the restricted field's flow agree with the original flow there. The Lie derivative therefore commutes with restriction to that chart. Locally every form is a sum of products of functions and coordinate differentials. The derivation rules therefore prove equality on every form in every chart. ∎

This is Tu, Theorem 20.10, with contraction in the first slot. For example, for $X = \partial_x$ and $\alpha = x\mathrm{d}y$, contraction gives zero while $\iota_X\mathrm{d}\alpha = \iota_X(\mathrm{d}x\wedge\mathrm{d}y) = \mathrm{d}y$. Thus $L_X\alpha = \mathrm{d}y$. Cartan's formula acts on forms. It does not replace the vector-field formula $L_XY = [X,Y]$.

For fixed α, the group law for an autonomous flow also gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\varphi_t^*\alpha = \varphi_t^* L_X\alpha.$$

Indeed, write $\varphi_{t+h}^* = \varphi_t^*\varphi_h^*$ and differentiate in h at zero. This identity holds wherever the indicated local maps exist.

A *time-dependent field* V_t has coefficients smooth jointly in $(t,p)\in I\times M$, where $I\subseteq\mathbb{R}$ is an open interval. Its evolution maps $\psi_{t,s}$ solve $\partial_t\psi_{t,s}(p) = (V_t)_{\psi_{t,s}(p)}$ with $\psi_{s,s}(p) = p$. Their local existence and joint smoothness follow by applying Theorem 9.1 to the field $(1, V_t)$ on $I\times M$. Uniqueness gives the local inverse $\psi_{s,t}$, so each evolution map is a diffeomorphism between its open domain and image. A *smooth family of forms* α_t has coefficients smooth in (t,p). Its derivative $\partial_t\alpha_t$ differentiates those coefficients in fixed, time-independent charts.

Proposition 11.4 (Time-dependent pullback · *proved given smooth local evolution maps*)**.** *Fix an initial time $s\in I$ and write $\psi_t = \psi_{t,s}$. On every region where these maps are defined,*

$$\frac{\mathrm{d}}{\mathrm{d}t}\psi_t^*\alpha_t = \psi_t^*\big(L_{V_t}\alpha_t + \partial_t\alpha_t\big) = \psi_t^*\big(\mathrm{d}\iota_{V_t}\alpha_t + \iota_{V_t}\mathrm{d}\alpha_t + \partial_t\alpha_t\big).$$

Proof. Fix t and define $\eta_h = \psi_{t+h}\circ\psi_t^{-1}$ on the current image, shrinking neighbourhoods and the range of h if necessary. Its initial value is the identity and $\partial_h\eta_h|_0 = V_t$. Joint smoothness gives the coordinate expansions

$$\eta_h(q) = q + hV_t(q) + O(h^2), \qquad D\eta_h(q) = I + hDV_t|_q + O(h^2).$$

These agree to first order with the autonomous flow of the fixed field V_t. The coordinate pullback formula uses just these first derivatives in h, so $\partial_h(\eta_h^*\alpha_t)|_0 = L_{V_t}\alpha_t$. Now $\psi_{t+h}^*\alpha_{t+h} = \psi_t^*\eta_h^*\alpha_{t+h}$. The family $\eta_h^*\alpha_{t+k}$ is jointly smooth in h and k. The chain rule along $h = k$ adds its two partial derivatives at zero: $L_{V_t}\alpha_t$ and $\partial_t\alpha_t$. The fixed pullback ψ_t^* carries their sum back

to the starting point. Cartan's formula gives the second equality. No autonomous group law for ψ_t was used. ∎

For example, $\psi_t(x,y) = (x + t^2 y, y)$ is generated from time zero by $V_t = 2ty\partial_x$. For $\alpha_t = tx\mathrm{d}y$ its pullback is $t(x + t^2 y)\mathrm{d}y$. The derivative is $(x + 3t^2 y)\mathrm{d}y$. Here $L_{V_t}\alpha_t = 2t^2 y\mathrm{d}y$ and $\partial_t \alpha_t = x\mathrm{d}y$. Pulling their sum back gives precisely that same derivative. Omitting the explicit time derivative would give a different answer.

11.4 Volume forms and divergence

On an m-manifold, a *volume form* μ is a smooth m-form that is nonzero at every point. Since the top exterior space is one-dimensional, there is a unique smooth function $\mathrm{div}_\mu X$ satisfying

$$L_X \mu = (\mathrm{div}_\mu X)\mu.$$

This function is the *divergence relative to* μ. Smoothness follows by dividing the local coefficient of $L_X\mu$ by the nowhere-zero coefficient of μ. A flow *preserves* a form if its pullback equals that form on every time map's domain. Such a form is called *invariant* under the flow.

Proposition 11.5 (Divergence and volume preservation · *proved given smooth local flows*). *In local coordinates, for $\mu = \rho\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^m$, with $\rho \neq 0$,*

$$\mathrm{div}_\mu X = \frac{1}{\rho}\sum_{i=1}^{m} \partial_i(\rho X^i).$$

The local flow of X preserves μ if and only if $\mathrm{div}_\mu X = 0$ everywhere.

Proof. Since there are no forms of degree $m+1$, Cartan gives $L_X\mu = \mathrm{d}\iota_X\mu$. The contraction formula gives

$$\iota_X\mu = \sum_i (-1)^{i-1}\rho X^i \mathrm{d}x^1 \wedge \cdots \wedge \widehat{\mathrm{d}x^i} \wedge \cdots \wedge \mathrm{d}x^m.$$

After differentiation, only the x^i derivative of the ith coefficient survives. Moving $\mathrm{d}x^i$ to its ordered place gives another $(-1)^{i-1}$. Thus the resulting top-form coefficient is $\sum_i \partial_i(\rho X^i)$. Dividing by ρ proves the formula. If divergence vanishes, the finite-time pullback derivative is zero. Along each allowed time interval it is constant and equals μ at time zero. Conversely, differentiating invariance at zero gives $L_X\mu = 0$, hence zero divergence. ∎

For Euclidean volume, $\rho = 1$ gives the usual sum $\sum_i \partial_i X^i$. For example, $X = x\partial_x + y\partial_y$ has divergence 2 relative to $\mathrm{d}x \wedge \mathrm{d}y$. Its flow multiplies that form by e^{2t}. On the punctured plane choose instead $\mu' = (x^2+y^2)^{-1}\mathrm{d}x \wedge \mathrm{d}y$. The same field gives

$$\mathrm{div}_{\mu'} X = (x^2+y^2)\left(\partial_x \frac{x}{x^2+y^2} + \partial_y \frac{y}{x^2+y^2}\right) = 0.$$

The two derivatives sum to $2/r^2 - 2r^2/r^4 = 0$, where $r^2 = x^2 + y^2$. Thus its flow preserves μ' while expanding Euclidean area. The choice of volume form matters.

11.5 Symplectic forms and Liouville's theorem

On a $2n$-manifold with $n \geq 1$, a *symplectic form* ω is a closed smooth 2-form that is *nondegenerate*: if $\omega_p(v, w) = 0$ for every $w \in T_pM$, then $v = 0$. For a smooth Hamiltonian function $H\colon M \to \mathbb{R}$, its *Hamiltonian field* X_H is determined by

$$\iota_{X_H}\omega = \mathrm{d}H.$$

Nondegeneracy makes $v \mapsto \omega_p(v, \cdot)$ an isomorphism. Its inverse has smooth matrix entries in charts, so the field exists uniquely and is smooth. The equation $\iota_{X_H}\omega = \mathrm{d}H$ is the sign convention of *Theoretical Mechanics*, notation §1, recalled here unchanged.

Corollary 11.6 (Liouville · *proved given smooth local flows*)**.** *For a symplectic form and its Hamiltonian field,*

$$L_{X_H}\omega = 0, \qquad L_{X_H}(\omega^n) = 0.$$

Consequently its local flow preserves both the symplectic form and the volume form $\omega^n/n!$.

Proof. Cartan's formula and the defining Hamiltonian equation give $L_{X_H}\omega = \mathrm{d}^2H + \iota_{X_H}\mathrm{d}\omega$. Both terms vanish because $\mathrm{d}^2 = 0$ and ω is closed. The derivation rule then gives zero on the n-fold wedge power ω^n. The finite-time pullback identity proves preservation.

To verify that the top wedge is nonzero, choose tangent vectors a, b with $\omega(a, b) = 1$. Every tangent vector v splits into a component $\omega(v, b)a - \omega(v, a)b$ in their plane and a component orthogonal to that plane for ω, meaning that its pairing with both a and b is zero. The restriction to the complementary subspace is nondegenerate, since a vector orthogonal to it and to the plane is orthogonal to the whole space. Repeat in the complement. In the resulting dual basis α^j, β^j, $\omega = \sum_j \alpha^j \wedge \beta^j$. Its nth power is $n!\alpha^1 \wedge \beta^1 \wedge \cdots \wedge \alpha^n \wedge \beta^n$, which is nonzero. Thus the preserved top form is indeed a volume form. ∎

11.6 Hamiltonian motion and an incompressible fluid

On $T^*\mathbb{R}^n$, write the position coordinates as q^i and the cotangent coefficients as p_i. Here θ denotes the canonical 1-form. The symplectic form is ω. Their coordinate expressions are

$$\theta = p_i \mathrm{d}q^i, \qquad \omega = \sum_i \mathrm{d}q^i \wedge \mathrm{d}p_i.$$

The positions q^i translate the lower coordinate labels of *Theoretical Mechanics*. The momenta retain their covector indices.

Example 11.7 (The canonical signs)**.** Differentiating gives $\mathrm{d}\theta = \sum_i \mathrm{d}p_i \wedge \mathrm{d}q^i = -\omega$. For $X = \dot{q}^i\partial_{q^i} + \dot{p}_i\partial_{p_i}$, contraction gives

$$\iota_X\omega = \sum_i (\dot{q}^i \mathrm{d}p_i - \dot{p}_i \mathrm{d}q^i).$$

Equating this with $\mathrm{d}H$ gives Hamilton's equations $\dot{q}^i = \partial H/\partial p_i$ and $\dot{p}_i = -\partial H/\partial q^i$. The form is closed because its coefficients are constant, and the displayed contraction shows nondegeneracy. Liouville therefore proves preservation of this form and of its top wedge volume.

Recall the canonical *Poisson bracket* of smooth functions:

$$\{f, g\} = \sum_i \left(\frac{\partial f}{\partial q^i}\frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial g}{\partial q^i} \right).$$

The displayed Hamiltonian field acts on a smooth function h by $X_H h = \{h, H\}$, as its coordinate coefficients show. For one degree of freedom, take $f = q^2/2$ and $g = p^2/2$. Their fields are $X_f = -q\partial_p$ and $X_g = p\partial_q$. The coordinate bracket is $[X_f, X_g] = -q\partial_q + p\partial_p$. The fixed Poisson bracket gives $\{f, g\} = qp$, whose field is $X_{qp} = q\partial_q - p\partial_p$. Thus $[X_f, X_g] = -X_{\{f,g\}}$, computed using Lecture 9's bracket $[X, Y]h = X(Yh) - Y(Xh)$.

The harmonic oscillator with $H = (q^2 + p^2)/2$ has field $p\partial_q - q\partial_p$. Its complete flow is

$$\varphi_t(q, p) = (q\cos t + p\sin t, -q\sin t + p\cos t).$$

Differentiation gives $(\dot{q}, \dot{p}) = (p(t), -q(t))$ and the initial value is (q, p). From $(1, 0)$ the flow reaches $(0, -1)$ at time $\pi/2$. The phase-plane motion is clockwise, as shown in the figure; the origin is fixed.

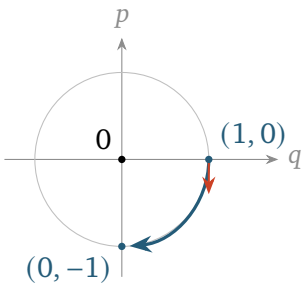


Figure 11.2: Positive-time harmonic-oscillator motion is clockwise in the (q, p) plane. The orbit through $(1, 0)$ reaches $(0, -1)$ after a quarter period; the origin is stationary.

For a fluid in Euclidean space, let v_t be its smooth velocity field. Constant-density *incompressibility* is the condition $\sum_i \partial_i v_t^i = 0$ at every point and time. With Euclidean volume $\mu = \mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^m$, this is exactly $L_{v_t}\mu = 0$. The time-dependent pullback formula, with $\partial_t \mu = 0$, therefore proves $\psi_t^* \mu = \mu$ wherever fluid trajectories exist. The moving fluid preserves this chosen local volume element.

Self-check. For a smooth function $f : \mathbb{R} \to \mathbb{R}$ and $X = x\partial_x$, compute $L_X(f\mathrm{d}x)$.

Answer. Cartan gives $\mathrm{d}\iota_X(f\mathrm{d}x) = \mathrm{d}(xf) = (xf' + f)\mathrm{d}x$. The other term is zero because every 2-form on $\mathbb{R}$ vanishes.

Summary. Contraction inserts a vector in the first slot and is a graded antiderivation. The Lie derivative of forms differentiates flow pullback. Cartan expresses it as $\mathrm{d}\iota_X + \iota_X\mathrm{d}$. A time-dependent form contributes its explicit time derivative as well. Zero divergence relative to a chosen volume form is equivalent to local flow preservation of that form. A Hamiltonian flow preserves its symplectic form and the associated top wedge volume. Lecture 12 integrates forms and relates their exterior derivatives to boundaries.

Reading. Tu, *An Introduction to Manifolds*, §20.3–§20.5, especially Theorem 20.10, pp. 229–230.

Lecture 12

Orientation, integration and Stokes' theorem

Objectives. *An orientation specifies the sign of a top-degree form. It lets us integrate forms on a manifold. Stokes' theorem identifies the integral of an exterior derivative with the integral on the boundary, using outward-first orientation. We prove it in every dimension and apply it to the flux of the Coulomb form.*

By the end of this lecture you will be able to:

- Determine orientability using charts and top forms.
- Define an integral independent of the oriented charts and partition of unity.
- Prove Stokes' theorem with its support and orientation hypotheses.
- Compute Coulomb flux by removing a ball around its singularity.

12.1 Orientations and oriented charts

For an n-dimensional real vector space with $n \geq 1$, two ordered bases have the same *orientation* if their change-of-basis determinant is positive. There are two classes. A *frame* on an open set is an ordered list of smooth tangent fields that is a basis at every point. An *orientation of a manifold* chooses a class in each tangent space, locally represented by a smooth frame. A manifold is *orientable* if it admits such a choice, and *oriented* when a choice has been made.

An *oriented atlas* has positive Jacobian determinants on all chart transitions. Its coordinate frames give compatible orientation classes: the change of frame uses the inverse Jacobian, whose determinant is also positive. Conversely, near any point an orientation is represented by a frame. Its determinant relative to a coordinate frame is continuous and nonzero. Shrink to a connected chart so that the sign is constant. If necessary replace x^1 by $-x^1$. The coordinate frame then represents the chosen orientation. Charts obtained this way have positive transitions. This proves the equivalence of orientations and oriented atlases.

A nowhere-zero top form μ selects the bases on which its value is positive. These choices are locally represented by coordinate frames, because its coefficient in a connected chart has constant sign. For oriented manifolds, a diffeomorphism is *orientation-preserving* when its differential sends positive bases to positive bases. In positive charts this means its Jacobian determinant is positive.

For example, on $S^n \subset \mathbb{R}^{n+1}$ the form

$$\mu_p(v_1, \dots, v_n) = \det[p\ v_1\ \cdots\ v_n] \qquad (v_j \in T_pS^n)$$

is smooth and nowhere zero: the radial vector p is independent of the n tangent basis vectors. It defines the outward orientation. For $n = 2$ it is the sphere area form of Lecture 10.

Lecture 10 computed the north-pole stereographic inverse pullback as $-4\mathrm{d}u\wedge\mathrm{d}v/(1+u^2+v^2)^2$. This chart reverses the outward orientation.

In dimension zero an orientation is a choice of sign $+1$ or -1 at each point. Integration of a function then means its signed sum. This convention includes the endpoints of an oriented interval.

12.2 Partitions of unity and orientability

A collection of sets is *locally finite* if every point has a neighbourhood meeting only finitely many of them. A smooth *partition of unity subordinate to an open cover* $\{U_i\}$ consists of nonnegative smooth functions ρ_i such that

$$\sum_i \rho_i = 1, \qquad \operatorname{supp}\rho_i \subset U_i, \qquad \{\operatorname{supp}\rho_i\} \text{ is locally finite.}$$

One may use several functions assigned to the same cover member. Local finiteness makes each sum finite near any point. It also implies that a compact set meets only finitely many supports: choose finitely many such neighbourhoods covering the compact set.

Theorem 12.1 (Partitions of unity · *sketched; normalization and use proved*)**.** *Every open cover of a smooth manifold admits a subordinate smooth partition of unity.*

Sketch of existence. Second countability and local compactness give a countable exhaustion by compact sets, each lying in the interior of the next. Refining the cover on the successive compact layers gives a locally finite cover by precompact coordinate balls with smaller balls still covering the manifold. Choose a smooth bump on each ball that is positive on its smaller ball and supported in the larger one. The coordinate construction uses the smooth function $e^{-1/t}$ for $t>0$, extended by zero for $t\le 0$. The locally finite sum $b=\sum_i b_i$ of these bumps is smooth and strictly positive. Thus $\rho_i=b_i/b$ is smooth and nonnegative, has the required support, and sums to one. The refinement and bump construction are the sketched steps. See Tu, Theorem 13.7 and Appendix C. ∎

Proposition 12.2 (Orientability criteria · *proved given partitions of unity*)**.** *A smooth manifold is orientable if and only if it has an oriented atlas, and if and only if it admits a nowhere-zero smooth top form.*

Proof. The atlas equivalence and the top-form-to-orientation implication were proved above. For the converse, take positively oriented charts and a subordinate partition. Extend $\rho_i\mathrm{d}x_i^1\wedge\cdots\wedge\mathrm{d}x_i^n$ by zero outside its chart. This is smooth because its support lies inside the chart. Their locally finite sum is smooth. Every summand is nonnegative on a positive tangent basis, and at least one has positive coefficient at every point. The sum therefore never vanishes. In dimension zero the constant function 1 is already a nowhere-zero 0-form, and every discrete manifold has an atlas. ∎

The strip model in the figure shows the transverse direction on a Möbius band.



Figure 12.1: The red edge arrows mark $(t, s) \mapsto (t + 1, -s)$. An upward transverse arrow at $t = 1$ therefore returns at $t = 0$ pointing down.

The open Möbius band is the quotient of $\mathbb{R} \times (-1, 1)$ by this identification. On a rectangle of width less than 1, the projection is injective and open: the inverse image of its image is the open union of all translated copies under the identification. The inverse restrictions give charts on the quotient. Their transitions are restrictions of integer powers of $(t, s) \mapsto (t+1, -s)$, hence smooth. Thus the projection is a local diffeomorphism. If the quotient had a nowhere-zero top form, its pullback would be $h(t, s)\mathrm{d}t \wedge \mathrm{d}s$ with h continuous, nonzero and of constant sign on the connected strip. Invariance under the identification gives $h(t + 1, -s) = -h(t, s)$, contradicting that constant sign. Hence it is not orientable.

For $\mathbb{R}\mathrm{P}^2$, the projection $\pi\colon S^2 \to \mathbb{R}\mathrm{P}^2$ is a local diffeomorphism by the ratio charts of Lecture 7. There it was called q; we write π here. On a hemisphere $\sigma X^i > 0$, with $\sigma \in \{+1, -1\}$, its coordinate map is $X \mapsto (X^j/X^i)_{j\neq i}$. Its smooth inverse is $y \mapsto \sigma Y/\|Y\|$, where Y has ith component 1 and remaining components y. This proves the local diffeomorphism assertion. A nonzero top form would pull back to $h\mu$ on S^2, where μ is its outward area form and h has constant nonzero sign. The antipodal map $a(p) = -p$ satisfies

$$(a^*\mu)_p(v, w) = \det[-p\ -v\ -w] = -\mu_p(v, w).$$

Since $\pi \circ a = \pi$, invariance forces $h(-p) = -h(p)$, again impossible. Thus $\mathbb{R}\mathrm{P}^2$ is not orientable.

12.3 Integrating a top form

Recall the ordinary change-of-variables theorem: under a smooth diffeomorphism $y = f(x)$, integration of a compactly supported function uses the factor $|\det Df|$. In positive coordinates the determinant itself is positive. This is the calculus input to integration of forms.

For $n \geq 1$, let ω be a smooth compactly supported n-form on an oriented n-manifold. On a positive chart $x\colon U \to x(U)$ write $(x^{-1})^*\omega = f\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n$. If its support lies in U, define its integral to be the ordinary integral of f on $x(U)$. For a general support, choose a positive atlas and a subordinate partition and set

$$\int_M \omega = \sum_i \int_{U_i} \rho_i \omega.$$

Only finitely many terms meet the compact support. Each has compact support inside its chart, so every term is defined.

Proposition 12.3 (Independence of choices · *proved given partitions of unity and change of variables*)**.** *The integral is independent of the positive atlas and subordinate partition. Reversing the orientation reverses its sign.*

Proof. On overlapping positive charts, the determinant rule for top-form pullback and ordinary change of variables give the same chart integral. For two partitions ρ_i and χ_j subordinate to positive charts, insert $\sum_j \chi_j = 1$ into each integral for the first partition. The result is the finite double sum of the integrals of $\rho_i \chi_j \omega$. Each term is supported in the chart overlap and has the same value in either chart system. Interchanging the sums and using $\sum_i \rho_i = 1$ gives the second partition's answer. This proves independence. For the opposite orientation, reflect one coordinate. Its derivative has determinant -1, whereas the absolute Jacobian in ordinary integration is 1. The top-form coefficient changes sign, proving the last assertion. ∎

For computations, let $D \subset \mathbb{R}^n$ be a compact parameter region, and let $F\colon D \to M$ be smooth with $F(D) = M$. Here smoothness on D means local smooth extension to open neighbourhoods in $\mathbb{R}^n$. Suppose F restricts to an orientation-preserving diffeomorphism $\operatorname{int} D \to M \setminus E$, where $E = F(\partial D)$. Assume ∂D has Lebesgue measure zero and E has measure zero in every manifold chart. The latter condition is chart-independent by change of variables. Then integration is computed by pullback:

$$\int_M \omega = \int_D F^* \omega.$$

Indeed, insert a subordinate partition ρ_i on M. For each summand, ordinary change of variables on $\operatorname{int} D$ gives the integral of $\rho_i \omega$ on $M \setminus E$. The two null boundaries add zero. Summing the finitely many nonzero chart integrals proves the formula. These are the hypotheses behind the angular rectangle used below; see also Tu, §23.4, Definition 23.10.

A *density* has local expressions $f|\mathrm{d}x^1 \cdots \mathrm{d}x^n|$ that transform with the absolute Jacobian. Densities can be integrated without orientation because the ordinary change-of-variables factor has the same law. A top form uses a signed determinant, so its signed integral requires a chosen orientation. The two transformation laws must be distinguished on nonorientable manifolds.

12.4 Manifolds with boundary

The upper and lower closed half-spaces are

$$\mathbb{H}^n = \{x \in \mathbb{R}^n : x^n \geq 0\}, \qquad \mathbb{H}^n_- = \{x \in \mathbb{R}^n : x^n \leq 0\}.$$

A smooth *manifold with boundary* is Hausdorff and second countable, with charts onto relatively open subsets of either half-space. Allowing either side permits the coordinate reflection needed for positive charts also in dimension one. For example, at the right endpoint of the oriented interval $[0, 1]$, the positive chart $u = x - 1$ maps into $(-\infty, 0]$. Chart transitions and their inverses extend smoothly to neighbourhoods in $\mathbb{R}^n$ near each point. The *boundary* ∂M consists of chart points with $x^n = 0$; the others form the interior. This geometric boundary differs from Lecture 6's boundary operator on chains. The set of boundary points is independent of the chart. Indeed, the chain rule on interior points and continuity make transition derivatives invertible also at boundary points. The inverse function theorem then makes their extensions local diffeomorphisms. Such a map cannot send a half-neighbourhood to a full interior neighbourhood: its inverse would have to take points on both sides of the source hyperplane into the half-space. Restricting charts to $x^n = 0$ gives charts of dimension $n - 1$ on ∂M. The restricted transitions make them smoothly compatible. A transition's last component vanishes on the boundary, so its tangential derivatives vanish there. Its

normal derivative is nonzero by invertibility. It sends inward vectors to inward vectors because it sends interior points to interior points. Thus the outward transverse direction is chart-independent, even when the two charts use opposite half-spaces.

At a boundary point, T_pM is the ambient n-dimensional space modeled by a half-space chart, with changes given by the transition Jacobians. Its vectors need not be velocities of two-sided curves remaining in M. In particular the outward vector $-\partial_n$ is allowed in the upper model. Functions and forms are smooth when their chart coefficients extend smoothly to open sets of $\mathbb{R}^n$. Their derivatives at the boundary are independent of extension, since extensions agree on the interior and all their derivatives agree there and extend continuously to the boundary. The same half-chart construction gives tangent changes and exterior differentiation. Partitions of unity and integration extend using restrictions of bumps to half-balls. The preceding normalization and independence proofs then apply.

On an oriented manifold the *induced boundary orientation* is *outward first*: a boundary basis $(v_1, \dots, v_{n-1})$ is positive if $(\nu, v_1, \dots, v_{n-1})$ is positive in T_pM, where ν is outward transverse. Changing ν by a tangent vector or multiplying it by a positive scalar does not change this sign. This proves independence of that choice; no metric is required. For the standard orientation on $\mathbb{H}^n$, take $\nu = -\partial_n$. The boundary orientation is represented by

$$\iota_{-\partial_n}(\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n) = (-1)^n \mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^{n-1} \quad \text{on } x^n = 0.$$

On $\mathbb{H}^n_-$ the outward vector is $+\partial_n$, so the boundary orientation is $(-1)^{n-1}\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^{n-1}$. For $n = 1$ the upper model gives the negative initial endpoint, and the lower model gives the positive final endpoint. The figure compares the upper half-plane with an interval.

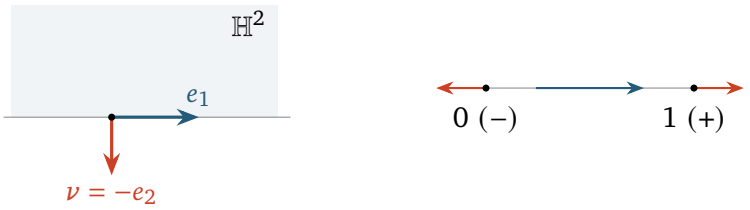


Figure 12.2: On the upper half-plane, the outward vector points down. The boundary tangent points right. On the positively oriented interval, the induced endpoint signs are − and +.

The sphere orientation above is exactly the boundary orientation of the unit ball. The usual Möbius band with its boundary is also nonorientable, since an orientation would restrict to its nonorientable interior. A *closed manifold* means a compact manifold without boundary.

12.5 Stokes' theorem in every dimension

Theorem 12.4 (Stokes · *proved given partitions of unity*)**.** *Let M be an oriented smooth n-manifold with boundary, where $n \geq 1$. For a smooth compactly supported $(n-1)$-form ω,*

$$\int_M \mathrm{d}\omega = \int_{\partial M} \omega.$$

The right side means restriction by the boundary inclusion, with the outward-first orientation. In particular the statement applies to every smooth $(n-1)$-form when M is compact.

Proof. First work on $\mathbb{H}^n$ with compact support. Write the form as

$$\omega = \sum_{i=1}^{n} (-1)^{i-1} f_i \mathrm{d}x^1 \wedge \cdots \wedge \widehat{\mathrm{d}x^i} \wedge \cdots \wedge \mathrm{d}x^n, \qquad \mathrm{d}\omega = \Big(\sum_i \partial_i f_i\Big) \mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n.$$

The derivative formula follows because repeated coordinate factors vanish; the two omission signs cancel for the surviving derivative. Fubini's theorem and the fundamental theorem of calculus integrate $\partial_i f_i$ first in x^i. For $i < n$ the endpoints are outside compact support, so each contribution is zero. The normal contribution is

$$\int_{\mathbb{H}^n} \mathrm{d}\omega = \int_{\mathbb{R}^{n-1}} \int_0^{\infty} \partial_n f_n(x', x^n)\, \mathrm{d}x^n\, \mathrm{d}x' = -\int_{\mathbb{R}^{n-1}} f_n(x', 0)\, \mathrm{d}x'.$$

On the boundary only the term $i = n$ restricts nontrivially. Its coefficient is $(-1)^{n-1} f_n$. The boundary orientation has the additional sign $(-1)^n$, so its integral is also $-\int f_n(x', 0)\mathrm{d}x'$. For $n = 1$, the same equality uses the negative boundary point and the ordinary endpoint formula. This proves the half-space case in every dimension. On $\mathbb{R}^n$ all coordinate integrations have two endpoints outside support, so the integral of $\mathrm{d}\omega$ is zero, matching its empty boundary. On $\mathbb{H}^n_-$, integrating the normal coordinate from $-\infty$ to 0 gives $+\int_{\mathbb{R}^{n-1}} f_n(x', 0)\mathrm{d}x'$. Its boundary orientation has sign $(-1)^{n-1}$. Multiplying by the same sign in the restricted form gives $+f_n$, so Stokes holds there too. For $n = 1$ this gives the positive final endpoint.

Now choose positively oriented interior and boundary charts on M and a subordinate partition ρ_i. Each $\rho_i \omega$ has compact support inside its chart. In coordinates it extends by zero across the artificial chart edges to a smooth compactly supported form on $\mathbb{R}^n$, $\mathbb{H}^n$ or $\mathbb{H}^n_-$. The local result therefore applies to it. Sum these equalities. Local finiteness and compact support make the sum finite. On M, the product rule gives

$$\sum_i \mathrm{d}(\rho_i \omega) = \Big(\sum_i \mathrm{d}\rho_i\Big) \wedge \omega + \Big(\sum_i \rho_i\Big) \mathrm{d}\omega = \mathrm{d}\omega.$$

Here $\sum_i \rho_i = 1$ and its differential is zero. The restricted partition also sums to one on the boundary. Thus the summed local equalities give the claimed global equality. ∎

Tu, Theorem 23.12, pp. 269–271, glues in every dimension but computes locally only on $\mathbb{H}^2$. His Problem 23.4 treats $\mathbb{R}^n$ and $\mathbb{H}^n$; their proofs are supplied above.

The support hypothesis matters on a noncompact manifold. On $\mathbb{R}$, the improper integral of $\mathrm{d}(\tanh x)$ is $1 - (-1) = 2$, while the boundary is empty. The function has no compact support. The nonzero limiting values at infinity remain in the endpoint formula.

12.6 Divergence and equal fluxes

Here a *compact smooth region* in $\mathbb{R}^n$ means a compact n-manifold with boundary embedded in $\mathbb{R}^n$. For an oriented manifold S, write $-S$ for the opposite orientation.

Corollary 12.5 (Consequences of Stokes · *proved given partitions of unity*)**.** *For a smooth vector field v on a compact smooth region $K \subset \mathbb{R}^3$, with Euclidean volume $\mu = \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z$,*

$$\int_K (\operatorname{div} v)\mu = \int_{\partial K} \iota_v \mu.$$

On a closed oriented manifold the integral of an exact top form is zero. If an oriented compact 3-manifold has boundary $S_1 \sqcup (-S_2)$ and a closed 2-form β is smooth on it, then $\int_{S_1} \beta = \int_{S_2} \beta$.

Proof. The first statement follows by applying Stokes to $\iota_v\mu$: Lecture 11 gives $\mathrm{d}\iota_v\mu = (\operatorname{div} v)\mu$. With an outward unit normal ν and a positively oriented orthonormal boundary basis, $\mu(v, \cdot, \cdot)$ is $(v \cdot \nu)$ times $\iota_\nu\mu$, the induced boundary area form. Thus this is the ordinary normal-flux divergence theorem. The empty boundary gives the second statement. Closedness gives $\int \mathrm{d}\beta = 0$; the two induced boundary orientations give $\int_{S_1} \beta - \int_{S_2} \beta = 0$, proving the third. ∎

12.7 Coulomb flux around a missing point

The Coulomb flux form uses Lecture 10's ambient β divided by r^3; these forms agree on the unit sphere. On $\mathbb{R}^3 \setminus \{0\}$ set $r = \sqrt{x^2 + y^2 + z^2}$ and define the Coulomb flux form by

$$\beta = \frac{x\mathrm{d}y \wedge \mathrm{d}z + y\mathrm{d}z \wedge \mathrm{d}x + z\mathrm{d}x \wedge \mathrm{d}y}{r^3}.$$

Let K be a compact smooth region with outward-oriented boundary $S = \partial K$ and $0 \notin S$. We compute the outward flux of β through S. When the origin lies inside K, a removed small ball has surface S_ε. Its induced inner normal points into the ball, as the section of the region in the figure shows.

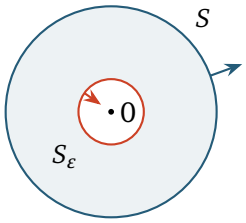


Figure 12.3: The region between S and the small sphere has an inner boundary with normal pointing into the removed ball. Its induced inner orientation is opposite to the ball's outward orientation.

Example 12.6 (Outward Coulomb flux)**.** The derivative of β has volume coefficient

$$\partial_x \frac{x}{r^3} + \partial_y \frac{y}{r^3} + \partial_z \frac{z}{r^3} = \frac{3}{r^3} - \frac{3(x^2 + y^2 + z^2)}{r^5} = 0.$$

Thus β is closed on its domain. On the unit sphere it is the area form of Lecture 10, so angular coordinates give

$$\int_{S^2} \beta = \int_0^{2\pi} \int_0^{\pi} \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\phi = 4\pi.$$

Here θ is the polar angle and ϕ the azimuth. The coordinate seam and poles have measure zero and do not change this integral. On the sphere of radius $\varepsilon > 0$, radial scaling contributes ε^3 in the numerator and ε^3 in the denominator, so its outward flux is again 4π.

If $0 \notin K$, Stokes applies directly to the closed form on K and gives zero flux. If 0 is inside K, remove an open ball of radius ε whose closure lies in the interior of K. The remaining compact region has boundary $S \sqcup (-S_\varepsilon)$. The form is smooth and closed on this region. The equal-flux consequence gives $\int_S \beta = \int_{S_\varepsilon} \beta = 4\pi$. Thus the outward Coulomb flux detects whether the original region contains the origin.

Self-check. Compute the area of the unit sphere using its angular area form.

Answer. The integral is $\int_0^{2\pi}\int_0^{\pi}\sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi = 2\pi[-\cos\theta]_0^{\pi} = 4\pi$.

Summary. Orientations fix signs of top forms. Partitions of unity define a chart-independent integral. Stokes applies to compactly supported forms on oriented manifolds with boundary, using outward-first boundary orientation. Its local proof is the fundamental theorem of calculus in the normal coordinate. Closed forms have equal fluxes across two oriented surfaces that together, with the second orientation reversed, bound a compact region. Removing a ball around the Coulomb singularity gives flux 4π. Lecture 13 studies closed forms modulo exact forms.

Reading. Tu, *An Introduction to Manifolds*, §13 and Appendix C; §§21–23, especially Theorem 23.12, pp. 269–271.

Lecture 13

De Rham cohomology

Objectives. *A closed form of positive degree has local primitives, forms whose exterior derivative is the given form, but a global primitive may not exist. De Rham cohomology records this obstruction. We prove the local assertion by integrating along a contraction. We then compare homotopic maps and compute the cohomology of spheres by joining two open sets. The Coulomb form gives a nonzero class with a measurable period.*

By the end of this lecture you will be able to:

- Distinguish closed forms, exact forms and their cohomology classes.
- Construct a primitive of a closed positive-degree form on a star-shaped open set.
- Prove smooth homotopy invariance using the homotopy operator.
- Compute sphere cohomology from Mayer–Vietoris and normalize its area class.

13.1 Closed forms modulo exact forms

All manifolds in this lecture are smooth and without boundary. All vector spaces and cohomology groups have real coefficients. Recall from Lecture 10 that a form α is *closed* if $\mathrm{d}\alpha = 0$ and *exact* if $\alpha = \mathrm{d}\eta$ for some form η. The identity $\mathrm{d}^2 = 0$ makes every exact form closed. A *cochain complex* is a sequence of vector spaces with linear maps $\mathrm{d}\colon C^k \to C^{k+1}$ whose successive composites are zero. The spaces $\Omega^k(M)$ with exterior differentiation form such a complex. We put $\Omega^k(M) = 0$ for $k < 0$.

The *de Rham cohomology* in degree k is the quotient

$$H^k(M) = \frac{\ker(\mathrm{d}\colon \Omega^k(M) \to \Omega^{k+1}(M))}{\operatorname{im}(\mathrm{d}\colon \Omega^{k-1}(M) \to \Omega^k(M))}.$$

The *cohomology class* $[\alpha]$ of a closed form consists of all closed forms differing from α by an exact form. In positive degree, $[\alpha] = 0$ means precisely that α has a global primitive. In degree zero, only the zero function represents the zero class. The superscript distinguishes cohomology from Lecture 6's homology. The only exact form of degree zero is zero. A function is closed exactly when all its partial derivatives vanish. It is then constant on connected coordinate balls, hence locally constant. Conversely a locally constant function has zero derivative. In particular, $H^0(M) = \mathbb{R}$ when M is connected and nonempty.

For example, on the punctured plane the angular form is

$$\alpha_{\mathrm{ang}} = \frac{-y\mathrm{d}x + x\mathrm{d}y}{x^2 + y^2}.$$

Lecture 10 proved that this form is closed and has integral 2π around the counter-clockwise unit circle. An exact form has zero integral around that circle by Stokes. Hence $[\alpha_{\mathrm{ang}}] \neq 0$.

More generally, a *period* of a closed k-form is its integral over a closed oriented k-manifold mapped smoothly into M, using pullback. Such a mapped manifold is a geometric cycle. Stokes makes every period of an exact form zero, so periods depend only on the class. These geometric cycles suffice here. Lecture 14 relates forms to the singular cycles of Lecture 6.

A smooth map $f\colon M \to N$ induces a linear map $f^*\colon H^k(N) \to H^k(M)$ by $f^*[\alpha] = [f^*\alpha]$. Indeed, pullback commutes with d, so it preserves closed forms and exact differences. Pullback reverses composition: $(g \circ f)^* = f^* g^*$, and the identity map induces the identity. The wedge product also descends to classes. For closed β, $(\mathrm{d}\eta) \wedge \beta = \mathrm{d}(\eta \wedge \beta)$. For closed α of degree k, $\alpha \wedge \mathrm{d}\xi = (-1)^k \mathrm{d}(\alpha \wedge \xi)$. Changing either representative therefore changes the product only by an exact form. We write $H^*(M)$ for all the degrees together.

13.2 The homotopy operator and the Poincaré lemma

A *smooth homotopy* between smooth maps F_0 and F_1 from M to N is a smooth map $F\colon [0,1] \times M \to N$ with $F_t(p) = F(t,p)$ and those endpoint maps. Smoothness at the endpoints means local smooth extension, as in Lecture 12. No inverse for any F_t is required. For a k-form α on N, the pulled-back form decomposes uniquely as

$$F^*\alpha = \mathrm{d}t \wedge a_t + b_t,$$

where a_t and b_t are forms on M of degrees $k-1$ and k. Here a_t is the restriction of $\iota_{\partial_t} F^*\alpha$ to the slice at time t, and $b_t = F_t^*\alpha$. Define the *homotopy operator* by integrating its coefficients:

$$K_F\alpha = \int_0^1 a_t \,\mathrm{d}t \quad (k \geq 1), \qquad K_F f = 0 \quad (k = 0).$$

The integral is smooth. Near each point, compactness of the time interval permits differentiating the smooth coefficients under the integral. The intrinsic definition by contraction makes the chart definitions agree.

Proposition 13.1 (Homotopy operator identity)**.** *For every smooth homotopy F and every differential form α,*

$$F_1^*\alpha - F_0^*\alpha = \mathrm{d}K_F\alpha + K_F\mathrm{d}\alpha.$$

Proof. Write d_M for differentiation in M. The product rule gives

$$\mathrm{d}(F^*\alpha) = \mathrm{d}t \wedge (\partial_t b_t - \mathrm{d}_M a_t) + \mathrm{d}_M b_t.$$

The minus sign comes from moving d_M past $\mathrm{d}t$. Since $\mathrm{d}F^*\alpha = F^*\mathrm{d}\alpha$, the time-containing coefficient on the right is the one integrated in $K_F\mathrm{d}\alpha$. Thus

$$K_F\mathrm{d}\alpha = \int_0^1 \partial_t b_t \,\mathrm{d}t - \mathrm{d}_M \int_0^1 a_t \,\mathrm{d}t = b_1 - b_0 - \mathrm{d}_M K_F\alpha.$$

The fundamental theorem of calculus gives the endpoint difference. This is the claimed identity. For functions, the same computation has $a_t = 0$ and is the ordinary endpoint formula. ∎

Two maps of complexes whose difference is $\mathrm{d}K + K\mathrm{d}$ are called *cochain homotopic*. The displayed identity constructs that relation for the two endpoint pullbacks.

An open set $U \subset \mathbb{R}^n$ is *star-shaped about* 0 if $0 \in U$ and $tx \in U$ for every $x \in U$ and $t \in [0,1]$. The figure shows the segments used by its contraction.

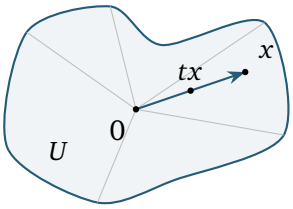


Figure 13.1: In a star-shaped set, each segment from 0 to x stays inside U. The contraction is given by $F(t,x) = tx$.

For this homotopy the operator has the explicit formula, for $k \geq 1$,

$$(K\alpha)_x(v_1,\dots,v_{k-1}) = \int_0^1 t^{k-1}\alpha_{tx}(x,v_1,\dots,v_{k-1})\,\mathrm{d}t. \tag{13.1}$$

Indeed, the time derivative of F is x, while each spatial tangent vector is multiplied by t. This gives the power t^{k-1}. Near a fixed $x \in U$, compactness of its segment and openness of U give $\varepsilon > 0$ and a neighbourhood of x such that $tx' \in U$ for every point x' in that neighbourhood and $t \in (-\varepsilon, 1+\varepsilon)$. This gives the required local smooth extension in time. Differentiating the coefficients under the integral therefore defines a smooth form on U.

Theorem 13.2 (Poincaré lemma). *On a star-shaped open set, every closed k-form with $k \geq 1$ is exact. Such a set has degree-zero cohomology consisting of the constant functions.*

Proof. After translating the star centre, use $F(t,x) = tx$. The map F_1 is the identity. The map F_0 is constant, so its differential is zero and its pullback on positive-degree forms is zero. For closed α the operator identity consequently gives $\alpha = \mathrm{d}K\alpha$. In degree zero it instead gives $f - f(0) = K\mathrm{d}f$. A closed function is therefore constant. ∎

For example, $\alpha = 2x\mathrm{d}x + 2y\mathrm{d}y$ on $\mathbb{R}^2$ has $K\alpha = \int_0^1 2t(x^2+y^2)\,\mathrm{d}t = x^2 + y^2$. The punctured plane is not star-shaped. Its radial paths end at a missing point, so the homotopy does not have its claimed target there. A formal insertion into α_{ang} would even give zero, because $-yx + xy = 0$. It cannot supply a primitive of that nonzero form.

13.3 Homotopy invariance

Theorem 13.3 (Homotopy invariance · *proved given cited smoothing; smooth homotopies proved directly*). *Smoothly homotopic maps induce the same maps on de Rham cohomology. The same holds for smooth maps that are continuously homotopic. Smooth manifolds with the same continuous homotopy type have isomorphic cohomology in every degree.*

Proof. For a smooth homotopy and a closed form α, the operator identity gives $F_1^*\alpha - F_0^*\alpha = \mathrm{d}K_F\alpha$. The two pullbacks therefore represent the same class.

For the continuous case we use the following smooth approximation results. Every continuous map between smooth manifolds is homotopic to a smooth map. Two homotopic smooth maps admit a smooth homotopy. These are cited from Lee, *Introduction to Smooth Manifolds*, second edition, Theorems 6.26 and 6.29, pp. 141–143. Choose a smooth representative of a continuous map's homotopy class and use its pullback. Two choices induce the same map by the second cited result and the smooth proof above. Composition and identities are preserved because compositions of smooth representatives represent the continuous compositions. If continuous maps $f\colon M \to N$ and $g\colon N \to M$ are homotopy inverses, their

composites represent identity maps. Contravariance then gives $f^*g^* = \mathrm{id}$ and $g^*f^* = \mathrm{id}$ on the respective cohomology spaces. Thus these maps are inverse isomorphisms. ∎

A contractible manifold consequently has the cohomology of a point: $\mathbb{R}$ in degree zero and zero in positive degrees. For $n \geq 1$, the punctured space $\mathbb{R}^n \setminus \{0\}$ retracts onto S^{n-1} by $r(x) = x/\|x\|$. With i the inclusion, $r \circ i = \mathrm{id}$. A smooth homotopy from the identity to $i \circ r$ is

$$F(t,x) = \left((1-t)^2 + \frac{t^2}{\|x\|}\right) x.$$

The coefficient is positive for every real t, since its two nonnegative summands never vanish together. Hence the homotopy stays in the punctured space. It has the required endpoints. This proves $H^*(\mathbb{R}^n \setminus \{0\}) \cong H^*(S^{n-1})$.

13.4 Exact sequences and Mayer–Vietoris

A sequence of vector spaces and linear maps is *exact* at a term if the image of the incoming map equals the kernel of the outgoing map. A *short exact sequence* has the form $0 \to A \to B \to C \to 0$. Its first nonzero map is injective and its last is surjective.

Let $M = U \cup V$ be an open cover and set $W = U \cap V$. Restriction gives a short exact sequence of form spaces in each degree:

$$0 \longrightarrow \Omega^k(M) \xrightarrow{i} \Omega^k(U) \oplus \Omega^k(V) \xrightarrow{j} \Omega^k(W) \longrightarrow 0, \qquad j(a,b) = b|_W - a|_W.$$

The map i restricts a global form to both sets. It is injective because these sets cover M. A pair is in $\ker j$ exactly when its forms agree on the overlap, which is precisely when they glue to a global form. To prove surjectivity, choose a partition $\rho_U + \rho_V = 1$ subordinate to the cover, supplied by Theorem 12.1. For $\zeta \in \Omega^k(W)$ take

$$a = -\operatorname{ext}_U(\rho_V \zeta), \qquad b = \operatorname{ext}_V(\rho_U \zeta).$$

Here ext means extension by zero from the overlap. This extension is smooth: near a point of $U \setminus V$, the support condition makes ρ_V zero, and similarly on $V \setminus U$. Their difference on W is $(\rho_U + \rho_V)\zeta = \zeta$. All restriction and difference maps commute with differentiation.

For a closed overlap form ζ, choose any lifts with $j(a,b) = \zeta$. Then $\mathrm{d}b - \mathrm{d}a = \mathrm{d}\zeta = 0$ on W. The derivatives glue to a global closed form c of degree $k+1$. The *connecting homomorphism* is

$$\delta_{\mathrm{MV}} \colon H^k(W) \longrightarrow H^{k+1}(M), \qquad \delta_{\mathrm{MV}}[\zeta] = [c], \quad c|_U = \mathrm{d}a, \quad c|_V = \mathrm{d}b.$$

There is no additional sign. The subscript identifies this as the Mayer–Vietoris map. Different lifts differ by restrictions of a global form η, so their resulting c differs by $\mathrm{d}\eta$. If ζ changes by $\mathrm{d}\xi$, lift ξ to (a', b') and replace (a,b) by $(a + \mathrm{d}a', b + \mathrm{d}b')$. The resulting c is unchanged. In degree zero there is no such exact change. These observations prove well-definedness. Adding or scaling lifts proves linearity.

Theorem 13.4 (Mayer–Vietoris · *exactness sketched; connecting map and overlap exactness proved given partitions of unity*)**.** *The induced maps and connecting homomorphisms form an exact sequence*

$$\begin{aligned} 0 \longrightarrow H^0(M) \xrightarrow{i^*} H^0(U) \oplus H^0(V) \xrightarrow{j^*} H^0(W) \xrightarrow{\delta_{\mathrm{MV}}} H^1(M) \longrightarrow \cdots, \\ \cdots \longrightarrow H^k(M) \xrightarrow{i^*} H^k(U) \oplus H^k(V) \xrightarrow{j^*} H^k(W) \xrightarrow{\delta_{\mathrm{MV}}} H^{k+1}(M) \longrightarrow \cdots. \end{aligned} \tag{13.2}$$

Proof at the overlap term and sketch elsewhere. If $\zeta = b - a$ for closed lifts, its connecting form is zero. Hence $\operatorname{im} j^* \subset \ker \delta_{\mathrm{MV}}$. Conversely, if $\delta_{\mathrm{MV}}[\zeta] = 0$, its glued form is $c = \mathrm{d}\eta$ for a global η. Subtract the restrictions of η from both lifts. Their derivatives now vanish and their difference remains ζ. Thus $[\zeta] \in \operatorname{im} j^*$, proving exactness at $H^k(W)$.

At the pair term, an exact difference of closed representatives can be removed by lifting its primitive through j. The adjusted representatives agree and glue. At the global term, local primitives of a class restricting to zero have a closed difference on W whose connecting class is the original class. These are the other exactness arguments; their representative and degree-zero details are assigned to Sheet 13. This is the cochain construction of Tu, §25–§26, especially (26.1), pp. 288–291. ∎

13.5 Cohomology of spheres

Remove the south pole to obtain $U \subset S^n$ and the north pole to obtain V, for $n \geq 1$. Each set is diffeomorphic to $\mathbb{R}^n$ by stereographic projection. Their overlap has the product description in the figure.

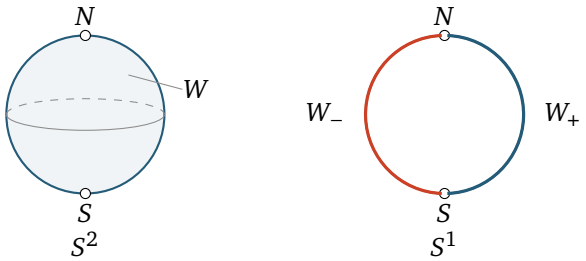


Figure 13.2: Each open set omits one pole. Their overlap is the whole sphere except the two marked poles, an open cylinder for S^2. For S^1 it has two components.

Write a sphere point as $(x', z) \in \mathbb{R}^n \times \mathbb{R}$. On W, the map $(x', z) \mapsto (x'/\sqrt{1 - z^2}, z)$ is a diffeomorphism onto $S^{n-1} \times (-1, 1)$, with inverse $(u, z) \mapsto (\sqrt{1 - z^2}u, z)$. Contracting the second coordinate to zero retracts W onto S^{n-1}. The two contractible cover members have zero positive-degree cohomology.

Theorem 13.5 (Sphere computation · *proved given Mayer–Vietoris exactness*)**.** *For $n \geq 1$ and $k \geq 0$,*

$$H^k(S^n) \cong \begin{cases} \mathbb{R}, & k = 0 \text{ or } k = n, \\ 0, & \text{otherwise.} \end{cases}$$

For S^0, degree zero is $\mathbb{R}^2$ and all higher degrees vanish. For $n \geq 1$, punctured $\mathbb{R}^n$ has the cohomology of S^{n-1}.

Proof. The space S^0 is two points. A function assigns one value to each point, and all positive-degree forms vanish. This proves the initial case. For $n = 1$, the overlap has two contractible components. The beginning of Mayer–Vietoris is

$$0 \longrightarrow H^0(S^1) \longrightarrow \mathbb{R}^2 \xrightarrow{(a,b)\mapsto(b-a,b-a)} \mathbb{R}^2 \longrightarrow H^1(S^1) \longrightarrow 0.$$

The middle map has a one-dimensional kernel and image. Its cokernel therefore also has dimension one. Exactness gives both claimed groups. Higher degrees vanish because the circle has dimension one.

For $n \geq 2$, the overlap is connected, so its degree-zero group is $\mathbb{R}$. The difference map $\mathbb{R}^2 \to \mathbb{R}$ is surjective with diagonal kernel. Exactness gives $H^0(S^n) = \mathbb{R}$ and $H^1(S^n) = 0$. For $k \geq 2$, the zero positive-degree groups of U and V give

$$0 \longrightarrow H^{k-1}(W) \xrightarrow{\delta_{\mathrm{MV}}} H^k(S^n) \longrightarrow 0.$$

Thus $H^k(S^n) \cong H^{k-1}(S^{n-1})$. Induction gives precisely one nonzero positive-degree group, in degree n. The explicit retraction in § 13.3 proves the punctured-space assertion. In particular, punctured $\mathbb{R}$ has two-dimensional degree-zero cohomology, whereas punctured $\mathbb{R}^n$ for $n \geq 2$ is connected. ∎

13.6 The area class and local primitives on the sphere

On the outward-oriented unit sphere let β be the area form, the restriction of Lecture 12's Coulomb flux form. We compute its class using the ordered cover above: $U = S^2 \setminus \{\text{south}\}$ and $V = S^2 \setminus \{\text{north}\}$. Write the local 1-forms as a_U and a_V, and their common derivative as c. Their difference and derivatives are related as in the diagram.

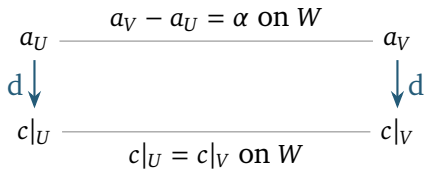


Figure 13.3: Local 1-forms differ by a closed form on W. Their derivatives agree and glue to the 2-form representing the connecting class.

Example 13.6 (The normalized sphere class)**.** The sphere computation gives $H^2(S^2) = \mathbb{R}$ and $H^1(S^2) = 0$. The form $\beta/(4\pi)$ is closed, since every 3-form on a surface vanishes. Its period is

$$\int_{S^2} \frac{\beta}{4\pi} = \frac{1}{4\pi} \int_0^{2\pi} \int_0^{\pi} \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\phi = 1.$$

It is therefore not exact and its class spans $H^2(S^2)$.

We can also compute the sign of the connecting map. On W, the globally defined angular 1-form is $\alpha = \mathrm{d}\phi/(2\pi)$, where $\mathrm{d}\phi$ denotes the angular form, not the differential of a globally defined real angle. Choose lifts

$$a_U = -\frac{x\mathrm{d}y - y\mathrm{d}x}{4\pi(1+z)}, \qquad a_V = \frac{x\mathrm{d}y - y\mathrm{d}x}{4\pi(1-z)}.$$

Their denominators do not vanish on their respective domains, so these forms are smooth also at the opposite poles. On the overlap, $x\mathrm{d}y - y\mathrm{d}x = (1-z^2)\mathrm{d}\phi$. Consequently $a_U = -(1-z)\mathrm{d}\phi/(4\pi)$ and $a_V = (1+z)\mathrm{d}\phi/(4\pi)$, giving $a_V - a_U = \alpha$. Their common derivative is

$$\mathrm{d}a_U = \mathrm{d}a_V = \frac{\mathrm{d}z \wedge \mathrm{d}\phi}{4\pi} = -\frac{\sin\theta \, \mathrm{d}\theta \wedge \mathrm{d}\phi}{4\pi} = -\frac{\beta}{4\pi}.$$

Equality extends over each pole by smoothness. Thus $\delta_{\mathrm{MV}}[\mathrm{d}\phi/(2\pi)] = -[\beta/(4\pi)]$, with the frozen order of U and V. The positive period-one class is $[\beta/(4\pi)]$. A nonzero period obstructs joining its local primitives to a global one. Write β_C for the closed Coulomb form of Lecture 12 on $\mathbb{R}^3 \setminus \{0\}$. Its restriction to S^2 is the area form β. Restriction is an isomorphism on cohomology, inverse to the radial retraction pullback. It sends $[\beta_C/(4\pi)]$ to $[\beta/(4\pi)]$. Hence $[\beta_C/(4\pi)]$ spans $H^2(\mathbb{R}^3 \setminus \{0\})$, with sphere period one.

Real coefficients lose torsion information. A torsion element is one annihilated by a positive integer. No real vector space has a nonzero torsion element. In particular, Lecture 4's order-two loop in $\mathrm{SO}(3)$ cannot be detected by a closed real 1-form's period. Use smooth loops constant near their endpoints, so their concatenation is smooth. Periods of a closed 1-form are invariant under smooth loop homotopy by Stokes on the cylinder. Continuous homotopies can be smoothed by the cited theorem. Periods add under loop composition, so twice that loop's period is zero and its real period is zero. This does not impose any integer normalization on the sphere class. Lecture 22 obtains that extra information from a bundle.

Self-check. Find $H^1(\mathbb{R}^2 \setminus \{0\})$ and a generator of period one on the counter-clockwise unit circle.

Answer. Radial retraction gives the cohomology of S^1, so this space is one-dimensional. The class $[\alpha_{\mathrm{ang}}/(2\pi)]$ has period one by Lecture 10. It is nonzero and therefore spans the space.

Summary. De Rham cohomology consists of closed forms modulo exact forms, over $\mathbb{R}$. The homotopy operator proves local exactness in positive degree on star-shaped sets and invariance under smooth homotopy. Cited smoothing extends invariance to continuous homotopy type. Mayer–Vietoris computes sphere cohomology from two contractible sets. On S^2, the area class $[\beta/(4\pi)]$ has period one. Lecture 14 relates periods to degree and singular homology.

Reading. Tu, *An Introduction to Manifolds*, §24–§29, especially §26, pp. 288–291, and §29, pp. 311–317. Lee, *Introduction to Smooth Manifolds*, second edition, Theorems 6.26 and 6.29, pp. 141–143.

Lecture 14

Degree, Poincaré–Hopf and de Rham's theorem

Objectives. *The integral of a normalized top form assigns a number to a smooth map between nonempty closed connected oriented manifolds of the same positive dimension. We prove that this number, the degree, is an integer and is unchanged under homotopy. Degree obstructs a nonvanishing smooth tangent field on an even sphere and proves the smooth fixed-point theorem. Two cited theorems then relate vector-field zeros and periods to the homology of Part I.*

By the end of this lecture you will be able to:

- Define degree with its dimension, orientation and compactness hypotheses.
- Compute degree as a signed count over a regular value.
- Prove the even-sphere hairy-ball theorem for smooth fields and the smooth Brouwer theorem.
- Compute the two local indices of the specified field on S^2.

14.1 Integration and top cohomology

Recall that a closed manifold is compact and has no boundary. All manifolds below are smooth. Cohomology has real coefficients. On an oriented n-manifold every n-form is closed, since all $(n+1)$-forms vanish. For a closed manifold, Stokes makes integration zero on exact forms. Integration therefore defines a linear map on $H^n(M)$.

Theorem 14.1 (Top cohomology · *sketched*)**.** *For a nonempty closed connected oriented n-manifold M,*

$$I_M \colon H^n(M) \longrightarrow \mathbb{R}, \qquad I_M[\alpha] = \int_M \alpha$$

is an isomorphism. Thus two top forms have equal integrals exactly when they differ by an exact form.

Sketch. For $n = 0$, connected M is a point and integration is its signed value, an isomorphism. For $n \geq 1$, a positive bump function in one positive chart gives a top form of positive integral. Rescaling it gives integral one, proving surjectivity. For injectivity the local input is this compact-support version of the Poincaré lemma: on $\mathbb{R}^n$, a compactly supported top form of integral zero has a compactly supported primitive.

The local construction uses induction and the fundamental theorem of calculus. For $n = 1$, integrate its coefficient from $-\infty$; zero total integral makes the primitive zero also beyond the support. For $n > 1$, write the coefficient as $f(x', u)$, set $h(x') = \int_{\mathbb{R}} f(x', u)\,\mathrm{d}u$, and

choose a compactly supported smooth function $\rho(u)$ of integral one. Split the coefficient as $(f - h\rho) + h\rho$. The first term has zero integral in every u fibre, so integrating in u gives a compactly supported primitive, with the sign from moving $\mathrm{d}u$ past $\mathrm{d}x'$. The second term has a primitive by the induction hypothesis applied to $h\mathrm{d}x'$, whose integral is $\int f = 0$, and then wedged with $\rho\mathrm{d}u$.

On M, choose a finite cover by positive coordinate balls diffeomorphic to $\mathbb{R}^n$. A partition of unity splits a top form into ball-supported pieces. Connectedness permits ordering the balls so that each after the first meets an earlier ball. In each chosen overlap take a top form of integral one. Starting with the last piece, subtract its integral times this overlap form and add the same form to the earlier piece. Repeating backwards makes every piece have integral zero, since the total is zero. The local construction now supplies primitives which extend by zero and sum to a global primitive. The cover and transfer details are the sketched steps. See Lee, *Introduction to Smooth Manifolds*, second edition, Theorems 17.30–17.31, pp. 454–455. ∎

For example, on outward S^2, the form $\beta/(4\pi)$ from Lecture 13 has integral one. A 2-form α of integral $a \in \mathbb{R}$ consequently satisfies $[\alpha] = a[\beta/(4\pi)]$.

14.2 The degree of a map

Let M and N be nonempty closed connected oriented manifolds of the same dimension $n \geq 1$. Let $f\colon M \to N$ be smooth. Choose a top form ν on N with integral one. The *degree* of f is

$$\deg f = \int_M f^*\nu.$$

It compares the orientations and will count inverse images with signs.

Theorem 14.2 (Degree and homotopy · *proved given top cohomology and cited smoothing*)**.** *The degree is independent of ν. For every top form α on N,*

$$\int_M f^*\alpha = (\deg f) \int_N \alpha.$$

Homotopic smooth maps have the same degree. The identity has degree one, and a constant map has degree zero when $n \geq 1$.

Proof. Two choices of ν differ by $\mathrm{d}\eta$, by the top-cohomology theorem. Their pullbacks differ by $\mathrm{d}f^*\eta$, whose integral is zero by Stokes. For an arbitrary α, write $a = \int_N \alpha$. The form $\alpha - a\nu$ has integral zero, so it is exact. Pulling back and integrating gives the claimed formula.

For smoothly homotopic maps, the homotopy operator of Proposition 13.1 makes the pullbacks of ν differ by an exact form. Their integrals coincide. A continuous homotopy between smooth maps can be smoothed by Lee's Theorem 6.29, recalled in Lecture 13, so the same conclusion holds. The identity pulls ν back to itself. A constant map has zero differential, hence pulls a positive-degree form back to zero. ∎

For example, the reflection $(x, y) \mapsto (x, -y)$ on S^1 reverses the angular form, so its degree is -1. The degree also multiplies under composition. For a smooth map g from N to another such manifold of the same dimension, choose a top form ν of integral one on its target. Apply the integral formula to $g^*\nu$ and then to ν to obtain $\deg(g \circ f) = (\deg g)(\deg f)$.

Closedness matters. The map $x \mapsto x/2$ from $[0,1]$ to itself pulls $\mathrm{d}x$ back to $\mathrm{d}x/2$. The quotient of the two integrals is $1/2$, but the interval has boundary, so this quotient is not a degree under our definition.

14.3 Integrality and regular values

A point $y \in N$ is a *regular value* of f if $\mathrm{d}f_x \colon T_xM \to T_yN$ is surjective for every $x \in f^{-1}(y)$. An empty inverse image satisfies this condition. In equal dimensions each such differential is invertible. Its *local degree* is $+1$ if it preserves orientation and -1 if it reverses orientation. Write this sign as $\operatorname{sgn}\det \mathrm{d}f_x$, using positive bases in both tangent spaces. An inverse branch is a neighbourhood mapped diffeomorphically onto a small target neighbourhood W. The diagram names three such branches V_1, V_2 and V_3 and shows their signs. The proof below constructs the common neighbourhood W.

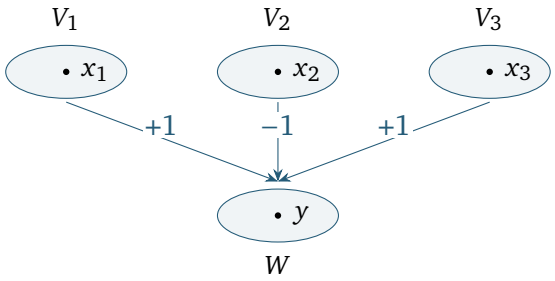


Figure 14.1: Three inverse branches over a small neighbourhood W contribute signs $+1$, -1 and $+1$. Their total contribution is one.

Proposition 14.3 (Signed count · *proved given top cohomology; regular values exist by cited Sard*)**.** *For every regular value $y \in N$, the inverse image is finite and*

$$\deg f = \sum_{x \in f^{-1}(y)} \operatorname{sgn}\det \mathrm{d}f_x .$$

In particular the degree is an integer, independent of the chosen regular value.

Proof. The inverse function theorem isolates each inverse-image point. The fibre is closed in compact M, hence compact. Its isolating neighbourhoods give an open cover with a finite subcover, so the fibre is finite. Write its points as $x_1, \dots, x_m$ when it is nonempty. Choose disjoint neighbourhoods U_i on which f is a diffeomorphism onto a neighbourhood of y. The complement $K = M \setminus \bigcup_i U_i$ is compact and has no point mapping to y. Its image is compact and closed. Choose a small connected coordinate ball W about y inside every $f(U_i)$ and disjoint from $f(K)$. The sets $V_i = U_i \cap f^{-1}(W)$ are then all the inverse branches, each diffeomorphic to W. On a connected branch the orientation sign is constant, since its nonzero Jacobian determinant is continuous.

Choose a smooth top form ν supported in W with integral one. Change of variables on each branch gives

$$\int_M f^*\nu = \sum_{i=1}^{m} \int_{V_i} f^*\nu = \sum_{i=1}^{m} \operatorname{sgn}\det \mathrm{d}f_{x_i} .$$

If the fibre is empty, choose W disjoint from the compact image $f(M)$. The pullback of such a supported ν is zero, giving degree zero and the empty sum. This proves the formula for every regular value.

Sard's theorem says that the critical values of a smooth map have measure zero in the target. We cite Lee, Theorem 6.10, p. 129. Here a critical value is the image of a point with nonsurjective differential. A nonempty coordinate ball has positive measure, so regular values exist. The formula at any of them proves integrality. Since it computes the already defined degree, it also proves independence of the regular value. ∎

14.4 Antipodal maps, tangent fields and fixed points

A fixed point of a map f means a point x with $f(x) = x$. Let μ be the outward area form on S^n, characterized in Lecture 12 by $\mu_p(v_1, \dots, v_n) = \det[p\ v_1\ \cdots\ v_n]$.

Corollary 14.4 (Applications of degree · *proved given top cohomology*)**.** *For $n \geq 1$, the antipodal map $a\colon S^n \to S^n$, given by $a(x) = -x$, has degree $(-1)^{n+1}$. Every smooth tangent field on S^{2m}, with $m \geq 1$, has a zero. Every smooth map from the closed unit ball $\overline{B}^d$ to itself has a fixed point, for $d \geq 1$.*

A nonvanishing unit tangent field would give the semicircle in the figure at every sphere point.

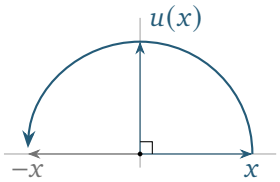


Figure 14.2: In the plane spanned by the orthogonal unit vectors x and $u(x)$, the unit semicircle joins x to $-x$ through $u(x)$.

Proof. The derivative of the antipodal map sends every tangent vector to its negative. Thus $a^*\mu = (-1)^{n+1}\mu$, because the determinant has $n+1$ columns. The integral definition gives its degree.

Suppose a tangent field X on S^{2m} never vanishes and normalize it to $u(x) = X_x/\|X_x\|$. Tangency gives $x \cdot u(x) = 0$. The map $H\colon [0,1] \times S^{2m} \to S^{2m}$ given by

$$H(t,x) = \cos(\pi t)x + \sin(\pi t)u(x)$$

has squared norm $\cos^2(\pi t) + \sin^2(\pi t) = 1$. It is a smooth homotopy from the identity to the antipodal map. Their degrees would be equal, contradicting $1 \neq -1$. This proves the *hairy-ball theorem* for even spheres.

For the ball of dimension $d \geq 2$, suppose a smooth f has no fixed point. The map $G\colon \overline{B}^d \to S^{d-1}$ given by $G(x) = (x - f(x))/\|x - f(x)\|$ is smooth. Its boundary restriction g has degree zero: for a normalized top form ν on S^{d-1}, Stokes gives $\int_{S^{d-1}} g^*\nu = \int_{\overline{B}^d} \mathrm{d}G^*\nu = 0$, since $\mathrm{d}\nu = 0$. On the boundary the formula

$$H(t,x) = \frac{x - tf(x)}{\|x - tf(x)\|}$$

is a smooth homotopy from the identity to g. For $t < 1$ its denominator is nonzero because $\|tf(x)\| \leq t < 1 = \|x\|$. At $t = 1$ it is nonzero by the assumption. Homotopy invariance gives $\deg g = 1$, a contradiction. For $d = 1$, the function $f(x) - x$ is nonnegative at -1 and nonpositive at 1, so the intermediate value theorem gives a zero. ∎

The fixed-point conclusion is the *Brouwer theorem*. It also holds for continuous maps. That extension is cited here from Lee, Theorem 17.39, pp. 459–460, for dimensions $d \geq 2$. Its proof uses Theorem 17.38 and smooth approximation into manifolds, Theorem 6.26. For $d = 1$ the intermediate value argument above applies directly to continuous maps. Sheet 14 develops the approximation argument.

14.5 The index of a vector-field zero

An isolated zero of a vector field is a zero with a neighbourhood containing no other zero. In dimension $n \geq 2$, choose coordinates centred at such a zero p and a small outward-oriented coordinate sphere S_ε^{n-1}. Write the coordinate components of the field as $X(x) \in \mathbb{R}^n$. Its *index* is

$$\operatorname{ind}_p X = \deg\left(x \longmapsto \frac{X(x)}{\|X(x)\|} : S_\varepsilon^{n-1} \longrightarrow S^{n-1}\right).$$

Changing the small radius gives homotopic maps through spheres avoiding zeros. Independence of the coordinate chart is cited from Milnor, *Topology from the Differentiable Viewpoint*, §6, Lemma 1, pp. 33–35. In dimension one use half the difference of the component's signs at the right and left endpoints of a small coordinate interval centred at the zero. In dimension zero each point has index one. For a closed manifold define its Betti numbers by $b_k(M) = \dim_{\mathbb{R}} H_k(M;\mathbb{R})$ and its Euler characteristic by $\chi(M) = \sum_{k=0}^{\dim M} (-1)^k b_k(M)$. These dimensions are finite by the cited finite-dimensionality of de Rham groups (Lee, Problem 18-8, p. 489, an exercise with hints) and the comparison below. For a finite Δ-complex model, Theorems 6.2 and 6.3 identify this with the count of Lecture 6.

Theorem 14.5 (Poincaré–Hopf · *cited*)**.** *For a smooth vector field with isolated zeros on a closed manifold M,*

$$\sum_{p : X_p = 0} \operatorname{ind}_p X = \chi(M).$$

Here χ is the Euler characteristic. The theorem is cited from Milnor, §6, p. 35.

On the unit sphere write $p = (x, y, z)$ and take $X_p = e_3 - z p$. This field provides the explicit test in the figure.

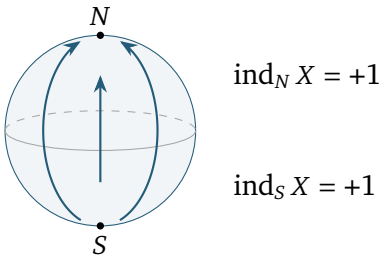


Figure 14.3: The field $X_p = e_3 - z p$ on S^2 points toward the north pole. It vanishes at the north pole N and the south pole S. Each zero has index +1.

The field is tangent because $p \cdot X_p = z - z\,\|p\|^2 = 0$. A zero requires $e_3 = z p$, hence $p = \pm e_3$. Near either pole use the coordinates $p = (x, y, \sigma\sqrt{1 - x^2 - y^2})$, where $\sigma = +1$ at the north pole and $\sigma = -1$ at the south pole. The coordinate components are $(-zx, -zy)$. On the small circle $(x, y) = \varepsilon(\cos\phi, \sin\phi)$, their normalization is $-\sigma(\cos\phi, \sin\phi)$. Both signs give one counter-clockwise turn as ϕ runs from 0 to 2π. Thus both indices are +1, and their

sum is $2 = \chi(S^2)$ from Lecture 6. A sink in two dimensions has index +1 here, not −1. For comparison, the planar source (x, y) has index +1, while the saddle $(x, -y)$ normalizes to $(\cos\phi, -\sin\phi)$ and has index −1.

14.6 Periods and singular homology

Recall from Lecture 6 that a singular cycle is a chain with zero boundary, and that $H_k(M;\mathbb{R})$ denotes homology with real coefficients. For a smooth singular simplex $\sigma\colon \Delta^k \to M$, integration means $\int_{\Delta^k} \sigma^*\alpha$, extended linearly to chains. A smooth simplex here extends smoothly to a neighbourhood of its parameter simplex. Arbitrary continuous singular simplices cannot be integrated this way.

Theorem 14.6 (De Rham · *cited*)**.** *For every smooth manifold M and $k \geq 0$, integration induces an isomorphism*

$$H^k(M) \longrightarrow \mathrm{Hom}_{\mathbb{Z}}(H_k(M;\mathbb{Z}),\mathbb{R}), \qquad [\alpha] \longmapsto \left([c] \longmapsto \int_c \alpha\right).$$

Here $\mathrm{Hom}_{\mathbb{Z}}$ *means homomorphisms of abelian groups. This includes replacing continuous cycles by smooth representatives. The cited construction uses Stokes on simplices, including their corners, and shows independence of all representatives. See Lee, Theorem 18.14, p. 484, with the comparison construction in Chapter 18.*

Thus the periods over all cycles determine a cohomology class completely. The same target is the real dual of $H_k(M;\mathbb{R})$. Indeed, real singular cochains are real-linear functionals on the real chain complex, equivalently additive real-valued functionals on integer chains.

A closed cochain vanishes on boundaries, so it restricts to a functional on homology. Every such functional extends from cycles to all chains by choosing a basis. A closed cochain restricts to zero on cycles exactly when it factors through the boundary map. Extending that factor to the preceding chain space makes it a coboundary.

For integer chains the same restriction and factorization argument uses a cited extension property in place of basis extension: every homomorphism from a subgroup of an abelian group to $\mathbb{R}$ extends to the whole group. This property is called injectivity of $\mathbb{R}$ and is stated in Lee, p. 473. Thus both homology duals compute the cohomology of the same real-valued cochain complex. This proves their identification given that cited property.

In finite dimension it gives $\dim H^k(M) = b_k(M)$. The real coefficients matter: an integer homology class annihilated by $m \geq 1$ has period P with $mP = 0$, hence $P = 0$. Differential forms therefore do not recover torsion classes.

14.7 Radial flux and complex power maps

Let $K \subset \mathbb{R}^3$ be a compact smooth region with connected boundary $S = \partial K$, oriented outward, and assume $0 \notin S$. The *radial Gauss map* $G\colon S \to S^2$ is given by $G(x) = x/\|x\|$. It records the direction from the origin. This map differs from the map assigning the outward unit normal to a surface point. The figure shows the two positions of the origin.



$0 \in \operatorname{int} K$ $\qquad$ $0 \notin K$

Figure 14.4: Sections of regions with the origin inside and outside. The radial map assigns the direction of the displayed ray to a boundary point.

Example 14.7 (Degree of radial flux)**.** Let β_{S^2} be the outward area form on the target sphere and let $r = \|x\|$. For tangent vectors v and w at $x \in S$,

$$\mathrm{d}G_x(v) = \frac{v}{r} - \frac{x(x \cdot v)}{r^3}, \qquad (G^*\beta_{S^2})_x(v, w) = \frac{\det[x\ v\ w]}{r^3}.$$

Indeed, the target form is the determinant with first column x/r. Every additional term parallel to x in the other columns gives zero. This pullback is precisely Lecture 12's Coulomb flux form restricted to S. Since the target area is 4π, its flux calculation gives

$$\deg G = \frac{1}{4\pi}\int_S G^*\beta_{S^2} = \begin{cases} 1, & 0 \in \operatorname{int} K, \\ 0, & 0 \notin K. \end{cases}$$

The flux is therefore an integer multiple of the sphere's area.

For complex maps use $\mathbb{C}\mathrm{P}^1$ with its complex orientation: if $z = u + \mathrm{i}v$ is a coordinate, the ordered real coordinates (u, v) are positive. The transition $w = 1/z$ has positive real Jacobian away from zero. This orientation agrees with the outward Bloch sphere of Lecture 7: $z = \tan(\theta/2)e^{\mathrm{i}\phi}$ gives its ordinary angular parametrization. Equivalently, the Bloch map pulls the outward area form back to $4\mathrm{d}u \wedge \mathrm{d}v/(1 + u^2 + v^2)^2$, which is positive.

Example 14.8 (Powers on the complex projective line)**.** For $n \in \mathbb{Z} \setminus \{0\}$ consider the map $f_n(z) = z^n$ on $\mathbb{C}\mathrm{P}^1$. For $n > 0$ it fixes zero and infinity. For $n < 0$ it exchanges them. Using $w = 1/z$ near infinity, in the source or target as appropriate, the expression near zero or near infinity is a positive power $w^{|n|}$ or $z^{|n|}$. Thus the extended map is smooth at zero and at infinity.

Fix a finite nonzero target value. Its inverse image has $|n|$ distinct points, by the polar formula for complex roots. At each such point the complex derivative nz^{n-1} is nonzero. Multiplication by a complex number $a + \mathrm{i}b$ has real derivative matrix

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}, \qquad \det = a^2 + b^2 > 0.$$

Each inverse-image point therefore has local degree +1. The signed-count formula gives $\deg f_n = |n|$. In particular, $z \mapsto 1/z$ has degree +1 on $\mathbb{C}\mathrm{P}^1$. On the equatorial circle the same formula has degree n instead: a negative exponent reverses its one-dimensional orientation. Conjugation $z \mapsto \bar{z}$ has real matrix $\operatorname{diag}(1, -1)$, so it reverses the complex orientation and has sphere degree -1.

Self-check. Identify the degree of a smooth map $f\colon S^1 \to S^1$ with Lecture 3's winding number. In particular, compute it for $f(e^{i\phi}) = e^{in\phi}$.

Answer. Use the normalized angular form $\alpha_{\mathrm{ang}}/(2\pi)$ on the target. A smooth angular lift ψ of the loop $\phi \mapsto f(e^{i\phi})$ has $\psi(2\pi) - \psi(0) = 2\pi\,\mathrm{wind}(f)$ by Lecture 3. Its pullback is $\psi'(\phi)\mathrm{d}\phi/(2\pi)$, so integration gives $\deg f = \mathrm{wind}(f)$. For the power map the lift is $\psi(\phi) = n\phi$, and the degree is n for every $n \in \mathbb{Z}$.

Summary. On a nonempty closed connected oriented manifold, top cohomology is measured by integration. Between such manifolds of equal positive dimension, degree is a homotopy invariant integer, computed by signed inverse images at a regular value. It gives the even-sphere hairy-ball theorem for smooth fields and the smooth Brouwer theorem. Poincaré–Hopf relates zero indices to the Euler characteristic. De Rham's theorem identifies cohomology with real period functionals on homology. Part II adds a metric, connections and physical fields to these topological constructions.

Reading. Lee, *Introduction to Smooth Manifolds*, second edition, Theorems 17.30–17.31, pp. 454–455; Theorem 17.35, pp. 457–458; Theorems 17.38–17.39, pp. 459–460; Theorem 18.14, p. 484. Milnor, *Topology from the Differentiable Viewpoint*, §6, pp. 32–35 (online scan).

Part II

Geometry and Topology in Physics

Lecture 15

Metrics, the Hodge star and Maxwell's equations

Objectives. *A metric measures tangent vectors and selects a volume form once an orientation is chosen. The Hodge star then turns a form into a form of complementary degree. On spacetime this operation packages Maxwell's four equations into two equations for forms. The monopole flux shows why closed field forms need not have global potentials.*

By the end of this lecture you will be able to:

- Construct a Riemannian metric and lower an index using its bilinear form.
- Compute the Hodge star with the metric, orientation and signature specified.
- Derive all four Maxwell equations and charge conservation from the form equations.
- Use a nonzero sphere period to rule out a global monopole potential.

15.1 Metrics and the identification of vectors with covectors

Let M be a smooth n-manifold. A *Riemannian metric* g is a smoothly varying positive-definite symmetric bilinear form g_p on each tangent space T_pM. A *pseudo-Riemannian metric* requires nondegeneracy instead of positive definiteness. Nondegeneracy means that $g_p(v, w) = 0$ for all $w \in T_pM$ implies $v = 0$. We use g for the metric throughout Part II.

By Sylvester's law of inertia, a nondegenerate symmetric form admits a basis with diagonal entries $+1$ and -1. Such a basis is called *pseudo-orthonormal*. The two counts are independent of the basis. This is the signature result imported from *Linear Algebra for Physics*, Lecture 11, the example *Signature of a form*. Write its signature as $(n - s, s)$, with s negative directions. For a smooth nondegenerate metric, the signature is constant on each connected component: eigenvalues vary continuously and cannot cross zero. Here a *Lorentzian metric* in dimension $n \geq 2$ has signature $(1, n - 1)$.

In coordinates, with $i, j \in \{1, \dots, n\}$, $g = g_{ij}\,\mathrm{d}x^i \otimes \mathrm{d}x^j$. Its inverse matrix is written (g^{ij}). For $X \in T_pM$ and $\alpha \in T_p^*M$, nondegeneracy gives the mutually inverse *musical maps*

$$X^\flat = g(X, \cdot), \qquad (X^\flat)_i = g_{ij}X^j, \qquad (\alpha^\sharp)^i = g^{ij}\alpha_j.$$

These really lower and raise an index using a metric. Lecture 7's translation of sibling coordinate labels did not do this. For example, in one tangent space take the following metric matrix and vector components. The resulting lower components are

$$(g_{ij}) = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \qquad (X^i) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \qquad ((X^\flat)_i) = \begin{pmatrix} 4 \\ 7 \end{pmatrix}.$$

The vector has squared length $4+14 = 18$. The figure shows how a positive metric determines its unit vectors.

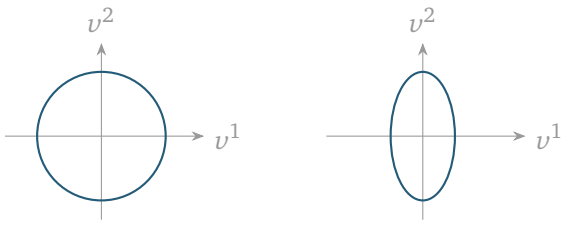


Figure 15.1: At a point, Euclidean unit vectors form a circle. For $g = 4\mathrm{d}x \otimes \mathrm{d}x + \mathrm{d}y \otimes \mathrm{d}y$ they form an ellipse with horizontal radius $1/2$.

Proposition 15.1 (Existence and induced metrics · *proved given partitions of unity*)**.** *Every smooth manifold admits a Riemannian metric. An immersion into a Riemannian manifold induces a Riemannian metric by pullback.*

Proof. On each coordinate domain use $g_a = \sum_i \mathrm{d}x_a^i \otimes \mathrm{d}x_a^i$. Choose a subordinate partition ρ_a from Theorem 12.1 and put $g = \sum_a \rho_a g_a$, extending each summand by zero. Supports make the extensions smooth. Local finiteness makes the sum smooth. For nonzero v, every summand $\rho_a g_a(v, v)$ is nonnegative, and at least one is positive because $\sum_a \rho_a = 1$. Thus the metric is positive definite. For an immersion f, the differential is injective, so $(f^*g)(v, v) = g(\mathrm{d}f(v), \mathrm{d}f(v)) > 0$ for $v \neq 0$. Smoothness and symmetry are preserved by pullback. ∎

An *isometry* is a diffeomorphism $f\colon (M, g_M) \to (N, g_N)$ with $f^*g_N = g_M$. A pseudo-Riemannian metric can restrict degenerately: for the matrix $\mathrm{diag}(1, -1, -1, -1)$, the vector $(1, 1, 0, 0)$ spans a line on which the restricted form is zero. So the induced-metric assertion used positivity.

15.2 Volume forms and the Hodge star

On an oriented pseudo-Riemannian n-manifold, the metric volume form is, in positive coordinates,

$$\mu = \sqrt{|\det(g_{ij})|}\, \mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n.$$

The defining normalization is $\mu = \theta^1 \wedge \cdots \wedge \theta^n$ when $\theta^1, \ldots, \theta^n$ is dual to a positive pseudo-orthonormal basis. To justify the coefficient, choose such a basis with metric matrix $S = \mathrm{diag}(\varepsilon_1, \ldots, \varepsilon_n)$, where $\varepsilon_i = \pm 1$. If the coordinate basis has columns A in that basis, its metric matrix is $A^T S A$. Hence $|\det g| = (\det A)^2$. The dual bases satisfy $\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n = (\det A)^{-1}\theta^1 \wedge \cdots \wedge \theta^n$. Since $\det A > 0$, the coefficient gives the required normalization. Under a positive coordinate change the square root of the absolute metric determinant contributes the absolute Jacobian and the top wedge contributes its inverse. The expressions therefore agree. They are smooth because the determinant never vanishes. This is the chosen volume form μ of Lecture 11, now fixed by metric and orientation. Without orientation the same coefficient defines a density.

For $0 \le k \le n$, the inverse metric gives a bilinear pairing of covectors. Its induced pairing on decomposable k-forms is

$$\langle \alpha_1 \wedge \cdots \wedge \alpha_k, \beta_1 \wedge \cdots \wedge \beta_k \rangle_g = \det\big[g^{-1}(\alpha_i, \beta_j)\big]_{i,j=1}^{k},$$

extended bilinearly. Expanding each covector in a basis identifies this determinant with the bilinear extension of its values on basis wedges. On functions it is ordinary multiplication. This uses Lecture 10's determinant wedge, with no additional normalization from the fermionic wedge of *Quantum Mechanics I*. The pairing is nondegenerate: in a pseudo-orthonormal dual basis the basis wedges are orthogonal and each has norm squared +1 or −1.

Theorem 15.2 (Hodge star). *Let (M, g) be an oriented pseudo-Riemannian n-manifold and $0 \le k \le n$. There is a unique pointwise real-linear operator, smooth in position, $\star\colon \Omega^k(M) \to \Omega^{n-k}(M)$ such that*

$$\alpha \wedge \star\beta = \langle \alpha, \beta \rangle_g \mu \qquad (\alpha, \beta \in \Omega^k(M)).$$

If the metric has s negative directions, then on k-forms

$$\star\star\alpha = (-1)^{k(n-k)+s}\alpha.$$

Proof. The wedge pairing between degrees k and $n-k$ is nondegenerate: pairing with a basis wedge selects the coefficient of its complementary wedge. It therefore represents the functional $\alpha \mapsto \langle \alpha, \beta \rangle_g \mu$ uniquely. The coordinate linear system uses smooth inverse-metric coefficients and nonzero volume coefficient, so its solution varies smoothly.

For the sign, use a positively oriented pseudo-orthonormal dual basis $\theta^1, \dots, \theta^n$ at one point. For an increasing index set I, write θ^I for its wedge, I^c for its ordered complement, $\varepsilon_I = \prod_{i \in I} \varepsilon_i$, and σ_I for the sign of the permutation (I, I^c). The defining equation gives

$$\star\theta^I = \varepsilon_I \sigma_I \theta^{I^c}.$$

The two metric signs multiply to $\prod_i \varepsilon_i = (-1)^s$. Interchanging the blocks I and I^c uses $k(n-k)$ transpositions, so $\sigma_I \sigma_{I^c} = (-1)^{k(n-k)}$. Applying the formula twice proves the square identity on every basis wedge, hence on every form. ∎

For example, in oriented Euclidean $\mathbb{R}^3$, $\star dx = dy \wedge dz$ and $\star(dy \wedge dz) = dx$. The complementary plane is shown in the figure.

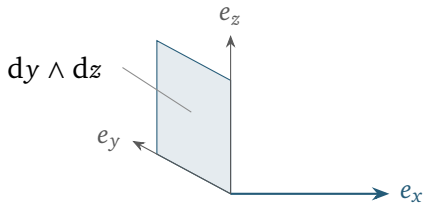


Figure 15.2: In Euclidean three-space, the covector dx pairs with the x direction. Its star is the oriented area form $dy \wedge dz$ on the complementary plane.

Reversing orientation negates μ and therefore negates the star. For a compactly supported real form α, the integral $\int_M \alpha \wedge \star\alpha$ is unchanged: the integrand and the integration orientation both change sign. The pairing need not be positive in indefinite signature. For complex forms extend the star complex-linearly and use $\bar{\alpha} \wedge \star\beta$ for the Hermitian pairing, antilinear in its first argument.

15.3 Maxwell's equations as equations for forms

Use coordinates $(x^0, x^1, x^2, x^3) = (t, x, y, z)$ on an open subset of spacetime $\mathbb{R}^4$. Greek indices run from 0 to 3 and spatial indices from 1 to 3. Take the Minkowski metric and orientation

$$\eta = \operatorname{diag}(1, -1, -1, -1), \qquad \mu = \mathrm{d}t \wedge \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z.$$

This is the metric recalled from *Symmetries in Physics*, Lecture 17, *Minkowski space from physics*. Units have $c = 1$ and rationalized sources. The metric has $s = 3$, so $\star^2 = -1$ on 2-forms. A real 2-form satisfying $\star F = F$ or $\star F = -F$ must therefore be zero. Euclidean four-space instead has $\star^2 = +1$ on 2-forms.

Let the electric field $\mathbf{E}$, magnetic field $\mathbf{B}$, charge density ρ and spatial current $\mathbf{J}$ be smooth fields. Their Cartesian spatial components use the Euclidean spatial metric. They are not obtained by lowering spacetime indices with η. The real *field-strength 2-form* and the two *current forms* are

$$F_{\mathrm{em}} = E_i \mathrm{d}x^i \wedge \mathrm{d}t + B_x \mathrm{d}y \wedge \mathrm{d}z + B_y \mathrm{d}z \wedge \mathrm{d}x + B_z \mathrm{d}x \wedge \mathrm{d}y,$$
$$j = \rho \mathrm{d}t - J_i \mathrm{d}x^i, \qquad \mathcal{J} = \star j.$$

The subscripts em distinguish these real electromagnetic forms from later connection forms.

Theorem 15.3 (Maxwell's equations in forms). *On the specified spacetime, with the fields and units above, Maxwell's equations*

$$\operatorname{div}\mathbf{B} = 0, \quad \partial_t \mathbf{B} + \operatorname{curl}\mathbf{E} = 0, \qquad \operatorname{div}\mathbf{E} = \rho, \quad \operatorname{curl}\mathbf{B} - \partial_t \mathbf{E} = \mathbf{J}$$

are equivalent to

$$\mathrm{d}F_{\mathrm{em}} = 0, \qquad \mathrm{d}\star F_{\mathrm{em}} = \mathcal{J}.$$

These equations imply charge conservation, $\partial_t \rho + \operatorname{div}\mathbf{J} = 0$. When these equations hold, every point has a neighbourhood with $F_{\mathrm{em}} = \mathrm{d}A_{\mathrm{em}}$ for a real 1-form A_{em}.

Proof. Use the star formula on the coordinate basis to obtain

$$\star F_{\mathrm{em}} = B_i \mathrm{d}t \wedge \mathrm{d}x^i + E_x \mathrm{d}y \wedge \mathrm{d}z + E_y \mathrm{d}z \wedge \mathrm{d}x + E_z \mathrm{d}x \wedge \mathrm{d}y,$$
$$\mathcal{J} = \rho \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z - J_x \mathrm{d}t \wedge \mathrm{d}y \wedge \mathrm{d}z + J_y \mathrm{d}t \wedge \mathrm{d}x \wedge \mathrm{d}z - J_z \mathrm{d}t \wedge \mathrm{d}x \wedge \mathrm{d}y.$$

For a spatial vector V, write the spatial 2-form $\mathcal{B}(V) = V_x \mathrm{d}y \wedge \mathrm{d}z + V_y \mathrm{d}z \wedge \mathrm{d}x + V_z \mathrm{d}x \wedge \mathrm{d}y$. Differentiation gives

$$\mathrm{d}F_{\mathrm{em}} = (\operatorname{div}\mathbf{B})\mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z + \mathrm{d}t \wedge \mathcal{B}(\partial_t \mathbf{B} + \operatorname{curl}\mathbf{E}),$$
$$\mathrm{d}\star F_{\mathrm{em}} = (\operatorname{div}\mathbf{E})\mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z + \mathrm{d}t \wedge \mathcal{B}(\partial_t \mathbf{E} - \operatorname{curl}\mathbf{B}).$$

For instance, the coefficient of $\mathrm{d}t \wedge \mathrm{d}y \wedge \mathrm{d}z$ in the first equation is $\partial_t B_x + \partial_y E_z - \partial_z E_y$. The other cyclic coefficients give the other components of the curl. Independence of the four basis 3-forms gives the four stated equations. Applying d to the source equation gives $\mathrm{d}\mathcal{J} = 0$. Differentiating the displayed current form gives $\mathrm{d}\mathcal{J} = (\partial_t \rho + \operatorname{div}\mathbf{J})\mu$, which proves conservation. Finally choose a star-shaped coordinate neighbourhood and apply Theorem 13.2 to the closed 2-form. Its primitive is the required local potential. ∎

Write the local potential in terms of its scalar and spatial parts as

$$A_{\mathrm{em}} = -\phi \mathrm{d}t + A_i \mathrm{d}x^i.$$

Taking its derivative gives $E_i = -\partial_i \phi - \partial_t A_i$ and $\mathbf{B} = \operatorname{curl} \mathbf{A}$, where $\mathbf{A} = (A_x, A_y, A_z)$. For a smooth real function χ, the *gauge transformation* $A_{\mathrm{em}} \mapsto A_{\mathrm{em}} + \mathrm{d}\chi$ leaves F_{em} unchanged because $\mathrm{d}^2\chi = 0$. Its component form is $A_i \mapsto A_i + \partial_i \chi$ and $\phi \mapsto \phi - \partial_t \chi$. This recalls *Theoretical Mechanics*, Lecture 3, *The charged particle*: the same transformation changes the particle Lagrangian by a total derivative.

Source signs must be translated together. Nakahara's Example 7.16 uses signature $-+++$.[1] The lower potential $\phi \mathrm{d}t - A_i \mathrm{d}x^i$ of Landau–Lifshitz has the opposite sign to A_{em} at matched units.[2]

15.4 The monopole obstruction

Let $r = \sqrt{x^2 + y^2 + z^2}$ and let $\Phi_{\mathrm{m}} \in \mathbb{R} \setminus \{0\}$ be the magnetic flux. A static *magnetic monopole* field on $\mathbb{R} \times (\mathbb{R}^3 \setminus \{0\})$ has magnetic field $\mathbf{B} = \Phi_{\mathrm{m}}(x, y, z)/(4\pi r^3)$ and zero electric field. Its field 2-form is

$$F_{\mathrm{em}} = \frac{\Phi_{\mathrm{m}}}{4\pi} \frac{x\,\mathrm{d}y \wedge \mathrm{d}z + y\,\mathrm{d}z \wedge \mathrm{d}x + z\,\mathrm{d}x \wedge \mathrm{d}y}{r^3}.$$

Proposition 15.4 (No global monopole potential · *proved given partitions of unity*)**.** *This form is closed and has period Φ_{m} on the outward unit sphere in each fixed-time slice. It has local potentials but no global 1-form potential on the punctured spacetime.*

Proof. The form is time-independent, with no time differential. Its spatial part is the Coulomb flux form of § 12.7 times $\Phi_{\mathrm{m}}/(4\pi)$. Lecture 12 proved closedness and sphere period 4π for that form, giving the stated period. Local potentials follow from the Poincaré lemma. If $F_{\mathrm{em}} = \mathrm{d}A_{\mathrm{em}}$ globally, restriction to a fixed-time sphere and Stokes would give zero period, contradicting $\Phi_{\mathrm{m}} \neq 0$. ∎

No integer condition has been imposed on Φ_{m}. Lecture 22 studies the additional condition from a bundle and a charged particle.

15.5 Harmonic representatives

On a closed oriented Riemannian manifold, real forms of equal degree have the integrated pairing $(\alpha, \beta)_{L^2} = \int_M \alpha \wedge \star\beta$. The *codifferential* $\delta \colon \Omega^k(M) \to \Omega^{k-1}(M)$ is the formal adjoint of exterior differentiation, characterized by $(\mathrm{d}\alpha, \beta)_{L^2} = (\alpha, \delta\beta)_{L^2}$. This operator differs from the connecting map δ_{MV} of Lecture 13. The *Hodge Laplacian* is $\Delta = \mathrm{d}\delta + \delta\mathrm{d}$. A form is *harmonic* if $\Delta\alpha = 0$. Existence of this adjoint and its local formula are supplied by the star and integration by parts; see Nakahara, Definition 7.6 and Theorem 7.5. Sheet 15 derives that formula and computes the sphere Laplacian.

Theorem 15.5 (Hodge representatives · *cited*)**.** *Every de Rham class on a closed oriented Riemannian manifold has a unique harmonic representative.*

[1] For fixed orientation, metric reversal multiplies the k-form pairing and its star by $(-1)^k$. Thus the star on 2-forms is unchanged, while Nakahara's current is $j_{\mathrm{src}} = -j$. It follows that $\star_{\mathrm{src}} j_{\mathrm{src}} = \mathcal{J}$. His matrix (5.73), with its $(0,3)$ entry corrected to $-E_z$ by antisymmetry, represents F_{em}. His source equation is $\mathrm{d}^\dagger F_{\mathrm{em}} = j_{\mathrm{src}}$. Definition 7.6 gives $\mathrm{d}^\dagger = \star_{\mathrm{src}} \mathrm{d} \star_{\mathrm{src}}$ on Lorentzian 2-forms. Apply the source star and use its square +1 on 3-forms to obtain $\mathrm{d}\star F_{\mathrm{em}} = \mathcal{J}$.

[2] At matched units, $A^{\mathrm{LL}} = -A_{\mathrm{em}}$ and $F^{\mathrm{LL}} = -F_{\mathrm{em}}$. To convert instead to Gaussian fields with $c = 1$, set $F_{\mathrm{G}} = \sqrt{4\pi} F_{\mathrm{em}}$ and $\mathcal{J}_{\mathrm{G}} = \mathcal{J}/\sqrt{4\pi}$. Then $\mathrm{d}\star F_{\mathrm{G}} = 4\pi \mathcal{J}_{\mathrm{G}}$. The Gaussian lower-potential convention $F_{\mathrm{G}}^{\mathrm{LL}} = -F_{\mathrm{G}}$ therefore gives $\mathrm{d}\star F_{\mathrm{G}}^{\mathrm{LL}} = -4\pi \mathcal{J}_{\mathrm{G}}$. This converts the field sign and source normalization together.

The analytic existence theorem is cited from Nakahara, Theorem 7.8, §7.9.6. The closedness hypothesis is inherited there from Theorem 7.7. The statement does not apply to the Lorentzian spacetime used above.

15.6 The spacetime star and vacuum duality

The six basis 2-forms give a direct way to use the signature convention. For example, $dt \wedge dx$ has norm squared -1 and $(dt \wedge dx) \wedge (dy \wedge dz) = \mu$. Its star is therefore $-dy \wedge dz$. The same rule gives the table:

α	$\star\alpha$	α	$\star\alpha$
$dt \wedge dx$	$-dy \wedge dz$	$dy \wedge dz$	$dt \wedge dx$
$dt \wedge dy$	$+dx \wedge dz$	$dx \wedge dz$	$-dt \wedge dy$
$dt \wedge dz$	$-dx \wedge dy$	$dx \wedge dy$	$dt \wedge dz$

Example 15.6 (An electric field and its magnetic dual)**.** Take a constant electric field $\mathbf{E} = E_0 e_x$, with $E_0 \neq 0$, and zero magnetic field on Minkowski space. Its potential and field are

$$A_{\mathrm{em}} = E_0 x\, dt, \qquad F_{\mathrm{em}} = E_0 dx \wedge dt.$$

All field derivatives vanish, so Maxwell's equations hold with zero sources. The table gives $\star F_{\mathrm{em}} = E_0 dy \wedge dz$. This is the field of a constant magnetic field $\mathbf{B}' = E_0 e_x$ and zero electric field. A potential for it is $E_0 y\, dz$. Thus the transformation $F_{\mathrm{em}} \mapsto \star F_{\mathrm{em}}$ changes this electric solution into a magnetic solution.

For any source-free field, this transformation is *vacuum duality*. Indeed, $dF_{\mathrm{em}} = 0$ and $d\star F_{\mathrm{em}} = 0$ imply $d(\star F_{\mathrm{em}}) = 0$ and $d\star(\star F_{\mathrm{em}}) = -dF_{\mathrm{em}} = 0$. The transformed components are $\mathbf{E}' = -\mathbf{B}$ and $\mathbf{B}' = \mathbf{E}$. Applying duality twice gives $-F_{\mathrm{em}}$, rather than F_{em}.

Self-check. In oriented Euclidean $\mathbb{R}^3 \setminus \{0\}$, take $\mathbf{E}^\flat = (x dx + y dy + z dz)/r^3$. Compute $\star\mathbf{E}^\flat$.

Answer. The Euclidean rules give

$$\star\mathbf{E}^\flat = \frac{x\, dy \wedge dz + y\, dz \wedge dx + z\, dx \wedge dy}{r^3}.$$

This is the Coulomb flux form called β in Lecture 12. Here the spatial Euclidean star is used, not the four-dimensional spacetime star.

Summary. A Riemannian metric exists by partitions of unity and identifies tangent vectors with covectors. A metric and orientation define volume and the Hodge star. Its square depends on degree, dimension and the number of negative metric directions. With signature $+---$, Maxwell's equations are $dF_{\mathrm{em}} = 0$ and $d\star F_{\mathrm{em}} = \mathcal{J}$. They imply charge conservation and local potentials. Monopole flux prevents a global potential. On closed oriented Riemannian manifolds, the cited Hodge theorem selects one harmonic form in each cohomology class. Lecture 16 uses a metric to differentiate tangent fields and define geodesics.

Reading. Nakahara, *Geometry, Topology and Physics*, second edition, §7.1 and §7.9, especially (7.172)–(7.177), Example 7.16, and Theorems 7.7–7.8. Tu, *Differential Geometry*, Theorem 1.12, pp. 6–7, and §16.4–§16.5, pp. 132–135.

Lecture 16

Connections and geodesics

Objectives. *Differentiating a tangent field involves both its components and its coordinate basis. A connection accounts for this change. A metric selects one connection by compatibility and zero torsion. Its parallel transport preserves inner products, and its geodesics have parallel velocity.*

By the end of this lecture you will be able to:

- Construct the Levi-Civita connection from a nondegenerate metric.
- Compute parallel transport and geodesic equations in coordinates.
- Find a conserved geodesic quantity from a Killing field.
- Explain why the exponential is onto for a compact connected Lie group.

16.1 Affine connections

Let M be a smooth n-manifold without boundary and let $\mathfrak{X}(M)$ denote its smooth vector fields, as in Lecture 9. An *affine connection* is an $\mathbb{R}$-bilinear map $\nabla\colon \mathfrak{X}(M)\times\mathfrak{X}(M)\to\mathfrak{X}(M)$ that satisfies, for $f\in C^\infty(M)$,

$$\nabla_{fX}Y = f\nabla_X Y, \qquad \nabla_X(fY) = X(f)Y + f\nabla_X Y.$$

The field $\nabla_X Y$ is the *covariant derivative* of Y in direction X. The second rule prevents this operation from being bilinear over functions. For example, on Cartesian $\mathbb{R}^n$, the usual directional derivative $\nabla_X Y = X^i\partial_i Y^j\partial_j$ satisfies both rules.

The rules make the derivative local. If Y vanishes near p, choose a smooth cutoff f supported there with $f=1$ near p. Then $fY=0$, and the Leibniz rule at p gives $\nabla_X Y|_p = 0$. The first rule proves the same locality in X. Thus local fields can be extended with cutoffs before taking their derivative.

In coordinates $x^1,\dots,x^n$, write $\partial_i = \partial/\partial x^i$. The *Christoffel symbols* Γ_{ij}^k are defined by

$$\nabla_{\partial_i}\partial_j = \Gamma_{ij}^k\partial_k, \qquad (\nabla_X Y)^k = X^i(\partial_i Y^k + \Gamma_{ij}^k Y^j).$$

Repeated indices range from 1 to n. The second formula follows from the two connection rules and shows that the symbols determine the connection. In a chart, this expression uses only X_p, Y_p and the first derivatives of Y at p.

The *torsion* is $T(X,Y) = \nabla_X Y - \nabla_Y X - [X,Y]$. It is a tensor: the derivative term in $\nabla_Y(fX) = Y(f)X + f\nabla_Y X$ cancels the derivative term in $[fX,Y] = f[X,Y] - Y(f)X$. Thus $T(fX,Y) = fT(X,Y)$. Antisymmetry gives linearity over functions in the other slot. To see why function-linearity gives a pointwise tensor, restrict locally using cutoffs. Expanding $Z = Z^i\partial_i$ in any function-linear slot gives $A(Z)(p) = Z^i(p)A(\partial_i)(p)$, which depends only on Z_p.

The connection is *torsion-free* when $T = 0$. Since coordinate fields commute, this is equivalent to $\Gamma^k_{ij} = \Gamma^k_{ji}$.

For a pseudo-Riemannian metric g, a *metric connection* satisfies

$$Xg(Y,Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z) \qquad (X,Y,Z \in \mathfrak{X}(M)).$$

This identity differentiates the inner product using the connection.

16.2 The Levi-Civita connection

Theorem 16.1 (Levi-Civita). *Every pseudo-Riemannian manifold has a unique torsion-free metric connection. It is called the* Levi-Civita connection. *It satisfies the Koszul formula*

$$\begin{aligned} 2g(\nabla_X Y, Z) = {} & Xg(Y,Z) + Yg(X,Z) - Zg(X,Y) \\ & - g(X,[Y,Z]) + g(Y,[Z,X]) + g(Z,[X,Y]). \end{aligned}$$

Its coordinate symbols are

$$\Gamma^k_{ij} = \frac{1}{2} g^{k\ell}(\partial_i g_{\ell j} + \partial_j g_{\ell i} - \partial_\ell g_{ij}).$$

Proof. For uniqueness, write metric compatibility in the directions X, Y and Z. Add the first two identities and subtract the third. Replace $\nabla_Y X$ by $\nabla_X Y - [X,Y]$, and do the same for the other reversed pairs. The connection terms that remain are $2g(\nabla_X Y, Z)$, giving Koszul's formula. Nondegeneracy determines $\nabla_X Y$ uniquely.

For existence, denote the right side by $\mathcal{K}(X,Y,Z)$. Expand derivatives by the product rule and brackets by $[X, fY] = X(f)Y + f[X,Y]$. The derivative terms cancel to give

$$\begin{aligned} \mathcal{K}(X,Y,fZ) &= f\mathcal{K}(X,Y,Z), \\ \mathcal{K}(fX,Y,Z) &= f\mathcal{K}(X,Y,Z), \\ \mathcal{K}(X,fY,Z) &= f\mathcal{K}(X,Y,Z) + 2X(f)g(Y,Z). \end{aligned}$$

For example, in the first identity the extra terms are $X(f)g(Y,Z) + Y(f)g(X,Z) - Y(f)g(X,Z) - X(f)g(Y,Z) = 0$. The cutoff and coordinate-expansion argument of § 16.1 therefore makes $\mathcal{K}(X,Y,\cdot)/2$ a smooth covector field. Apply the inverse metric to define $\nabla_X Y$. The remaining identities give linearity in X and the Leibniz rule in Y. Additivity and real bilinearity follow directly from the formula.

Finally, exchanging the indicated slots in $\mathcal{K}$ gives

$$\begin{aligned} \mathcal{K}(X,Y,Z) - \mathcal{K}(Y,X,Z) &= 2g([X,Y],Z), \\ \mathcal{K}(X,Y,Z) + \mathcal{K}(X,Z,Y) &= 2Xg(Y,Z). \end{aligned}$$

The first identity proves zero torsion by nondegeneracy. The second proves metric compatibility. Substituting coordinate fields in Koszul, whose brackets vanish, gives $2g_{k\ell}\Gamma^k_{ij} = \partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}$. Multiplication by the inverse metric yields the stated symbols. ∎

For a constant metric matrix the symbols vanish in those coordinates. In particular, Minkowski geodesics will be straight lines. For an embedded surface in Euclidean three-space, the connection is $\nabla_X Y = \mathrm{pr}_{TM}(D_X Y)$, the tangential projection of the ambient derivative. Indeed, projection preserves the derivative rules. The ambient product rule gives metric compatibility because normal components pair to zero with tangent fields. The ambient identity $D_X Y - D_Y X = [X,Y]$ gives zero torsion. Uniqueness identifies this connection.

16.3 Parallel transport and geodesics

A *field along a curve* $\gamma\colon [a,b] \to M$ assigns $V(t) \in T_{\gamma(t)}M$ smoothly. In a coordinate segment its derivative is

$$\frac{DV}{dt} = (\dot V^k + \Gamma^k_{ij}(\gamma(t))\dot x^i V^j)\partial_k.$$

Equivalently, define this expression as $\dot v^j \partial_j + v^j \nabla_{\dot\gamma}\partial_j$ for $V = v^j\partial_j$ along the curve. In its first argument, the connection uses only the value of the direction field at the point. Write a local smooth basis as a row $e = (e_1, \dots, e_n)$, called a *frame*. For a new frame $e' = eA$ and coefficient columns $v = Av'$, define C by $\nabla_{\dot\gamma} e = eC$. The Leibniz rule and the coefficient change give

$$\nabla_{\dot\gamma}(eA) = e(\dot A + CA),$$
$$e(A\dot v' + \dot A v' + CAv') = e(\dot v + Cv).$$

Here $\dot A$ is the derivative along γ; the Leibniz rule supplies it. Thus the derivative is independent of the frame, including at repeated points of the curve. For $V = Y \circ \gamma$ it equals $\nabla_{\dot\gamma} Y$. A field is *parallel* when $DV/dt = 0$.

Proposition 16.2 (Parallel transport · *proved given linear ODE existence and uniqueness*)**.** *For a smooth curve $\gamma\colon [a,b] \to M$ and $v \in T_{\gamma(a)}M$, there is a unique parallel field with $V(a) = v$. The map $P_\gamma(v) = V(b)$ is a linear isomorphism. For a metric connection it preserves the metric.*

Proof. The coordinate equation is a linear ODE $\dot V = -C(t)V$, where $C^k{}_j = \Gamma^k_{ij}\dot x^i$. On a compact coordinate segment its smooth coefficients are bounded, so linear ODE existence and uniqueness give a solution throughout that segment. A finite subdivision places each closed segment in a chart. Propagate the initial value through the subdivision. Coordinate independence and uniqueness glue the solutions. The same uniqueness proves linearity. Reversing the curve gives the inverse map. In coordinates, metric compatibility reads $\partial_i g_{k\ell} = \Gamma^m_{ik} g_{m\ell} + \Gamma^m_{i\ell} g_{km}$. Differentiate $g_{k\ell}V^k W^\ell$ along the curve and substitute this identity. It gives the along-curve product rule. For parallel fields,

$$\frac{d}{dt} g(V,W) = g(DV/dt, W) + g(V, DW/dt) = 0.$$

Thus endpoint inner products agree. ∎

This is the ODE input already used in Theorem 9.1. The polar-coordinate figure shows that parallel fields can have changing components.

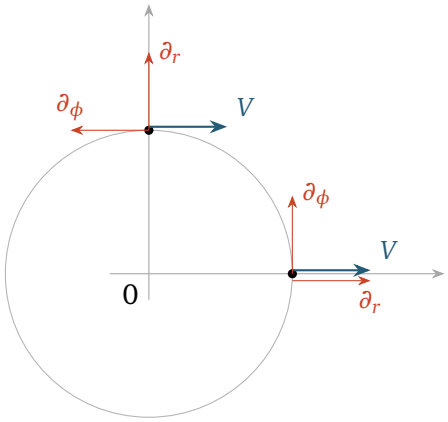


Figure 16.1: The same Cartesian vector $(1,0)$ at two points of the unit circle. Its polar components change from $(1,0)$ at $(1,0)$ to $(0,-1)$ at $(0,1)$. The arrows ∂_r and ∂_ϕ show the changing basis.

A *geodesic* is a parametrized curve with $D\dot\gamma/dt = 0$. Its coordinate equation is

$$\ddot x^k + \Gamma^k_{ij}\dot x^i \dot x^j = 0.$$

The local second-order ODE theorem gives a unique maximal geodesic γ_v with $\gamma_v(0) = p$ and $\dot\gamma_v(0) = v \in T_pM$. Here *maximal* means that its interval of definition admits no extension. The geodesic equation is a first-order field on TM, with bundle components $(\dot x^k, \dot v^k) = (v^k, -\Gamma^k_{ij}v^i v^j)$. The open smooth flow domain in Theorem 9.1 makes the *connection exponential* a smooth map $\exp_p \colon \mathcal{D}_p \to M$, given by $\exp_p(v) = \gamma_v(1)$. Its domain $\mathcal{D}_p \subset T_pM$ consists of the initial velocities for which the geodesic exists through time 1. The domain is open as the time-1 slice of that flow domain, restricted to T_pM. It contains zero because the constant curve is a geodesic. The connection is *geodesically complete* if every maximal geodesic is defined on all of $\mathbb{R}$; then $\mathcal{D}_p = T_pM$. For the Levi-Civita connection, $g(\dot\gamma,\dot\gamma)$ is constant by metric compatibility. In the Riemannian case, its square root is the speed.

The *energy* of a curve on $[a,b]$ is

$$E(\gamma) = \frac{1}{2}\int_a^b g(\dot\gamma,\dot\gamma)\,dt.$$

A curve is *stationary* when $\frac{d}{ds}\big|_0 E(\gamma_s) = 0$ for every smooth family γ_s with $\gamma_0 = \gamma$ and fixed endpoints.

Proposition 16.3 (Energy equations · *sketched; coordinate equivalence proved*)**.** *The Levi-Civita geodesics are precisely the stationary curves of the energy with fixed endpoints.*

The Euler–Lagrange equation, recalled from *Theoretical Mechanics*, Lecture 2, *Lagrange's equations*, is $d(\partial L/\partial \dot x^k)/dt - \partial L/\partial x^k = 0$. For $L = \frac12 g_{ij}\dot x^i\dot x^j$, its left side is

$$g_{k\ell}\ddot x^\ell + \partial_i g_{kj}\dot x^i\dot x^j - \frac12 \partial_k g_{ij}\dot x^i\dot x^j = g_{k\ell}(\ddot x^\ell + \Gamma^\ell_{ij}\dot x^i\dot x^j).$$

Nondegeneracy gives the geodesic equation. The variational step differentiates a fixed-endpoint variation and integrates its velocity term by parts. The endpoint term vanishes; the remaining integral pairs the displayed expression with the variation field. Sheet 16 supplies those details. No minimizing assertion follows in indefinite signature.

On the unit sphere, tangential projection gives $D\dot\gamma/dt = \ddot\gamma + |\dot\gamma|^2\gamma$. Indeed, differentiating $\gamma\cdot\gamma = 1$ twice gives $\gamma\cdot\ddot\gamma = -|\dot\gamma|^2$. For $v \in T_pS^2$ with $c = |v| > 0$, the unique solution is

$$\gamma_v(t) = \cos(ct)p + \frac{\sin(ct)}{c}v.$$

It lies on the sphere, has the prescribed initial data, and satisfies $\ddot\gamma = -c^2\gamma$. Its image is a *great circle*, the intersection with a plane through the origin. For $v = 0$ the geodesic is constant. In Minkowski coordinates, the zero Christoffel symbols instead give $\gamma_v(t) = p + tv$ for every initial vector, including zero and *null vectors*, which satisfy $\eta(v,v) = 0$.

16.4 Killing fields and conserved quantities

For a field K with local flow φ_t, define the metric Lie derivative by $L_Kg = \frac{d}{dt}\big|_0 \varphi_t^*g$. A *Killing field* satisfies $L_Kg = 0$. The flow law gives $\frac{d}{dt}\varphi_t^*g = \varphi_t^*(L_Kg)$, as for forms in Lecture 11. Thus the local flow of a Killing field consists of isometries, and conversely. For local fields Y and Z, put $Y_t = (\varphi_{-t})_*Y$ and $Z_t = (\varphi_{-t})_*Z$. The definition of tensor pullback gives $g(Y,Z)\circ\varphi_t = (\varphi_t^*g)(Y_t,Z_t)$. Differentiate at zero. The product rule and Theorem 9.3 give $Kg(Y,Z) = (L_Kg)(Y,Z) + g([K,Y],Z) + g(Y,[K,Z])$. Rearranging proves $(L_Kg)(Y,Z) = Kg(Y,Z) - g([K,Y],Z) - g(Y,[K,Z])$. Metric compatibility and zero torsion turn this into

$$(L_Kg)(Y,Z) = g(\nabla_YK,Z) + g(Y,\nabla_ZK).$$

Hence K is Killing exactly when the bilinear form $(Y,Z)\mapsto g(\nabla_YK,Z)$ is antisymmetric.

Proposition 16.4 (Killing conservation)**.** *Along a Levi-Civita geodesic γ, every Killing field K gives the constant quantity $g(K,\dot\gamma)$.*

Proof. Metric compatibility and the geodesic equation give

$$\frac{d}{dt}g(K,\dot\gamma) = g(\nabla_{\dot\gamma}K,\dot\gamma) = \frac{1}{2}(L_Kg)(\dot\gamma,\dot\gamma) = 0.$$ ∎

For example, a constant translation field in Minkowski space is Killing because the metric is constant. Its pairing with $\dot\gamma = v$ is constant, as the straight-line solution also shows.

16.5 Bi-invariant metrics and the group exponential

On a Lie group G, write $\ell_h(g) = hg$ and $r_h(g) = gh$. A metric is *bi-invariant* if every left and right translation is an isometry. Recall from Lecture 9 that each $X \in \mathfrak{g} = T_eG$ defines a complete left-invariant field X^L, whose identity curve $c_X(t)$ is a one-parameter subgroup. The *group exponential* $\exp\colon \mathfrak{g}\to G$ is given by $\exp X = c_X(1)$. Scaling the ODE gives $c_X(t) = \exp(tX)$, and the flow of X^L is $r_{\exp(tX)}$. This map differs in definition from Exp_p.

Theorem 16.5 (Bi-invariant geodesics · *proved given smooth ODE uniqueness*)**.** *For a bi-invariant pseudo-Riemannian metric, $\nabla_{X^L}Y^L = \frac{1}{2}[X,Y]^L$. The geodesics through the identity are exactly $t\mapsto\exp(tX)$, and $\mathrm{Exp}_e = \exp$ on all of $\mathfrak{g}$.*

Proof. The inner products of left-invariant fields are constant. Right translations are isometries, so X^L is Killing. The metric Lie derivative identity above therefore gives $g([X^L,Y^L],Z^L) + g(Y^L,[X^L,Z^L]) = 0$. Using this identity in the three bracket terms of Koszul gives

$$2g(\nabla_{X^L}Y^L,Z^L) = g([X^L,Y^L],Z^L).$$

The fields Z^L span each tangent space, so nondegeneracy proves the formula. Along c_X, its velocity is X^L, hence its acceleration is $\frac{1}{2}[X,X]^L = 0$. Uniqueness of the geodesic ODE gives all identity geodesics. Completeness of X^L gives their full real domains and the exponential equality. ∎

For compact G, a bi-invariant Riemannian metric exists. Here is a construction using Lecture 12's integration. A left-translated basis of $\mathfrak{g}$ gives a global frame and a positive, left-invariant volume form μ, normalized to have integral 1. Let b be any positive inner product on $\mathfrak{g}$. Define Ad_h to be the differential at the identity of $g \mapsto hgh^{-1}$. Average by

$$B(X,Y) = \int_G b(\mathrm{Ad}_{g^{-1}} X, \mathrm{Ad}_{g^{-1}} Y)\, \mu(g).$$

This is positive definite: a nonzero vector has a strictly positive integrand. Conjugation composes, so substituting $g = hk$ and using left-invariance of volume gives $B(\mathrm{Ad}_h X, \mathrm{Ad}_h Y) = B(X,Y)$. Extend B by left translation to a smooth metric. The identity $r_h \circ \ell_g = \ell_{gh} \circ (g' \mapsto h^{-1}g'h)$ shows that right translation acts on left-translated components by $\mathrm{Ad}_{h^{-1}}$. The Ad-invariance of B proves that the metric is bi-invariant.

For a connected Riemannian manifold, define a curve's length by $\ell(\gamma) = \int_a^b \sqrt{g(\dot\gamma,\dot\gamma)}\, dt$. The distance is the infimum of lengths of piecewise smooth curves joining its endpoints. A *minimizing geodesic* attains that infimum. The cited Hopf–Rinow theorem states that completeness of this distance is equivalent to geodesic completeness, and that on a complete connected Riemannian manifold every two points are joined by a minimizing geodesic (Tu, Theorem 16.6, p. 131). The distance induces the manifold topology (Lee, Lemma 6.2, pp. 94–95), so compactness makes it complete. Reparametrize the joining geodesic to $[0,1]$ by a linear change of time. Thus a compact connected G has every element $g = \mathrm{Exp}_e v = \exp v$ for some $v \in \mathfrak{g}$. This surjectivity conclusion is proved given integration and Hopf–Rinow; it does not assert injectivity of the exponential.

16.6 The round sphere in coordinates

For the unit sphere, the spherical parametrization is

$$p(\theta,\phi) = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta), \qquad 0 < \theta < \pi.$$

Here ϕ lies in an angular chart. Dot products of its coordinate tangents give

$$(g_{ij}) = \begin{pmatrix} 1 & 0 \\ 0 & \sin^2\theta \end{pmatrix}.$$

The figure compares a great circle with a latitude.

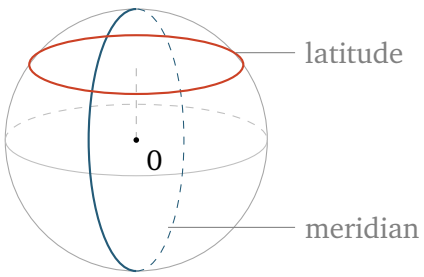


Figure 16.2: A meridian lies in a plane through the sphere's centre. A northern latitude lies in a plane parallel to the equator, away from the centre.

Example 16.6 (Sphere connection and angular momentum)**.** Only $\partial_\theta g_{\phi\phi} = 2\sin\theta\cos\theta$ is nonzero among the metric derivatives. The Christoffel formula gives

$$\Gamma^\theta_{\phi\phi} = -\frac{1}{2}\partial_\theta g_{\phi\phi} = -\sin\theta\cos\theta,$$
$$\Gamma^\phi_{\theta\phi} = \Gamma^\phi_{\phi\theta} = \frac{1}{2\sin^2\theta}\partial_\theta g_{\phi\phi} = \cot\theta.$$

All other symbols vanish. Substitution gives

$$\ddot\theta - \sin\theta\cos\theta\,\dot\phi^2 = 0, \qquad \ddot\phi + 2\cot\theta\,\dot\theta\dot\phi = 0.$$

For a nonconstant latitude $\theta = \theta_0$ with constant angular speed, the first equation requires $\sin\theta_0\cos\theta_0 = 0$. Within this chart only $\theta_0 = \pi/2$ works: the equator is a geodesic. The coordinate poles are points, not nonconstant latitude curves.

The field $K = \partial_\phi$ generates rotations about the vertical axis, which preserve the metric. Its conserved quantity and its derivative are

$$g(K,\dot\gamma) = \sin^2\theta\,\dot\phi, \qquad \frac{d}{dt}(\sin^2\theta\,\dot\phi) = \sin^2\theta(\ddot\phi + 2\cot\theta\,\dot\theta\dot\phi) = 0.$$

This is the axial angular momentum for the unit-mass energy Lagrangian. The coordinate singularity at a pole does not make the smooth connection singular.

A geodesic need not minimize length on its entire interval. The following great-circle arcs have the same endpoints.

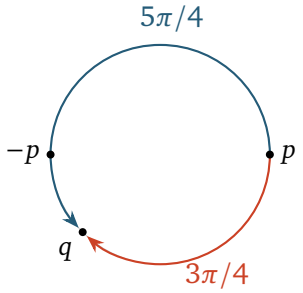


Figure 16.3: From p to q, the increasing-angle unit-circle arc has length $5\pi/4$. The decreasing-angle arc has length $3\pi/4$. The longer geodesic has passed the antipode $-p$.

The longer arc is a geodesic but cannot minimize length because the shorter arc joins the same endpoints. No local geodesic equation implies a global minimum.

Self-check. Compute the plane's Levi-Civita symbols in polar coordinates $r > 0$. Compare with Cartesian coordinates to test whether Christoffel symbols are a tensor.

Answer. The metric is $g = \mathrm{d}r\otimes\mathrm{d}r + r^2\mathrm{d}\phi\otimes\mathrm{d}\phi$. Its only nonzero derivative is $\partial_r g_{\phi\phi} = 2r$. The formula gives

$$\Gamma^r_{\phi\phi} = -r, \qquad \Gamma^\phi_{r\phi} = \Gamma^\phi_{\phi r} = 1/r.$$

All other symbols vanish. The Cartesian symbols are zero, whereas these are nonzero at the same points. A tensor with zero components in one basis has zero components in every basis. Thus Christoffel symbols are not tensor components.

Summary. A nondegenerate metric determines the torsion-free metric connection by Koszul's formula. Parallel transport solves a linear ODE and preserves the metric. Geodesics have parallel velocity and constant metric norm. Killing fields give conserved pairings with that velocity. For a bi-invariant metric the identity geodesics are one-parameter subgroups. Compact averaging and the cited Hopf–Rinow theorem make the group exponential onto for compact connected groups. Lecture 17 computes curvature from this connection.

Reading. Tu, *Differential Geometry*, §6, §13–§15, especially Problems 15.4–15.5, and Theorem 16.6, p. 131. Lee, *Riemannian Manifolds: An Introduction to Curvature*, Theorem 5.4, pp. 68–70, and Lemma 6.2, pp. 94–95. Nakahara, §7.2, §7.4 and §7.7.

Lecture 17

Curvature

Objectives. *Second covariant derivatives need not commute. Their difference, corrected for the bracket of the directions, defines curvature. We compute this tensor from Christoffel symbols and from differential forms in an orthonormal frame. The sphere fixes the sign used for surface geometry and gravitation.*

By the end of this lecture you will be able to:

- Compute curvature with the output and direction indices specified.
- Prove tensoriality, the metric antisymmetries and the first Bianchi identity.
- Use Cartan's equations to find the curvature of a surface.
- Distinguish zero curvature from zero Christoffel symbols in one chart.

17.1 The curvature tensor

Let M be a smooth manifold with an affine connection ∇. The *curvature operator* is defined on smooth fields by

$$R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z.$$

For the Cartesian connection on $\mathbb{R}^n$, its components are $XY(Z^k) - YX(Z^k) - [X,Y](Z^k) = 0$. The bracket correction removes the failure of the direction fields to commute.

Theorem 17.1 (Tensoriality of curvature)**.** *Curvature is linear over $C^\infty(M)$ in all three arguments. It therefore gives a smooth tensor of type $(1,3)$, called the* Riemann curvature tensor.

Proof. Real multilinearity is immediate from the connection rules. For $f \in C^\infty(M)$, the Leibniz rule and $[fX,Y] = f[X,Y] - Y(f)X$ give

$$R(fX,Y)Z = fR(X,Y)Z - Y(f)\nabla_X Z + Y(f)\nabla_X Z = fR(X,Y)Z.$$

Antisymmetry in X and Y gives linearity in Y. Expansion in the last slot gives

$$R(X,Y)(fZ) = fR(X,Y)Z + \big(XY(f) - YX(f) - [X,Y](f)\big)Z = fR(X,Y)Z.$$

The cross terms $Y(f)\nabla_X Z$ and $X(f)\nabla_Y Z$ cancel between the two second derivatives. By the cutoff and coordinate-expansion argument of § 16.1, the value therefore depends only on the three vectors at the point. Smoothness follows from the local coefficients of the connection. ∎

Write $\partial_\mu = \partial/\partial x^\mu$ in a chart. Greek indices here range from 1 to n. The index order is fixed by the action

$$R(\partial_\mu, \partial_\nu)\partial_\sigma = R^\rho{}_{\sigma\mu\nu}\partial_\rho.$$

Applying the connection product rule twice gives

$$R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu \Gamma^{\rho}_{\nu\sigma} - \partial_\nu \Gamma^{\rho}_{\mu\sigma} + \Gamma^{\rho}_{\mu\lambda}\Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\mu\sigma}.$$

For example, applying the product rule gives

$$\nabla_{\partial_\mu}\nabla_{\partial_\nu}\partial_\sigma = (\partial_\mu \Gamma^{\rho}_{\nu\sigma} + \Gamma^{\lambda}_{\nu\sigma}\Gamma^{\rho}_{\mu\lambda})\partial_\rho.$$

Subtract the reversed expression. The bracket term vanishes because coordinate fields commute. Curvature is a tensor even though the Christoffel symbols are not. For the Euclidean plane, Lecture 16 gives the polar symbols $\Gamma^{r}_{\phi\phi} = -r$ and $\Gamma^{\phi}_{r\phi} = \Gamma^{\phi}_{\phi r} = 1/r$. For example, the curvature component is

$$R^{r}{}_{\phi r\phi} = \partial_r(-r) - (-r)(1/r) = -1 + 1 = 0.$$

All curvature components vanish because the tensor is zero in Cartesian coordinates. A connection is *flat* when its curvature is zero. On the unit sphere, the calculation in Example 17.6 below gives $R^{\theta}{}_{\phi\theta\phi} = \sin^2\theta > 0$ in the spherical chart of Lecture 16. This is the sphere sign test for our operator convention.

17.2 Metric symmetries and the first Bianchi identity

Now let ∇ be the Levi-Civita connection of a pseudo-Riemannian metric. Lower the output index first: $R_{\rho\sigma\mu\nu} = g_{\rho\lambda}R^{\lambda}{}_{\sigma\mu\nu}$. For the proofs, abbreviate the four-field expression by $C(X,Y,Z,W) = g(R(X,Y)Z,W)$.

Proposition 17.2 (Curvature symmetries · *sketched; antisymmetries and first Bianchi proved*)**.** *The curvature satisfies*

$$\begin{aligned} C(X,Y,Z,W) &= -C(Y,X,Z,W) = -C(X,Y,W,Z), \\ R(X,Y)Z + R(Y,Z)X + R(Z,X)Y &= 0, \\ C(X,Y,Z,W) &= C(Z,W,X,Y). \end{aligned}$$

The middle equation is the first Bianchi identity.

Proof and sketch. The first antisymmetry follows from the operator definition. For the other one, apply $XY - YX - [X,Y]$ to $g(Z,W)$. It vanishes on this scalar function. Metric compatibility expands the result as

$$0 = g(R(X,Y)Z,W) + g(Z,R(X,Y)W),$$

with the two cross products of first derivatives cancelling. This proves antisymmetry in the last pair.

For Bianchi, sum cyclically over (X,Y,Z). Zero torsion first combines the second derivatives into $\sum_{\text{cyc}} \nabla_X[Y,Z]$. Relabel the bracket-direction terms to obtain

$$\sum_{\text{cyc}} R(X,Y)Z = \sum_{\text{cyc}} \big(\nabla_X[Y,Z] - \nabla_{[Y,Z]}X\big) = \sum_{\text{cyc}} [X,[Y,Z]] = 0.$$

The last equality is Jacobi's identity.

For pair exchange, write Bianchi for four cyclic permutations of the four fields. Add the identities and use the two antisymmetries to cancel terms. The remaining equality gives pair exchange after renaming the fields. This sketches the algebraic step; see Lee, Proposition 7.4, pp. 121–123. ∎

In the lowered-output convention, these identities give $R_{abcd} = -R_{bacd} = -R_{abdc}$ and $R_{abcd} = R_{cdab}$. The slot order must also be translated when comparing sources.[1] Changing the operator sign would negate every curvature value below; changing slot order alone can also introduce a sign.

17.3 Ricci, scalar and sectional curvature

The *Ricci tensor* is the trace $\mathrm{Ric}(X,Y) = \mathrm{tr}(Z \mapsto R(Z,Y)X)$. The *scalar curvature* is its metric trace. In coordinates,

$$\mathrm{Ric}_{\mu\nu} = R^{\rho}{}_{\mu\rho\nu}, \qquad R_{\mathrm{sc}} = g^{\mu\nu}\,\mathrm{Ric}_{\mu\nu}\,.$$

Pair exchange makes Ricci symmetric: in $\mathrm{Ric}_{ij} = g^{ab}R_{aibj}$, exchange the curvature pairs and then rename a and b. The symbol R_{sc} distinguishes scalar curvature from a sphere radius.

For a Riemannian metric and a tangent 2-plane $P \subset T_pM$, its *sectional curvature* is

$$K(P) = \frac{g(R(u,v)v,u)}{g(u,u)g(v,v) - g(u,v)^2}, \qquad P = \mathrm{span}\{u,v\}.$$

Here u and v are linearly independent. The denominator is positive. On changing the basis of P, antisymmetry in both pairs multiplies the numerator by the square of the basis-change determinant. The Gram determinant in the denominator has the same factor. Thus the quotient is independent of the basis. For orthonormal u and v, it is $g(R(u,v)v,u)$. In indefinite signature the same quotient defines sectional curvature only for nondegenerate planes, whose Gram determinant is nonzero.

17.4 Moving frames and Cartan's equations

A *local frame* $e = (e_1,\dots,e_n)$ is a smooth basis of tangent vectors at every point of an open set U. It is a row of fields. Its *dual coframe* is the column of 1-forms $\theta = (\theta^1,\dots,\theta^n)^T$ with $\theta^i(e_j) = \delta^i_j$. For a Riemannian metric, a frame is *orthonormal* when $g(e_i,e_j) = \delta_{ij}$. Gram–Schmidt applied smoothly to a coordinate frame supplies local orthonormal frames. The construction divides only by positive smooth lengths. The dual coframe then gives $g = \sum_i \theta^i \otimes \theta^i$. The figure compares coordinate vectors with unit frame vectors on a sphere.

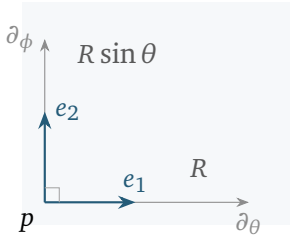


Figure 17.1: For a radius-R sphere, the spherical coordinate vectors have lengths R and $R\sin\theta$. Dividing by those lengths gives the orthonormal frame e_1 and e_2 on a chart away from the poles.

The *connection forms* $\omega^i{}_j$ and *curvature forms* $\Omega^i{}_j$ are specified by

$$\nabla_X e_j = e_i\omega^i{}_j(X), \qquad R(X,Y)e_j = e_i\Omega^i{}_j(X,Y).$$

[1] Lee uses $\mathrm{Rm}_{\mathrm{Lee}}(X,Y,Z,W) = C(X,Y,Z,W)$. Thus $R_{abcd} = \mathrm{Rm}_{\mathrm{Lee}}(\partial_c,\partial_d,\partial_b,\partial_a)$, with the same operator sign.

Linearity of the connection in its direction makes the first coefficients 1-forms. Tensoriality and direction antisymmetry make the second 2-forms. Write ∇V for the vector-valued 1-form $X \mapsto \nabla_X V$. The matrices act on coefficient columns: for $V = e v$, $\nabla V = e(\mathrm{d}v + \omega v)$. For matrices of forms, the wedge includes matrix multiplication: $(A \wedge B)^i{}_j = \sum_k A^i{}_k \wedge B^k{}_j$. The exterior derivative acts on each entry.

Theorem 17.3 (Cartan's structure equations)**.** *On a Riemannian manifold, the Levi-Civita connection in a local orthonormal frame satisfies*

$$\mathrm{d}\theta = -\omega \wedge \theta, \qquad \Omega = \mathrm{d}\omega + \omega \wedge \omega.$$

The connection matrix ω and curvature matrix Ω are skew-symmetric. Conversely, a skew matrix ω satisfying the first equation is the unique Levi-Civita connection matrix for this coframe.

Proof. Differentiate $g(e_i, e_j) = \delta_{ij}$. Metric compatibility gives $\omega^i{}_j + \omega^j{}_i = 0$. For any 1-form α, the expression $X\alpha(Y) - Y\alpha(X) - \alpha([X,Y])$ is function-linear in X and Y by the product and bracket rules. On coordinate fields it agrees with Lecture 10's definition of $\mathrm{d}\alpha$, so the expressions coincide. For any connection matrix in this frame, the same evaluation gives

$$\theta^i(T(X,Y)) = (\mathrm{d}\theta^i + \omega^i{}_j \wedge \theta^j)(X,Y).$$

Indeed, expand $Y = \theta^j(Y) e_j$ in $\nabla_X Y$ and subtract the reversed expression and $[X,Y]$. Zero torsion therefore gives the first structure equation.

Conversely, a smooth matrix ω defines a connection by $\nabla_X(ev) = e(Xv + \omega(X)v)$. The product rule verifies both connection axioms. If ω is skew, the extra terms in differentiating $u^T v$ cancel because $u^T(\omega(X)^T + \omega(X))v = 0$. Thus this connection is metric. The displayed torsion identity makes it torsion-free when the first equation holds. Uniqueness in Theorem 16.1 identifies it with the Levi-Civita connection, proving the converse and uniqueness of the skew solution.

For the second, expand $\nabla_X \nabla_Y e_j$ using the Leibniz rule and subtract its reversed expression and $\nabla_{[X,Y]} e_j$. The coefficient of e_i is

$$X\omega^i{}_j(Y) - Y\omega^i{}_j(X) - \omega^i{}_j([X,Y]) + \omega^i{}_k(X)\omega^k{}_j(Y) - \omega^i{}_k(Y)\omega^k{}_j(X).$$

The first terms are $\mathrm{d}\omega^i{}_j(X,Y)$ and the last terms are $(\omega \wedge \omega)^i{}_j(X,Y)$. This proves the curvature equation. The last-pair metric antisymmetry already proved gives $\Omega^i{}_j = -\Omega^j{}_i$. ∎

The curvature-form calculation used no metric hypothesis. Thus $\Omega = \mathrm{d}\omega + \omega \wedge \omega$ holds in any local frame for any affine connection. Differentiate a tensor by the connection product rule: $\nabla_X T$ differentiates the output and subtracts a term for the covariant derivative of each input. For curvature, this means

$$\begin{aligned}(\nabla_X R)(Y,Z)W = {} & \nabla_X(R(Y,Z)W) - R(\nabla_X Y, Z)W \\ & - R(Y, \nabla_X Z)W - R(Y,Z)\nabla_X W.\end{aligned}$$

Proposition 17.4 (Second Bianchi identity · *sketched; matrix-form identity proved*)**.** *For the Levi-Civita connection of a pseudo-Riemannian metric,*

$$\mathrm{d}\Omega + \omega \wedge \Omega - \Omega \wedge \omega = 0, \qquad \sum_{\mathrm{cyc}(X,Y,Z)} (\nabla_X R)(Y,Z) = 0.$$

Proof and sketch. The identity $\mathrm{d}^2\omega = 0$ gives $\mathrm{d}\Omega = \mathrm{d}\omega\wedge\omega - \omega\wedge\mathrm{d}\omega$. Substitute $\mathrm{d}\omega = \Omega - \omega\wedge\omega$; the cubic terms cancel. This proves the matrix equation. Evaluating it on three directions and using zero torsion cancels their bracket terms, giving the stated cyclic covariant identity. This sketches the conversion from forms to the covariant expression. Nakahara's (7.147b) states the matrix identity with the same action convention. ∎

17.5 Gaussian curvature of surfaces

On a Riemannian surface, the *Gaussian curvature* K is the sectional curvature of its tangent plane. Choose a positively oriented orthonormal frame on a chart. Write its connection entry as $\omega^1{}_2$. Skew-symmetry makes the matrix

$$\omega = \begin{pmatrix} 0 & \omega^1{}_2 \\ -\omega^1{}_2 & 0 \end{pmatrix}.$$

Cartan's equations become

$$\mathrm{d}\theta^1 = -\omega^1{}_2 \wedge \theta^2, \qquad \mathrm{d}\theta^2 = \omega^1{}_2 \wedge \theta^1, \qquad \mathrm{d}\omega^1{}_2 = K\theta^1 \wedge \theta^2.$$

For the last equality, the diagonal-free 2×2 matrix has $(\omega\wedge\omega)^1{}_2 = 0$. The coefficient of $\Omega^1{}_2$ on (e_1, e_2) is $g(R(e_1,e_2)e_2, e_1) = K$. Thus the curvature coefficient has the displayed positive sign. In terms of the frame, the same convention is $\nabla e_1 = -\omega^1{}_2 e_2$ and $\nabla e_2 = \omega^1{}_2 e_1$.

Proposition 17.5 (Surface curvature)**.** *On a Riemannian surface, the lowered-output curvature satisfies*

$$R_{abcd} = K(g_{ac}g_{bd} - g_{ad}g_{bc}), \qquad \mathrm{Ric} = Kg, \qquad R_{\mathrm{sc}} = 2K.$$

The round sphere of radius $R > 0$ has $K = 1/R^2$. The Poincaré half-plane with metric $(\mathrm{d}x\otimes\mathrm{d}x + \mathrm{d}y\otimes\mathrm{d}y)/y^2$ has $K = -1$. The torus $\mathbb{R}^2/(2\pi\mathbb{Z})^2$ with its quotient Euclidean metric has $K = 0$.

In two dimensions, antisymmetry in both pairs leaves a single independent curvature component in an orthonormal basis. Its value is K. Tensoriality therefore gives the displayed formula in any coordinates. Contracting its first and third indices with the inverse metric gives $\mathrm{Ric} = Kg$ and $R_{\mathrm{sc}} = 2K$. This is intrinsic; no embedding is required. Tu's Theorem 12.3, p. 90, identifies it with classical surface curvature.[2]

The sphere and half-plane calculations follow in the worked example. For the torus $\mathbb{R}^2/(2\pi\mathbb{Z})^2$, use the metric induced by the Euclidean plane under the quotient charts. Their transition maps are translations, so the metric agrees on overlaps. Its coordinate coefficients are constant, hence its Christoffel symbols and curvature vanish. The figure shows why the Cartesian frame also descends.

[2] Read Tu's entry by its action on the frame: $\nabla e_1 = -\omega^1{}_2 e_2$ and $\nabla e_2 = \omega^1{}_2 e_1$. His (12.9) then reads $\mathrm{d}\omega^1{}_2 = +K\theta^1\wedge\theta^2$.

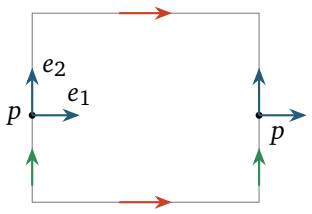


Figure 17.2: Opposite sides are identified by translations. The same Cartesian orthonormal frame matches across both pairs, giving the flat quotient metric.

This metric differs from the metric induced on a doughnut-shaped surface in $\mathbb{R}^3$.

17.6 The sphere by coordinates and by forms

Example 17.6 (Two curvature calculations)**.** On the sphere of radius $R > 0$, spherical coordinates give

$$g = R^2 \mathrm{d}\theta \otimes \mathrm{d}\theta + R^2 \sin^2\theta \mathrm{d}\phi \otimes \mathrm{d}\phi, \qquad 0 < \theta < \pi.$$

The constant scale does not change Lecture 16's symbols. Substitute them into the coordinate formula:

$$\begin{aligned} R^{\theta}{}_{\phi\theta\phi} &= \partial_\theta(-\sin\theta\cos\theta) - (-\sin\theta\cos\theta)\cot\theta \\ &= (\sin^2\theta - \cos^2\theta) + \cos^2\theta = \sin^2\theta. \end{aligned}$$

Lowering the output gives $R_{\theta\phi\theta\phi} = R^2\sin^2\theta$. Divide by the Gram determinant $R^4\sin^2\theta$ to obtain $K = 1/R^2$. In particular the unit sphere has positive curvature, fixing the operator sign.

For the forms calculation use

$$\theta^1 = R\mathrm{d}\theta, \qquad \theta^2 = R\sin\theta\mathrm{d}\phi.$$

Their derivatives are $\mathrm{d}\theta^1 = 0$ and $\mathrm{d}\theta^2 = R\cos\theta\mathrm{d}\theta \wedge \mathrm{d}\phi$. The first structure equation has the unique skew solution

$$\omega^1{}_2 = -\cos\theta\mathrm{d}\phi,$$

since $-\omega^1{}_2 \wedge \theta^2 = 0$ and $\omega^1{}_2 \wedge \theta^1 = R\cos\theta\mathrm{d}\theta \wedge \mathrm{d}\phi$. It follows that

$$\mathrm{d}\omega^1{}_2 = \sin\theta\mathrm{d}\theta \wedge \mathrm{d}\phi = \frac{1}{R^2}\theta^1 \wedge \theta^2.$$

Uniqueness of the Levi-Civita connection makes this the same curvature. Both routes give $\mathrm{Ric} = g/R^2$ and $R_{\mathrm{sc}} = 2/R^2$. These are smooth tensor identities on a dense coordinate domain, so continuity extends them, and $K = 1/R^2$, across the poles and angular seams.

On the half-plane $y > 0$, choose instead $\theta^1 = \mathrm{d}x/y$ and $\theta^2 = \mathrm{d}y/y$. The derivatives and solution of the first equation are

$$\mathrm{d}\theta^1 = \frac{\mathrm{d}x \wedge \mathrm{d}y}{y^2}, \qquad \mathrm{d}\theta^2 = 0, \qquad \omega^1{}_2 = -\frac{\mathrm{d}x}{y}.$$

Taking its derivative gives $\mathrm{d}\omega^1{}_2 = -\mathrm{d}x \wedge \mathrm{d}y/y^2 = -\theta^1 \wedge \theta^2$, so $K = -1$.

Self-check. Compute the intrinsic Gaussian curvature of the cylinder of radius $a > 0$.

Answer. The parametrization $p(u, z) = (a\cos u, a\sin u, z)$ induces $g = a^2 \mathrm{d}u \otimes \mathrm{d}u + \mathrm{d}z \otimes \mathrm{d}z$. Use $\theta^1 = a\mathrm{d}u$ and $\theta^2 = \mathrm{d}z$ on an angular chart. Both forms are closed, so the skew connection matrix $\omega = 0$ satisfies Cartan's first equation and is the matrix of the unique metric torsion-free connection. Its curvature is zero, hence $K = 0$. The cylinder bends in its embedding but has zero intrinsic curvature.

Summary. Curvature is the tensor $R(X, Y) = [\nabla_X, \nabla_Y] - \nabla_{[X,Y]}$. Metric compatibility and zero torsion give its antisymmetries and first Bianchi identity. Ricci and scalar curvature are its specified contractions. In a local orthonormal frame, Cartan's equations replace Christoffel computations by $\mathrm{d}\theta = -\omega \wedge \theta$ and $\Omega = \mathrm{d}\omega + \omega \wedge \omega$. On a surface, $\mathrm{d}\omega^1{}_2 = K\theta^1 \wedge \theta^2$. The sphere gives positive curvature, the half-plane negative curvature, and the flat quotient torus and cylinder zero curvature. Lecture 18 integrates the surface equation to compute parallel transport around loops.

Reading. Tu, *Differential Geometry*, §10–§12, especially Theorems 11.1 and 11.7 and Theorem 12.3, p. 90. Lee, *Riemannian Manifolds: An Introduction to Curvature*, Chapter 7, Propositions 7.1, 7.4 and 7.5. Nakahara, §7.3 and §7.8.

Lecture 18

Holonomy and Gauss–Bonnet

Objectives. *Parallel transport around a loop can rotate a tangent vector. On a surface, the rotation is the enclosed curvature, modulo a full turn. Keeping track of the boundary tangent and its corner turns gives the Gauss–Bonnet formula. Summing that formula over a closed surface relates total curvature to its Euler characteristic.*

By the end of this lecture you will be able to:

- Compute surface holonomy using a frame defined on the enclosed disk.
- Derive the angle excess and the boundary terms in local Gauss–Bonnet.
- Prove global Gauss–Bonnet given the cited topological inputs.
- Compute the latitude dependence of ideal Foucault precession.

18.1 Holonomy on an oriented surface

Let (M, g) be an oriented Riemannian surface with its Levi-Civita connection. Write $\mathrm{d}A$ for its metric area form. In a positive orthonormal frame $e = (e_1, e_2)$, recall Lecture 17's convention

$$\nabla e_1 = -\omega^1{}_2 e_2, \qquad \nabla e_2 = \omega^1{}_2 e_1, \qquad \mathrm{d}\omega^1{}_2 = K\mathrm{d}A.$$

The positive quarter-turn J is defined by $Je_1 = e_2$ and $Je_2 = -e_1$. It is independent of the positive orthonormal frame because plane rotations commute with this quarter-turn.

For a piecewise smooth loop γ based at p, its *holonomy* is the parallel-transport map $P_\gamma : T_pM \to T_pM$ of Lecture 16, composed along its smooth segments. It is a rotation. To see the angle and its sign, write a transported unit vector along a framed segment as

$$V = \cos\alpha\, e_1 + \sin\alpha\, e_2, \qquad \frac{DV}{dt} = (\dot\alpha - \omega^1{}_2(\dot\gamma))JV.$$

The derivative follows by the product rule and the two frame actions above. The angle α has a real lift along the interval by Lecture 3's circle lifting theorem. Parallel transport means $\dot\alpha = \omega^1{}_2(\dot\gamma)$. Thus its matrix between the positive frames at the segment endpoints is the rotation through $\alpha(b) - \alpha(a)$, measured counter-clockwise. The same construction works across finitely many curve segments. The figure illustrates the angle measured relative to a local frame.

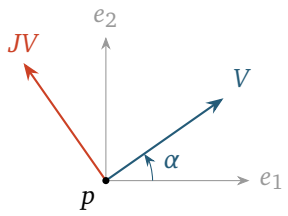


Figure 18.1: A unit vector has angle α from the first frame vector. Parallel transport changes this angle by the connection integral along the curve.

Theorem 18.1 (Surface holonomy · *proved given partitions of unity*)**.** *Let $D \subset M$ be a compact disk with smooth boundary, contained in an open set with one positive orthonormal frame. Parallel transport around ∂D, with its positive boundary orientation, rotates vectors by*

$$\Theta_D \equiv \int_D K \mathrm{d}A \pmod{2\pi}.$$

Proof. The frame agrees with itself when the loop returns to its starting point. Integrating the parallel-angle equation and using Stokes gives

$$\alpha(b) - \alpha(a) = \oint_{\partial D} \omega^1{}_2 = \int_D \mathrm{d}\omega^1{}_2 = \int_D K \mathrm{d}A.$$

A tangent-plane rotation records this real angle only modulo 2π. Linearity of transport extends the unit-vector conclusion to every vector. ∎

Reversing the loop negates the angle. A different positive frame along the loop changes the net lifted angle $\alpha(b) - \alpha(a)$ by an integer multiple of 2π. Indeed, its change-of-frame map is circle-valued, and its lifted angle has endpoint difference $2\pi m$ for some $m \in \mathbb{Z}$ by Theorem 3.7. The rotation is independent of that choice. Stokes requires a frame on the whole disk, not merely near its boundary.

18.2 Geodesic triangles and angle excess

A *geodesic triangle* bounds an embedded disk by three regular geodesic segments, with nondegenerate corners. Its interior angles α_1, α_2 and α_3 lie in $(0, 2\pi)$. The signed exterior turn at a vertex is $\varepsilon_i = \pi - \alpha_i \in (-\pi, \pi)$. It is the change from the incoming to the outgoing tangent, positive counter-clockwise in a positive tangent frame. The *angle excess* is $\alpha_1 + \alpha_2 + \alpha_3 - \pi$. The spherical triangle in the figure has positive excess.

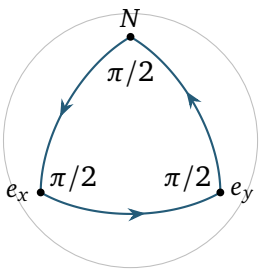


Figure 18.2: The three coordinate great-circle arcs bound an octant of the unit sphere. Each interior angle is $\pi/2$. Arrows give the positive boundary orientation, seen from outside.

We use the cited *turning-angle theorem*: for the positive boundary of a disk in one positive frame extending over the disk, the continuous changes of tangent angle along the sides plus the exterior turns add to 2π. See Tu, Theorem 17.4, pp. 142–143. A frame defined only on an annulus does not satisfy this hypothesis. For a proof in a chart, see Lee, Theorem 9.1 and Lemma 9.2, pp. 158–163.

Theorem 18.2 (Triangle excess · *proved given turning and partitions of unity*)**.** *If a geodesic triangle D lies in one framed neighbourhood as above, then*

$$\alpha_1 + \alpha_2 + \alpha_3 - \pi = \int_D K \mathrm{d}A.$$

Proof. Along each side, the unit tangent is parallel, so its frame angle changes by the integral of $\omega^1{}_2$ on that side. The turning theorem therefore gives

$$\oint_{\partial D} \omega^1{}_2 + \sum_i (\pi - \alpha_i) = 2\pi.$$

Stokes gives $\oint_{\partial D} \omega^1{}_2 = \int_D K \mathrm{d}A$. Rearranging proves the formula. For this piecewise smooth boundary, Stokes follows by rounding each corner in a coordinate ball of radius ϵ. Smoothness bounds the changed connection integral by $O(\epsilon)$ and the changed curvature integral by $O(\epsilon^2)$. The replacement arcs have length $O(\epsilon)$, including at concave corners. Apply smooth-boundary Stokes to the rounded disk and take $\epsilon \to 0$. ∎

For example, the unit-sphere octant has curvature integral

$$\int_0^{\pi/2} \int_0^{\pi/2} \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\phi = \frac{\pi}{2}.$$

Its three right angles give the same excess $3\pi/2 - \pi = \pi/2$. This triangle is contained in a framed chart obtained by removing a point outside the octant from the sphere.

18.3 The local Gauss–Bonnet formula

Let $T = \dot\gamma$ be the unit tangent of a smooth boundary segment parametrized by arc length s. Its signed *geodesic curvature* is

$$k_g = g(\nabla_T T, JT).$$

For a positive boundary, JT points into the disk. Differentiating $g(T,T) = 1$ makes its covariant acceleration perpendicular to T. Writing $T = \cos\zeta\, e_1 + \sin\zeta\, e_2$ in the angle derivative already computed gives

$$\nabla_T T = (\zeta' - \omega^1{}_2(T))JT, \qquad k_g = \zeta' - \omega^1{}_2(T).$$

Thus geodesic sides have zero geodesic curvature. The corner diagram distinguishes a jump in the tangent from curvature on a smooth side.

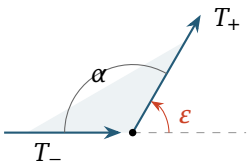


Figure 18.3: The exterior turn ε takes the incoming tangent to the outgoing tangent. The interior angle is $\alpha = \pi - \varepsilon$. A concave corner has negative exterior turn.

Theorem 18.3 (Local Gauss–Bonnet · *proved given turning and partitions of unity*)**.** *Let D be a compact embedded disk whose positive boundary has finitely many regular smooth sides and nondegenerate corners. Suppose one positive orthonormal frame is defined on a neighbourhood of all of D. Then*

$$\int_D K\,\mathrm{d}A + \oint_{\partial D} k_g\,\mathrm{d}s + \sum_i \varepsilon_i = 2\pi.$$

The side integrals exclude the corners; their turns occur in the finite sum.

Proof. Integrate $k_g = \zeta' - \omega^1{}_2(T)$ on each side. The turning theorem gives $\sum \Delta\zeta + \sum_i \varepsilon_i = 2\pi$. Summing the side integrals therefore gives

$$\oint_{\partial D} k_g\,\mathrm{d}s = 2\pi - \sum_i \varepsilon_i - \oint_{\partial D} \omega^1{}_2.$$

Use Stokes, with the same corner-rounding limit as above, to replace the last integral by $\int_D K\mathrm{d}A$. ∎

Consider a northern unit-sphere cap $0 \le \theta \le a$, where $0 < a < \pi$. Its positive boundary runs in increasing ϕ. Use the unit spherical frame $e_\theta = \partial_\theta$ and $e_\phi = (\sin\theta)^{-1}\partial_\phi$. The unit tangent is $T = e_\phi$ and $JT = -e_\theta$. Lecture 16's symbols give $\nabla_T T = -\cot a\, e_\theta$, so $k_g = \cot a$. The two integrals are

$$\int_D K\mathrm{d}A = 2\pi(1 - \cos a), \qquad \oint_{\partial D} k_g\,\mathrm{d}s = \cot a\,(2\pi \sin a) = 2\pi\cos a.$$

Their sum is 2π, with no corner contribution. A smooth frame on this cap is exhibited in the latitude example below.

18.4 Gauss–Bonnet on a closed surface

Theorem 18.4 (Gauss–Bonnet · *proved given cited topology and Stokes inputs*)**.** *Every closed oriented Riemannian surface satisfies*

$$\int_M K\mathrm{d}A = 2\pi\chi(M).$$

Proof. Use the cited existence of a finite smooth triangulation of a compact surface (Lee, discussion before Theorem 9.7, pp. 166–167). In Lee's definition, each curved triangle and its enclosed region lie in one coordinate chart, with regular sides and no cusps (p. 162). We also use the cited planar *Schoenflies theorem*: a simple closed plane curve and its enclosed region form a closed topological disk (Stillwell, §0.3.9, pp. 35–36). Gram–Schmidt on the connected chart neighbourhood of the face gives a positive orthonormal frame, changing one vector's sign if needed. Thus each closed face is a disk to which the local formula applies. The edges need not be geodesics.

To obtain the required Δ-complex, order the vertices and choose one parametrization for each edge, from its lower to its higher vertex. These give a homeomorphism from the boundary of each standard triangle to its face boundary. Identify the standard triangle with a closed unit disk by radially rescaling rays from its centroid. Schoenflies supplies a disk homeomorphism; it can be chosen to match that boundary map. Indeed, any difference

on the boundary circle is a homeomorphism $\psi\colon S^1 \to S^1$, which extends to the disk by $r\zeta \mapsto r\psi(\zeta)$, with the centre fixed. Composing with this extension matches the chosen boundary map. The face maps now agree on shared edges and give a finite Δ-complex structure on M.

Let V, E and F count vertices, edges and faces. Orient each face by the surface orientation and sum the local formula. Every shared edge has opposite boundary directions in its two faces. Reversing T leaves $\nabla_T T$ unchanged and negates JT, so it negates k_g. The two edge integrals cancel. At each vertex, the incident interior angles sum to 2π. Thus the sum of exterior turns over all face corners is $3\pi F - 2\pi V$. The figure displays these cancellations and the complete angle around a vertex.

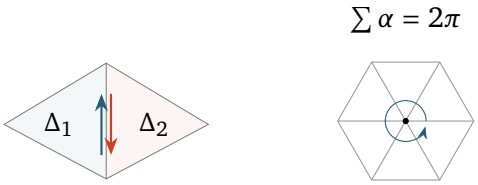


Figure 18.4: Adjacent faces traverse a shared edge in opposite directions. The incident face angles at an interior vertex fill one full turn.

The summed formula gives

$$\int_M K \mathrm{d}A = 2\pi F - (3\pi F - 2\pi V) = 2\pi V - \pi F = 2\pi(V - E + F).$$

The last equality uses $3F = 2E$, since each triangular side belongs to two faces on a surface without boundary. By Theorems 6.2 and 6.3, $V - E + F = \chi(M)$. For disconnected M, apply the argument to each component and add. ∎

The integral does not change when the metric changes, although the pointwise Gaussian curvature and the area form both change. A sphere therefore has total curvature 4π for every Riemannian metric. A torus has total curvature zero for every Riemannian metric, even when its curvature is not identically zero.

18.5 The Euler number of the tangent sphere

Recall from § 14.5 that the index of an isolated tangent-field zero is the winding of its normalized components on a small positive circle. The cited Poincaré–Hopf theorem, Theorem 14.5, states that the sum of these indices on a closed oriented surface is its Euler characteristic. In particular this sum is independent of the chosen field with isolated zeros. For the tangent bundle of S^2, call this common sum its *Euler number*.

> **Corollary 18.5** (The tangent-sphere Euler number · *sketched given Poincaré–Hopf*)**.** *The Euler number of TS^2 is $\chi(S^2) = 2$.*

The field $X(p) = e_3 - p_3 p$ from Lecture 14 has its two zeros at the poles, with index +1 at each. It supplies a field to which the definition applies. Poincaré–Hopf makes every such index sum equal to 2. A nowhere-zero tangent field would have empty index sum zero, contradicting this value. Hence there can be no global tangent frame on the sphere. A disk in one coordinate chart has a framed neighbourhood by Gram–Schmidt. The local curvature formula requires that the chosen frame extend over that disk, rather than only around its

boundary; the whole sphere admits no such frame. Lecture 22 expresses the same integer using changes between local frames.

18.6 Latitude holonomy and the Foucault pendulum

The *solid angle* of a region on the unit sphere is its area. Let the latitude be $\lambda \in (-\pi/2, \pi/2)$ and put $a = \pi/2 - \lambda$. Follow its circle once in increasing ϕ. The figure shows the northern cap and unit vectors obtained from the parallel equation.

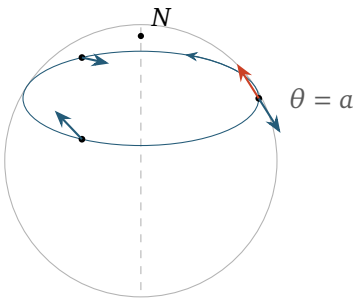


Figure 18.5: A latitude bounds the northern cap. At colatitude $a = \pi/3$, the arrows are projected values of the exact parallel field with frame angle $\alpha(\phi) = -\phi \cos a$. The red-orange arrow shows the final vector at the starting point after one full circuit.

Example 18.6 (Latitude rotation)**.** On the loop, the spherical frame has $\omega^1{}_2 = -\cos a \, \mathrm{d}\phi$. The parallel-angle equation gives

$$\Delta\alpha = -2\pi \cos a = -2\pi \sin \lambda.$$

The cap's solid angle is $\Omega_D = 2\pi(1 - \cos a) = 2\pi(1 - \sin \lambda)$. These real numbers differ by 2π and represent the same holonomy. The spherical frame itself is undefined at the north pole, so Stokes cannot be applied to that frame across the cap.

To use a frame on the full cap, define the rotated vectors by

$$e_{N,1} = \cos\phi \, e_\theta - \sin\phi \, e_\phi, \qquad e_{N,2} = \sin\phi \, e_\theta + \cos\phi \, e_\phi.$$

In Cartesian coordinates $p = (x, y, z)$ on the unit sphere, its columns are

$$E_N = \begin{pmatrix} 1 - x^2/(1+z) & -xy/(1+z) \\ -xy/(1+z) & 1 - y^2/(1+z) \\ -x & -y \end{pmatrix}.$$

The rotation makes it tangent, positive and orthonormal away from the poles. The displayed expression is smooth for $z > -1$ and at the north pole gives e_x and e_y. Continuity therefore preserves those properties there. Differentiating the rotated frame gives

$$\omega_N = \omega^1{}_2 + \mathrm{d}\phi = (1 - \cos\theta)\mathrm{d}\phi = \frac{x\mathrm{d}y - y\mathrm{d}x}{1+z}.$$

This is smooth at the north pole. Its integral around the latitude is $2\pi(1 - \cos a)$, so Stokes in this frame gives the cap's curvature integral. At the equator the holonomy is zero modulo 2π, while the cap integral is 2π. Holonomy alone does not retain that full turn.

For the physical interpretation, use the small-swing Coriolis model of an ideal pendulum on a slowly rotating sphere.[1] Let $\nu_\oplus > 0$ be the sphere's rotation frequency and $\nu_0 > 0$ the pendulum's natural frequency, with $\nu_\oplus \ll \nu_0$. In a positive east–north frame, write the small displacement as $\xi = \xi_E + \mathrm{i}\xi_N$. Define the local Coriolis rate $b = \nu_\oplus \sin\lambda$. Keeping the linear Coriolis term gives

$$\ddot{\xi} + 2\mathrm{i}b\dot{\xi} + \nu_0^2\xi = 0.$$

Substituting $\xi = e^{-\mathrm{i}bt}w$ cancels the first-derivative terms and gives

$$\ddot{w} + (\nu_0^2 + b^2)w = 0.$$

An oscillation along a fixed line in the w plane therefore has slow orientation angle $-bt$. Its rate agrees with parallel transport:

$$\dot{\alpha} = -\cos a\,\dot{\phi} = -\nu_\oplus \sin\lambda = -b.$$

Over one rotation $2\pi/\nu_\oplus$, the change is $-2\pi\sin\lambda$. It is clockwise in the northern hemisphere and counter-clockwise in the southern. An oriented swing direction records angles modulo 2π; the physical unoriented swing line records them modulo π. For a spin-1/2 ground state aligned with a field direction traversing this latitude, Lecture 24 derives the Berry phase $-\Omega_D/2$ modulo 2π.

Self-check. Check global Gauss–Bonnet for the round sphere and the flat quotient torus.

Answer. On a sphere of radius R, curvature $1/R^2$ times area $4\pi R^2$ gives $4\pi = 2\pi\chi(S^2)$. On the flat torus $K = 0$, so the integral is $0 = 2\pi\chi(T^2)$, using $\chi(T^2) = 0$ from Lecture 6.

Summary. Surface transport has angle $\int_D K\mathrm{d}A$ modulo 2π when one positive orthonormal frame covers the disk. The turning theorem adds the boundary tangent and corner turns, giving local Gauss–Bonnet. Using the cited topology inputs and summing over the triangulation yields $\int_M K\mathrm{d}A = 2\pi\chi(M)$ on every closed oriented Riemannian surface. The tangent sphere has Euler number 2 by the cited Poincaré–Hopf theorem. Latitude transport gives the ideal Foucault angle $-2\pi\sin\lambda$ modulo 2π. Lecture 19 uses curvature and geodesics in a Lorentzian spacetime.

Reading. Tu, *Differential Geometry*, §17, especially Theorems 17.4–17.5 and 17.9, pp. 142–146. Lee, *Riemannian Manifolds: An Introduction to Curvature*, Ch. 9, pp. 158–168, including the smooth triangulation discussion. Stillwell, *Classical Topology and Combinatorial Group Theory*, §0.3.9, pp. 35–36 (Schoenflies theorem).

[1] See Fitzpatrick, *Newtonian Dynamics*, Foucault Pendulum, for the linear Coriolis model, and Dray, *The Foucault Pendulum* (1998), for its slowly changing direction and tangential-projection interpretation.

Lecture 19

Lorentzian geometry and general relativity

Objectives. *A Lorentzian metric specifies elapsed time and the paths of freely falling particles. Einstein's equation relates its curvature to matter. We recover Newton's acceleration in a weak field and verify the Schwarzschild vacuum metric. Its conserved quantities then reduce a massive particle's motion to an orbit equation.*

By the end of this lecture you will be able to:

- Distinguish timelike, null and spacelike tangent vectors in signature $+---$.
- Recover Newton's equation from a weak static metric and slow motion.
- Explain the action and conservation statements in Einstein's equation.
- Verify Schwarzschild vacuum curvature and derive the massive orbit equation.

19.1 Lorentzian metrics and proper time

A *Lorentzian spacetime* is a smooth four-dimensional manifold with a metric of signature $(+,-,-,-)$. At each point an orthonormal frame identifies the metric with $\eta = \operatorname{diag}(1,-1,-1,-1)$. This recalls the flat interval in *Symmetries in Physics*, Lecture 17, *Minkowski space from physics*. Here the metric may vary from point to point. Assume the spacetime is oriented, so its metric volume form is defined.

A nonzero tangent vector v is *timelike* if $g(v,v) > 0$, *null* if $g(v,v) = 0$, and *spacelike* if $g(v,v) < 0$. The null vectors form the *light cone*. A *time orientation* is a continuous choice of one of the two timelike cones as future. We assume such a choice. In an orthonormal frame adapted to this choice, future vectors have positive time component. The figure shows one space direction and the time direction.

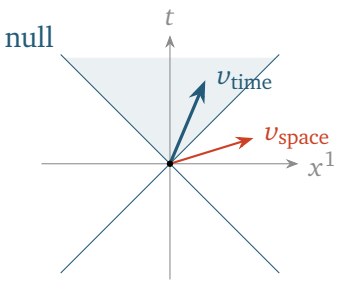


Figure 19.1: The null lines bound the future timelike cone in a tangent plane. The classification uses the metric at the point, rather than the drawing's Euclidean angles.

Use $c = 1$ and coordinates $x^0 = t$. The *proper time* along a future timelike curve $x(\sigma)$ is defined by

$$\tau(\sigma_1) - \tau(\sigma_0) = \int_{\sigma_0}^{\sigma_1} \sqrt{g_{\mu\nu} \frac{dx^\mu}{d\sigma} \frac{dx^\nu}{d\sigma}}\, d\sigma.$$

Its *four-velocity* $U = dx/d\tau$ satisfies $g(U, U) = 1$. The physical free-fall postulate says that an unforced test particle follows a timelike geodesic of the Levi-Civita connection. A light ray follows a null geodesic, using an affine parameter in place of proper time. These are physical inputs. The existence and equation of these geodesics are the mathematical results of Lecture 16.

19.2 Newton's equation from a weak metric

Let $\Phi = \epsilon\varphi(x^1, x^2, x^3)$ be a smooth static potential, where ϵ is a small parameter. Derivatives of Φ count as first order in ϵ. In spatial Cartesian coordinates consider

$$ds^2 = (1 + 2\Phi)dt^2 - (1 - 2\Phi) \sum_{i=1}^{3} (dx^i)^2.$$

Assume $|\Phi| < 1/2$, so the signature is Lorentzian. Define the Euclidean spatial gradient by

$$\nabla_E \Phi = (\partial_1 \Phi, \partial_2 \Phi, \partial_3 \Phi).$$

The figure shows a trajectory in its leading Newtonian approximation.

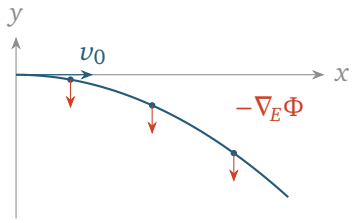


Figure 19.2: For a locally uniform downward acceleration, the leading Newtonian trajectory is $x = v_0 t$, $y = -at^2/2$, with $v_0 > 0$ and $a > 0$. This curve represents the weak-field, slow-motion limit.

Proposition 19.1 (Weak-field free fall · *proved*)**.** *For a future timelike geodesic with small coordinate speed, the coordinate-time acceleration to first order in the potential and lowest order in speed is*

$$\frac{d^2 x^i}{dt^2} = -\partial_i \Phi, \qquad i = 1, 2, 3.$$

Proof. Write $v^\mu = dx^\mu/dt$, so $v^0 = 1$. Changing from an affine parameter to t in the geodesic equation gives

$$\frac{d^2 x^i}{dt^2} = -\Gamma^i_{\mu\nu} v^\mu v^\nu + v^i \Gamma^0_{\mu\nu} v^\mu v^\nu.$$

A nonzero timelike tangent has nonzero time component in this metric, since a tangent with zero time component has negative norm. Thus division by the square of that component is valid. The affine time equation eliminates the second derivative of t when the spatial equation is divided by the square of its first derivative. The Christoffel formula gives

$$\Gamma^i_{00} = \frac{\partial_i \Phi}{1 - 2\Phi} = \partial_i \Phi + O(\epsilon^2), \qquad \Gamma^0_{0i} = \frac{\partial_i \Phi}{1 + 2\Phi}.$$

The static diagonal metric has $\Gamma^0_{00} = \Gamma^0_{ij} = \Gamma^i_{0j} = 0$. Write $\mathbf{v} = (v^1, v^2, v^3)$ for the coordinate spatial velocity. All remaining terms in the spatial equation contain at least two spatial velocity factors. The acceleration is $-\partial_i \Phi + O(\epsilon^2) + O(\epsilon |\mathbf{v}|^2)$, which proves the stated approximation. ∎

For an attractive point source of mass $m > 0$, $\Phi = -G_N m / r$, where $G_N > 0$ is Newton's constant. Its negative gradient points toward the source.

19.3 Einstein's equation and its conservation identity

Recall Lecture 17's definitions $\mathrm{Ric}_{\mu\nu} = R^{\rho}{}_{\mu\rho\nu}$ and $R_{\mathrm{sc}} = g^{\mu\nu} \mathrm{Ric}_{\mu\nu}$. The *Einstein tensor* is

$$G_{\mu\nu} = \mathrm{Ric}_{\mu\nu} - \frac{1}{2} R_{\mathrm{sc}} g_{\mu\nu}.$$

Take action integrals on a relatively compact coordinate domain. All variations vanish near its boundary. Let S_{m} be a matter action and let $\mu = \sqrt{|\det g|}\, \mathrm{d}t \wedge \mathrm{d}x^1 \wedge \mathrm{d}x^2 \wedge \mathrm{d}x^3$ in positive coordinates. Its symmetric *stress–energy tensor* $T_{\mu\nu}$ is defined by the response to compactly supported inverse-metric variations, with matter fixed:

$$\delta S_{\mathrm{m}} = \frac{1}{2} \int T_{\alpha\beta}\, \delta g^{\alpha\beta}\, \mu.$$

The physical normalization gives $T_{00} = \rho$ for matter at rest in an orthonormal frame, where ρ is energy density. For pressureless matter of rest density ρ and unit four-velocity U, the model is $T_{\mu\nu} = \rho U_\mu U_\nu$. Write $F_{\mu\nu}$ for the components of F_{em} in this paragraph. For the field of Lecture 15, define the normalized Maxwell action by

$$S_{\mathrm{em}} = -\frac{1}{2} \int F_{\mathrm{em}} \wedge \star F_{\mathrm{em}} = -\frac{1}{4} \int F_{\rho\sigma} F^{\rho\sigma} \mu.$$

This is Nakahara's (7.210b) in the present units. Vary the inverse metric while fixing the potential, hence the covariant field components. The inverse-metric factors and the determinant variation give

$$\delta(F_{\rho\sigma} F^{\rho\sigma}) = 2 F_{\alpha\lambda} F_\beta{}^{\lambda} \delta g^{\alpha\beta}, \qquad \delta\mu = -\frac{1}{2} g_{\alpha\beta} \delta g^{\alpha\beta} \mu.$$

Substitution in the action and comparison with the stress definition give

$$T_{\mu\nu} = -F_{\mu\lambda} F_\nu{}^{\lambda} + \frac{1}{4} g_{\mu\nu} F_{\alpha\beta} F^{\alpha\beta}.$$

In an orthonormal frame, $F_{0\lambda} F_0{}^{\lambda} = -|\mathbf{E}|^2$ and $F_{\alpha\beta} F^{\alpha\beta} = 2(|\mathbf{B}|^2 - |\mathbf{E}|^2)$. Hence $T_{00} = (|\mathbf{E}|^2 + |\mathbf{B}|^2)/2$ is nonnegative.

Theorem 19.2 (Einstein's equation · *cited variation; algebraic consequence proved*)**.** *The Einstein–Hilbert action in these conventions is*

$$S_{\mathrm{grav}} = -\frac{1}{16\pi G_N} \int R_{\mathrm{sc}} \mu.$$

For compactly supported metric variations, the cited variation formula is

$$\delta \int R_{\mathrm{sc}} \mu = \int G_{\alpha\beta} \delta g^{\alpha\beta} \mu.$$

Stationarity of $S_{\mathrm{grav}} + S_{\mathrm{m}}$ therefore gives

$$\mathrm{Ric}_{\mu\nu} - \frac{1}{2} R_{\mathrm{sc}} g_{\mu\nu} = 8\pi G_N T_{\mu\nu}.$$

The variation formula is Nakahara, §7.10.2, equations (7.203)–(7.214), translated to our signature and matter convention. It discards a divergence because the variation has compact support. The displayed coefficient follows by setting $-G_{\mu\nu}/(16\pi G_N) + T_{\mu\nu}/2 = 0$. These equations are a physical field law. The action supplies their variational form.

The signature translation is explicit. Replacing a metric by its negative leaves its Christoffels, curvature operator and Ricci tensor unchanged. The scalar curvature reverses sign, so the Einstein tensor is unchanged. Nakahara's $-+++$ Einstein–Hilbert sign consequently becomes the minus sign above. For stress, his (7.211) varies the lower metric. Converting to his inverse metric contributes a minus sign by (7.203). Our inverse metric is its negative, giving a second minus sign. The two cancel, yielding our positive inverse-metric variation with the same lowered stress tensor.[1] Carroll's free notes use the same curvature index order as ours.[2]

For the weak metric above, the additional first-order symbols are

$$\Gamma^i_{jk} = -\delta^i_k \partial_j \Phi - \delta^i_j \partial_k \Phi + \delta_{jk} \partial_i \Phi, \qquad \Gamma^\mu_{\mu i} = -2\partial_i \Phi.$$

Here and in the next calculation terms of order ϵ^2 are omitted. The quadratic Christoffel terms do not contribute at first order. Staticity removes time derivatives and gives $\mathrm{Ric}_{0i} = 0$. Thus $\mathrm{Ric}_{ij} = \partial_k \Gamma^k_{ji} - \partial_j \Gamma^\mu_{\mu i}$: the mixed second derivatives cancel, leaving $\delta_{ij} \sum_k \partial_k^2 \Phi$. The full contraction gives

$$\mathrm{Ric}_{00} = \nabla_E^2 \Phi, \qquad \mathrm{Ric}_{ij} = \delta_{ij} \nabla_E^2 \Phi, \qquad R_{\mathrm{sc}} = -2\nabla_E^2 \Phi, \qquad G_{00} = 2\nabla_E^2 \Phi.$$

Here $\nabla_E^2 = \sum_i \partial_i^2$ is the Euclidean spatial Laplacian. For example, at this order $\mathrm{Ric}_{00} = \partial_i \Gamma^i_{00}$. The 00 field equation for slowly moving matter gives $\nabla_E^2 \Phi = 4\pi G_N \rho$, the positive Newtonian Poisson equation. This fixes the Einstein sign independently of a source's notation.

The conservation identity follows from the covariant second Bianchi identity of Proposition 17.4, whose conversion from forms was sketched. Here is its full contraction. Use the lowered tensor $R_{ab\mu\nu}$ of Lecture 17. Metric compatibility and the two pair antisymmetries give

$$\begin{aligned} 0 &= g^{a\mu} g^{b\lambda} \left(\nabla_\lambda R_{ab\mu\nu} + \nabla_\mu R_{ab\nu\lambda} + \nabla_\nu R_{ab\lambda\mu} \right) \\ &= 2\nabla^\mu \mathrm{Ric}_{\mu\nu} - \partial_\nu R_{\mathrm{sc}}. \end{aligned}$$

The three contractions are respectively $\nabla^\mu \mathrm{Ric}_{\mu\nu}$, $\nabla^\mu \mathrm{Ric}_{\mu\nu}$ and $-\partial_\nu R_{\mathrm{sc}}$. Thus $\nabla^\mu G_{\mu\nu} = 0$. Einstein's equation implies $\nabla^\mu T_{\mu\nu} = 0$. This is a local covariant conservation statement. It alone does not assert a globally conserved energy on an arbitrary spacetime.

19.4 The Schwarzschild vacuum metric

Now use geometric units $G_N = c = 1$. Let $M > 0$ denote the mass parameter, equal to $G_N m/c^2$ before this unit choice. On the exterior $r > 2M$, set $f(r) = 1 - 2M/r$. The *Schwarzschild*

[1] Use (7.211) for this translation. The inverse-metric prefactor in (7.212) needs a minus sign to agree with it.

[2] Carroll, *Lecture Notes on General Relativity* (1997), (3.67), (3.90)–(3.96), pp. 76, 81–82. His metric has the opposite sign. His Schwarzschild geodesic formulas (7.33)–(7.48), pp. 172–174, translate by this metric reversal.

metric is

$$\mathrm{d}s^2 = f\,\mathrm{d}t^2 - f^{-1}\mathrm{d}r^2 - r^2(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2).$$

The angular expression is the round-sphere metric, with usual overlapping charts at its coordinate poles.

Proposition 19.3 (Schwarzschild vacuum · *proved with exact algebra check*)**.** *The exterior Schwarzschild metric has* $\mathrm{Ric} = 0$*. The fields* ∂_t *and* ∂_ϕ *are Killing fields. Along an affinely parametrized geodesic, the quantities*

$$E = f\dot{t}, \qquad L = r^2\sin^2\theta\,\dot{\phi}$$

are constant.

Proof. First compute for any smooth positive function $f(r)$. Up to symmetry in the two lower indices, the nonzero Christoffels are

$$\begin{array}{lll} \Gamma^t_{tr} = f'/(2f), & \Gamma^r_{tt} = ff'/2, & \Gamma^r_{rr} = -f'/(2f), \\ \Gamma^r_{\theta\theta} = -rf, & \Gamma^r_{\phi\phi} = -rf\sin^2\theta, & \Gamma^\theta_{r\theta} = 1/r, \\ \Gamma^\phi_{r\phi} = 1/r, & \Gamma^\theta_{\phi\phi} = -\sin\theta\cos\theta, & \Gamma^\phi_{\theta\phi} = \cot\theta. \end{array}$$

Substitute them into Lecture 17's Ricci contraction,

$$\mathrm{Ric}_{\mu\nu} = \partial_\rho\Gamma^\rho_{\nu\mu} - \partial_\nu\Gamma^\rho_{\rho\mu} + \Gamma^\rho_{\rho\lambda}\Gamma^\lambda_{\nu\mu} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\rho\mu}.$$

For example, $\Gamma^\rho_{\rho r} = 2/r$ and the tt entry is

$$\mathrm{Ric}_{tt} = \frac{(f')^2 + ff''}{2} + \frac{ff'}{r} - \frac{(f')^2}{2} = f\left(\frac{f''}{2} + \frac{f'}{r}\right).$$

The full substitution gives zero off-diagonal entries and

$$\mathrm{Ric}_{rr} = -f^{-1}\left(\frac{f''}{2} + \frac{f'}{r}\right), \qquad \mathrm{Ric}_{\theta\theta} = 1 - f - rf', \qquad \mathrm{Ric}_{\phi\phi} = \sin^2\theta\ \mathrm{Ric}_{\theta\theta}\,.$$

For $f = 1 - 2M/r$, the combinations are $f''/2 + f'/r = -2M/r^3 + 2M/r^3 = 0$ and $1 - f - rf' = 2M/r - 2M/r = 0$. Thus every Ricci component is zero. Smoothness extends this tensor equality across the angular chart poles.

The metric coefficients are independent of t and ϕ. The coordinate formula for the Lie derivative gives $\mathcal{L}_{\partial_t} g = \mathcal{L}_{\partial_\phi} g = 0$. Lecture 16's Killing conservation theorem makes $g(\partial_t, \dot{\gamma}) = f\dot{t}$ and $-g(\partial_\phi, \dot{\gamma}) = r^2\sin^2\theta\dot{\phi}$ constant. ∎

The divergence of f^{-1} at $r = 2M$ is a coordinate singularity. Indeed, put $v = t + r_*(r)$, where $r_*' = f^{-1}$. Substitution of $\mathrm{d}t = \mathrm{d}v - f^{-1}\mathrm{d}r$ gives

$$\mathrm{d}s^2 = f\,\mathrm{d}v^2 - 2\mathrm{d}v\,\mathrm{d}r - r^2(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2).$$

Its determinant is $-r^4\sin^2\theta$, nonzero at $r = 2M$ away from the angular coordinate poles. These coordinates extend the metric there. In the Schwarzschild black-hole extension, this surface is the future *event horizon*: the boundary of the region from which future-directed light signals can reach arbitrarily distant observers. That global causal identification is cited, rather than inferred from a diverging coefficient; see Carroll, §7, pp. 183–185.

19.5 The massive orbit equation

Consider a future timelike Schwarzschild geodesic parametrized by proper time. Rotations of the angular coordinates are isometries because they preserve the round-sphere factor. Rotate them so the initial position and tangential velocity lie in the equatorial plane. The θ equation has the solution $\theta = \pi/2$, $\dot\theta = 0$ for these initial data. Uniqueness of the geodesic equation keeps the orbit in that plane. The constants E and L are its energy and angular momentum per unit rest mass.

Proposition 19.4 (Schwarzschild orbits · *proved*)**.** *A proper-time-parametrized equatorial timelike geodesic obeys*

$$\dot r^2 + f\left(1 + \frac{L^2}{r^2}\right) = E^2.$$

If $L \neq 0$, use ϕ as parameter and put $u = 1/r$. Then, with primes now denoting derivatives with respect to ϕ,

$$u'' + u = \frac{M}{L^2} + 3Mu^2.$$

Proof. The normalization $g(\dot\gamma, \dot\gamma) = 1$ gives

$$1 = f\dot t^2 - f^{-1}\dot r^2 - r^2\dot\phi^2 = \frac{E^2 - \dot r^2}{f} - \frac{L^2}{r^2}.$$

Rearranging gives the radial energy equation. To retain turning points, use the radial geodesic equation itself:

$$\ddot r = -\frac{ff'}{2}\dot t^2 + \frac{f'}{2f}\dot r^2 + rf\dot\phi^2 = -\frac{M}{r^2} + \frac{L^2}{r^3} - \frac{3ML^2}{r^4}.$$

The last equality substitutes the constants and the normalization just proved. It never divides by $\dot r$. Since $\dot\phi = Lu^2 \neq 0$, the chain rule gives $\dot r = -Lu'$ and $\ddot r = -L^2u^2u''$. Substitute in the radial equation and divide by $-L^2u^2$ to obtain the result. ∎

The Newtonian orbit equation lacks the term $3Mu^2$. For radial motion $L = 0$, the radial energy equation still applies, but ϕ cannot serve as parameter.

19.6 A circular orbit and its effective potential

Define the *effective potential* for fixed L by

$$U_{\mathrm{eff}}(r) = f(r)\left(1 + \frac{L^2}{r^2}\right).$$

The radial equation is $\dot r^2 + U_{\mathrm{eff}} = E^2$ and $\ddot r = -U'_{\mathrm{eff}}/2$. The figure plots an exact example of this potential.

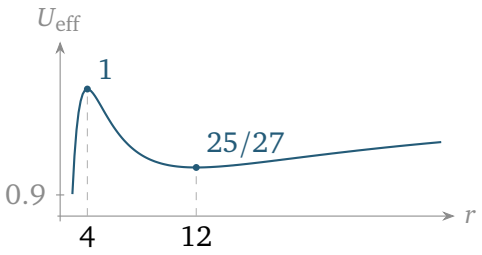


Figure 19.3: For $M = 1$ and $L = 4$, the potential has a maximum at $r = 4$ and a minimum at $r = 12$. The plot uses $2.9 \le r \le 30$ and a truncated vertical axis.

Example 19.5 (A stable circular orbit)**.** Find a massive circular orbit at radius $r_0 = 12M$. For a general circular radius $r_0 > 3M$, solving $U'_{\text{eff}}(r_0) = 0$ and $E^2 = U_{\text{eff}}(r_0)$ gives

$$L^2 = \frac{Mr_0^2}{r_0 - 3M}, \qquad E^2 = \frac{(r_0 - 2M)^2}{r_0(r_0 - 3M)}.$$

With the angular momentum held fixed, the second derivative there is

$$U''_{\text{eff}}(r_0) = \frac{2M(r_0 - 6M)}{r_0^3(r_0 - 3M)}.$$

It is positive for $r_0 > 6M$, so the radial linearization has a restoring force. At $r_0 = 12M$ the constants are $L = 4M$ and $E^2 = 25/27$. This gives the potential minimum marked in the figure after setting $M = 1$.

Self-check. Recover the inward Newtonian acceleration from Γ^r_{tt} far from the Schwarzschild source.

Answer. The table gives $\Gamma^r_{tt} = ff'/2 = M(r - 2M)/r^3 = M/r^2 + O(M^2/r^3)$. For slow motion, the geodesic equation contributes $-\Gamma^r_{tt}$ to the coordinate-time radial acceleration. Its leading value is $-M/r^2$, the attractive Newtonian result.

Summary. A weak static Lorentzian metric gives Newton's equation for slow free fall. The cited Einstein–Hilbert variation gives Einstein's equation with positive $8\pi G_N T_{\mu\nu}$ in our conventions. Contracting the second Bianchi identity gives covariant conservation. The Schwarzschild exterior has zero Ricci tensor. Its Killing constants, unit timelike normalization and radial geodesic equation give the massive orbit equation $u'' + u = M/L^2 + 3Mu^2$ in geometric units, for $L \neq 0$. Lecture 20 describes fields that require several local frames.

Reading. Nakahara, §7.10.1–2 and Example 7.15. Carroll, *Lecture Notes on General Relativity* (1997), §4, especially pp. 104–116, and §7, pp. 172–185.

Lecture 20

Fibre bundles

Objectives. *A vector field assigns a vector in a different tangent space at each point. A bundle describes these fibres and the changes between their local coordinates. We reconstruct bundles from transition functions and prove the global-section criterion for principal bundles. The Hopf bundle and the tangent sphere show how local products can fail to give a global product.*

By the end of this lecture you will be able to:

- Construct a smooth vector or principal bundle from a smooth cocycle.
- Relate local frames, section components and transition functions.
- Prove the principal-bundle section criterion and identify the Hopf bundle.
- Distinguish the tangent sphere's frame winding from its component winding.

20.1 Local products and fibres

A smooth *fibre bundle* consists of a smooth surjection $\pi_E : E \to M$ and a manifold F, its *typical fibre*. Every point of the *base space* M has an open neighbourhood U_i and a diffeomorphism

$$\varphi_i : U_i \times F \longrightarrow \pi_E^{-1}(U_i), \qquad \pi_E(\varphi_i(x, v)) = x.$$

This is a *local trivialization*. The manifold E is the *total space*. The *fibre at* x is $E_x = \pi_E^{-1}(x)$. All manifolds here are Hausdorff and second countable. A *bundle isomorphism* over M is a diffeomorphism of total spaces commuting with the projections and preserving the specified fibre structure. A bundle is *trivial* if it is isomorphic to $M \times F$. The figure shows a local product only.

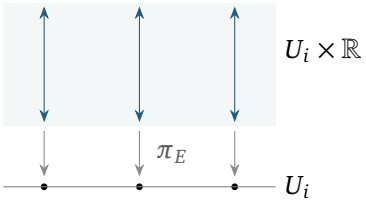


Figure 20.1: A local trivialization gives fibre coordinates over U_i. The projection sends each vertical fibre to its base point.

A real or complex *vector bundle of rank* r has vector-space fibres and local trivializations linear on each fibre, with typical fibre $V_{\text{rep}} = \mathbb{R}^r$ or $\mathbb{C}^r$. A rank-one vector bundle is a *line bundle*. For example, TM of Lecture 8 is a rank-dim M real vector bundle.

A *section over an open set* $U \subset M$ is a smooth map $s : U \to E$ such that $\pi_E \circ s = \mathrm{id}_U$. It is a *global section* when $U = M$. A *local frame* of a rank-r vector bundle is a row of r smooth sections that form a basis in each fibre of its domain.

Let G be a Lie group. A *principal G-bundle* is a bundle $\pi_E : P \to M$ with a smooth right action of G that is free and transitive on each fibre. Its local trivializations are required to satisfy $\varphi_i(x, ab) = \varphi_i(x, a)b$.

20.2 Transition functions and reconstruction

For a vector bundle, comparing trivializations defines smooth *transition functions* $g_{ij} : U_i \cap U_j \to \mathrm{GL}(r, \mathbb{K})$, where $\mathbb{K}$ is $\mathbb{R}$ or $\mathbb{C}$, by

$$\varphi_j(x, v) = \varphi_i(x, g_{ij}(x)v).$$

Thus the coordinates of the same vector satisfy $v_i = g_{ij}v_j$. The *structure group* is the group of allowed transition matrices, usually $\mathrm{GL}(r, \mathbb{R})$ or $\mathrm{GL}(r, \mathbb{C})$. For a principal bundle, equivariance instead gives

$$\varphi_j(x, a) = \varphi_i(x, g_{ij}(x)a), \qquad g_{ij} : U_i \cap U_j \to G.$$

Indeed, compare the trivializations at the group identity and then act on the right by a. This also shows that the principal transitions are smooth. For a vector bundle, the standard bases give local row frames e_i with $e_j = e_i g_{ij}$. On triple overlaps the *cocycle condition* is

$$g_{ij}g_{jk} = g_{ik}, \qquad g_{ii} = I, \qquad g_{ji} = g_{ij}^{-1}.$$

For a principal bundle, replace the identity matrix I by the group identity e. The figure follows the coordinate maps, whose directions reverse the frame relation.

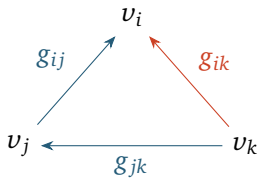


Figure 20.2: A fibre-coordinate column passes from region k to j and then to i by $g_{ij}g_{jk}$. Direct passage uses g_{ik}. All maps refer to the same base point in a triple overlap.

Proposition 20.1 (Cocycle reconstruction · *proved*)**.** *Smooth transition matrices satisfying the cocycle condition on an open cover of M determine a smooth vector bundle, up to bundle isomorphism. The same construction with fibre a Lie group G and transition functions acting by left multiplication constructs a principal bundle. Let $h_i : U_i \to \mathrm{GL}(r, \mathbb{K})$ be smooth frame changes. For a principal bundle take smooth $h_i : U_i \to G$ instead. In this case replace frames by local sections in the following change law. Under changes $e'_i = e_i h_i$, the transitions become*

$$g'_{ij} = h_i^{-1} g_{ij} h_j.$$

Conversely, transitions related in this way give isomorphic bundles. Different covers can be compared on their common refinement.

Proof. Take the disjoint union of $U_i \times V_{\mathrm{rep}}$ and identify

$$(x, v)_j \sim (x, g_{ij}(x)v)_i.$$

Identity, inverse and product in the cocycle make this an equivalence relation. Project its quotient to M by the common base point. Over U_i, each class has a unique representative in

the ith copy. The corresponding local coordinate map sends a representative in copy j to $(x, g_{ij}(x)v)$. It is a homeomorphism because the preimage of an open set in one copy is open in every other copy, since the transition map is a diffeomorphism. The quotient topology therefore agrees with these product charts. Their changes are smooth and linear on fibres.

Points with different base points have disjoint neighbourhoods lifted from M. Points with the same base point can be separated in one product chart. Thus the total space is Hausdorff. Second countability of M gives a countable subcover. Choose a basis element inside a cover member at each point and choose one member per basis element. The corresponding product charts, each second countable, give a countable basis for the total space. Hence the quotient is a smooth manifold.

The same argument with V_{rep} replaced by G gives smooth charts $U_i \times G$. The action $(x, a) \cdot b = (x, ab)$ is independent of the chart because left transition multiplication commutes with right multiplication. It is free and transitive on each fibre, as required in the principal definition.

Finally, $e'_j = e'_i h_i^{-1} g_{ij} h_j$, proving the displayed change law. For an isomorphism between the two constructions, use $(x, v)_i \mapsto (x, h_i^{-1} v)'_i$. The change law makes this map agree on overlaps. Its inverse uses h_i, so it is a smooth bundle isomorphism. If another bundle has trivializations φ_i with these transitions, send the quotient class $[(x, v)_i]$ to $\varphi_i(x, v)$. The defining transition relation makes this independent of its representative. In product charts it is the identity, so it is a smooth bundle isomorphism, equivariant in the principal case. This proves uniqueness. Restricting trivializations from two covers to their pairwise intersections gives a common cover. The same chartwise map compares the reconstructions on this refinement. ∎

This is the smooth reconstruction of Nakahara, §9.2.2, with our frame convention. Naber's Theorem 4.3.4, pp. 223–224, is its continuous principal-bundle counterpart, with the same transition convention.

20.3 Sections and frames of a vector bundle

In a local frame, write $s = e_i \sigma_i$. Its coefficient columns satisfy

$$\sigma_i = g_{ij} \sigma_j.$$

Conversely, smooth columns with this compatibility glue to a section. The zero columns always give a global zero section.

A global frame exists exactly when the vector bundle is trivial. Indeed, a frame defines the isomorphism $(x, v) \mapsto e(x)v$. Its inverse is smooth in local coordinates because the frame matrix is invertible. A trivialization supplies the converse frame by its standard basis. For a line bundle, one nowhere-zero section is a frame. For higher rank, one section need not give an entire basis.

20.4 Principal and associated bundles

The local sections $s_i(x) = \varphi_i(x, e)$ obey $s_j = s_i g_{ij}$. The group acts on the right, while g_{ij} changes fibre coordinates by multiplication on the left.

> **Proposition 20.2** (The section criterion · *proved*)**.** *A principal bundle is trivial if and only if it has a global section.*

Proof. A global section s gives a map $\Psi\colon M \times G \to P$ by $\Psi(x,a) = s(x)a$. Freeness and transitivity make it bijective and equivariant. In a local trivialization write $s(x) = s_i(x)b_i(x)$. Then $\Psi(x,a)$ has local coordinate $b_i(x)a$. Its inverse has coordinate $b_i(x)^{-1}a$, so both maps are smooth. Conversely, a trivialization gives the section $x \mapsto (x,e)$. ∎

This is Nakahara, Theorem 9.2. Vector-bundle triviality requires a full frame, as above.

For an oriented Riemannian n-manifold, its *oriented orthonormal frame bundle* consists of all positive orthonormal frames in its tangent spaces. Right multiplication of a row frame by $\mathrm{SO}(n)$ is free and transitive on each fibre. Local Gram–Schmidt frames give equivariant product charts. This makes it a principal $\mathrm{SO}(n)$-bundle, even when no global tangent frame exists.

Remark 20.3 (Associated bundles)**.** Let $\rho\colon G \to \mathrm{GL}(V_{\mathrm{rep}})$ be a smooth representation. The *associated bundle* is

$$P \times_G V_{\mathrm{rep}} = (P \times V_{\mathrm{rep}})/\!\sim, \qquad (pa, v) \sim (p, \rho(a)v), \qquad [p,v] \longmapsto \pi_E(p).$$

Its local trivializations are $(x,v) \mapsto [s_i(x), v]$. Their transitions are $\rho(g_{ij})$, so reconstruction gives a smooth vector bundle. Sections correspond to smooth *equivariant maps* $f\colon P \to V_{\mathrm{rep}}$ with

$$f(pa) = \rho(a)^{-1} f(p).$$

The section is $x \mapsto [p, f(p)]$ for any p over x: $[pa, f(pa)] = [p, \rho(a)f(pa)] = [p, f(p)]$. Conversely, a section uniquely determines these fibre coordinates $f(p)$. In a product chart they are $\rho(a)^{-1} f(s_i(x))$, which proves smoothness and the converse. The homogeneous case is the construction in *Lie Groups*, Lecture 26, *From equivariant functions to sections*.

20.5 The Hopf bundle

For a closed subgroup H of a Lie group G, Theorem 8.5 gives the smooth quotient G/H and a submersion $\pi_E\colon G \to G/H$. The inverse function theorem gives local submersion coordinates (Lee, *Introduction to Smooth Manifolds*, second edition, Theorem 4.5). Choose base coordinates y and supplement the independent differentials of $y \circ \pi_E$ by complementary coordinates ζ. The map $(y \circ \pi_E, \zeta)$ has invertible differential, hence is a local coordinate system. In these coordinates $\pi_E(y,\zeta) = y$. Fixing ζ gives a smooth local section s near each base point. The maps

$$U \times H \longrightarrow \pi_E^{-1}(U), \qquad (x,h) \longmapsto s(x)h$$

are diffeomorphisms. Indeed, their inverse sends a to $(aH, s(aH)^{-1}a)$. The second entry lies in the embedded subgroup H and is smooth there. In submanifold charts its normal coordinates vanish. Thus $H \to G \to G/H$ is a principal H-bundle.

Let $N = [1:0]$ and $S = [0:1]$ be the north and south poles of the Bloch sphere. The projective coordinate $z = b/a$ is defined on $U_N = \mathbb{C}\mathrm{P}^1 \setminus \{S\}$. The coordinate $w = a/b = 1/z$ is defined on $U_S = \mathbb{C}\mathrm{P}^1 \setminus \{N\}$. Unit representatives of their lines are

$$p_N(z) = \frac{(1,z)^T}{\sqrt{1+|z|^2}}, \qquad p_S(w) = \frac{(w,1)^T}{\sqrt{1+|w|^2}}.$$

The *tautological line bundle* is the set

$$\mathcal{T} = \{(\ell, v) \in \mathbb{C}\mathrm{P}^1 \times \mathbb{C}^2 : v \in \ell\}, \qquad (\ell, v) \longmapsto \ell.$$

Its local coordinates are $v = cp_N(z)$ or $v = cp_S(w)$, with $c \in \mathbb{C}$. The inverse coefficient is $c = p_N^\dagger v$ or $c = p_S^\dagger v$, respectively. These maps give smooth product charts with linear overlap maps. On the overlap, $p_S = p_N|z|/z$. Orient the equator $|z| = 1$ by increasing ϕ, as the boundary of the northern cap. On this positive equator $z = e^{i\phi}$, the frame transition is $g_{NS} = e^{-i\phi}$, with winding -1. This sign follows from the stated frame order.

Theorem 20.4 (The Hopf bundle · *proved given the cited homogeneous-space and inverse function theorems*). *The map* $\mathrm{SU}(2) \to \mathrm{SU}(2)/\mathrm{U}(1) \cong \mathbb{C}\mathrm{P}^1 \cong S^2$ *is a nontrivial principal* $\mathrm{U}(1)$*-bundle. Under* $\mathrm{SU}(2) \cong S^3$*, it sends a unit column* $(a, b)^T$ *to its line* $[a : b]$*. Its associated complex line bundle for the standard* $\mathrm{U}(1)$ *representation is the tautological line bundle.*

Proof. Recall Lecture 8's explicit identification

$$(a, b) \longmapsto \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

Right multiplication by $\mathrm{diag}(u, u^{-1})$, for $u \in \mathrm{U}(1)$, multiplies the first column by u. Its orbits are exactly the unit representatives of each complex line. The preceding quotient argument proves the principal structure.

To prove that S^n is simply connected for $n \in \{2, 3\}$, subdivide a loop into hemispherical pieces using Lemma 3.5. On each piece replace it by the normalized straight segment between its endpoints. Both curves have positive scalar product with the hemisphere centre, so their normalized straight interpolation is a homotopy fixing those endpoints. The resulting polygon lies in finitely many two-dimensional linear subspaces. Their union misses a point of the sphere. For each subspace choose a nonzero linear functional vanishing on it and evaluate these on $(1, t, \dots, t^n)$. Each gives a nonzero polynomial, so a parameter outside their finite root sets gives a vector outside the union. Normalize it to obtain the omitted sphere point. Stereographic projection from that point puts the polygon in $\mathbb{R}^n$, where straight contraction to its base point fixes that base point. Pasting the piece homotopies proves the assertion for $n = 2$ and $n = 3$. Both spheres are path-connected by short arcs, joining antipodes via a third point.

Loops and based homotopies in a product are precisely pairs of loops and homotopies. Thus $\pi_1(S^2 \times S^1) \cong \mathbb{Z}$, whereas $\pi_1(S^3) = 0$. A trivial Hopf bundle would give a diffeomorphism between these total spaces, which is impossible.

The map $[p, v] \mapsto ([p], vp)$ identifies the associated line with $\mathcal{T}$. It is well-defined because $(pu, v) \sim (p, uv)$. Its inverse exists on each fibre, and in the local unit representatives above both maps are product-coordinate identity maps. Hence it is a smooth bundle isomorphism. ∎

The figure separates the phase fibre from its fixed base line.

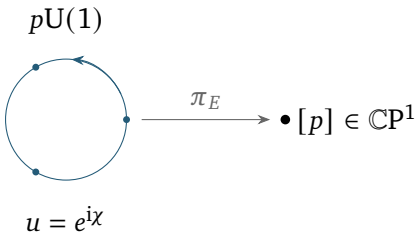


Figure 20.3: All unit vectors pu, with $u \in \mathrm{U}(1)$, lie over the same line $[p]$. The circle records their relative phase.

Proposition 20.5 (Clutching classification · *cited*)**.** *The underlying topological principal* $\mathrm{U}(1)$*-bundles over* S^2*, up to continuous equivariant bundle isomorphism, are classified by the winding of the transition* g_{NS} *on the positive equator. Every integer occurs.*

This is Naber, Theorem 4.4.3, pp. 231–232. That theorem concerns continuous bundles. No smooth classification proof is claimed here. Its integer invariant already distinguishes the smooth examples. On the overlap $z \neq 0$, the transitions $(z/|z|)^n$ give smooth representatives for every $n \in \mathbb{Z}$.

20.6 The tangent sphere and its transition winding

Use the oriented complex charts $z = x + \mathrm{i}y$ and $w = u + \mathrm{i}v = 1/z$ of Lecture 7 on $S^2 \cong \mathbb{C}\mathrm{P}^1$. Multiplication by i on tangent vectors is defined by $\mathrm{i}\partial_x = \partial_y$. The change $w = 1/z$ has complex-linear differential and preserves this rule. The real tangent planes therefore form a complex line bundle.

The figure displays the two angle lifts computed in the example.

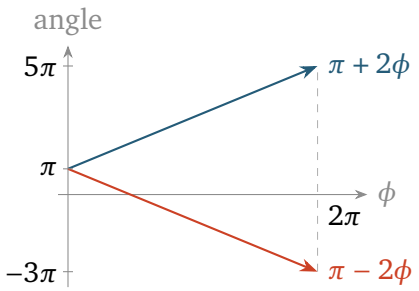


Figure 20.4: The frame angle increases by 4π during one equatorial circuit. The inverse component angle decreases by 4π.

Example 20.6 (The tangent-sphere transition)**.** Find its frame transition and its component transition around the equator. Use complex frames $e_z = \partial_x$ and $e_w = \partial_u$, where a complex coefficient $a + \mathrm{i}b$ means the real vector $a\partial_x + b\partial_y$ in the first frame. The coordinate derivative and its inverse are

$$\frac{\mathrm{d}w}{\mathrm{d}z} = -z^{-2}, \qquad \frac{\mathrm{d}z}{\mathrm{d}w} = -z^2.$$

The chain rule gives the frame relation and the component relation

$$e_w = e_z(-z^2), \qquad \sigma_w = (-z^{-2})\sigma_z.$$

The real plane rotation through β is

$$\mathrm{Rot}(\beta) = \begin{pmatrix} \cos\beta & -\sin\beta \\ \sin\beta & \cos\beta \end{pmatrix}.$$

Identify a complex section coefficient with the real column $(\mathrm{Re}\,\sigma, \mathrm{Im}\,\sigma)^T$. On $z = e^{\mathrm{i}\phi}$ the corresponding real matrices are

$$g_{zw} = -\mathrm{Rot}(2\phi), \qquad \sigma_w = -\mathrm{Rot}(-2\phi)\sigma_z.$$

The frame loop has lifted angle $\pi + 2\phi$ and winding +2. The component loop has angle $\pi - 2\phi$ and winding −2. The Bloch map is $(2x, 2y, 1 - |z|^2)/(1 + |z|^2)$. For the round metric,

differentiating it gives

$$g = \frac{4(\mathrm{d}x^2 + \mathrm{d}y^2)}{(1 + |z|^2)^2}.$$

Normalizing the coordinate frames therefore replaces $-z^2$ by its phase $-z^2/|z|^2$ on the overlap. Its equatorial winding is still +2. A new unit frame on either cap changes this winding by zero, because its boundary phase extends across the disk and is null-homotopic by Lecture 3. Consequently the oriented orthonormal frame bundle is nontrivial by the cited classification. A nowhere-zero tangent field would give a unit frame and trivialize it, contradicting its winding. The integer 2 agrees with $\chi(S^2)$ in Lecture 18.

Self-check. Describe the Möbius real line bundle over S^1 and show it is nontrivial.

Answer. Take $[0, 2\pi] \times \mathbb{R}$ and identify $(2\pi, v)$ with $(0, -v)$. On two overlapping base arcs the transition is +1 on one overlap component and −1 on the other. A nowhere-zero section would lift to a continuous real function with $f(2\pi) = -f(0)$, impossible by the intermediate value theorem. A trivial line bundle has a nowhere-zero section, so this bundle is nontrivial. Its zero section still exists. Putting −1 on both overlap components instead gives a trivial bundle. Change the frame on one whole arc by −1 to make both transitions +1.

Summary. Transition functions reconstruct smooth vector and principal bundles. Frames transform by $e_j = e_i g_{ij}$, while section components transform inversely. A principal bundle is trivial exactly when it has a global section. The Hopf bundle $S^3 \to S^2$ is nontrivial and its associated standard line is tautological. The tangent sphere has frame winding +2, opposite to its component winding. Lecture 21 describes connections and transport in these local frames.

Reading. Nakahara, §9.2–9.4, especially Theorem 9.2. Naber, §§4.2–4.4, especially Theorems 4.3.4 and 4.4.3.

Lecture 21

Gauge fields: connections, curvature and holonomy

Objectives. *A bundle has local frames, but differentiating their coefficients depends on the frame. A connection corrects this derivative and specifies parallel transport. We derive its change law and show how curvature measures transport around a small loop. A flat connection can still have nontrivial transport around a hole. The Aharonov–Bohm phase is this effect for a charged particle.*

By the end of this lecture you will be able to:

- Derive the connection and curvature laws under a change of frame.
- Compute parallel transport with the correct multiplication order.
- Relate a small counter-clockwise loop to its curvature.
- Compute the flat Aharonov–Bohm holonomy and its gauge-invariant phase.

21.1 Covariant derivatives and minimal coupling

Let $E \to M$ be a smooth rank-r complex vector bundle. Write $\Gamma(E)$ for its smooth sections and $\Omega^k(M;E)$ for its E-valued k-forms. In a frame, the latter are columns of ordinary k-forms. Recall Lecture 20's row frame e and coefficient column σ, with $s = e\sigma$. A *connection* is a complex-linear map

$$\nabla\colon \Gamma(E) \longrightarrow \Omega^1(M;E), \qquad \nabla(fs) = \mathrm{d}f \otimes s + f\nabla s \quad \text{for } f \in C^\infty(M,\mathbb{C}).$$

We first construct a *bump function*, a smooth function equal to one near a chosen point $p \in M$, with support inside a chosen neighbourhood U. Take a coordinate ball centred at p with $\overline{B_R} \subset U$ and $0 < r < R$. Define $\rho(t) = e^{-1/t}$ for $t > 0$ and $\rho(t) = 0$ for $t \le 0$. Every positive-side derivative is $e^{-1/t}$ times a polynomial in $1/t$, by induction. All tend to zero at zero, so ρ is smooth. In the ball's coordinates the desired function is

$$f(x) = \frac{\rho(R^2 - |x|^2)}{\rho(R^2 - |x|^2) + \rho(|x|^2 - r^2)}.$$

The denominator is positive. The function is one on B_r and zero outside B_R. Its support is compact inside the chart, so extension by zero is smooth on M.

The connection is local. If s vanishes near p, choose such an f supported where $s = 0$. The Leibniz rule applied to $fs = 0$ gives $(\nabla s)_p = 0$. A section defined on U can instead be multiplied by a bump supported in U, equal to one near p, and extended by zero. Locality makes its derivative at p independent of the extension. This defines the local operator with

the same Leibniz rule and commutes with restriction. In a local frame, define the matrix of one-forms A by $\nabla e = eA$. The Leibniz rule gives

$$\nabla s = eD\sigma, \qquad D\sigma = \mathrm{d}\sigma + A\sigma.$$

The matrix A is the local *gauge potential*. It is a matrix of one-forms, rather than a single global matrix on a nontrivial bundle. Lecture 16's tangent connection is the real version of this definition, with real fibres, real linearity and real-valued functions.

A *Hermitian metric* is a smoothly varying positive Hermitian inner product on each fibre, antilinear in its first argument. A connection is *unitary* if it differentiates this inner product by the product rule. A *unitary frame* is an orthonormal basis in every fibre of its domain. In such a frame that rule is equivalent to $A^\dagger = -A$. The covariant product rule differs from the ordinary derivative of $\sigma^\dagger\tau$ by the term $\sigma^\dagger(A^\dagger + A)\tau$. Here σ and τ are coefficient columns of two local sections. For a line bundle, A is therefore imaginary. For an atlas with transitions in a matrix group G, its *G-frames* are the right translates $e_i a$, with $a \in G$. Require A to take values in its Lie algebra in these frames. This is the setting of the transport statements below.

For charge $q \in \mathbb{R}$, the physical potential of Lecture 15 is the real one-form $A_{\mathrm{em}} = -\phi\mathrm{d}t + \mathbf{A}\cdot\mathrm{d}\mathbf{x}$. The connection convention and its component derivatives are

$$A_{\mathrm{conn}} = -\frac{\mathrm{i}q}{\hbar}A_{\mathrm{em}}, \qquad D_t = \partial_t + \frac{\mathrm{i}q}{\hbar}\phi, \qquad D_i = \partial_i - \frac{\mathrm{i}q}{\hbar}A_i.$$

Here $\hbar > 0$ remains explicit. Thus $-\mathrm{i}\hbar D_i = -\mathrm{i}\hbar\partial_i - qA_i$. Replacing ordinary derivatives by these derivatives is *minimal coupling*. The real electromagnetic potential and the imaginary connection potential are distinct. For a Hermitian matrix potential A_{phys} written with $D = \mathrm{d} - \mathrm{i}g_{\mathrm{YM}}A_{\mathrm{phys}}$, the corresponding matrix is $A_{\mathrm{conn}} = -\mathrm{i}g_{\mathrm{YM}}A_{\mathrm{phys}}$. The coupling g_{YM} is a scalar, never a gauge map.

21.2 Frame changes and gauge transformations

Let $h\colon U \to \mathrm{GL}(r,\mathbb{C})$ be a smooth change of frame, $e' = eh$. This is the map h, not Planck's constant. The same section has coefficients $\sigma' = h^{-1}\sigma$. Differentiating the new frame gives

$$\nabla(eh) = e(Ah + \mathrm{d}h) = ehA', \qquad A' = h^{-1}Ah + h^{-1}\mathrm{d}h. \tag{21.1}$$

The extra derivative compensates for the changing frame. The following diagram expresses the resulting covariance.

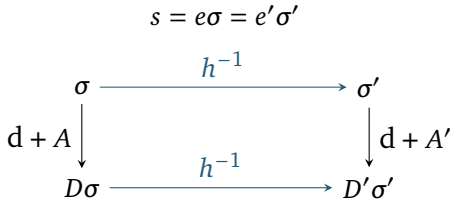


Figure 21.1: Both a section's coefficient column and its covariant derivative change by h^{-1}. The connection law makes the square commute.

Indeed, $\mathrm{d}(h^{-1}) = -h^{-1}(\mathrm{d}h)h^{-1}$ implies $D'\sigma' = h^{-1}D\sigma$ after the two derivative terms cancel. Conversely, local matrices obeying this change law on overlaps define a connection, because

these local derivatives agree as bundle-valued one-forms. This is Nakahara's compatibility law (10.9), translated to row frames.

The G-frames of E form a principal bundle by Lecture 20's cocycle construction. A *gauge transformation* is an equivariant smooth automorphism of this principal bundle over the identity of M. In local sections s_i, write it as $s_i(x)a \mapsto s_i(x)u_i(x)a$. The right-hand group coordinate is a. Since $s_j = s_i g_{ij}$, compatibility is precisely

$$u_j = g_{ij}^{-1} u_i g_{ij}.$$

A chosen global trivialization identifies these transformations with maps $M \to G$. For an abelian group the compatibility law reduces to $u_i = u_j$, even on a nontrivial bundle. For a nonabelian group, arbitrary global matrix maps need not satisfy it. A passive frame change describes the same connection. On E, the corresponding automorphism acts on columns by $\sigma_i \mapsto u_i \sigma_i$. Transporting the connection so that $D_i'(u_i\sigma_i) = u_i D_i \sigma_i$ gives

$$A_i' = u_i A_i u_i^{-1} + u_i \mathrm{d}(u_i^{-1}).$$

This follows by differentiating $u_i\sigma_i$ and cancelling $\mathrm{d}u_i$. It is the frame law with $h = u_i^{-1}$, the active description.

For electromagnetism take a real smooth function χ and $h = e^{-\mathrm{i}q\chi/\hbar}$. The frame law then reads

$$A'_{\mathrm{em}} = A_{\mathrm{em}} + \mathrm{d}\chi, \qquad \psi' = e^{\mathrm{i}q\chi/\hbar}\psi, \qquad D'\psi' = e^{\mathrm{i}q\chi/\hbar} D\psi.$$

Here ψ is the charged section's coefficient, which transforms inversely to its frame. These equations include the charge sign and agree with the minimal-coupling derivatives.

21.3 Curvature and the Bianchi identity

For matrix-valued forms, wedge multiplication includes matrix multiplication: $(A \wedge B)_{ab} = \sum_c A_{ac} \wedge B_{cb}$. Extend D to coefficient-valued forms by $D\eta = \mathrm{d}\eta + A \wedge \eta$. Its square on a coefficient column is multiplication by the *curvature two-form*

$$F = \mathrm{d}A + A \wedge A, \qquad F_{ij} = \partial_i A_j - \partial_j A_i + [A_i, A_j].$$

The components mean $F = \frac{1}{2} F_{ij} \mathrm{d}x^i \wedge \mathrm{d}x^j$. For an endomorphism-valued k-form B, the *covariant exterior derivative* is

$$DB = \mathrm{d}B + A \wedge B - (-1)^k B \wedge A.$$

This derivative acts on both the output and input of an endomorphism.

Theorem 21.1 (Local connection laws · *proved*)**.** *A connection has local form $D = \mathrm{d} + A$ and the frame law (21.1). Its curvature transforms by $F' = h^{-1}Fh$ and satisfies the Bianchi identity*

$$DF = \mathrm{d}F + A \wedge F - F \wedge A = 0.$$

Proof. The first assertions follow from the Leibniz and frame calculations above. Expanding $D^2\sigma$ cancels $-A \wedge \mathrm{d}\sigma$ against $A \wedge \mathrm{d}\sigma$, leaving $F\sigma$. To check curvature covariance explicitly, put $B = h^{-1}Ah$ and introduce the *Maurer–Cartan form* $\vartheta_{\mathrm{MC}} = h^{-1}\mathrm{d}h$. The product rule and the inverse derivative give

$$\mathrm{d}B = h^{-1}(\mathrm{d}A)h - \vartheta_{\mathrm{MC}} \wedge B - B \wedge \vartheta_{\mathrm{MC}}, \qquad \mathrm{d}\vartheta_{\mathrm{MC}} = -\vartheta_{\mathrm{MC}} \wedge \vartheta_{\mathrm{MC}}.$$

Since $A' = B + \vartheta_{\mathrm{MC}}$, substituting these identities into $\mathrm{d}A' + A' \wedge A'$ leaves only $h^{-1}(\mathrm{d}A + A \wedge A)h$. Finally,

$$\mathrm{d}F = \mathrm{d}A \wedge A - A \wedge \mathrm{d}A, \qquad A \wedge F - F \wedge A = A \wedge \mathrm{d}A - \mathrm{d}A \wedge A.$$

Adding gives the Bianchi identity. These are Nakahara's (10.38)–(10.47) in the present matrix notation. ∎

For U(1) the scalar one-form obeys $A \wedge A = 0$, so $F_{\mathrm{conn}} = -(\mathrm{i}q/\hbar)F_{\mathrm{em}}$ and the Bianchi identity is $\mathrm{d}F = 0$. The real source three-form $\mathcal{J}$ of Lecture 15 corresponds to $\mathcal{J}_{\mathrm{conn}} = -(\mathrm{i}q/\hbar)\mathcal{J}$ in connection units. The connection-valued source $\mathcal{J}_{\mathrm{conn}}$ is distinct from the real current. For a Hermitian matrix potential in the preceding convention, instead

$$F_{\mathrm{conn}} = -\mathrm{i}g_{\mathrm{YM}}(\mathrm{d}A_{\mathrm{phys}} - \mathrm{i}g_{\mathrm{YM}}A_{\mathrm{phys}} \wedge A_{\mathrm{phys}}).$$

The commutator term remains in the nonabelian theory. Within its matrix Lie algebra, the potential's value at a point can be changed arbitrarily by prescribing $\mathrm{d}h$ there with $h = I$. With $h = I$ there, curvature is unchanged. A general frame change conjugates it.

21.4 Parallel transport and a small square

A section along a piecewise smooth path $\gamma\colon [0,1] \to M$ is *parallel* if its covariant derivative along the path vanishes. In one local frame put $a(t) = A_{\gamma(t)}(\dot\gamma(t))$. Its coefficient equation is $\dot\sigma = -a(t)\sigma$. The *transport matrix* $U(t)$ sends the initial column to this column. Its initial value is I. Its equation is $\dot U = -a(t)U$. The coefficient equation gives the matrix equation column by column. Linear ODE existence and uniqueness give a solution throughout the compact parameter interval. The inverse solves $\dot V = Va(t)$, so $\frac{\mathrm{d}}{\mathrm{d}t}(VU) = 0$ and transport is invertible. On successive bundle charts, change coefficients at crossings and compose these solutions. The frame covariance proved above and uniqueness make this independent of subdivisions.

Iterating $U(t) = I - \int_0^t a(s)U(s)\mathrm{d}s$ gives

$$U(t) = I + \sum_{n \geq 1} (-1)^n \int_{0 < t_1 < \cdots < t_n < t} a(t_n) \cdots a(t_1)\mathrm{d}t_1 \cdots \mathrm{d}t_n.$$

The nth term has norm at most $K^n t^n / n!$ when $\|a\| \leq K$. The convergent series is the *path-ordered exponential*. Later times stand on the left. In a unitary frame for a unitary connection, $\frac{\mathrm{d}}{\mathrm{d}t}(U^\dagger U) = 0$, so transport is unitary. For scalar A this reduces to $U(\gamma) = \exp(-\int_\gamma A)$ in a frame covering the path. This is the U(1) transport in *Lie Groups*, Lecture 27, with the same sign.

The *holonomy* $\mathrm{Hol}(\gamma)$ is the transport around a based loop. Coefficient covariance $D'\sigma' = h^{-1}D\sigma$ and uniqueness show that changing the base frame conjugates it. Thus its trace, the *Wilson loop*, is invariant. A loop crossing several frames requires their transition matrices as well.

Theorem 21.2 (Parallel transport · *proved*). *Parallel transport is the unique solution*

$$\dot U(t) = -a(t)U(t), \qquad U(0) = I.$$

Under a change of frame it becomes

$$U'(\gamma) = h(\gamma(1))^{-1} U(\gamma) h(\gamma(0)).$$

For the counter-clockwise coordinate square of side ϵ based at 0,

$$\mathrm{Hol}(\square_\epsilon) = I - \epsilon^2 F_{xy}(0) + O(\epsilon^3).$$

Proof. The preceding construction proves existence, uniqueness and invertibility. The endpoint formula follows by differentiating $h(\gamma(t))^{-1}U(t)h(\gamma(0))$.

For the square, write $C = A_x(0)$, $D = A_y(0)$, $C_x = \partial_x A_x(0)$, $D_y = \partial_y A_y(0)$, $P = \partial_y A_x(0)$ and $Q = \partial_x A_y(0)$. Starting along positive x, integrate each edge equation through second order. The series bound makes all terms with at least three integrals $O(\epsilon^3)$:

$$\begin{aligned} U_1 &= I - \epsilon C + \frac{\epsilon^2}{2}(C^2 - C_x) + O(\epsilon^3),\\ U_2 &= I - \epsilon D + \epsilon^2(D^2/2 - Q - D_y/2) + O(\epsilon^3),\\ U_3 &= I + \epsilon C + \epsilon^2(C^2/2 + C_x/2 + P) + O(\epsilon^3),\\ U_4 &= I + \epsilon D + \frac{\epsilon^2}{2}(D^2 + D_y) + O(\epsilon^3). \end{aligned}$$

For example, on the second edge $a(t) = D + \epsilon Q + tD_y + O(\epsilon^2)$, for $0 \le t \le \epsilon$. The first integral contributes $-\epsilon D - \epsilon^2 Q - \epsilon^2 D_y/2$. The iterated integral contributes $\epsilon^2 D^2/2$. The other edges follow from their signed tangents in the same way. The successive transport acts on the left, giving

$$U_4U_3U_2U_1 = I + \epsilon^2(P - Q - [C, D]) + O(\epsilon^3) = I - \epsilon^2 F_{xy}(0) + O(\epsilon^3). \qquad \blacksquare$$

The figure fixes the traversal order used in this calculation.

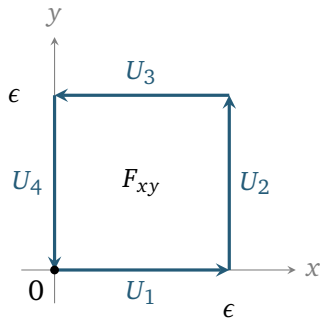


Figure 21.2: The path starts at 0 and traverses positive x, positive y, negative x, then negative y. Its transport is $U_4U_3U_2U_1$.

Recall that $\gamma \cdot \delta$ traverses γ first. Uniqueness therefore gives

$$U(\gamma \cdot \delta) = U(\delta)U(\gamma), \qquad U(\gamma^{-1}) = U(\gamma)^{-1}.$$

Transport reverses the order of the path product.

21.5 Flat connections and the Aharonov–Bohm phase

A connection is *flat* if $F = 0$. This condition removes local loop curvature but need not remove holonomy around a hole.

Theorem 21.3 (Flat holonomy · *sketched; proved for U(1)*)**.** *For a flat connection, based-loop holonomy depends only on the based homotopy class. After choosing a base G-frame, the map*

$$[\gamma] \longmapsto \mathrm{Hol}(\gamma)^{-1}$$

is a homomorphism $\pi_1(M, x_0) \to G$, where G is the matrix structure group. For each based loop its conjugacy class is independent of the chosen base frame.

Proof for U(1). In a small ball with a unitary frame, $\mathrm{d}A = 0$. By Theorem 13.2, write $A = \mathrm{d}f$, with imaginary f. For endpoints p_0 and p_1 in this ball, transport is $\exp(f(p_0) - f(p_1))$. Equivalently, Stokes gives unit transport around every loop bounding a disk in that ball. The endpoint formula also defines local transport on continuous paths. Subdividing into these balls extends it to continuous paths, agreeing with ODE transport on piecewise smooth paths. Refinements do not change it, since endpoint factors cancel. The local frames $e_i e^{-f_i}$ have zero connection form. On a connected overlap they therefore differ by a constant phase, by the frame law. Either local formula gives the same fibre map.

Given a continuous based homotopy, the Lebesgue lemma subdivides its parameter square into rectangles whose images lie in these balls. Choose a unit basis in the fibre at every grid vertex. Each rectangle has boundary product 1 by the local endpoint formula. Multiply the scalar boundary products. Every internal edge cancels with its reverse. The two outer side paths are constant at the base point and have transport 1. The bottom and top transports are therefore equal. Finally, taking inverses in the first-path-first composition rule gives the claimed homomorphism. This argument uses no global trivialization of the bundle. ∎

For a general matrix group, the variation of transport gives the corresponding sketch. Let $H(s,t)$ be a smooth based homotopy and $U(s,t)$ its transport from $t = 0$ to t. In a pullback frame fixed along both endpoint edges, the based boundary conditions give

$$U(s,1)^{-1}\partial_s U(s,1) = -\int_0^1 U(s,t)^{-1} F(H_s, H_t) U(s,t)\,\mathrm{d}t.$$

To obtain it, differentiate $\partial_t U = -A(H_t)U$ in s and combine $\partial_s U + A(H_s)U$ before integrating in t. Patching frames and the passage to continuous homotopies are omitted here. If $F = 0$, the formula makes the endpoint transport constant in s.

The angular one-form on the punctured plane is global although its angle is not:

$$\mathrm{d}\theta = \frac{-y\,\mathrm{d}x + x\,\mathrm{d}y}{x^2 + y^2}, \qquad M = \mathbb{R}^2 \setminus \{0\}.$$

The ideal thin solenoid has flux Φ positive out of the page. The two paths in the figure have the same endpoints.

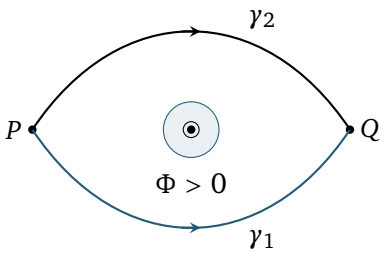


Figure 21.3: Both paths go from P to Q outside the solenoid. The loop $\gamma_1 \cdot \gamma_2^{-1}$ is counterclockwise and encloses positive flux.

Example 21.4 (The Aharonov–Bohm connection)**.** Compute the holonomy on the trivial complex line over M with

$$A_{\mathrm{em}} = \frac{\Phi}{2\pi}\mathrm{d}\theta, \qquad A = -\frac{\mathrm{i}q\Phi}{2\pi\hbar}\mathrm{d}\theta.$$

Differentiation gives $\mathrm{d}(\mathrm{d}\theta) = 0$ on M, so $F = 0$ there. Let γ be the counter-clockwise unit circle, with $\theta(t) = 2\pi t$ and circulation 2π. Hence

$$\mathrm{Hol}(\gamma) = \exp\left(-\oint_\gamma A\right) = e^{\mathrm{i}q\Phi/\hbar}.$$

The loop does not bound a disk in M, so Stokes does not set this circulation to zero. The bundle is trivial, but the connection need not be gauge equivalent to the zero connection. Put $\alpha = q\Phi/(2\pi\hbar)$. On the angular cover $h = e^{\mathrm{i}\alpha\theta}$ would cancel A. It descends to a single-valued map on M exactly when $\alpha \in \mathbb{Z}$. For an integer n, the genuine gauge map $h = e^{\mathrm{i}n\theta}$ changes α to $\alpha - n$ and leaves the holonomy unchanged. Thus these flat connections are distinguished by α modulo $\mathbb{Z}$.

The loop $\gamma_1 \cdot \gamma_2^{-1}$ winds once counter-clockwise, so its angular-form integral is also 2π by Lecture 3. Scalar transport gives $U(\gamma_1 \cdot \gamma_2^{-1}) = U(\gamma_1)U(\gamma_2)^{-1}$. In the ideal interference model, the ratio of the two transport phases is $U(\gamma_1)U(\gamma_2)^{-1} = e^{\mathrm{i}q\Phi/\hbar}$. For amplitudes c_1 and c_2 this changes the interference term to $2\,\mathrm{Re}(c_1\bar{c}_2 e^{\mathrm{i}q\Phi/\hbar})$. The phase is defined modulo 2π. The signed particle charge is q. This is the phase recalled in *Quantum Mechanics II*, Lecture 3, the physics box at the end of §3.4, *The momenta P_θ*, when forward hops run counter-clockwise. Nakahara, §10.5.3, (10.95)–(10.100), gives the physical minimal-coupling and interference calculation with $\hbar = 1$. His normalized imaginary-potential notation is translated here by the explicit charge ledger, rather than identifying his $\mathrm{i}A$ directly with our A_{conn}.

21.6 A constant nonabelian connection

The coefficient column σ is distinct from the Pauli matrices σ_x, σ_y and σ_z, recalled explicitly as

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Example 21.5 (Curvature from noncommuting transport)**.** Find the curvature and small-square holonomy of the connection on $\mathbb{R}^2 \times \mathbb{C}^2$ given, for $a, b \in \mathbb{R}$, by

$$A = C\mathrm{d}x + D\mathrm{d}y, \qquad C = -\frac{\mathrm{i}a}{2}\sigma_z, \qquad D = -\frac{\mathrm{i}b}{2}\sigma_x.$$

Both coefficient matrices are traceless and anti-Hermitian, so their transport lies in $\mathrm{SU}(2)$. Indeed, unitarity follows as above and $\frac{\mathrm{d}}{\mathrm{d}t}\det U = -\operatorname{tr}(A(\dot\gamma))\det U = 0$. The exterior derivative is zero, but $[\sigma_z, \sigma_x] = 2\mathrm{i}\sigma_y$ gives

$$F = [C, D]\mathrm{d}x \wedge \mathrm{d}y = -\frac{\mathrm{i}ab}{2}\sigma_y \mathrm{d}x \wedge \mathrm{d}y.$$

It is nonzero when $ab \neq 0$. The constant edge equations have exact solutions, so

$$\mathrm{Hol}(\square_\epsilon) = e^{\epsilon D}e^{\epsilon C}e^{-\epsilon D}e^{-\epsilon C} = I + \frac{\mathrm{i}ab\epsilon^2}{2}\sigma_y + O(\epsilon^3).$$

The coefficient is $-F_{xy}$, as the transport theorem requires. Thus vanishing ordinary derivatives of a nonabelian potential do not imply flatness.

Self-check. Show that a pure gauge $A = h^{-1}\mathrm{d}h$ is flat and compute its transport.

Answer. The inverse derivative gives $\mathrm{d}(h^{-1}\mathrm{d}h) = -h^{-1}\mathrm{d}h \wedge h^{-1}\mathrm{d}h$, which cancels $A \wedge A$. This is the Maurer–Cartan identity for $\vartheta_{\mathrm{MC}} = h^{-1}\mathrm{d}h$. Differentiating $U(t) = h(\gamma(t))^{-1}h(\gamma(0))$

verifies the transport equation and its initial condition. If h is global, every closed loop has holonomy I. The Aharonov–Bohm cancellation uses only a local h unless $\alpha \in \mathbb{Z}$.

Summary. A connection differentiates sections independently of their chosen frame. Its matrices change by $A' = h^{-1}Ah + h^{-1}\mathrm{d}h$. Curvature is $F = \mathrm{d}A + A \wedge A$, with $F' = h^{-1}Fh$ and $DF = 0$. Transport solves $\dot{U} = -A(\dot{\gamma})U$ and multiplies later segments on the left. A counter-clockwise small square has $\mathrm{Hol} = I - \epsilon^2 F_{xy} + O(\epsilon^3)$. A flat connection can retain nontrivial holonomy around a hole, as the Aharonov–Bohm phase $e^{\mathrm{i}q\Phi/\hbar}$ shows. Lecture 22 relates curvature flux to the transition winding of a line bundle.

Reading. Nakahara, §10.1.3–4, §10.3.4–5, §10.4.2 and §10.5.1, 10.5.3.

Lecture 22

Chern classes and the Dirac monopole

Objectives. *A connection can change while its bundle stays fixed. We extract a cohomology class from its curvature that survives this change. On a sphere its integral equals the transition winding and is therefore an integer. For a charged particle in a monopole field, this integer gives Dirac's quantization condition. The tangent sphere supplies a second example through Gauss–Bonnet.*

By the end of this lecture you will be able to:

- Prove that a line bundle's curvature class is independent of its connection.
- Recover its sphere Chern number from the oriented equatorial transition.
- Derive Dirac quantization from the two smooth monopole potentials.
- Relate the tangent sphere and the degree-n line to their Chern numbers.

22.1 The curvature class of a line bundle

Let $E \to M$ be a complex line bundle with a Hermitian fibre metric and a unitary connection. In a unitary frame e_i, the connection's imaginary potential A_i has curvature $F_i = \mathrm{d}A_i$. Lecture 21's frame law on overlaps is

$$e_j = e_i g_{ij}, \qquad A_j = A_i + g_{ij}^{-1}\mathrm{d}g_{ij}, \qquad g_{ij} \colon U_i \cap U_j \to \mathrm{U}(1).$$

Locally write $g_{ij} = e^{\mathrm{i}\chi}$. Then $\mathrm{d}(g_{ij}^{-1}\mathrm{d}g_{ij}) = \mathrm{d}(\mathrm{i}\mathrm{d}\chi) = 0$. Thus the F_i agree and define a global imaginary two-form F. It is closed because $\mathrm{d}F_i = \mathrm{d}^2 A_i = 0$.

For a smooth map $f \colon N \to M$, the *pullback bundle* $f^*E \to N$ has fibre $E_{f(y)}$ at $y \in N$. Pull back the product charts to $f^{-1}(U_i)$. Their transitions $g_{ij} \circ f$ satisfy the same cocycle, so Lecture 20 makes this a smooth bundle.

A *characteristic class* assigns a cohomology class to a bundle, invariant under bundle isomorphism and compatible with pullback. Here the *real first Chern class* is

$$c_1(E) = \left[\frac{\mathrm{i}}{2\pi}F\right] \in H^2_{\mathrm{dR}}(M;\mathbb{R}).$$

The form inside the brackets is real since F is imaginary. We use de Rham classes throughout this lecture. Torsion classes, those killed by a nonzero integer, are not detected by these real forms. The integer sphere computation below needs no torsion comparison.

Theorem 22.1 (Chern–Weil · *cited; line-bundle first Chern class proved given partitions of unity*)**.** *Every complex line bundle has a Hermitian metric and a unitary connection. The class $c_1(E)$ is independent of both choices and is a characteristic class. More generally, on a unitary rank-r bundle, the forms $\mathrm{tr}(F^k)$ are closed and their classes in $H^{2k}_{\mathrm{dR}}(M;\mathbb{C})$ are independent of the connection, for every integer $k \geq 1$. Here powers use matrix wedge multiplication. This general assertion is cited.*

Proof for the line-bundle class. Lecture 12's cited partitions of unity supply the auxiliary choices. Take a locally finite partition ψ_i subordinate to product charts U_i. For $v, w \in E_x$, the weighted sum of their standard chartwise Hermitian products is

$$\langle v, w \rangle = \sum_i \psi_i(x) \langle v, w \rangle_i.$$

It is positive definite since the nonnegative weights sum to one. Their supports and local finiteness make the sum smooth. Normalize the local frames for this metric. In each unit frame let $\nabla^{(i)}$ be the connection with potential zero. The sum $\nabla s = \sum_i \psi_i \nabla^{(i)} s$ defines a global connection. Each summand extends smoothly by zero. The sum is locally finite. The Leibniz and metric product rules are affine in the connection, so their weighted sums give the same rules because $\sum_i \psi_i = 1$. Thus this connection is unitary.

Now compare connections unitary for possibly different metrics. Use the same complex frames for both, without requiring them to be unitary. The line frame law also holds for $h \in \mathbb{C}^\times$. Locally write $h = e^{b+\mathrm{i}\chi}$ with b and χ real. Then $h^{-1}\mathrm{d}h = \mathrm{d}b + \mathrm{i}\mathrm{d}\chi$ is closed, so curvature is unchanged in any such frame. Write the two potentials as $A_i^{(0)}$ and $A_i^{(1)}$ in these common frames. Their difference $B_i = A_i^{(1)} - A_i^{(0)}$ has no transition derivative term, so $B_j = B_i$. These complex one-forms glue to a global B. Both curvatures are imaginary. Taking real parts in their exact difference gives

$$F^{(1)} - F^{(0)} = \mathrm{d}B, \qquad \frac{\mathrm{i}}{2\pi} F^{(1)} - \frac{\mathrm{i}}{2\pi} F^{(0)} = \mathrm{d}\,\mathrm{Re}\left(\frac{\mathrm{i}}{2\pi} B\right).$$

Their real de Rham classes coincide. For a smooth map $f\colon N \to M$, the pullback bundle has pulled-back local potentials. Its curvature is f^*F because pullback commutes with d. Thus $c_1(f^*E) = f^* c_1(E)$. An arbitrary complex bundle isomorphism transports the metric and connection together. Independence of both choices then proves invariance even if the isomorphism does not preserve a previously chosen target metric. ∎

The general statement is Nakahara, Theorem 11.1 and (11.20)–(11.21). No Riemannian metric or orientation on the base entered this definition. For a closed oriented surface Σ, the bundle's *first Chern number* is

$$C_1(E; \Sigma) = \frac{\mathrm{i}}{2\pi} \int_\Sigma F.$$

Changing the surface orientation reverses this number. Stokes makes it independent of the representative curvature in the same class.

22.2 Integrality on the sphere

Give S^2 its outward orientation, equal to the complex orientation of $\mathbb{C}\mathrm{P}^1$. Let D_N and D_S be the closed northern and southern hemispheres. Their common equator C has increasing azimuth ϕ as ∂D_N. The southern boundary is $-C$.

Theorem 22.2 (The sphere Chern number · *proved*). *Let $E \to S^2$ be a Hermitian line bundle with a unitary connection. Choose smooth unitary frames over neighbourhoods of D_N and D_S. In the frame order $e_S = e_N g_{NS}$,*

$$C_1(E; S^2) = \frac{\mathrm{i}}{2\pi} \int_{S^2} F = \frac{1}{2\pi\mathrm{i}} \oint_C g_{NS}^{-1} \mathrm{d} g_{NS} = \mathrm{wind}(g_{NS}) \in \mathbb{Z}.$$

The figure represents the hemispheres as oriented disks.

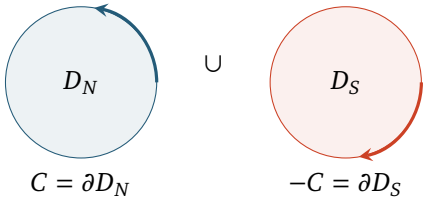


Figure 22.1: The two hemispheres induce opposite orientations on their common equator. Adding their Stokes formulas leaves $\oint_C (A_N - A_S)$. Both disks are drawn as seen from above the north pole.

Proof. Such cap frames exist. Identify the sphere minus the opposite pole with $\mathbb{R}^2$. Choose one unit vector in the fibre over its coordinate origin and transport it along every radial segment to its endpoint. Lecture 21 gives unique unitary transport. Smooth dependence of linear ODE solutions on the endpoint gives a smooth frame. Locally in the endpoint, a finite chain of trivializations covers these compact segments, so this smooth-dependence argument also applies when no initial global frame is given. Each resulting chart contains its entire closed hemisphere.

In these frames, Stokes and the boundary orientations give

$$\frac{\mathrm{i}}{2\pi}\int_{S^2} F = \frac{\mathrm{i}}{2\pi}\left(\int_{D_N} \mathrm{d}A_N + \int_{D_S} \mathrm{d}A_S\right) = \frac{\mathrm{i}}{2\pi}\oint_C (A_N - A_S) = \frac{1}{2\pi\mathrm{i}}\oint_C g_{NS}^{-1}\mathrm{d}g_{NS}.$$

The last equality uses $A_S = A_N + g_{NS}^{-1}\mathrm{d}g_{NS}$ and $-\mathrm{i} = 1/\mathrm{i}$. Lecture 3's circle lifting theorem identifies the last integral with the integer winding. This proves the result for every such bundle, rather than only the monopole potentials. ∎

A different cap frame changes the transition by phases extending over the disks. Each such boundary phase has winding zero, so the integer is unchanged. Lecture 20's cited topological clutching classification applies to the principal $\mathrm{U}(1)$ bundle of unitary frames. It adds that equal integers give continuously isomorphic underlying bundles and that every integer occurs. The tautological line has transition $e^{-\mathrm{i}\phi}$, hence Chern number -1.

22.3 The Dirac monopole

Let $q \neq 0$ be the signed particle charge and Φ_{m} the magnetic flux. On a sphere enclosing the monopole, the real electromagnetic curvature is

$$F_{\mathrm{em}} = \frac{\Phi_{\mathrm{m}}}{4\pi}\sin\theta\,\mathrm{d}\theta\wedge\mathrm{d}\phi, \qquad \int_{S^2} F_{\mathrm{em}} = \Phi_{\mathrm{m}}.$$

The *Wu–Yang potentials* on the two pole-deleted charts are

$$A_{\mathrm{em}}^{(N)} = \frac{\Phi_{\mathrm{m}}}{4\pi}(1-\cos\theta)\mathrm{d}\phi, \quad A_{\mathrm{em}}^{(S)} = -\frac{\Phi_{\mathrm{m}}}{4\pi}(1+\cos\theta)\mathrm{d}\phi,$$
$$A_N = -\frac{\mathrm{i}n}{2}(1-\cos\theta)\mathrm{d}\phi, \qquad A_S = +\frac{\mathrm{i}n}{2}(1+\cos\theta)\mathrm{d}\phi,$$

where $n = q\Phi_{\mathrm{m}}/(2\pi\hbar)$ is initially a real parameter. The ledger is $A = -(\mathrm{i}q/\hbar)A_{\mathrm{em}}$ and $F = -(\mathrm{i}q/\hbar)F_{\mathrm{em}}$. The imaginary connection potential includes $q/\hbar$ and is distinct from the real potential.

The polar coordinate ϕ fails at the poles, but these potentials are smooth on their claimed charts. For ambient unit-sphere coordinates (x, y, z),

$$(1 - \cos\theta)\mathrm{d}\phi = \frac{x\mathrm{d}y - y\mathrm{d}x}{1+z}, \qquad (1 + \cos\theta)\mathrm{d}\phi = \frac{x\mathrm{d}y - y\mathrm{d}x}{1-z}.$$

The first denominator is nonzero off the south pole, the second off the north pole. Both give the same smooth curvature $F = -(\mathrm{i}n/2)\sin\theta\mathrm{d}\theta \wedge \mathrm{d}\phi$.

Corollary 22.3 (Dirac quantization · *proved*)**.** *The Wu–Yang potentials define a unitary line-bundle connection for charge $q \neq 0$ exactly when*

$$q\Phi_{\mathrm{m}} = 2\pi\hbar n, \qquad n \in \mathbb{Z}.$$

Its sphere Chern number is n.

Proof. On the overlap, $A_S - A_N = \mathrm{i}n\mathrm{d}\phi$. The frame transition must therefore solve $g_{NS}^{-1}\mathrm{d}g_{NS} = \mathrm{i}n\mathrm{d}\phi$. It has the form $ce^{\mathrm{i}n\phi}$ for a constant $c \in \mathrm{U}(1)$. It is single-valued around the equator exactly when $e^{2\pi\mathrm{i}n} = 1$, which holds exactly for $n \in \mathbb{Z}$. For every such integer choose $g_{NS} = (z_{\mathbb{C}}/|z_{\mathbb{C}}|)^n$ on the full overlap, where $z_{\mathbb{C}} = \tan(\theta/2)e^{\mathrm{i}\phi}$ is the complex chart coordinate. Lecture 20 reconstructs its smooth bundle. The connection law glues the two potentials. The sphere theorem gives the bundle's Chern number n. ∎

Here the ambient height z and the complex coordinate $z_{\mathbb{C}}$ are distinct. This is a conditional charge-quantization argument, not an assertion that monopoles exist. The potentials follow Nakahara, §10.5.2, with its imaginary-potential and north/south transition choices translated by the displayed ledger and frame order.

The cap integral need not be an integer. For a northern cap of polar radius $\alpha \in [0, \pi]$, the same curvature gives

$$\frac{\mathrm{i}}{2\pi}\int_{D_\alpha} F = \frac{n}{2}(1 - \cos\alpha).$$

The figure plots $n = 1$. Its hemisphere value is $1/2$, although the sphere number is 1.

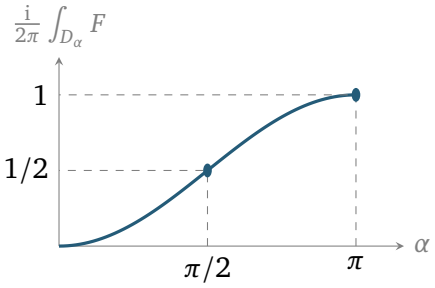


Figure 22.2: For the charge-one curvature, cap flux varies continuously with cap size. Only the full closed-sphere integral is constrained to the integer 1.

22.4 The tangent sphere as a complex line

On an oriented Riemannian surface Σ, let J rotate a positive orthonormal frame (e_1, e_2) by $Je_1 = e_2$ and $Je_2 = -e_1$. Positive orthonormal frame changes commute with this rotation, so it defines multiplication by i on every tangent plane. Its Hermitian fibre product is $\langle u, v\rangle = g(u, v) + \mathrm{i}g(Ju, v)$ for $p \in \Sigma$ and $u, v \in T_p\Sigma$. Lecture 17's Levi-Civita connection

has $\nabla e_1 = -\omega^1{}_2 e_2$ and $\mathrm{d}\omega^1{}_2 = K\mathrm{d}\mathcal{A}$, where $\mathrm{d}\mathcal{A}$ denotes the oriented area form. Its skew connection matrix commutes with J, hence the connection is complex-linear and unitary on this complex line.

Proposition 22.4 (The tangent Chern number · *proved*)**.** *For a closed oriented Riemannian surface Σ, its tangent complex line has*

$$A = -\mathrm{i}\omega^1{}_2, \qquad F = -\mathrm{i}K\mathrm{d}\mathcal{A}, \qquad C_1(T\Sigma;\Sigma) = \frac{1}{2\pi}\int_\Sigma K\mathrm{d}\mathcal{A} = \chi(\Sigma).$$

In particular, $C_1(TS^2;S^2) = 2$.

Proof. The complex frame is e_1, with $e_2 = \mathrm{i}e_1$. Thus $\nabla e_1 = -\mathrm{i}\omega^1{}_2 e_1$, proving the connection formula. Differentiate and use Lecture 17's surface structure equation to obtain the curvature. Multiplication by $\mathrm{i}/(2\pi)$ gives $K\mathrm{d}\mathcal{A}/(2\pi)$. Lecture 18's Gauss–Bonnet theorem evaluates its integral as $\chi(\Sigma)$. For the sphere the Euler characteristic is 2, as required. ∎

This agrees with Lecture 20's tangent-frame winding +2. The inverse component transition winds −2. It is not the transition used in the Chern-number theorem.

22.5 Higher Chern classes

For a unitary rank-r bundle, use the defining $r \times r$ matrix curvature F. The frame law $F' = h^{-1}Fh$ and conjugation invariance of the determinant make the following form global. Even-degree forms commute as scalar entries, so its *total Chern form* is defined by

$$c(F) = \det\left(I + \frac{\mathrm{i}F}{2\pi}\right) = 1 + c_1(F) + c_2(F) + \cdots + c_r(F).$$

The component $c_j(F)$ has degree $2j$ and vanishes when $2j > \dim M$. The corresponding cohomology classes are the *Chern classes*. The invariant-polynomial form of Chern–Weil, Nakahara Theorem 11.1, gives their closedness and connection independence. It applies to each determinant coefficient, not only the traces of powers. For two connections, their difference $B = A^{(1)} - A^{(0)}$ is an endomorphism-valued one-form. With $A_t = A^{(0)} + tB$ and curvature F_t, the cited relative transgression is

$$\mathrm{tr}\big((F^{(1)})^k\big) - \mathrm{tr}\big((F^{(0)})^k\big) = \mathrm{d}\left(k\int_0^1 \mathrm{tr}(B \wedge F_t^{k-1})\mathrm{d}t\right).$$

The trace expression inside is global, although the individual potentials are local. Nakahara's (11.20)–(11.21) supplies this assertion. Expanding the determinant through degree four gives

$$c_1(F) = \frac{\mathrm{i}}{2\pi}\,\mathrm{tr}\,F, \qquad c_2(F) = \frac{\mathrm{tr}(F\wedge F) - \mathrm{tr}\,F \wedge \mathrm{tr}\,F}{8\pi^2}.$$

This uses the elementary determinant coefficient $((\mathrm{tr}\,X)^2 - \mathrm{tr}(X^2))/2$ with $X = \mathrm{i}F/(2\pi)$. For an $\mathrm{SU}(r)$ connection, $\mathrm{tr}\,F = 0$, so $c_2(F) = \mathrm{tr}(F\wedge F)/(8\pi^2)$. Lecture 23 uses this normalization for its second Chern number.

22.6 The degree-n holomorphic line

In a complex chart $z = x + \mathrm{i}y$, a smooth function is holomorphic if $\partial_{\bar z} f = 0$, where $\partial_{\bar z} = \frac{1}{2}(\partial_x + \mathrm{i}\partial_y)$. A *holomorphic line bundle* has local frames with holomorphic transition functions. The $(0,1)$ *part* of a complex one-form is its $\mathrm{d}\bar z$ term. Write $\bar\partial f = (\partial_{\bar z} f)\mathrm{d}\bar z$. Use the projective coordinates z and $w = 1/z$ of Lecture 20. Here z denotes the complex coordinate rather than the ambient height. Define the *degree* of a line over $\mathbb{C}\mathrm{P}^1$ to be its Chern number in the complex orientation. Define the *degree-n line* $O(n)$ for $n \in \mathbb{Z}$ by holomorphic frames with

$$e_S^{\mathrm{hol}} = e_N^{\mathrm{hol}} z^n .$$

The diagram shows their relation to unitary frames.

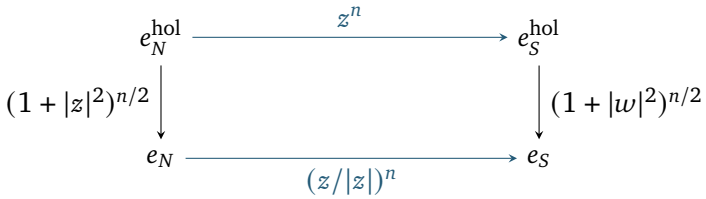


Figure 22.3: Positive real rescalings turn holomorphic frames into unitary frames. Their transition is the phase $(z/|z|)^n$, with winding n.

Example 22.5 (The monopole line as $O(n)$)**.** Identify the Wu–Yang monopole line with $O(n)$, for $n \in \mathbb{Z}$. Equip these frames with squared norms $\|e_N^{\mathrm{hol}}\|^2 = (1+|z|^2)^{-n}$ and $\|e_S^{\mathrm{hol}}\|^2 = (1+|w|^2)^{-n}$. They agree on the overlap since $|w| = 1/|z|$ and $e_S^{\mathrm{hol}} = e_N^{\mathrm{hol}} z^n$. The unitary frames are

$$e_N = (1+|z|^2)^{n/2} e_N^{\mathrm{hol}}, \qquad e_S = (1+|w|^2)^{n/2} e_S^{\mathrm{hol}}.$$

Their transition is $(z/|z|)^n$. In the northern unit frame the Wu–Yang connection is

$$A_N = -\frac{n}{2}\frac{\bar z \mathrm{d}z - z\mathrm{d}\bar z}{1+|z|^2}.$$

Changing to $e_N^{\mathrm{hol}} = (1+|z|^2)^{-n/2} e_N$ gives

$$A_N^{\mathrm{hol}} = A_N - \frac{n}{2}\mathrm{d}\log(1+|z|^2) = -\frac{n\bar z}{1+|z|^2}\mathrm{d}z.$$

In the southern holomorphic frame the same calculation gives

$$A_S^{\mathrm{hol}} = -\frac{n\bar w}{1+|w|^2}\mathrm{d}w = A_N^{\mathrm{hol}} + n\frac{\mathrm{d}z}{z} \quad \text{on the overlap.}$$

Both expressions have no antiholomorphic differential term and are smooth at their own chart origin. Thus the $(0,1)$ part of the covariant derivative is ordinary $\bar\partial$ in these frames. A local section is holomorphic exactly when its holomorphic-frame coefficient satisfies $\partial_{\bar z} f = 0$. This equips the monopole line with the displayed holomorphic structure. Its Chern number is n by the equatorial winding theorem. For the coefficient ψ in the unit frame the same equation is $(\partial_{\bar z} + nz/[2(1+|z|^2)])\psi = 0$. This local equation will reappear in Lecture 28 with its stated spinor twist.

The cited holomorphic classification on $\mathbb{C}\mathrm{P}^1$ says that every holomorphic line is isomorphic to exactly one $O(n)$, with n its degree. See Gunning, *Function Theory on Compact Riemann Surfaces*, Theorem 2.2, (2.3) and Corollary 2.3, pp. 34–35. This is a holomorphic result in addition to Lecture 20's continuous clutching classification. The tautological frames $(1, z)^T$ and $(w, 1)^T$ have holomorphic transition z^{-1}. The tangent coordinate frames have holomorphic transition $-z^2$ by Lecture 20. Their degrees and the cited classification identify the tautological line with $O(-1)$ and the tangent line with $O(2)$.

Self-check. Show that a trivial Hermitian line bundle over a closed oriented surface has Chern number zero.

Answer. Choose a global unit frame, so the connection has a global one-form A and $F = \mathrm{d}A$. Stokes gives $\int_\Sigma F = \int_{\partial\Sigma} A = 0$ because $\partial\Sigma$ is empty.

Summary. A line bundle's real first Chern class is $[\mathrm{i}F/(2\pi)]$ and is independent of the connection. On the outward sphere its integral is the winding of the frame transition $e_S = e_N g_{NS}$ along the northern boundary. The monopole condition is $q\Phi_\mathrm{m} = 2\pi\hbar n$ with $n \in \mathbb{Z}$. The tangent sphere has Chern number 2, while $O(n)$ has number n. A given complex bundle's class needs no base metric or orientation. The tangent example uses an oriented metric to define its complex structure. Signed surface integrals need the orientation. Lecture 23 uses the second Chern form to study nonabelian fields.

Reading. Nakahara, §10.5.2 and §11.1–11.2. Tu, *Differential Geometry*, §25.8, p. 235. Gunning, Theorem 2.2, (2.3) and Corollary 2.3, pp. 34–35, for the holomorphic classification.

Lecture 23

Yang–Mills theory and instantons

Objectives. *Lecture 21 supplied the geometry of a gauge field. We now choose an action and derive its field equation. For an* SU(2) *connection extending over the four-sphere, the curvature also gives an integer. This integer bounds the action and identifies solutions of a first-order equation. The BPST field fixes the orientation and normalization in a concrete example.*

By the end of this lecture you will be able to:

- Derive the vacuum Yang–Mills equation with the fundamental trace normalization.
- Prove the local Chern–Simons identity and distinguish it from global exactness.
- Complete the Euclidean action square and identify its equality condition.
- Relate BPST duality and Chern number to the chosen orientation.

23.1 A connection on the principal bundle

Lecture 20's principal bundle $P \to M$ carries a right action of a Lie group G. Its Lie algebra is $\mathfrak{g}$. For $\xi \in \mathfrak{g}$, the *fundamental vertical vector* at $p \in P$ is $\xi_p^{\#} = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_0 p \exp(t\xi)$. It is tangent to the fibre.

Remark 23.1 (Principal connections)**.** A *principal connection* is a smooth $\mathfrak{g}$-valued one-form ω_P on P with

$$\omega_P(\xi^{\#}) = \xi, \qquad R_a^* \omega_P = \mathrm{Ad}_{a^{-1}}\, \omega_P \quad (a \in G),$$

where $R_a(p) = pa$ and $\mathrm{Ad}_a\, \xi = a\xi a^{-1}$ for a matrix group. Its kernel gives the horizontal tangent spaces, complementary to the vertical tangent spaces. In a local section $s_i : U_i \to P$, the potential is $A_i = s_i^* \omega_P$. For $s_j = s_i g_{ij}$ these potentials obey Lecture 21's frame law. A representation $\rho : G \to \mathrm{GL}(V)$ induces on the associated bundle the local derivative $\mathrm{d} + \rho_*(A_i)$, where ρ_* is the representation's differential at the identity. Thus one principal connection supplies the potentials in all representations. These intrinsic-to-local facts are cited from Tu, §28.1, §29.4 and §31.3–31.5.

Gauge transformations are bundle automorphisms over the identity that commute with the right action. Lecture 21 represents them by compatible maps $u_j = g_{ij}^{-1} u_i g_{ij}$. A global map $M \to G$ suffices in a chosen trivialization. The passive frame change $e' = eh$ has $A^h = h^{-1}Ah + h^{-1}\mathrm{d}h$. All calculations below use this convention.

23.2 The Yang–Mills action and equation

Use a connection for SU(r), with $r \geq 2$, in its fundamental representation. Choose anti-Hermitian generators T^a normalized by $\mathrm{tr}(T^a T^b) = -\delta^{ab}/2$. For SU(2), these are $T^a = -\mathrm{i}\sigma_a/2$

with the Pauli matrices of Lecture 21. The real basis from *Lie Groups and Their Representation Theory* is related to these generators by $(\mathrm{i}\sigma_z, \mathrm{i}\sigma_y, \mathrm{i}\sigma_x) = (-2T^3, -2T^2, -2T^1)$. That basis has cyclic bracket factor 2. The generators T^a used here have cyclic bracket factor one. In Minkowski coordinates $x^0 = t$, the metric has signature $(+---)$ and the volume orientation is $\mathrm{d}t \wedge \mathrm{d}x^1 \wedge \mathrm{d}x^2 \wedge \mathrm{d}x^3$. The real component coefficients are defined by

$$A = T^a A^a_\mu \mathrm{d}x^\mu, \qquad F = \mathrm{d}A + A \wedge A = \frac{1}{2} T^a F^a_{\mu\nu} \mathrm{d}x^\mu \wedge \mathrm{d}x^\nu.$$

Here a runs over the $r^2 - 1$ generators. Spacetime indices run from 0 to 3. The coupling $g_{\mathrm{YM}} > 0$ is a scalar. Assume the displayed action integrals converge, or work on a bounded region with variations supported in its interior. The *Yang–Mills action* is the following metric-dependent functional.

Theorem 23.2 (The field equation · *proved*)**.** *The Minkowski action*

$$S_M[A] = \frac{1}{g_{\mathrm{YM}}^2} \int_M \operatorname{tr}(F \wedge \star F) = -\frac{1}{4 g_{\mathrm{YM}}^2} \int_M F^a_{\mu\nu} F^{a\mu\nu} \operatorname{vol}$$

is gauge invariant. Its stationary points under arbitrary smooth compactly supported connection variations satisfy $D\star F = 0$. For an endomorphism-valued k-form B, the derivative here is $DB = \mathrm{d}B + A \wedge B - (-1)^k B \wedge A$.

Proof. The Hodge pairing of two two-forms contracts their components with a factor $1/2$. Multiplying this by $\operatorname{tr}(T^a T^b) = -\delta^{ab}/2$ gives the coefficient $-1/4$. On overlaps and under a frame change, $F^h = h^{-1} F h$. The star acts only on differential forms, so conjugation and cyclicity of the trace make the integrand global and gauge invariant.

The difference of two connections transforms by conjugation, with no derivative term. Such Lie-algebra-valued forms are called *adjoint-valued*. Let $a = \delta A$ be a compactly supported adjoint-valued one-form. Differentiate $\mathrm{d}(A + \varepsilon a) + (A + \varepsilon a)^2$ at $\varepsilon = 0$. The result is $\delta F = \mathrm{d}a + A \wedge a + a \wedge A = Da$. Symmetry of the two-form pairing gives

$$\delta S_M = \frac{2}{g_{\mathrm{YM}}^2} \int_M \operatorname{tr}(Da \wedge \star F),$$
$$\operatorname{tr}(Da \wedge \star F) = \mathrm{d} \operatorname{tr}(a \wedge \star F) + \operatorname{tr}(a \wedge D\star F).$$

The second line follows from the graded product rule. The connection commutators cancel inside the trace. Stokes removes the compactly supported exact term. The remaining pairing with every a vanishes exactly when $D\star F = 0$. The trace pairing is nondegenerate on $\mathfrak{su}(r)$, so it detects every component. ∎

With matter, define the connection-valued current three-form $\mathcal{J}_{\mathrm{conn}}$ by $\delta S_{\mathrm{matter}} = -(2/g_{\mathrm{YM}}^2) \int_M \operatorname{tr}(a \wedge \mathcal{J}_{\mathrm{conn}})$. The same variation then gives $D\star F = \mathcal{J}_{\mathrm{conn}}$.

For U(1) the generator is $-\mathrm{i}$, whose trace square is -1. Use the different prefactor $1/(2g^2)$, where $g = q/\hbar \neq 0$ is signed. Lecture 21's physical ledger gives

$$A = -\frac{\mathrm{i}q}{\hbar} A_{\mathrm{em}}, \qquad F = -\frac{\mathrm{i}q}{\hbar} F_{\mathrm{em}}, \qquad S_M = \frac{1}{2g^2} \int_M F \wedge \star F = -\frac{1}{2} \int_M F_{\mathrm{em}} \wedge \star F_{\mathrm{em}}.$$

Let $\mathcal{J} = \star j$ be Lecture 15's real current and set $\mathcal{J}_{\mathrm{conn}} = -(\mathrm{i}q/\hbar)\mathcal{J}$. For this prefactor, with $a = \delta A$, define the matter current by

$$\delta S_{\mathrm{matter}} = -\frac{1}{g^2} \int_M a \wedge \mathcal{J}_{\mathrm{conn}} = \int_M \delta A_{\mathrm{em}} \wedge \mathcal{J}.$$

The equation $\mathrm{d}{\star}F = \mathcal{J}_{\mathrm{conn}}$ becomes Lecture 15's $\mathrm{d}{\star}F_{\mathrm{em}} = \mathcal{J}$. The real current and the imaginary connection current are distinct.

The Hermitian-potential dictionary is $A = -\mathrm{i}A_{\mathrm{Tong}}$ and $F = -\mathrm{i}F_{\mathrm{Tong}}$ for Tong's unrescaled matrices. His rescaling $A_{\mathrm{phys}} = A_{\mathrm{Tong}}/g_{\mathrm{YM}}$ instead gives $A = -\mathrm{i}g_{\mathrm{YM}}A_{\mathrm{phys}}$. Tong, §2.1, uses the positive Hermitian trace in his component action.

23.3 Euclidean action and duality

Replace the Minkowski metric by a positive Riemannian metric on an oriented four-manifold. The *Euclidean action* is

$$S_E[A] = -\frac{1}{g_{\mathrm{YM}}^2}\int_M \operatorname{tr}(F \wedge {\star}F).$$

The minus sign makes the density nonnegative. In an orthonormal form basis, the matrix pairing is $-\operatorname{tr}(X^2) \geq 0$ for anti-Hermitian X. We use Euclidean time $\tau = x^4$ and spatial-first orientation $\mathrm{d}x^1 \wedge \mathrm{d}x^2 \wedge \mathrm{d}x^3 \wedge \mathrm{d}x^4$ on $\mathbb{R}^4$. It is the negative of the time-first orientation.

Lecture 15 gives ${\star}^2 = 1$ on Euclidean two-forms. A curvature is *self-dual* if $F = {\star}F$ and *anti-self-dual* if $F = -{\star}F$. Its orthogonal parts $F_+ = (F + {\star}F)/2$ and $F_- = (F - {\star}F)/2$ obey ${\star}F_+ = F_+$ and ${\star}F_- = -F_-$. Either duality equation implies the vacuum field equation, since $D{\star}F = \pm DF = 0$ by Lecture 21's Bianchi identity. Real Minkowski two-forms have ${\star}^2 = -1$. Imposing either real duality equation there forces $F = 0$.

Reversing orientation changes the star's sign and also changes integration's sign. Their product leaves S_E unchanged. The action depends on the metric through the star. An integral of a fixed four-form instead changes sign with orientation.

23.4 Chern–Simons and the second Chern number

For matrix-valued forms of degrees p and q, cyclicity becomes $\operatorname{tr}(B \wedge C) = (-1)^{pq}\operatorname{tr}(C \wedge B)$. This follows by interchanging the scalar differential forms in each matrix entry. In a local frame define the *Chern–Simons three-form* by

$$\mathrm{CS}(A) = \operatorname{tr}\left(A \wedge \mathrm{d}A + \frac{2}{3}A \wedge A \wedge A\right).$$

Theorem 23.3 (The local transgression · *sketched; local identities proved*)**.** *The local forms satisfy*

$$\mathrm{d}\,\mathrm{CS}(A) = \operatorname{tr}(F \wedge F), \qquad \mathrm{CS}(\vartheta_{\mathrm{MC}}) = -\frac{1}{3}\operatorname{tr}(\vartheta_{\mathrm{MC}}^3), \qquad \vartheta_{\mathrm{MC}} = h^{-1}\mathrm{d}h.$$

For a general frame change, the local gauge law is

$$\mathrm{CS}(A^h) = \mathrm{CS}(A) - \mathrm{d}\operatorname{tr}(\mathrm{d}hh^{-1} \wedge A) - \frac{1}{3}\operatorname{tr}(\vartheta_{\mathrm{MC}}^3).$$

The resulting global characteristic-class interpretation is cited.

Proof of the local identities. Write matrix wedge products as juxtaposition. The product rule gives $\mathrm{d}\operatorname{tr}(A\mathrm{d}A) = \operatorname{tr}((\mathrm{d}A)^2)$. For the cubic term it gives

$$\mathrm{d}\operatorname{tr}(A^3) = \operatorname{tr}(\mathrm{d}A\,A^2 - A\,\mathrm{d}A\,A + A^2\mathrm{d}A) = 3\operatorname{tr}(\mathrm{d}A\,A^2).$$

Moving the first A past the remaining total degree three changes sign. This proves the middle term's sign and also gives $\operatorname{tr}(A^4) = -\operatorname{tr}(A^4) = 0$. Consequently

$$\operatorname{tr}(F^2) = \operatorname{tr}((\mathrm{d}A)^2 + 2\mathrm{d}A\,A^2) = \mathrm{d}\,\mathrm{CS}(A).$$

The Maurer–Cartan equation is $\mathrm{d}\vartheta_{\mathrm{MC}} = -\vartheta_{\mathrm{MC}}^2$. Substitution into the definition gives $(-1 + 2/3)\operatorname{tr}(\vartheta_{\mathrm{MC}}^3)$, which is the claimed pure-gauge value. ∎

For the general law set $B = h^{-1}Ah$. Expand $\mathrm{CS}(B+\vartheta_{\mathrm{MC}})$ using $\mathrm{d}B = h^{-1}\mathrm{d}Ah - \vartheta_{\mathrm{MC}}B - B\vartheta_{\mathrm{MC}}$. Graded trace cyclicity groups the mixed terms into $-\mathrm{d}\operatorname{tr}(\vartheta_{\mathrm{MC}} \wedge B) = -\mathrm{d}\operatorname{tr}(\mathrm{d}hh^{-1} \wedge A)$. The unmixed terms give the two Chern–Simons forms above. This outlines the expansion. Nakahara, §11.5, supplies the general transgression framework. The local potential and its Chern–Simons form need not be global. In particular, the local identity does not make every closed-manifold Chern number zero.

For an SU(2) bundle over an oriented closed four-manifold, Lecture 22 gives its *second Chern number*

$$k = \int_M c_2(F) = \frac{1}{8\pi^2}\int_M \operatorname{tr}(F \wedge F).$$

It is independent of the connection and metric. Its sign uses the orientation. The *Chern character* is the trace exponential $\mathrm{ch}(F) = \operatorname{tr}\exp(\mathrm{i}F/(2\pi))$, expanded in wedge powers. The quadratic exponential term gives its degree-four component on an SU(2) bundle:

$$\mathrm{ch}_2(F) = -\frac{1}{8\pi^2}\operatorname{tr}(F \wedge F) = -c_2(F).$$

Thus our k is minus the integrated second Chern character. Nakahara's (11.54)–(11.55) and (12.90) use the latter as their instanton number.

On S^4 the integer value of our Chern number can be read from an equatorial transition. Represent the two hemispheres by an inside and an outside four-ball as in the figure.

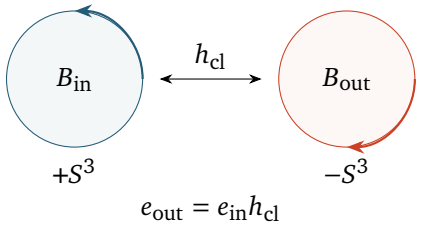


Figure 23.1: Two local Stokes integrals have opposite equatorial orientations. The frame order is $e_{\mathrm{out}} = e_{\mathrm{in}}h_{\mathrm{cl}}$. The equator is oriented as ∂B_{in}.

Theorem 23.4 (The clutching integer · *sketched; global construction cited*). *Let* $h_{\mathrm{cl}} \colon S^3 \to \mathrm{SU}(2)$ *glue the frames in the figure's order. Orient* SU(2) *as the outward unit sphere under* $(a_1, a_2, a_3, a_4) \mapsto a_4 I + \mathrm{i}a_a\sigma_a$. *Then* $k = -\deg h_{\mathrm{cl}} \in \mathbb{Z}$.

Sketch. Use local frames over the two balls, and Stokes with their opposite boundaries. The gauge law and the vanishing integral of an equatorial exact form give

$$k = \frac{1}{8\pi^2}\oint_{S^3}\bigl(\mathrm{CS}(A_{\mathrm{in}}) - \mathrm{CS}(A_{\mathrm{out}})\bigr) = \frac{1}{24\pi^2}\oint_{S^3}\operatorname{tr}\bigl((h_{\mathrm{cl}}^{-1}\mathrm{d}h_{\mathrm{cl}})^3\bigr).$$

For the identity unit-quaternion map, at $a_4 = 1$ the Maurer–Cartan form is $\mathrm{i}\sigma_a\mathrm{d}a_a$. The Pauli products give cubic trace $12\mathrm{d}a_1 \wedge \mathrm{d}a_2 \wedge \mathrm{d}a_3$. The outward sphere volume there is its negative,

since the outward normal is last in the ordered ambient coordinates. Both forms are left invariant, so everywhere $\operatorname{tr}(\vartheta_{\mathrm{MC}}^3) = -12\,\mathrm{vol}_{S^3}$. The sphere volume is $4\pi\int_0^\pi \sin^2\chi\,\mathrm{d}\chi = 2\pi^2$. Thus the normalized three-form $-\operatorname{tr}(\vartheta_{\mathrm{MC}}^3)/(24\pi^2)$ has integral one. Lecture 14's degree formula identifies the displayed integral as minus the degree. The global two-ball construction is the cited input from Nakahara, §10.5.5. ∎

On a region with boundary, the integral of the Chern form need not be an integer, just as the cap flux in Lecture 22 need not be integral.

23.5 The action bound and BPST

For an adjoint-valued two-form B, write $\|B\|^2 = -\int_M \operatorname{tr}(B\wedge\star B)$. This is nonnegative and vanishes only for $B = 0$.

Theorem 23.5 (The instanton bound · *proved*). *For an* SU(2) *connection over Euclidean* S^4,

$$S_E \geq \frac{8\pi^2}{g_{\mathrm{YM}}^2}|k|.$$

Equality holds exactly for a self-dual or anti-self-dual curvature. In our negative-trace convention, a nonzero self-dual field has $k < 0$, *and a nonzero anti-self-dual field has* $k > 0$.

Proof. The Euclidean star is an isometry with square one on two-forms. Expand the square for either $s = +1$ or $s = -1$ to obtain

$$-\operatorname{tr}(F\wedge\star F) = -\frac{1}{2}\operatorname{tr}\big((F - s\star F)\wedge\star(F - s\star F)\big) - s\operatorname{tr}(F\wedge F).$$

After integration this gives $g_{\mathrm{YM}}^2 S_E = \frac{1}{2}\|F - s\star F\|^2 - s8\pi^2 k$. Choose s so that $-sk = |k|$. Nonnegativity proves the bound. Equality requires $F = s\star F$. Conversely, that equation removes the square and gives equality. For $k = 0$ equality forces $F = 0$, which obeys both duality equations. For a nonzero dual field, $8\pi^2 k = -s\|F\|^2$, proving the stated signs. ∎

An *instanton* here is a smooth finite-action Euclidean connection with self-dual or anti-self-dual curvature. On S^4, or under the stated extension from $\mathbb{R}^4$, the bound makes it an action minimum within its fixed integer sector. The *BPST instanton* is the following charge-one SU(2) example.

Proposition 23.6 (BPST normalization · *sketched*). *For size* $\lambda > 0$ *on spatial-first* $\mathbb{R}^4$, *the regular BPST field is anti-self-dual, extends over the conformal compactification* S^4, *and has* $k = +1$ *and* $S_E = 8\pi^2/g_{\mathrm{YM}}^2$. *Reflection of one coordinate gives its self-dual partner with* $k = -1$ *and the same action.*

The extension over the compactification is a cited part of the BPST construction in Nakahara, §10.5.5, and Tong, §2.3.1. We assume this asymptotic regularity when treating an $\mathbb{R}^4$ field as a connection on S^4. Finite action alone is not used as a proof of that extension. The Euclidean star on two-forms is unchanged by a conformal metric rescaling, because its scale factor in dimension four is the zeroth power. Thus stereographic compactification preserves the duality equation and action. Give $S^4 \subset \mathbb{R}^5$ its outward orientation. Let $e_1,\dots,e_5$ be the standard basis of $\mathbb{R}^5$ and put $r^2 = \sum_{\mu=1}^4 (x^\mu)^2$. Use the inverse south-pole stereographic chart $X\colon \mathbb{R}^4 \to S^4 \setminus \{-e_5\}$, given by

$$X(x) = \frac{(2x^1, 2x^2, 2x^3, 2x^4, 1 - r^2)}{1 + r^2}.$$

At $x = 0$, the outward normal is e_5 and the four coordinate tangent vectors are $2e_i$. The ordered determinant is +16. The inverse $x^i = X^i/(1 + X^5)$, for $i \in \{1, 2, 3, 4\}$, shows that this is a smooth chart. Since $\mathbb{R}^4$ is connected, it preserves the spatial-first orientation everywhere. Consequently the computed charge on $\mathbb{R}^4$ is also the charge on this outward S^4.

The normalized radial contribution for this example is plotted in the figure.

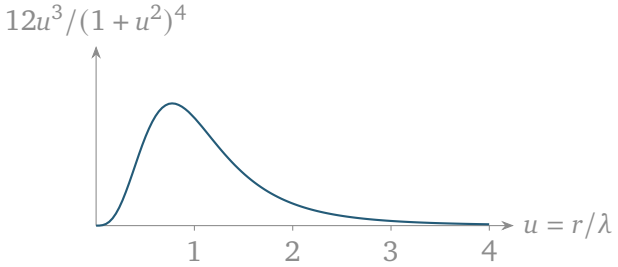


Figure 23.2: For $u = r/\lambda$, the BPST Chern-number contribution per du is $12u^3/(1 + u^2)^4$. Its integral over $[0, \infty)$ is 1. The plot shows the initial interval $0 \le u \le 4$.

23.6 A charge-one field and a topological term

Example 23.7 (The BPST action)**.** Compute the charge and action of the regular field with $\lambda > 0$. Recall $r^2 = \sum_{\mu=1}^{4} (x^\mu)^2$, with spatial indices $a, b, c \in \{1, 2, 3\}$. The potential is smooth at the origin and is given by

$$A = \frac{\mathrm{i}\sigma_a \left(x^4 \mathrm{d}x^a - x^a \mathrm{d}x^4 + \epsilon_{abc} x^b \mathrm{d}x^c\right)}{r^2 + \lambda^2}.$$

Differentiation and the Pauli commutator give its curvature

$$F = \frac{2\mathrm{i}\lambda^2 \sigma_a}{(r^2 + \lambda^2)^2} \left(\frac{1}{2} \epsilon_{abc} \mathrm{d}x^b \wedge \mathrm{d}x^c - \mathrm{d}x^a \wedge \mathrm{d}x^4 \right).$$

For example, the coefficient pair multiplying σ_1 is $\mathrm{d}x^2 \wedge \mathrm{d}x^3 - \mathrm{d}x^1 \wedge \mathrm{d}x^4$. The spatial-first star interchanges these two basis forms, so their difference is anti-self-dual. The other cyclic pairs have the same property. Taking the fundamental trace gives

$$\operatorname{tr}(F \wedge F) = -\operatorname{tr}(F \wedge \star F) = \frac{48\lambda^4}{(r^2 + \lambda^2)^4} \operatorname{vol}_{\mathbb{R}^4}.$$

Radial integration, followed by $w = r^2/\lambda^2$, yields

$$\int_{\mathbb{R}^4} \operatorname{tr}(F \wedge F) = 96\pi^2 \lambda^4 \int_0^\infty \frac{r^3 \mathrm{d}r}{(r^2 + \lambda^2)^4} = 48\pi^2 \int_0^\infty \frac{w \mathrm{d}w}{(1 + w)^4} = 8\pi^2.$$

The last integral is $1/6$, by writing $w = (1 + w) - 1$ and integrating the two powers. Consequently $k = 1$ and $S_E = 8\pi^2/g_{\mathrm{YM}}^2$, independent of the size.

For $r > 0$, define the asymptotic unit-quaternion map $h_\infty = (x^4 I + \mathrm{i}x^a \sigma_a)/r$. The same potential is $A = [r^2/(r^2 + \lambda^2)] h_\infty^{-1} \mathrm{d}h_\infty$. The map has degree +1 in the stated orientation. At infinity the pure-gauge Chern–Simons integral therefore gives $k = -\int_{S^3} \operatorname{tr}((h_\infty^{-1}\mathrm{d}h_\infty)^3)/(24\pi^2) = +1$. The clutching map has the inverse order. The change of frame $h_{\mathrm{cl}} = h_\infty^{-1}$ makes the outer asymptotic potential zero. Its degree is -1, so the clutching formula gives the same charge. Lecture 4 wrote $U = aI - \mathrm{i}v^a \sigma_a$. Our oriented quaternion coordinates there are $(a_4, \mathbf{a}) = (a, -\mathbf{v})$. Reversing to time-first orientation makes this same field self-dual and changes k to -1.

For a fixed real angle θ_{YM}, the *theta term* is $S_\theta = \theta_{\mathrm{YM}} \int_M \operatorname{tr}(F \wedge F)/(8\pi^2) = \theta_{\mathrm{YM}} k$. Locally its density is $\theta_{\mathrm{YM}} \mathrm{d}\,\mathrm{CS}(A)/(8\pi^2)$. Its variation is $\delta \operatorname{tr}(F \wedge F) = 2\mathrm{d} \operatorname{tr}(a \wedge F)$, using $DF = 0$. It therefore leaves the field equation unchanged for compactly supported variations, or on a closed manifold. On a manifold with boundary, boundary conditions still matter.

Self-check. Show directly that $\operatorname{tr}(F \wedge \star F)$ is invariant under a frame change.

Answer. The transformed product is $h^{-1}(F \wedge \star F)h$, since the star commutes with matrix conjugation. Its trace equals the original trace by cyclicity.

Summary. The fundamental anti-Hermitian trace fixes the Minkowski prefactor and gives $D\star F = 0$. For U(1) the different trace normalization recovers Maxwell's action. Locally $\mathrm{d}\,\mathrm{CS}(A) = \operatorname{tr}(F \wedge F)$. The Chern–Simons form need not be global. On S^4, $k = -\deg h_{\mathrm{cl}}$ in the stated two-frame order. The Euclidean bound is $S_E \geq 8\pi^2|k|/g_{\mathrm{YM}}^2$, saturated exactly by dual fields. Regular spatial-first BPST is anti-self-dual with $k = +1$. The action needs the metric and is orientation independent. The Chern number has the opposite dependence. Lecture 24 returns to line-bundle connections generated by quantum eigenstates.

Reading. Tu, *Differential Geometry*, §28.1, §29.4 and §31.3–31.5. Nakahara, §10.5.4–10.5.5, §11.3.1 (11.54)–(11.55), §11.5 and §12.6 (12.90). Tong, *Gauge Theory*, §2.1 and §2.3.1, pp. 28–30 and 52–55. In Nakahara (10.124)–(10.125), the equatorial north-pole volume is $+\mathrm{d}x^1 \wedge \mathrm{d}x^2 \wedge \mathrm{d}x^3$. Our outward volume form at that pole is its negative. Tong calls his time-first BPST field self-dual with $\nu = +1$. Our spatial-first orientation and translated anti-Hermitian field give anti-self-duality and $k = +1$. Use the orientation, potential and Chern-character dictionaries together.

Lecture 24

Berry's phase

Objectives. *A slowly cycled Hamiltonian can return an eigenstate with a measurable relative phase. Part of this phase depends on the elapsed time. Another part depends on the path of the parameters. We identify the second part with line-bundle holonomy. For the aligned spin-1/2 ground state, it is minus half the signed solid angle enclosed by the field-direction loop. The eigenvalue gap states where this description applies.*

By the end of this lecture you will be able to:

- State the gap and slow-evolution hypotheses of the cited adiabatic theorem.
- Construct the connection and curvature of a smooth simple eigenline.
- Compute a closed-cycle geometric phase with the correct gauge convention.
- Derive the spin-1/2 phase and its ground-state Chern number -1.

24.1 Slow evolution and the gap

Recall the finite-dimensional quantum notation from *Quantum Mechanics I*. A pure state is a complex line in $\mathbb{C}^N$. A unit representative is written as a ket $|\psi\rangle$. Its conjugate transpose is the bra $\langle\psi|$, a linear functional. The inner product $\langle\phi|\psi\rangle = \phi^\dagger\psi$ is antilinear in its first argument. Multiplying a representative by a phase leaves its line unchanged. The Hamiltonian is a Hermitian matrix. Evolution satisfies $i\hbar\,\partial_t\,|\psi\rangle = H(t)\,|\psi\rangle$.

Let $s \in [0,1]$ describe a fixed path of external parameters. Executing it in a duration $T > 0$ means $s = t/T$ for $0 \le t \le T$. Write the evolving vector as $\psi_T(s)$. The *adiabatic limit* is $T \to \infty$ with this path fixed. An eigenvalue is *simple* if its eigenspace has dimension one. Its *gap* is its distance to the other eigenvalues.

Theorem 24.1 (Adiabatic evolution · *cited*)**.** *Let $H(s)$ be a smooth family of Hermitian $N \times N$ matrices on $[0,1]$. Let $E_n(s)$ be a smooth simple eigenvalue separated from all other eigenvalues by a fixed $\delta > 0$. Start in a unit vector of its eigenspace and solve $i\hbar\,\psi_T'(s) = TH(s)\psi_T(s)$. There is a unique smooth unit vector $v(s)$ in that eigenspace with $v(0) = \psi_T(0)$ and $\langle v(s)|v'(s)\rangle = 0$. For each fixed $s \in [0,1]$, the cited theorem gives*

$$\left\| \psi_T(s) - \exp\left(-\frac{iT}{\hbar}\int_0^s E_n(u)\,du\right) v(s) \right\| \longrightarrow 0 \quad \text{as } T \to \infty.$$

This is the finite-dimensional form of Kato's adiabatic theorem. See Barry Simon, p. 2167, for the gap assumption and phase separation, and p. 2168 for the transport condition on the limit. The exponential is the *dynamical phase factor*.

If $H(1) = H(0)$ and the chosen eigenline returns to itself, then $v(1) = e^{i\gamma(C)}v(0)$ for a phase defined modulo 2π. This scalar is the *geometric phase factor*, or *Berry phase factor*. It can be measured relative to a reference arm after accounting for the dynamical phase.

The theorem is a limit statement. A changing eigenvector need not solve the full Schrödinger equation exactly at finite T.

The figure separates a gap condition from the choice of eigenvectors.

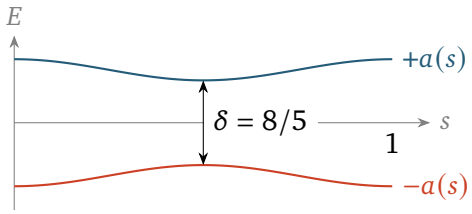


Figure 24.1: The diagonal family $H(s) = \mathrm{diag}(-a(s), a(s))$, with $a(s) = 1 + \cos(2\pi s)/5$, has a gap at least $8/5$. Its eigenspaces are constant, so its geometric phase is zero.

24.2 Eigenlines and their connection

Let $\mathcal{U}$ be an open region in a smooth parameter manifold, with a smooth family $H\colon \mathcal{U} \to \mathrm{Herm}(N)$ and a chosen smooth simple eigenvalue $E_n(R)$. Here $\mathrm{Herm}(N)$ is the real vector space of Hermitian matrices. The *eigenline bundle* is the subset of $\mathcal{U} \times \mathbb{C}^N$ given by

$$L_n = \{(R, w) : R \in \mathcal{U},\ H(R)w = E_n(R)w\}.$$

Its fibre over R is the eigenspace, including its zero vector. Smoothness of the selected branch is an explicit hypothesis here. It holds for the separated branches used below. Along a smooth parameter path $R(s)$, the adiabatic theorem applies to $H(R(s))$ when its eigenvalue has the stated uniform gap.

Theorem 24.2 (The Berry connection · *proved geometrically; adiabatic interpretation given the cited theorem*)**.** *The eigenlines form a smooth Hermitian line bundle. For a smooth unit eigenvector $|n_i\rangle$ on $U_i \subset \mathcal{U}$, the* Berry connection *has imaginary local one-form*

$$\mathcal{A}_i = \langle n_i | \mathrm{d}n_i \rangle\,.$$

These forms define the projected derivative $\nabla\xi = P\mathrm{d}\xi$, where $P(R)$ is the orthogonal projection onto the eigenline. Its parallel transport is the vector $v(s)$ in the adiabatic theorem. Thus its holonomy around a cycle C is $e^{i\gamma(C)}$.

Proof. The matrix $H(R) - E_n(R)I$ has rank $N-1$. Its adjugate, the transposed cofactor matrix, has a nonzero column at any fixed point. Its columns lie in the kernel of the original matrix, by the identity $B\,\mathrm{adj}(B) = \det(B)I$ for every square matrix B. Choose such a column. It stays nonzero nearby and is smooth in R. Divide it by its positive norm to obtain a smooth unit eigenvector $|n_i\rangle$. Its scalar multiples give a local product chart for L_n. On overlaps write $|n_j\rangle = |n_i\rangle\, g_{ij}$, where $g_{ij} = \langle n_i | n_j \rangle \in \mathrm{U}(1)$. These smooth transitions satisfy the cocycle law, so Lecture 20's charts give the bundle.

The projection is $P = |n_i\rangle\langle n_i|$ and is unchanged by this unit phase. It is therefore global. For a section $\xi = |n_i\rangle\, c$, direct differentiation gives

$$P\mathrm{d}\xi = |n_i\rangle\,\big(\mathrm{d}c + \langle n_i|\mathrm{d}n_i\rangle\, c\big).$$

The ordinary product rule gives the connection Leibniz rule because $P\xi = \xi$. Self-adjointness of P similarly gives the Hermitian metric product rule. Differentiating $\langle n_i | n_i \rangle = 1$ shows

$\mathcal{A}_i + \overline{\mathcal{A}_i} = 0$, so the potential is imaginary. Finally, for a vector $v(s)$ in the eigenline, its projection of $v'(s)$ is $v(s)\,\langle v(s)|v'(s)\rangle$ when v is unit. Thus the adiabatic theorem's condition is exactly $\nabla_{\partial_s} v = 0$. Uniqueness of Lecture 21's transport identifies the two vectors and their endpoint phase. ∎

The bundle and connection depend only on the eigenspaces. The dynamical phase also uses the eigenvalues and traversal time.

24.3 Curvature, gauge changes and holonomy

A different unit vector on one chart has the form $|n'\rangle = |n\rangle\, h$, with $h\colon U \to \mathrm{U}(1)$. Differentiating the product and using unit norm gives

$$\mathcal{A}' = \mathcal{A} + h^{-1}\mathrm{d}h, \qquad \mathcal{A}' = \mathcal{A} + \mathrm{i}\mathrm{d}\chi \quad \text{if } h = e^{\mathrm{i}\chi}.$$

This is Lecture 21's passive line-frame law. The coefficient of a fixed section changes by $c' = h^{-1}c$. The *real Berry potential* is $A_{\mathrm{B}} = \mathrm{i}\mathcal{A}$. It differs from the imaginary connection form used in the transport equation.

The *Berry curvature* is the global two-form $\mathrm{d}\mathcal{A}$. Differentiating the local connection yields the full local formula

$$\mathrm{d}\mathcal{A} = \langle \mathrm{d}n| \wedge |\mathrm{d}n\rangle = \sum_{a=1}^{N} \mathrm{d}\overline{n^a} \wedge \mathrm{d}n^a.$$

Here the bra–ket wedge denotes the componentwise sum on the right. The term with $\mathrm{d}^2 n$ vanishes. The gauge law shows that the curvature is unchanged on overlaps. Thus Lecture 22 supplies the real class $c_1(L_n) = [\mathrm{i}\mathrm{d}\mathcal{A}/(2\pi)]$.

In a periodic unit frame along a closed cycle, the scalar transport equation gives

$$e^{\mathrm{i}\gamma(C)} = \exp\left(-\oint_C \mathcal{A}\right), \qquad \gamma(C) = \mathrm{i}\oint_C \mathcal{A} = \oint_C A_{\mathrm{B}} \pmod{2\pi}.$$

Such a frame exists along the cycle even when none exists on all of $\mathcal{U}$. Indeed, transport a unit vector along the parameter interval to get $v(s)$. If $v(1) = e^{\mathrm{i}\alpha} v(0)$, the vector $e^{-\mathrm{i}\alpha s} v(s)$ is periodic. Alternatively, use local frames and the transition factors when changing charts, as in Lecture 21. Under any single-valued gauge map on the cycle, $\oint_C h^{-1}\mathrm{d}h = 2\pi\mathrm{i}\,\mathrm{wind}(h)$. The geometric phase changes by an integer multiple of 2π, leaving its factor invariant. No global real logarithm of h is required.

If one frame covers an oriented compact spanning surface Σ with $\partial\Sigma = C$, Stokes gives $\gamma(C) = \mathrm{i}\int_\Sigma \mathrm{d}\mathcal{A}$ modulo 2π. The surface and frame hypotheses matter, as they did for the Aharonov–Bohm calculation. Berry's original equation (6) uses our $\mathrm{i}\oint \mathcal{A}$. Tong's current QHE notes instead use a real potential $A_{\mathrm{Tong}} = -\mathrm{i}\mathcal{A} = -A_{\mathrm{B}}$. His factor $\exp(-\mathrm{i}\oint A_{\mathrm{Tong}})$ is the same holonomy.

24.4 A spin-$1/2$ in a rotating field

Recall the Pauli matrices and the spin operators S_a, for $a \in \{x, y, z\}$:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad S_a = \frac{\hbar}{2}\sigma_a.$$

The symbol σ_a denotes these matrices, distinct from Lecture 21's section-coefficient column. Their products obey $\sigma_a\sigma_b = \delta_{ab}I + \mathrm{i}\epsilon_{abc}\sigma_c$. Use an energy-scaled field $\mathbf{B} = b\mathbf{n}$, with $b > 0$ and $|\mathbf{n}| = 1$. The Hamiltonian is $H = -\mathbf{B}\cdot\boldsymbol{\sigma}$. Its eigenvalues are $-b$ and $+b$ because $(\mathbf{n}\cdot\boldsymbol{\sigma})^2 = I$. Its gap is $2b$ everywhere on the direction sphere.

For a smooth loop $C\colon [0,1] \to S^2$ based at $p = C(0) = C(1)$, choose a continuous filling $f\colon [0,1]^2 \to S^2$, smooth on each rectangle of a finite subdivision, with

$$f(s,0) = C(s), \qquad f(s,1) = f(0,u) = f(1,u) = p.$$

Lecture 20's hemisphere, polygon and stereographic construction supplies such a filling, smooth on each rectangle of a finite subdivision. Orient the square by $\mathrm{d}s \wedge \mathrm{d}u$. Recall the outward unit-sphere area form β from Lecture 12. The *signed solid angle* of this filling is

$$\Omega_f(C) = \int_{[0,1]^2} f^*\beta.$$

The proof below shows that its class modulo 4π is independent of the filling. We denote this class by $\Omega(C)$.

Proposition 24.3 (The spin eigenline · *proved*). *For* $\mathbf{n} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$*, the aligned ground-state eigenline is Lecture 20's tautological line on* $\mathbb{C}\mathrm{P}^1$*. On the northern chart it has*

$$|n_N\rangle = \begin{pmatrix} \cos(\theta/2) \\ e^{\mathrm{i}\phi}\sin(\theta/2) \end{pmatrix}, \quad \mathcal{A}_N = \frac{\mathrm{i}}{2}(1-\cos\theta)\mathrm{d}\phi, \quad \mathrm{d}\mathcal{A} = \frac{\mathrm{i}}{2}\sin\theta\mathrm{d}\theta\wedge\mathrm{d}\phi.$$

In the outward orientation its Chern number is -1*. For an oriented loop, the geometric phase is* $\gamma(C) = -\Omega(C)/2$ *modulo* 2π*, where the solid angle is defined modulo* 4π*.*

Proof. The direction matrix is

$$\mathbf{n}\cdot\boldsymbol{\sigma} = \begin{pmatrix} \cos\theta & e^{-\mathrm{i}\phi}\sin\theta \\ e^{\mathrm{i}\phi}\sin\theta & -\cos\theta \end{pmatrix}.$$

Multiplying by the displayed vector gives $|n_N\rangle$, so $H|n_N\rangle = -b|n_N\rangle$. Its norm is one. In $z = \tan(\theta/2)e^{\mathrm{i}\phi}$ it is $(1,z)^T/\sqrt{1+|z|^2}$, the tautological unit frame of Lecture 20. The inner product with its θ derivative is $-\cos(\theta/2)\sin(\theta/2)/2 + \sin(\theta/2)\cos(\theta/2)/2 = 0$. The ϕ derivative gives $\mathrm{i}\sin^2(\theta/2) = \mathrm{i}(1-\cos\theta)/2$. Differentiating gives the curvature stated above.

The southern unit vector is $|n_S\rangle = e^{-\mathrm{i}\phi}|n_N\rangle$. It is smooth at the south pole and has $\mathcal{A}_S = \mathcal{A}_N - \mathrm{i}\mathrm{d}\phi$. Thus its frame transition has winding -1. Equivalently, direct curvature integration gives

$$\frac{\mathrm{i}}{2\pi}\int_{S^2}\mathrm{d}\mathcal{A} = -\frac{1}{4\pi}\int_0^{2\pi}\int_0^{\pi}\sin\theta\mathrm{d}\theta\mathrm{d}\phi = -1.$$

To compute the phase for a general loop, pull the eigenline and connection back to the filling square. Fix a unit vector over p. From the top edge, transport it down each vertical segment. Smooth dependence of transport on parameters gives a unit frame $w(s,u)$, continuous on the square and smooth on each rectangle of the finite subdivision. The top and side edges map constantly to p, so the frame is the same fixed vector on all three. In particular, its bottom restriction is a periodic frame along C.

Let $\mathcal{A}_f = \langle w|\mathrm{d}w\rangle$ be its connection form on the square. On each rectangle, its curvature is $\mathrm{d}\mathcal{A}_f = f^*\mathrm{d}\mathcal{A}$. The tangential restrictions on shared edges agree, so internal boundary integrals cancel. The top and side integrals vanish because their frame is constant. The

positively oriented bottom edge runs in increasing s. Stokes on the rectangles therefore gives

$$\gamma(C) = \mathrm{i}\int_0^1 \mathcal{A}_f(\partial_s)(s,0)\mathrm{d}s = \mathrm{i}\int_{[0,1]^2} f^*\mathrm{d}\mathcal{A} = -\frac{1}{2}\Omega_f(C) \pmod{2\pi}.$$

Two fillings f and g compute the same intrinsic holonomy. Hence $\exp[-\mathrm{i}(\Omega_f(C)-\Omega_g(C))/2] = 1$, so their signed areas differ by an integer multiple of 4π. This proves both the solid-angle convention and the claimed phase formula. ∎

The figure fixes the orientation of a latitude and its northern cap.

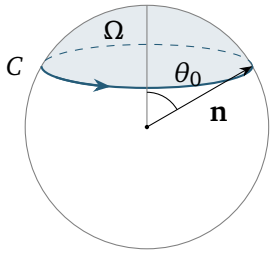


Figure 24.2: Increasing azimuth orients the latitude as the boundary of the northern cap. The cap has solid angle $\Omega = 2\pi(1-\cos\theta_0)$. Its spin ground-state geometric phase is $-\Omega/2$.

Example 24.4 (A closed field rotation)**.** Let $b > 0$ be fixed, choose $0 < \theta_0 < \pi$, and rotate the field with $\phi(t) = 2\pi t/T$ for $0 \le t \le T$. Start in $|n_N(\theta_0, 0)\rangle$ and take the adiabatic limit. The northern unit frame is periodic on this path, so

$$\gamma(C) = \mathrm{i}\int_0^{2\pi} \frac{\mathrm{i}}{2}(1-\cos\theta_0)\mathrm{d}\phi = -\pi(1-\cos\theta_0).$$

The energy is $-b$, so the dynamical factor is $e^{+\mathrm{i}bT/\hbar}$. The final state is asymptotic to

$$e^{+\mathrm{i}bT/\hbar}e^{-\mathrm{i}\pi(1-\cos\theta_0)}\,|n_N(\theta_0,0)\rangle\,.$$

For $\theta_0 = \pi/3$ the geometric factor is $-\mathrm{i}$. The southern potential instead gives a phase larger by 2π, hence the same factor. Using the same cap, Lecture 18's tangent-vector transport angle is Ω. The spin phase is its negative half before either angle is reduced modulo 2π. At the equator the vector holonomy is a full turn while the spin phase factor is -1. The orthogonal excited vector $(-e^{-\mathrm{i}\phi}\sin(\theta/2), \cos(\theta/2))^T$ has $\mathcal{A}_{\mathrm{exc}} = -\mathrm{i}(1-\cos\theta)\mathrm{d}\phi/2$. Its curvature and Chern number have the opposite sign, giving $+1$.

24.5 The degeneracy and general spin

For $\mathbf{B} \neq 0$, the ground-state projector is $P(\mathbf{B}) = (I + \mathbf{B}\cdot\boldsymbol{\sigma}/|\mathbf{B}|)/2$. It depends only on the field direction. Pulling the sphere curvature back to $\mathbb{R}^3 \setminus \{0\}$ gives

$$\mathrm{d}\mathcal{A} = \frac{\mathrm{i}}{2|\mathbf{B}|^3}\left(B^1\mathrm{d}B^2\wedge\mathrm{d}B^3 + B^2\mathrm{d}B^3\wedge\mathrm{d}B^1 + B^3\mathrm{d}B^1\wedge\mathrm{d}B^2\right).$$

The real curvature $\mathrm{d}A_{\mathrm{B}} = \mathrm{i}\mathrm{d}\mathcal{A}$ points inward. Its flux through an outward-oriented sphere is -2π. The figure shows this parameter-space monopole.

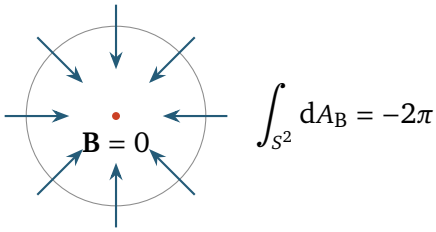


Figure 24.3: The real Berry curvature points inward in field-parameter space. Any outward sphere around the excluded degeneracy has $\int dA_B = -2\pi$ and ground-line Chern number -1.

At $\mathbf{B} = 0$ the two eigenvalues coincide. Every line is then an eigenline, so the selected ground line and its projector are not determined. For the identically zero Hamiltonian the exact evolution is constant. It does not force a prescribed changing eigenline to be followed. The gap hypothesis prevents this ambiguity.

Proposition 24.5 (Spin j · *sketched*)**.** *Let $j > 0$ be an integer or half-integer. Write the dimensionless spin matrices as $J_a = S_a/\hbar$, with $[J_a, J_b] = i\epsilon_{abc}J_c$. On the outward direction sphere, for the eigenline of $\mathbf{n}\cdot\mathbf{J}$ of weight $m \in \{-j, -j+1, \ldots, j\}$,*

$$\mathcal{A}_N = im(1-\cos\theta)d\phi, \qquad \gamma(C) = -m\Omega(C) \pmod{2\pi}, \qquad C_1 = -2m.$$

Sketch. Choose $J_z|j,m\rangle = m|j,m\rangle$ and rotate the weight vector by

$$|n_N\rangle = e^{im\phi}e^{-i\phi J_z}e^{-i\theta J_y}|j,m\rangle\,.$$

The phase makes this vector single-valued in ϕ, since $m - j \in \mathbb{Z}$. The angular-momentum commutators give $e^{i\theta J_y}J_z e^{-i\theta J_y} = J_z\cos\theta - J_x\sin\theta$. In a weight vector, $\langle j,m|J_x|j,m\rangle = \langle j,m|J_y|j,m\rangle = 0$ and $\langle j,m|J_z|j,m\rangle = m$. Thus the θ coefficient is zero and the ϕ coefficient is $im - im\cos\theta$. Differentiate, integrate over the sphere, and apply the transport formula to get the stated values. The rotation and weight identities are the standard angular-momentum input recalled from *Quantum Mechanics I*. Berry, §4, pp. 50–51, obtains the same spin phase using ladder operators. ∎

The quantity $2m$ is an integer, so changing the solid angle by 4π does not change the phase factor. For the ground line of $H = -\kappa\mathbf{B}\cdot\mathbf{J}$ with $\kappa > 0$, the weight is $m = j$ and the degeneracy is again at $\mathbf{B} = 0$.

Self-check. For a smooth open path $C\colon [0,1] \to \mathcal{U}$, explain how $i\int_C \mathcal{A}$ changes under $|n\rangle \mapsto e^{i\chi}|n\rangle$, for a single-valued real χ on the path.

Answer. It changes by $-[\chi(C(1)) - \chi(C(0))]$. Thus this integral alone is not an intrinsic open-path phase. An endpoint reference must also transform if an open-path interference measurement is specified. For a closed cycle the holonomy is intrinsic and its phase is defined modulo 2π.

Summary. A smooth simple eigenline has the unitary connection $\mathcal{A} = \langle n|\mathrm{d}n\rangle$ and curvature $\langle \mathrm{d}n| \wedge |\mathrm{d}n\rangle$. The cited gapped adiabatic theorem identifies its holonomy with the geometric factor after removing the dynamical phase. The imaginary connection and real potential satisfy $A_{\mathrm{B}} = \mathrm{i}\mathcal{A}$. For $H = -\mathbf{B}\cdot\boldsymbol{\sigma}$ with $\mathbf{B} \neq 0$, the ground line has Chern number -1 on an outward direction sphere and $\gamma = -\Omega/2$ modulo 2π. The excluded degeneracy is the source of this parameter-space curvature. Lecture 25 integrates the same curvature over the momentum torus of a gapped band.

Reading. Berry, *Quantal phase factors accompanying adiabatic changes*, §2 and §4, pp. 46–48 and 50–51. Simon, *Holonomy, the quantum adiabatic theorem and Berry's phase*, pp. 2167–2168. Nakahara, §10.6. Tong, *The Quantum Hall Effect*, current §1.5.1–1.5.2, pp. 28–34, with $A_{\mathrm{Tong}} = -A_{\mathrm{B}}$.

Lecture 25

Topological band theory

Objectives. *Lecture 24 attached a line bundle to an isolated simple eigenvalue. For a periodic crystal, the parameters are momenta on a torus. The normalized curvature integral on that torus is an integer. For a two-band model we compute it as a degree, then use the cited Hall-response theorem to relate this integer to a measured conductance.*

By the end of this lecture you will be able to:

- Construct a Bloch eigenline bundle for a smooth isolated simple band.
- Prove integrality of its normalized curvature flux on the oriented momentum torus.
- Compute the lower-band Chern number and gap closings of the QWZ model.
- State the filled-band Hall formula with its physical hypotheses and component convention.

25.1 Bloch Hamiltonians and the momentum torus

Consider a square lattice with cell label $r \in \mathbb{Z}^2$ and N orbitals in each cell. Use lattice spacing one and a fixed orthonormal orbital basis. A one-particle wavefunction has a column $\psi_r \in \mathbb{C}^N$ at each cell. A translation-invariant finite-range Hamiltonian is given by

$$(H\psi)_r = \sum_{R\in\mathbb{Z}^2} T_R \psi_{r+R}, \qquad T_{-R} = T_R^\dagger,$$

where only finitely many hopping matrices T_R are nonzero. The adjoint condition makes the operator Hermitian. Recall the positive-phase Bloch convention from *Quantum Mechanics I*, Lecture 8. Substitution of $\psi_r = e^{\mathrm{i}\mathbf{k}\cdot r} v$ gives the *Bloch Hamiltonian*

$$H(\mathbf{k}) = \sum_R T_R e^{\mathrm{i}\mathbf{k}\cdot R}.$$

It is a smooth Hermitian $N \times N$ matrix. Adding 2π to either momentum component leaves this matrix unchanged. Thus its parameter space is the *Brillouin torus* $\mathbb{T}^2 = \mathbb{R}^2/(2\pi\mathbb{Z})^2$, oriented by $\mathrm{d}k_x \wedge \mathrm{d}k_y$.

On an $L \times L$ periodic lattice, the allowed momenta have components $2\pi a/L$, with $a \in \{0, \dots, L-1\}$. The finite geometric sum makes Bloch waves with distinct momenta orthogonal. There are L^2 such momenta, each paired with the N orbital columns. The resulting Bloch waves form a basis of the NL^2-dimensional one-particle space. The substitution above therefore gives its full block decomposition. The continuous torus describes the same matrix family as the system size grows. The infinite-lattice Bloch ansatz is not a normalized square-summable state.

A *band* is an energy branch $E_n(\mathbf{k})$ of this family. We call a chosen smooth simple band *isolated* when, at every momentum, its distance from every other eigenvalue is at least a

fixed $\delta > 0$. This is a separation at the same momentum. It does not yet assert an energy interval separating occupied and empty states at all momenta.

Proposition 25.1 (The Bloch bundle · *proved*)**.** *The eigenspaces of a smooth isolated simple band form a Hermitian line bundle $L_n \to \mathbb{T}^2$. For local unit eigenvectors $v_i(\mathbf{k})$, its connection is $\mathcal{A}_i = v_i^\dagger \mathrm{d}v_i$. Its* band Chern number *is*

$$C_n = \frac{\mathrm{i}}{2\pi} \int_{\mathbb{T}^2} \mathrm{d}\mathcal{A}.$$

Proof. The periodic matrix family is defined on the torus itself. Lecture 24's adjugate construction supplies smooth local eigenvectors there. Their unit-phase transitions give the line bundle and the projected Hermitian connection. Its curvature is global and imaginary, so the displayed integral is real. The next proposition proves that it is an integer. ∎

The square in the figure is a parameter domain. Its edge frames need not agree.

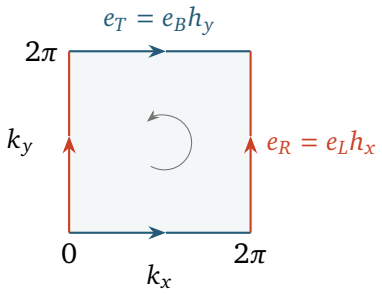


Figure 25.1: Opposite sides represent the same momenta. The circular arrow gives the positive torus orientation. The two unit-phase functions compare the chosen square frame on each pair of edges.

25.2 Integral flux and the two-band degree

A *smooth family* of Hermitian line bundles with connection here means one Hermitian line bundle with a smooth unitary connection on $\mathbb{T}^2 \times [0,1]$, restricted to the momentum slices.

Proposition 25.2 (Integrality on the torus · *proved*)**.** *For a Hermitian line bundle with a unitary connection on the oriented torus, the number $\mathrm{i}\int_{\mathbb{T}^2} F/(2\pi)$ is an integer. It is unchanged in a smooth family as defined above.*

Proof. Pull the bundle and connection back along $\mathbb{R}^2 \to \mathbb{T}^2$. Transport a fixed unit vector at the origin along radial segments. Lecture 21's smooth parameter dependence gives a smooth unit frame on $\mathbb{R}^2$. Restrict it to the square $[0, 2\pi]^2$ and call its connection form A. Write $x = k_x$ and $y = k_y$ in this argument. After identifying equal torus fibres, the edge frames satisfy

$$e(2\pi, y) = e(0, y) h_x(y), \qquad e(x, 2\pi) = e(x, 0) h_y(x),$$

with smooth maps $h_x, h_y \colon [0, 2\pi] \to \mathrm{U}(1)$. Lecture 3's phase lifting on an interval gives real smooth functions χ_x and χ_y with $h_x = e^{\mathrm{i}\chi_x}$ and $h_y = e^{\mathrm{i}\chi_y}$. Smoothness follows locally by a smooth inverse of the exponential covering. The frame law gives $A_R - A_L = \mathrm{i}\mathrm{d}\chi_x$ along

the vertical edges and $A_T - A_B = \mathrm{i}\mathrm{d}\chi_y$ along the horizontal edges. With $\Delta\chi = \chi(2\pi) - \chi(0)$, Stokes therefore gives

$$\int_{\mathbb{T}^2} F = \mathrm{i}(\Delta\chi_x - \Delta\chi_y), \qquad \frac{\mathrm{i}}{2\pi}\int_{\mathbb{T}^2} F = \frac{\Delta\chi_y - \Delta\chi_x}{2\pi}.$$

The right and top comparisons give the same frame at the last corner, so

$$h_y(0)h_x(2\pi) = h_x(0)h_y(2\pi).$$

Substitution of their lifted phases makes $\Delta\chi_x - \Delta\chi_y$ an integer multiple of 2π. This proves integrality. Neither edge phase is required to be a closed loop. In such a family, its global curvature restricts smoothly to the momentum slices, so the curvature integral varies continuously. An integer-valued continuous function on an interval is constant, proving the last claim. ∎

For a smooth Hamiltonian family whose chosen band stays isolated, the adjugate construction is smooth jointly in momentum and deformation parameter. The eigenline and projected connection of the chosen band give such a family.

For an example of integrality, the square potential $A = -\mathrm{i}n x\,\mathrm{d}y/(2\pi)$, with $n \in \mathbb{Z}$, has $h_x(y) = e^{-\mathrm{i}ny}$ and $h_y(x) = 1$. Its flux is $-2\pi\mathrm{i}n$, so its Chern number is n. The edge phase at $y = 2\pi$ agrees with its value at zero precisely because n is an integer.

Now take two orbitals per cell and a Hamiltonian $H(\mathbf{k}) = \mathbf{d}(\mathbf{k})\cdot\boldsymbol{\sigma}$, where $\mathbf{d}\colon \mathbb{T}^2 \to \mathbb{R}^3 \setminus \{0\}$ is smooth. The Pauli product identity gives energies $-|\mathbf{d}|$ and $+|\mathbf{d}|$. Define the direction map $\widehat{\mathbf{d}} = \mathbf{d}/|\mathbf{d}|$.

Theorem 25.3 (The lower band as a degree · *proved*). *For the lower band and the outward orientation on S^2,*

$$\mathrm{d}\mathcal{A} = -\frac{\mathrm{i}}{2}\widehat{\mathbf{d}}\cdot(\partial_{k_x}\widehat{\mathbf{d}} \times \partial_{k_y}\widehat{\mathbf{d}})\mathrm{d}k_x \wedge \mathrm{d}k_y, \qquad C = \deg(\widehat{\mathbf{d}}).$$

Proof. The lower eigenspace of $\widehat{\mathbf{d}}\cdot\boldsymbol{\sigma}$ is the excited eigenspace of Lecture 24's $-\mathbf{n}\cdot\boldsymbol{\sigma}$. That sphere line has curvature $-\mathrm{i}\beta/2$, where β is the outward area form. The band line is its pullback by $\widehat{\mathbf{d}}$. Pullback of the projected connection pulls back its curvature, giving the displayed formula. Lecture 14's degree identity gives

$$C = \frac{1}{4\pi}\int_{\mathbb{T}^2} \widehat{\mathbf{d}}^{\,*}\beta = \deg(\widehat{\mathbf{d}}),$$

because $\int_{S^2}\beta = 4\pi$. ∎

The upper band has the opposite curvature and Chern number. For more than two orbitals, the curvature definition still applies, but this particular map to a direction sphere is unavailable.

25.3 The Qi–Wu–Zhang model

The *Qi–Wu–Zhang model*, abbreviated QWZ, has direction vector

$$\mathbf{d} = (\sin k_x, \sin k_y, u + \cos k_x + \cos k_y), \qquad u \in \mathbb{R}.$$

Its Hamiltonian is the matrix

$$H(\mathbf{k}) = \begin{pmatrix} u + \cos k_x + \cos k_y & \sin k_x - \mathrm{i} \sin k_y \\ \sin k_x + \mathrm{i} \sin k_y & -u - \cos k_x - \cos k_y \end{pmatrix}.$$

The parameter u is an energy parameter, distinct from Lecture 19's inverse radius. In the lattice convention of the first section its only possibly nonzero hopping matrices are $T_0 = u\sigma_z$, $T_{(1,0)} = (\sigma_z - \mathrm{i}\sigma_x)/2$ and $T_{(0,1)} = (\sigma_z - \mathrm{i}\sigma_y)/2$, together with their adjoints. Thus it is a finite-range Hermitian lattice system.

The figure marks the momenta at which both transverse components vanish.

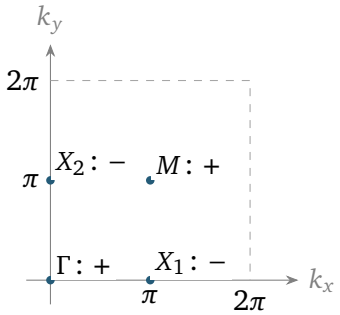


Figure 25.2: The four distinct vertical points in one torus cell. The signs are the local orientation signs used when the corresponding mass is positive.

The next figure records the lower-band integer. The missing endpoints are gap closings.

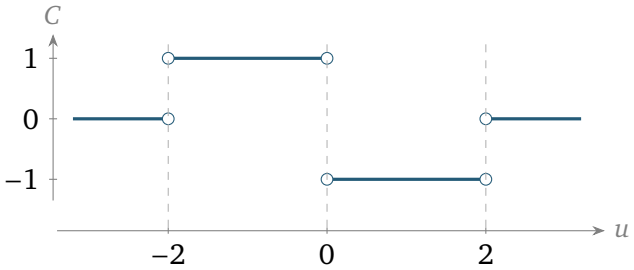


Figure 25.3: The QWZ lower-band Chern number as u increases. The number is undefined at $u = -2$, $u = 0$ and $u = 2$.

Example 25.4 (The QWZ phase diagram · *proved*)**.** Find the lower-band Chern number and every gap closing as u varies. The energies are $E_- = -|\mathbf{d}|$ and $E_+ = |\mathbf{d}|$, so a gap closes exactly when $\sin k_x = \sin k_y = 0$ and $u + \cos k_x + \cos k_y = 0$. The relevant masses and transverse Jacobian signs are

point	(k_x, k_y)	d_z	$\cos k_x \cos k_y$
Γ	$(0, 0)$	$u + 2$	$+1$
X_1	$(\pi, 0)$	u	-1
X_2	$(0, \pi)$	u	-1
M	(π, π)	$u - 2$	$+1$

Hence the critical parameters are -2, 0 and 2. There are two distinct gap-closing points at zero. At every other parameter, compactness of the torus gives a positive minimum gap.

To compute the degree, count preimages of the north pole. A row contributes precisely when its mass d_z is positive. Near such a point the first two components of $\widehat{\mathbf{d}}$ are local sphere

coordinates. Their Jacobian determinant is $\cos k_x \cos k_y / d_z^2$ at the point. These coordinates have the outward sphere orientation at the north pole. All selected preimages are therefore regular, with the signs in the last column. Lecture 14's signed-count theorem gives

$$C = \begin{cases} 0, & u < -2, \\ +1, & -2 < u < 0, \\ -1, & 0 < u < 2, \\ 0, & u > 2. \end{cases}$$

For $u < -2$ there are no north preimages. In the next interval only Γ contributes. Crossing zero adds both negatively oriented points. Crossing two adds the positive contribution at M. Thus the simultaneous closing at zero changes the integer by -2.

These values use $C = (\mathrm{i}/2\pi) \int \mathrm{d}\mathcal{A}$. Asbóth–Oroszlány–Pályi's continuum convention uses $Q = -C$.[1] This sign difference is not a reversal of the Hamiltonian or the torus orientation.

The exact degree calculation requires no numerical mesh. An independent numerical check may use forward overlaps of neighbouring lower-band eigenvectors. With a counterclockwise plaquette, the product of normalized overlaps has argument approaching $\mathrm{Im} \int \mathrm{d}\mathcal{A}$. Thus the *negative* sum of these arguments, divided by 2π, approximates our C. At $u = -3, -1, 1, 3$, grids with 20×20 and 32×32 points give values close to $0, +1, -1, 0$, respectively, in agreement with a separate curvature integral. These numerical checks do not replace the degree proof.

25.4 Filled bands and the Hall response

At zero temperature, noninteracting electrons fill all states below the *Fermi energy* E_F. An insulator has an open energy interval around E_F containing no bulk states. This full energy gap is stronger than separation of a band at each momentum. A *Chern insulator* is a band insulator whose occupied bands have nonzero total Chern number.

Let $e > 0$ be the magnitude of the electron charge, so its signed charge is $-e$. Write Planck's constant as $h_{\mathrm{P}} = 2\pi\hbar$, distinct from a frame-change map. Fix the scalar Hall conductance by the transverse component convention

$$J_y = \sigma_{\mathrm{H}} E_x, \qquad J_x = -\sigma_{\mathrm{H}} E_y.$$

Theorem 25.5 (TKNN response · *cited*). *For a clean two-dimensional periodic noninteracting band insulator at zero temperature, with a bulk Fermi-energy gap and isolated occupied bands, the bulk linear Hall response is*

$$\sigma_{\mathrm{H}} = \frac{e^2}{h_{\mathrm{P}}} \sum_{n \text{ occupied}} C_n.$$

For one filled band the sum has one term.

[1] Their real potential is our $A_{\mathrm{B}} = \mathrm{i}\mathcal{A}$. The extra minus is in their definition $Q = -(1/2\pi) \int \mathrm{d}A_{\mathrm{B}}$, equations (2.38) and (2.41). Their equation (6.5) has the opposite integers, consistent with that continuum definition for the lower band. Their discrete definitions (2.9) and (2.17) instead use the negative sum of forward plaquette arguments divided by 2π, which approaches our C. These source definitions have different signs.

This is the filled-band response theorem of Thouless, Kohmoto, Nightingale and den Nijs (TKNN, 1982). The Kubo-to-curvature step is cited, not proved here. In their p. 406 equation (5), the antisymmetrized derivative integral has the coefficient $ie^2/(2\pi h_{\mathrm{P}})$, matching the curvature normalization above. The response components in this lecture are fixed by the displayed definition.

The sign can also be checked in semiclassical notation. Let the real curvature be $\mathrm{d}A_{\mathrm{B}} = \Omega_{\mathrm{B}}\mathrm{d}k_x \wedge \mathrm{d}k_y$. The wavepacket equations use $\hbar\dot{\mathbf{k}} = -e\mathbf{E}$ and anomalous velocity $-\dot{\mathbf{k}} \times (\Omega_{\mathrm{B}}\mathbf{e}_z)$. Multiplication of this anomalous velocity by charge $-e$ gives a current contribution proportional to $(-E_y, E_x)$. For a filled band the group-velocity integral vanishes by periodicity. The momentum density of states is $\mathrm{d}^2k/(2\pi)^2$, so the anomalous contribution gives

$$\mathbf{J} = \frac{e^2}{\hbar}\int_{\mathbb{T}^2}\frac{\mathrm{d}^2k}{(2\pi)^2}\Omega_{\mathrm{B}}(-E_y, E_x) = \frac{e^2}{h_{\mathrm{P}}}C(-E_y, E_x).$$

This checks the component sign and units of the cited formula. It is not a derivation of the Kubo theorem. For QWZ with only the lower band filled, the four insulating intervals have the conductances 0, $+e^2/h_{\mathrm{P}}$, $-e^2/h_{\mathrm{P}}$ and 0, in increasing u. Indeed, $\min|\mathbf{d}| > 0$ off the critical parameters, so $E_F = 0$ lies in the full gap between $-\min|\mathbf{d}|$ and $+\min|\mathbf{d}|$. At a critical parameter the gap hypothesis fails and the theorem does not assign that quantized value.

25.5 Gap closings and Dirac cones

A *Dirac cone* is a linear two-band crossing with local Hamiltonian

$$H_{\mathrm{D}}(\mathbf{q}) = v_x q_x \sigma_x + v_y q_y \sigma_y + m\sigma_z, \qquad v_x v_y \neq 0.$$

Here $\mathbf{q}$ measures displacement from the crossing momentum. The energies of this local Hamiltonian at $m = 0$ are $\pm\sqrt{v_x^2 q_x^2 + v_y^2 q_y^2}$. The mass m opens a gap. The figure shows the integrated lower-band curvature for unit positive velocities.

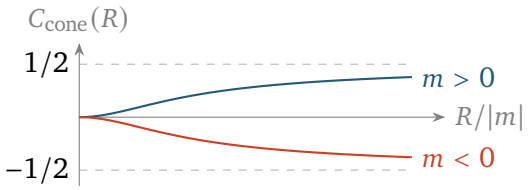


Figure 25.4: The continuum-cone contribution inside a disk of radius R, for $|m| = 1$. The limits are half-integers because the momentum plane is not a closed torus.

Proposition 25.6 (Changes at a closing · *sketched*). *A smooth deformation of an isolated band cannot change its Chern number without losing its isolation. When isolated linear Dirac crossings are the only gap closings, their signed unit changes add. For a cone whose mass increases through zero, the change is* $\operatorname{sgn}(v_x v_y)$.

Sketch. Away from a closing, the eigenline curvature varies smoothly, so the integer remains constant by Proposition 25.2. Near a cone, the degree density gives the plane contribution

$$C_{\mathrm{cone}} = \frac{1}{4\pi}\int_{\mathbb{R}^2}\frac{m v_x v_y \mathrm{d}q_x \mathrm{d}q_y}{(v_x^2 q_x^2 + v_y^2 q_y^2 + m^2)^{3/2}} = \frac{1}{2}\operatorname{sgn}(v_x v_y)\operatorname{sgn}(m).$$

Indeed, rescale the two momentum axes and use polar coordinates. The remaining radial integral is $\int_0^\infty |m| r (r^2 + m^2)^{-3/2} \mathrm{d}r = 1$. Its change across zero is the claimed signed unit. The curvature outside small crossing neighbourhoods has the same limiting integral from either side. In the local rescaling limit each neighbourhood has the displayed unit jump. Disjoint neighbourhoods add their contributions. This local limiting argument sketches the jump. It does not assign an integer to an isolated continuum cone. ∎

For QWZ the velocities at a vertical point are $v_x = \cos k_x$ and $v_y = \cos k_y$. Each local mass increases with u. The jumps are therefore +1 at -2, $-1 - 1 = -2$ at zero and +1 at two, in agreement with the full degree proof.

Self-check. Explain why the QWZ lower-band number is zero at $u = 3$, without counting preimages.

Answer. The component $d_z = 3 + \cos k_x + \cos k_y$ is at least one. Thus the direction map lies in the northern hemisphere and contracts there to the north pole. Its degree is zero by Lecture 14's homotopy invariance.

Summary. An isolated simple Bloch band defines a line bundle on the momentum torus. Its normalized curvature integral is an integer, fixed during any smooth deformation preserving band isolation. For $H = \mathbf{d} \cdot \boldsymbol{\sigma}$ with $\mathbf{d} \neq 0$, the lower-band number is $C = \deg(\widehat{\mathbf{d}})$. For QWZ its values are $0, +1, -1, 0$ in increasing u, with undefined numbers at $-2, 0, 2$. The cited filled-band theorem gives the Hall conductance in units of e^2/h_{P}. The same principle of a homotopy class obstructing deformation will classify defects in Lecture 26.

Reading. Asbóth–Oroszlány–Pályi, *A Short Course on Topological Insulators*, §2.1.3–2.1.4 for the discrete sign; §2.2.1, §2.2.4 and §6.1, pp. 80–82 for QWZ, with continuum $Q = -C$. Thouless et al., *Quantized Hall Conductance in a Two-Dimensional Periodic Potential*, pp. 405–408, especially equation (5) on p. 406.

Lecture 26

Higher homotopy groups and topological defects

Objectives. *A loop detects whether a line defect can be removed while the surrounding medium stays ordered. A sphere provides the corresponding test for a point defect. We extend the fundamental group to maps from higher-dimensional spheres. We then distinguish based charges from the free homotopy classes that describe local defects. This distinction identifies opposite point charges for unoriented axes, but preserves their sign for oriented magnetic directions.*

By the end of this lecture you will be able to:

- Prove the group laws and commutativity of higher homotopy groups.
- Use the cited sphere groups and fibration sequence in the stated cases.
- Choose the linking circle for a line defect and sphere for a point defect.
- Compute vortex, nematic and sphere-valued charges with the required basepoint qualification.

26.1 Maps from cubes and higher homotopy groups

Let X be a space with basepoint $x_0 \in X$. For $n \geq 1$, put $I^n = [0,1]^n$. The *higher homotopy group* $\pi_n(X, x_0)$ is initially the set of continuous maps $f\colon I^n \to X$ sending the entire boundary to x_0, modulo homotopies that keep that boundary at x_0. Collapsing the boundary identifies $I^n/\partial I^n$ with S^n. Thus the same classes are based maps from a sphere. For $n = 1$ this is Lecture 3's fundamental group. The symbol $\pi_0(X)$ instead denotes the set of path components. It is not generally a group.

Write $t' = (t^2, \dots, t^n)$. Concatenation in the first coordinate is given by

$$(f+g)(t^1, t') = \begin{cases} f(2t^1, t'), & 0 \leq t^1 \leq 1/2, \\ g(2t^1 - 1, t'), & 1/2 \leq t^1 \leq 1. \end{cases}$$

Both seam values are x_0, so pasting gives a continuous map relative to the boundary. For $n = 1$, this operation traverses f first, in the course's path convention.

Proposition 26.1 (The group structure · *proved*)**.** *Concatenation makes $\pi_n(X, x_0)$ a group for every $n \geq 1$. Its identity is the constant map and its inverse is represented by $f^-(t^1, t') = f(1 - t^1, t')$. For $n \geq 2$ the group is abelian.*

Proof. Pasting homotopies on the two halves shows that concatenation is well-defined on classes. The two bracketings of three maps occupy first-coordinate slabs of lengths $(1/4, 1/4, 1/2)$ and $(1/2, 1/4, 1/4)$. For a homotopy parameter $u \in [0,1]$, use slab lengths

$((1+u)/4, 1/4, (2-u)/4)$ and rescale the first coordinate on each slab affinely to the unit interval. All lengths are positive. Adjacent maps agree at their boundary value x_0. This continuous reparametrization proves associativity.

A constant first slab is a pause before f. Shrink its length a continuously from $1/2$ to zero, using the value x_0 for $t^1 \le a$ and $f((t^1-a)/(1-a), t')$ for $t^1 \ge a$. This is a homotopy relative to the cube boundary and proves the left unit law. Shrinking a final pause proves the right unit law. For $f + f^-$ the explicit retracing contraction is

$$K_u(t^1, t') = f\big((1-u)\min\{2t^1, 2-2t^1\}, t'\big).$$

At $u = 0$ it is the concatenation and at $u = 1$ it is the constant boundary value. For $f^- + f$, replace the first argument by $1-(1-u)\min\{2t^1, 2-2t^1\}$, ending at the other boundary face. This proves both inverse laws.

For $n \ge 2$, let $\star$ concatenate in the second coordinate. The same proof gives it the same identity, denoted 0 at the class level. Pasting four maps into quadrants gives the exact interchange identity shown in the figure:

$$(a+b)\star(c+d) = (a\star c)+(b\star d).$$

Here $a, b, c, d \in \pi_n(X, x_0)$. Its two specializations give

$$a+b = (a\star 0)+(0\star b) = (a+0)\star(0+b) = a\star b, \qquad a+b = (0\star a)+(b\star 0) = (0+b)\star(a+0) = b\star a.$$

The operations therefore agree and are commutative. ∎

The figure uses only the first two coordinates. The other coordinates stay fixed.

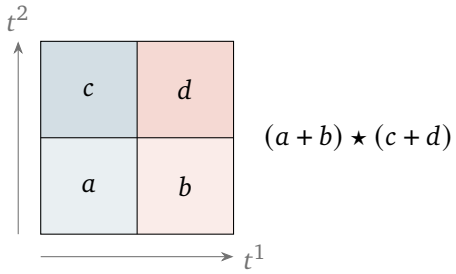


Figure 26.1: The four cube maps have the same placement whether one first joins horizontal pairs or first joins vertical pairs. All seam faces map to the basepoint.

For example, based maps $S^2 \to S^2$ add by joining two sphere maps at their basepoint. The next section identifies this group law with addition of degrees.

26.2 Sphere groups and the Hopf map

The degree from Lecture 14 is invariant under homotopy. The converse for sphere maps is an additional theorem: equal degrees imply based homotopy. We cite it here rather than infer it from the integral formula alone.

Theorem 26.2 (Homotopy groups of spheres · *cited*). *For every $n \ge 1$, the degree map gives $\pi_n(S^n) \cong \mathbb{Z}$. For $1 \le k < n$, the group $\pi_k(S^n)$ vanishes. The group $\pi_3(S^2)$ is infinite cyclic and is generated by the Hopf map $S^3 \to S^2$ of Lecture 20.*

The precise sources are Hatcher, Corollary 4.9, p. 349, Corollary 4.25, p. 361, and Example 4.51, p. 380. Hatcher's homological degree is the integer multiplier d on $H_n(S^n;\mathbb{Z}) \cong \mathbb{Z}$. This definition is given in §2.2, p. 134. This multiplier does not depend on a choice of generator. For smooth maps, Lecture 14's cited de Rham identification intertwines this action with pullback of forms: integrating a form over a composed simplex equals integrating its pullback over that simplex. Thus the homological degree agrees with the integral degree of Theorems 14.2 and 14.6. To verify this, let c be the period of a normalized volume form ν on a generator $[S]$. Its de Rham class is nonzero since $\int_{S^n} \nu = 1$, so injectivity of the pairing gives $c \neq 0$. The top-cohomology argument in Lecture 14 gives $[f^*\nu] = (\deg f)[\nu]$. Naturality and the vanishing of exact-form periods therefore give

$$dc = \int_{f_*[S]} \nu = \int_{[S]} f^*\nu = (\deg f)c.$$

Cancellation of c identifies the two degrees. For continuous maps, Hatcher's definition uses homology directly. Thus the identity sphere map has charge +1 and generates $\pi_n(S^n)$. Maps from a smaller-dimensional sphere cannot carry such a charge. The Hopf result also shows that higher homotopy groups need not vanish when the domain sphere is larger.

We will also use the covering-space result in Hatcher, Proposition 4.1, p. 342: a covering induces an isomorphism on π_n for $n \geq 2$. Sphere maps and their homotopies lift, which explains this result. The contractible cover $\mathbb{R} \to S^1$ gives $\pi_n(S^1) = 0$ for $n \geq 2$. The cover $S^2 \to \mathbb{R}\mathrm{P}^2$ gives $\pi_2(\mathbb{R}\mathrm{P}^2) \cong \mathbb{Z}$ after choosing a lifted basepoint. That choice will matter when the basepoint is allowed to move.

26.3 The fibration sequence and two consequences

A *fibration* is a continuous map $p\colon E \to B$ with the homotopy lifting property: for every space Y, a homotopy $Y \times [0,1] \to B$ with a chosen lift at its initial time has a continuous lift throughout that restricts to the chosen lift at time zero. For the theorem below it suffices to require this property for disks of every dimension. Fibre bundles have this property for disks by Hatcher's Proposition 4.48, p. 379. Choose a basepoint $b_0 \in B$ and a point e_0 in its fibre $F = p^{-1}(b_0)$. All homotopy groups below use these compatible basepoints.

Theorem 26.3 (The fibration sequence · *cited*)**.** *For a fibration over a path-connected base, the homotopy groups have a long exact sequence. Its relevant segment is*

$$\cdots \longrightarrow \pi_2(F) \longrightarrow \pi_2(E) \longrightarrow \pi_2(B) \xrightarrow{\partial_{\mathrm{fib}}} \pi_1(F) \longrightarrow \pi_1(E) \longrightarrow \pi_1(B) \longrightarrow \cdots .$$

Exactness means that the image of each arrow is the kernel of the next.

This is Hatcher's Theorem 4.41, p. 376. The symbol ∂_{fib} names the connecting homomorphism, distinct from a geometric boundary. Represent a based sphere by a square with its boundary collapsed. Lifting this square leaves a boundary loop in the fibre. Its construction and the general exactness proof are cited.

For the Hopf bundle, $F = S^1$, $E = S^3$ and $B = S^2$. The cited sphere groups give $\pi_2(S^3) = 0$ and $\pi_1(S^3) = 0$. The figure isolates the resulting segment.

$$0 = \pi_2(S^3) \longrightarrow \pi_2(S^2) \xrightarrow[\cong]{\partial_{\mathrm{fib}}} \pi_1(S^1) \longrightarrow \pi_1(S^3) = 0$$

Figure 26.2: Exactness makes the Hopf connecting map an isomorphism: its kernel is zero and its image is the entire fibre-loop group.

Hence $\pi_2(S^2) \cong \pi_1(S^1) \cong \mathbb{Z}$. This is a consequence of the cited sequence, not a second proof of that sequence.

More generally, let G be a simply connected Lie group with $\pi_2(G) = 0$ and let H be a closed Lie subgroup. Lecture 20 gives the principal bundle $H \to G \to G/H$. Exactness gives $\pi_2(G/H) \cong \pi_1(H)$ because both adjacent groups from G vanish. Here the basepoint of H is its identity, so its loops lie in its identity component. The assumption $\pi_2(G) = 0$ is explicit. Simple connectedness by itself is not this assumption.

26.4 Order parameters and moving basepoints

An *order parameter* records the local ordered state of a medium. Its allowed ordered values form the *order-parameter space* $\mathcal{M}$. When a symmetry group G acts transitively on the uniform ordered states, Lecture 8 identifies this space with G/H, where H fixes a reference state. The relevant descriptions in this lecture are a superfluid phase in S^1, a unit ferromagnetic direction in S^2, and a nematic director in $\mathbb{R}\mathrm{P}^2$. A director is an unoriented unit axis: $\mathbf{n}$ and $-\mathbf{n}$ describe the same state. The word *texture* means a continuous spatial configuration of the order parameter.

For an ordinary based charge, a chosen point of the domain sphere must have a prescribed image x_0. A deformation of the physical texture can move that image. We therefore need the effect of moving the basepoint. Let $\gamma\colon [0,1] \to \mathcal{M}$ run from x_0 to x_1. Rescale the domain of a cube map based at x_1 onto a smaller concentric cube. On each radial segment of the surrounding shell, insert γ, running from x_0 at the outer boundary to x_1 at the inner boundary. The figure depicts this construction.

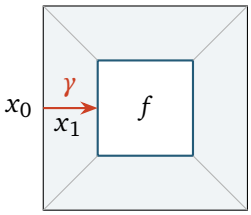


Figure 26.3: The inner cube carries a map based at x_1. The shell changes its basepoint to x_0 by the path γ. The outer boundary maps constantly to x_0.

Proposition 26.4 (The basepoint action · *sketched*)**.** *The shell construction gives an isomorphism $\beta_\gamma : \pi_n(\mathcal{M}, x_1) \to \pi_n(\mathcal{M}, x_0)$. For loops at x_0, it gives an action of $\pi_1(\mathcal{M}, x_0)$ on $\pi_n(\mathcal{M}, x_0)$. For $n = 1$ this is conjugation.*

Sketch. A homotopy of the path or inner map gives a homotopy of the shell map. Reversing the path gives the inverse isomorphism, since the retraced shell contracts. Two nested shells concatenate their paths. Consequently $\beta_{\gamma\eta} = \beta_\gamma \beta_\eta$ in the first-path-first convention. Joining inner cube maps shows that the induced maps preserve the group operation. For $n = 1$ the shell map is the path $\gamma f \gamma^{-1}$, as in Lecture 3. These constructions give the asserted action. They are described in Hatcher, pp. 341–342. ∎

Even though π_n is abelian for $n \geq 2$, this external action can be nontrivial. Commutativity of its own group law does not remove the basepoint issue.

26.5 Local defects and linking spheres

A *defect core* is a region where the chosen ordered description fails. Outside the cores the order parameter is a continuous map into $\mathcal{M}$. A small circle surrounding a line in three-dimensional space is a *linking circle*. A small sphere surrounding an isolated point is its *linking sphere*. The figure separates these two domains of the order-parameter map.

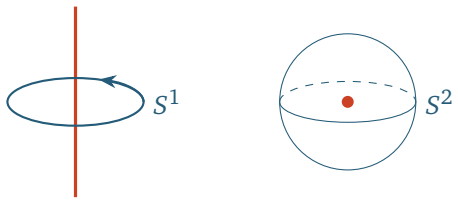


Figure 26.4: A circle links a line core. A sphere encloses a point core. The order parameter is required to remain in $\mathcal{M}$ on the linking domain throughout a deformation.

For a core of codimension c, the local transverse boundary is S^{c-1}. This explains which homotopy group occurs. The physical sphere's orientation is fixed. Reversing it can reverse an integer charge. A *topological defect* has a nontrivial linking homotopy class that obstructs removal of its core within $\mathcal{M}$.

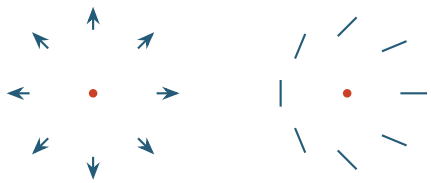
Theorem 26.5 (Local defect classes · *sketched*)**.** *Let $\mathcal{M}$ be a path-connected order-parameter manifold. Classify restrictions to a linking domain by continuous homotopy through maps into $\mathcal{M}$. Without a fixed image basepoint, line defects in three dimensions have the conjugacy classes of $\pi_1(\mathcal{M})$, and point defects have the orbits of $\pi_2(\mathcal{M})$ under the $\pi_1(\mathcal{M})$ action. A two-dimensional texture with a fixed uniform value at spatial infinity has a based class in $\pi_2(\mathcal{M})$.*

Sketch. The first assertions concern free homotopy classes of maps from S^1 and S^2. To convert a free homotopy to a based one, follow the moving image of the chosen domain point. Its path is precisely the shell change of basepoint. Thus representatives of two based classes are freely homotopic exactly when this action identifies them. For loops the action is conjugation, recovering Lecture 3's full loop result. For spheres this is its higher-dimensional version (Mermin, §VII.D, p. 631). For the texture, adding the point at infinity compactifies the oriented plane to S^2. The fixed limiting value supplies its target basepoint. This gives the stated based class. ∎

If the value at infinity is also allowed to move, the texture likewise needs action orbits. A nonzero linking class obstructs extension across the core within $\mathcal{M}$: an extension to the transverse disk would contract the boundary map radially. A zero class admits such an extension because a representative map is null-homotopic. This is a topological obstruction, not a claim about the energy or time needed to remove the core. The local boundary classes do not classify knotting or all configurations with several defects.

26.6 Vortices, nematics and magnetic textures

The figure compares an oriented phase field with an unoriented director field on a linking circle.



$e^{i\phi}$ $\qquad$ $[\cos(\phi/2) : \sin(\phi/2) : 0]$

Figure 26.5: The phase turns once on a superfluid linking circle. The nematic axis turns through a half-turn and already returns to the same unoriented axis.

Example 26.6 (Three order-parameter spaces)**.** Determine the local line and point classes for a superfluid, a uniaxial nematic and a Heisenberg ferromagnet.

1. A superfluid has a phase $e^{i\alpha} \in S^1$. On an oriented linking circle a vortex of winding $n \in \mathbb{Z}$ is represented by $\phi \mapsto e^{in\phi}$ for $0 \le \phi \le 2\pi$. Lecture 3 gives $\pi_1(S^1) = \mathbb{Z}$. This group is abelian, so its conjugacy classes are its individual integers. Based joining adds windings, which is also the local fusion rule for these free classes. Two vortices of winding +1 have total winding 2 and cannot cancel within S^1. Opposite windings have total zero and have no such obstruction. The group $\pi_2(S^1)$ vanishes, so there is no corresponding point charge.
2. A uniaxial nematic has $\mathcal{M} = \mathbb{R}\mathrm{P}^2$. Lecture 4 gives $\pi_1(\mathbb{R}\mathrm{P}^2) = \mathbb{Z}_2$. The nontrivial line defect is called a *disclination*. A representative director is $[\cos(\phi/2) : \sin(\phi/2) : 0]$. Its vector lift starts at a unit vector and ends at its negative. The director nevertheless closes, since antipodal vectors give the same axis. Joining two such loops gives a closed full-turn vector loop on S^2. That loop contracts because S^2 is simply connected. Thus the two nontrivial disclinations have total class zero and can cancel topologically. The chosen sphere cover gives $\pi_2(\mathbb{R}\mathrm{P}^2) \cong \mathbb{Z}$. Changing the lifted basepoint around the nontrivial director loop applies the antipodal deck map. That map has degree -1 on S^2, so its action sends n to $-n$. Hence free point-defect classes identify opposite integers. They can be labelled by $|n|$, including zero. There is no single signed addition law on these orbits without the joining-path data. This is the nematic qualification in Mermin, p. 631.
3. A Heisenberg ferromagnet has $\mathcal{M} = S^2$. The fundamental group of this target is zero, so there is no local line obstruction of this type. Its point classes are the signed integers $\pi_2(S^2) = \mathbb{Z}$, with no nontrivial basepoint action. The *hedgehog* is the field $\mathbf{n}(\mathbf{r}) = \mathbf{r}/|\mathbf{r}|$ outside the origin. Its restriction to an outward linking sphere is the identity direction map and has degree +1.
   A *skyrmion texture* is a nonsingular plane texture with a nonzero sphere degree and a fixed uniform value at infinity. For $a > 0$, an explicit degree-one example is

$$\mathbf{n}(x, y) = \frac{(2ax, 2ay, a^2 - x^2 - y^2)}{a^2 + x^2 + y^2}, \qquad \mathbf{n} \cdot (\partial_x \mathbf{n} \times \partial_y \mathbf{n}) = \frac{4a^2}{(a^2 + x^2 + y^2)^2}.$$

   Its norm is one. It tends to the south pole at infinity. For the density calculation, write the field's numerator as $\mathbf{d}$ and denominator as $D = a^2 + x^2 + y^2$. Differentiating

$\mathbf{n} = \mathbf{d}/D$ gives

$$\mathbf{n} \cdot (\partial_x \mathbf{n} \times \partial_y \mathbf{n}) = \frac{\mathbf{d} \cdot (\partial_x \mathbf{d} \times \partial_y \mathbf{d})}{D^3}, \qquad \mathbf{d} \cdot (\partial_x \mathbf{d} \times \partial_y \mathbf{d}) = 4a^2 D.$$

The terms proportional to $\mathbf{d}$ in either derivative drop out of the determinant. In inversion coordinates $X = x/(x^2 + y^2)$ and $Y = y/(x^2 + y^2)$, the same texture is

$$\mathbf{n} = \frac{(2aX, 2aY, a^2(X^2 + Y^2) - 1)}{1 + a^2(X^2 + Y^2)}.$$

It extends smoothly to $X = Y = 0$, the point at infinity. Its degree is

$$\frac{1}{4\pi} \int_{\mathbb{R}^2} \frac{4a^2 \mathrm{d}x\mathrm{d}y}{(a^2 + x^2 + y^2)^2} = \int_0^\infty \frac{2a^2 r \mathrm{d}r}{(a^2 + r^2)^2} = 1.$$

This is a texture, with no singular core in the plane. The degree labels its topology. No energy-minimizing field equation has been imposed.

Self-check. An ordered medium has $\mathcal{M} = \mathrm{SO}(3)$. Find its local line classes and the result of joining two nontrivial lines.

Answer. Lecture 4 gives $\pi_1(\mathrm{SO}(3)) = \mathbb{Z}_2$. It is abelian, so there are two free line classes. The sum of two nontrivial classes is zero, removing this local topological obstruction. This does not assert that a particular physical evolution makes the defects annihilate.

Summary. The groups π_n classify based sphere maps and are abelian for $n \geq 2$. The cited sphere results and fibration sequence compute the groups used here. Local free defect classes still require conjugacy for loops and the fundamental-group action for spheres. Superfluid vortices retain signed integer winding. Nematic disclinations have $\mathbb{Z}_2$ charge. Free nematic point charges identify n with $-n$. Sphere-valued hedgehogs and based plane textures have signed degree. Lecture 27 adds boundary conditions and energy to this topological classification.

Reading. Hatcher, §4.1, pp. 340–343 and 349; §4.2, Corollary 4.25, p. 361, Theorem 4.41, p. 376, Proposition 4.48, p. 379, and Example 4.51, p. 380. Mermin, *The topological theory of defects in ordered media*, §VII.D–E, pp. 630–632. Nakahara, §4.8–4.9.

Lecture 27

Solitons and topological charge

Objectives. *Lecture 26 classified maps into an order-parameter space. A field theory also assigns an energy to each configuration. We separate the boundary conditions that define a topological sector from the field equations that select stationary configurations. The kink supplies a complete energy-bound calculation. The gauge vortex shows how a phase winding becomes a quantized magnetic flux.*

By the end of this lecture you will be able to:

- Identify a sector from its stated continuous vacuum boundary data.
- Derive the kink bound and its saturating profile with the signed charge.
- Prove Derrick's obstruction for the stated linear-target scalar energy, including $d = 2$.
- Derive vortex flux quantization under the stated asymptotic conditions.

27.1 Vacuum boundary data and sectors

Recall that a classical field configuration assigns a value to each spatial point. For a scalar field $\phi\colon \mathbb{R}^d \to \mathbb{R}^N$, take a smooth potential $V \geq 0$. Its zero set $\mathcal{V} = V^{-1}(0)$ is the *vacuum manifold* when it is a manifold. A constant field in $\mathcal{V}$ has zero potential and gradient energy. A *static soliton* is a spatially localized, finite-energy stationary solution of the field equations. The boundary conditions below specify the allowed behaviour far from its localized region. A configuration carrying a nonzero topological charge is not automatically such a solution.

In one dimension, assume limits ϕ_- and ϕ_+ in $\mathcal{V}$ at $-\infty$ and $+\infty$. Call their path components $\mathcal{V}_-$ and $\mathcal{V}_+$. In dimensions two and three, assume a continuous angular limit $\phi_\infty\colon S^{d-1} \to \mathcal{V}$, with convergence uniform in direction. A *soliton sector* fixes the corresponding boundary homotopy data. For an evolving field, require that its limiting boundary map be jointly continuous in angle and time. In one dimension require each endpoint limit to be continuous in time. These are boundary assumptions. Finite energy alone does not imply them. The figure preserves the order of the one-dimensional endpoints.



Figure 27.1: The left and right vacuum components form an ordered pair. Interchanging the two ends can reverse a kink's charge.

Proposition 27.1 (Boundary sectors · *proved*)**.** *Under these boundary conditions, the one-dimensional datum is the ordered pair of path components of $\mathcal{V}$ containing ϕ_- and ϕ_+. In dimensions two and three it is the free homotopy class of ϕ_∞. An evolution with jointly continuous vacuum boundary data preserves these data.*

Proof. At either end in one dimension, the time-dependent limiting value is a path in $\mathcal{V}$. It cannot leave its path component. The two ends remain distinguished, giving an ordered pair. For $d = 2$ or $d = 3$, the jointly continuous boundary map is exactly a homotopy of maps $S^{d-1} \to \mathcal{V}$. Its free homotopy class is therefore fixed. Lecture 26 expresses these classes using conjugacy or the fundamental-group action when needed. ∎

A *topological charge* is an invariant used to label such a sector. An integer charge exists in the examples below, but the general sector datum need not be an integer. It constrains allowed deformations. Energy and the field equations determine whether a soliton exists.

27.2 The scalar kink and the Bogomolny bound

Take one real field on the line and constants $\mu, v > 0$. Define the superpotential W, potential V and static energy by

$$W(\phi) = \mu\left(v^2\phi - \frac{\phi^3}{3}\right), \qquad V(\phi) = \frac{1}{2}\big(W'(\phi)\big)^2, \qquad E = \int_{\mathbb{R}} \left(\frac{1}{2}(\phi')^2 + V(\phi)\right) \mathrm{d}x.$$

Here W' differentiates with respect to the field value, while ϕ' differentiates with respect to x. The vacua are $-v$ and $+v$. Assume the specified vacuum limits at both ends. The figure shows the increasing kink profile centred at zero.

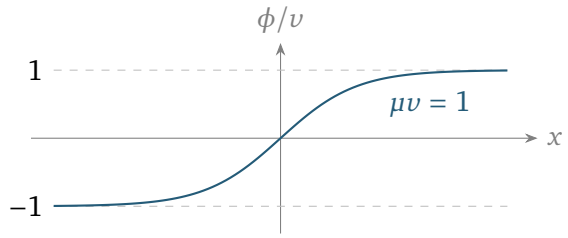


Figure 27.2: The exact kink $\phi/v = \tanh(\mu v x)$ joins $-v$ to $+v$. The increasing profile has charge +1.

Proposition 27.2 (The kink bound · *proved*)**.** *Every smooth finite-energy configuration with these limits satisfies $E \geq |W(\phi_+) - W(\phi_-)|$. In the sector $\phi_- = -v$, $\phi_+ = v$, equality is attained by*

$$\phi_{\mathrm{kink}}(x) = v\tanh\big(\mu v(x - x_0)\big), \qquad x_0 \in \mathbb{R}, \qquad E_{\mathrm{kink}} = \frac{4\mu v^3}{3}.$$

Its charge $Q = (\phi_+ - \phi_-)/(2v)$ is +1. The decreasing antikink has charge -1 and the same energy.

Proof. For either sign $s \in \{+1, -1\}$, completing the square gives

$$E = \frac{1}{2}\int_{\mathbb{R}} (\phi' - sW'(\phi))^2 \mathrm{d}x + s\big(W(\phi_+) - W(\phi_-)\big).$$

The last term follows from the chain rule on a finite interval and then the stated limits. Choose s so its product with this difference is nonnegative. The square is nonnegative, proving the bound. Equality requires $\phi' = sW'(\phi)$, the *Bogomolny equation*.

For the increasing sector, this equation is $\phi' = \mu(v^2 - \phi^2)$. Separating variables on $-v < \phi < v$ gives $\operatorname{artanh}(\phi/v) = \mu v(x - x_0)$, hence the displayed profile. Its derivative

is $\mu v^2 \operatorname{sech}^2(\mu v(x - x_0)) = W'(\phi)$. It has the required endpoint limits, so it saturates the bound. Direct endpoint evaluation gives $W(v) - W(-v) = 4\mu v^3/3$. Differentiating the first-order equation gives $\phi'' = W''W' = V'(\phi)$, the static field equation. Thus this saturating configuration is a solution, not only a trial field. Changing its sign gives the antikink and reverses the ordered vacuum pair and charge. ∎

The term *Bogomolny bound* means an energy bound fixed by boundary topological data. Equal endpoint vacua give charge zero. The bound then permits the vacuum configuration. The square completion proves a global energy minimum in each kink sector. It does not imply that every configuration in that sector is stationary.

27.3 Derrick's scaling obstruction

For a smooth linear-target field $\phi\colon \mathbb{R}^d \to \mathbb{R}^N$, use

$$E = E_2 + E_0, \qquad E_2 = \frac{1}{2}\int_{\mathbb{R}^d} |\nabla\phi|^2 \mathrm{d}^d x, \qquad E_0 = \int_{\mathbb{R}^d} V(\phi)\mathrm{d}^d x, \qquad V \geq 0.$$

The subscript counts derivatives in the energy term. A spatial dilation is $\phi_\lambda(x) = \phi(\lambda x)$ for $\lambda > 0$. The chain rule contributes λ^2 to the gradient term. The change of variables contributes λ^{-d} to both terms. Thus

$$E(\lambda) = \lambda^{2-d}E_2 + \lambda^{-d}E_0, \qquad E'(1) = (2-d)E_2 - dE_0.$$

The figure shows this expression with $E_2 = E_0 = 1$.

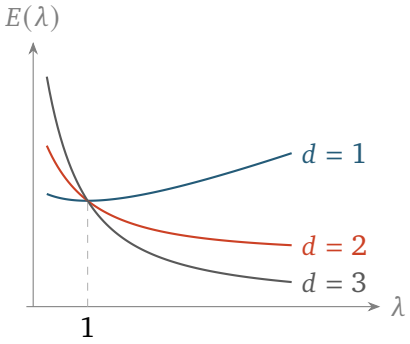


Figure 27.3: Dilation energy for the stated linear-target scalar model. Only the one-dimensional curve has a stationary point for positive E_2 and E_0.

Proposition 27.3 (Derrick's theorem · *proved*). *Let $V\colon \mathbb{R}^N \to [0,\infty)$ be smooth. Every smooth finite-energy static solution $\Delta\phi = \nabla V(\phi)$ of this model on $\mathbb{R}^d$, for $d \geq 2$, is a constant vacuum.*

Proof. **Step 1: the stress identity.** First justify the stationary scaling identity without assuming a variation with noncompact support is admissible. Use the Euclidean metric to raise indices and let $e = |\nabla\phi|^2/2 + V(\phi)$. The stress tensor and its divergence are

$$T^i{}_j = \partial^i\phi \cdot \partial_j\phi - \delta^i_j e, \qquad \partial_i T^i{}_j = (\Delta\phi - \nabla V(\phi)) \cdot \partial_j\phi = 0.$$

To see the cancellation, differentiate the first term by the product rule. Its second-derivative term $\partial^i\phi \cdot \partial_i\partial_j\phi$ is the derivative of $|\nabla\phi|^2/2$ and cancels that part of $\partial_j e$. The remaining potential derivative is $\nabla V \cdot \partial_j\phi$.

Step 2: the cutoff limit. Fix one smooth bump χ from Lecture 21, equal to one on the unit ball and zero outside the radius-two ball. Put $\chi_R(x) = \chi(x/R)$. The chain rule gives $|\nabla\chi_R| \le C/R$ with $C = \sup|\nabla\chi|$. Integrate the divergence of $T^i{}_j x^j \chi_R$ and use compact support to obtain

$$\int_{\mathbb{R}^d} \chi_R T^i{}_i \mathrm{d}^d x = -\int_{\mathbb{R}^d} T^i{}_j x^j \partial_i \chi_R \mathrm{d}^d x.$$

The absolute value of the right side is bounded by a constant times $\int_{R<|x|<2R}(|\nabla\phi|^2 + V)\mathrm{d}^d x$, which tends to zero by finite energy. The trace is integrable, so dominated convergence on the left gives $(2-d)E_2 - dE_0 = 0$. For $d > 2$, nonnegativity forces $E_2 = E_0 = 0$, hence a constant vacuum.

Step 3: dimension two. For $d = 2$ the identity gives $E_0 = 0$. Continuity and nonnegativity imply $V(\phi(x)) = 0$ everywhere. Each such field value minimizes V, so $\nabla V(\phi(x)) = 0$. The field equation becomes $\Delta\phi = 0$. Let u be any component of a derivative $\partial_i\phi$. It is harmonic and square-integrable. Its spherical mean about any point p is independent of radius: its derivative is the normalized outward flux of ∇u, zero by Stokes and $\Delta u = 0$. The mean tends to $u(p)$ at radius zero, so averaging the sphere means radially gives the ball mean. Cauchy–Schwarz therefore yields

$$|u(p)| = \left|\frac{1}{|B_R|}\int_{B_R(p)} u\,\mathrm{d}^d x\right| \le |B_R|^{-1/2}\|u\|_{L^2(\mathbb{R}^d)} \longrightarrow 0.$$

Thus all derivatives vanish and ϕ is a constant vacuum in this case too. ∎

A constrained sphere-valued sigma model is outside the linear-target hypothesis: its field equation is not $\Delta\phi = \nabla V$ for a potential on all of $\mathbb{R}^N$. Gauge fields and higher-derivative energies also have different scaling powers. They need not satisfy this obstruction. Our dilation is $\phi(\lambda x)$, as in Manton–Sutcliffe, equations (4.17)–(4.19), pp. 84–85. Inverting the scale parameter changes the displayed derivative sign but not the obstruction.

27.4 The abelian Higgs vortex and its flux

A *Higgs field* here is a charged complex scalar $\phi\colon \mathbb{R}^2 \to \mathbb{C}$. Let $q \neq 0$ be its signed charge and put $a = q/\hbar$. Use a smooth real potential A_{em} on the plane and Lecture 21's derivative $D = \mathrm{d} - \mathrm{i}aA_{\mathrm{em}}$. Write $A_{\mathrm{em}} = A_x\mathrm{d}x + A_y\mathrm{d}y$ and $B = \partial_x A_y - \partial_y A_x$. For $v > 0$ and $\lambda > 0$, the static abelian Higgs energy is

$$E = \int_{\mathbb{R}^2}\left[\frac{1}{2}B^2 + \frac{1}{2}\left(|D_x\phi|^2 + |D_y\phi|^2\right) + \frac{\lambda}{8}\left(|\phi|^2 - v^2\right)^2\right]\mathrm{d}x\mathrm{d}y.$$

Its vacuum field values have modulus v, a circle. Write $r = \sqrt{x^2+y^2}$. For $r = R$, write $\phi_R(\theta) = \phi(R\cos\theta, R\sin\theta)$. The symbol D_θ means the covariant derivative along the angular parameter, so $D_\theta\phi_R = \partial_\theta\phi_R - \mathrm{i}a(A_{\mathrm{em}})_\theta\phi_R$. It includes the radius factor relative to a unit tangent derivative.

Theorem 27.4 (Vortex flux · *sketched; flux proved*)**.** *Assume B is integrable, $|\phi_R| \to v$ uniformly, and $\int_0^{2\pi}|D_\theta\phi_R|\mathrm{d}\theta \to 0$ as $R \to \infty$. Let n be the winding of $\phi_R/|\phi_R|$ for large R. The total flux is*

$$\Phi = \int_{\mathbb{R}^2} B\mathrm{d}x\mathrm{d}y = \frac{2\pi n}{a} = \frac{2\pi\hbar n}{q}.$$

At critical coupling $\lambda = a^2$, finite-energy fields in this sector obey $E \ge \pi v^2|n|$.

Flux proof. The amplitude is bounded below by $v/2$ outside a sufficiently large disk. Thus the nonzero fields on any two large circles are homotopic through the intervening annulus, and their winding n agrees. On each circle, the imaginary part of the logarithmic derivative gives

$$a \oint_{r=R} A_{\mathrm{em}} = 2\pi n - \int_0^{2\pi} \operatorname{Im}\left(\frac{D_\theta \phi_R}{\phi_R}\right) \mathrm{d}\theta.$$

The last integral tends to zero by the amplitude lower bound and the assumed angular derivative integral. Stokes on the disk gives $\oint A_{\mathrm{em}} = \int_{r\le R} B \mathrm{d}x\mathrm{d}y$. Integrability of B gives the total flux in the limit, proving the formula. ∎

These asymptotic conditions are part of the sector. Finite energy alone was not used to infer pointwise limits. The magnetic flux now has the same integer origin as Lecture 3's winding and Lecture 22's quantization. Gauge maps in this discussion are smooth on the whole plane. Their restriction to a circle contracts through the disk to the value at its centre. Its winding is therefore zero, so these gauge changes preserve n.

To sketch the bound, let $j_i = \operatorname{Im}(\bar{\phi} D_i \phi)$, for $i \in \{x, y\}$, be the components of a real boundary-current one-form, distinct from the physical current of Lecture 15. Set $\operatorname{curl} j = \partial_x j_y - \partial_y j_x$. On a radius-$R$ circle its circulation satisfies

$$\left| \oint_{r=R} j \right| \le \sup_\theta |\phi_R| \int_0^{2\pi} |D_\theta \phi_R| \mathrm{d}\theta \longrightarrow 0.$$

The sector assumptions therefore remove this boundary term. Using $[D_x, D_y]\phi = -\mathrm{i}aB\phi$ gives $\operatorname{curl} j = 2\operatorname{Im}(\overline{D_x \phi} D_y \phi) - aB|\phi|^2$. For either sign $s \in \{+1, -1\}$, the resulting square completion at $\lambda = a^2$ is

$$E = \frac{1}{2} \int_{\mathbb{R}^2} \left[|(D_x + \mathrm{i}sD_y)\phi|^2 + \left(B + \frac{sa}{2}(|\phi|^2 - v^2) \right)^2 \right] \mathrm{d}x\mathrm{d}y + \frac{sav^2}{2}\Phi.$$

Obtain this identity first on disks using Stokes for the curl term. The boundary-current circulation tends to zero by the estimate above. Finite energy and integrability of the magnetic field give the limit. Choosing $s = \operatorname{sgn}(n)$ for $n \neq 0$ gives the bound. Solutions of the two first-order equations obtained by setting both squares to zero saturate the bound. Their existence requires an analytic input. Away from the zeros of ϕ, these first-order equations reduce to one nonlinear elliptic equation for $\log|\phi|^2$. The prescribed zeros supply its point-source conditions. Solving that boundary problem constructs vortices. The analytic existence step is cited from Manton–Sutcliffe, §7.8, pp. 199–200. Their normalization is $a = v = 1$, with critical $\lambda = 1$ and bound $\pi|n|$. In general their connection potential is aA_{em}. This potential rescaling alone does not equate their unit-coupling energy with the general-a energy printed here.

The figure shows a positive winding with $a > 0$, fixing the circulation arrows.

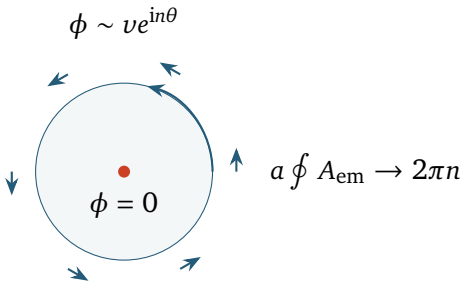


Figure 27.4: The asymptotic Higgs phase winds counterclockwise. For $a > 0$ and $n = 1$, its covariantly constant limit requires positive potential circulation and flux.

Example 27.5 (From asymptotic phase to flux)**.** Assume $\phi_R \to ve^{in\theta}$ with its first angular derivative, and $D_\theta \phi_R \to 0$ uniformly. The limiting derivative equation then gives $(A_{\mathrm{em}})_\theta \to n/a$ in this gauge. With B integrable, the flux proof needs only the integrated angular condition, and gives $\Phi = 2\pi n/a$. A negative signed charge reverses the magnetic flux for the same positive winding.

For a concrete smooth configuration of winding one, use

$$\phi(x,y) = \frac{v(x+\mathrm{i}y)}{\sqrt{1+x^2+y^2}}, \qquad A_{\mathrm{em}} = \frac{-y\mathrm{d}x + x\mathrm{d}y}{a(1+x^2+y^2)}, \qquad B = \frac{2}{a(1+x^2+y^2)^2}.$$

It has a Higgs zero at the origin and no singular potential there. On a radius-R circle, $\phi_R = vRe^{\mathrm{i}\theta}/\sqrt{1+R^2}$ and $(A_{\mathrm{em}})_\theta = R^2/[a(1+R^2)]$. Thus $D_\theta \phi_R = \mathrm{i}\phi_R/(1+R^2)$ tends uniformly to zero. The flux is

$$\Phi = \frac{4\pi}{a}\int_0^\infty \frac{r\mathrm{d}r}{(1+r^2)^2} = \frac{2\pi}{a}.$$

Direct differentiation of the displayed fields gives

$$D_x\phi = \frac{v}{(1+r^2)^{3/2}}, \qquad D_y\phi = \frac{\mathrm{i}v}{(1+r^2)^{3/2}}.$$

Thus the sum of the squared norms of these two derivatives is $2v^2/(1+r^2)^3$. Substitution of $t = 1+r^2$ makes the radial integrals of $r/(1+r^2)^p$ equal to $1/[2(p-1)]$, for $p \in \{2,3,4\}$. Each energy density term is integrable. The total energy is

$$E = \pi\left[\frac{2}{3a^2} + \frac{v^2}{2} + \frac{\lambda v^4}{8}\right].$$

This field demonstrates the finite-energy flux sector. It is a trial configuration, not an asserted solution of the vortex field equations.

27.5 The monopole charge

Remark 27.6 (The Yang–Mills–Higgs monopole)**.** In the SU(2) Yang–Mills–Higgs model, an adjoint Higgs field is a vector in the Lie algebra that transforms by conjugation. Its fixed-norm vacuum directions form $\mathrm{SU}(2)/\mathrm{U}(1) \cong S^2$. With a continuous nonzero Higgs limit at infinity, the normalized field defines a map from the outward spatial sphere to this vacuum sphere. Its degree is the topological monopole charge $Q \in \mathbb{Z}$, as in Lecture 26. Orient the Lie algebra by Lecture 23's basis T^1, T^2, T^3. The hedgehog with coefficients x^a/r in that basis restricts to the identity direction map between outward spheres, so it has $Q = +1$. The

remaining U(1) connection acts on charged line bundles at infinity. Lecture 22 then requires $q\Phi_{\mathrm{m}} = 2\pi\hbar n$ for each such charge convention. Relating that integer to the degree also requires specifying the chosen charged representation. The full 't Hooft–Polyakov ansatz and energy bound are on Sheet 27 (Manton–Sutcliffe, §8.2).

Self-check. For the scalar kink model, list the possible values of $Q = (\phi_+ - \phi_-)/(2v)$ for vacuum endpoints.

Answer. The values are +1 for $(-v, +v)$, −1 for $(+v, -v)$ and zero for equal endpoints. The endpoint pair is ordered.

Summary. Continuous vacuum boundary data define sectors that an evolution with jointly continuous vacuum boundary data cannot change. Finite energy alone does not supply those limits. The scalar kink saturates $E \geq |\Delta W|$ and has signed charge +1. Derrick's theorem excludes nonconstant smooth finite-energy static solutions of the stated nonnegative-potential linear-target scalar model for $d \geq 2$. The gauge vortex lies outside that model and has flux $2\pi\hbar n/q$ under the stated asymptotic conditions. Its critical-coupling bound follows by square completion. Existence uses the cited analytic input. Lecture 28 uses a quantized sphere flux to count chiral quantum zero modes.

Reading. Manton–Sutcliffe, §4.2, pp. 84–85; §5.1–5.2, pp. 110–113; §7.3, pp. 164–165; §7.8, pp. 197–200; §8.2.

Lecture 28

The index on the sphere

Objectives. *Lecture 22 associated an integer with the flux through a sphere. We now use that integer to count quantum zero modes of a Dirac operator. The spin connection changes the line bundle in which one chiral component lives. On the round sphere this turns the count into a polynomial calculation. The cited index theorem explains why the difference of the two chiral counts is topological.*

By the end of this lecture you will be able to:

- Specify the two chiral components and the sign of their index.
- Count the holomorphic sections of $O(m)$ for every $m \in \mathbb{Z}$.
- Compute both monopole zero-mode counts using the stated cited identifications.
- Distinguish the Dirac count from the spinless Landau-level count and state the anomaly convention.

28.1 Spinors and the canonical line on the sphere

Use the unit round sphere with its outward orientation. Identify it with $\mathbb{C}\mathrm{P}^1$ using Lecture 20's complex coordinates z and $w = 1/z$. The complex orientation agrees with this outward orientation. A *spinor field* is a section of a spinor bundle, the geometric bundle on which the Dirac equation acts. Its construction is a cited input here (Lawson–Michelsohn, Chapter II and Appendix D). On an oriented surface it has two complex line components S^+ and S^-. Their labels specify *chirality*, a grading with eigenvalues +1 and −1. Chirality here is a spinor grading, distinct from even and odd differential-form degree.

The *canonical bundle* K_{S^2} is the line of holomorphic cotangent vectors. Its local frames are $\mathrm{d}z$ and $\mathrm{d}w$. Since $\mathrm{d}w = -z^{-2}\mathrm{d}z$, absorbing the constant minus sign into one frame gives $K_{S^2} \cong O(-2)$ in Lecture 22's frame convention. We choose the positive spinor line to be its square root:

$$S^+ = K_{S^2}^{1/2} \cong O(-1), \qquad (K_{S^2}^{1/2})^{\otimes 2} \cong K_{S^2}.$$

This square-root choice is part of the cited spinor identification. We do not construct spin structures in this lecture.

Recall that $L_n = O(n)$ is the monopole line of Chern number $n \in \mathbb{Z}$. Twisting means taking the tensor product with this charged line. Put $E_n = S^+ \otimes L_n \cong O(n-1)$. The full twisted spinor bundle has the cited description

$$S \otimes L_n \cong \left(\Lambda^{0,0} \oplus \Lambda^{0,1}\right) \otimes E_n.$$

Here a $(0,0)$-form is a function and a $(0,1)$-form is locally a multiple of $\mathrm{d}\bar{z}$. The positive part is E_n. The negative part is $\Lambda^{0,1} \otimes E_n$. The figure records the twist that will change the polynomial count.

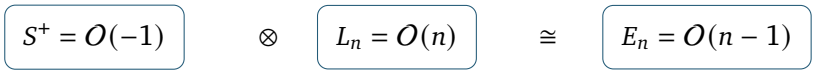


Figure 28.1: The positive spinor line has degree -1. Tensoring with the monopole line of degree n gives degree $n-1$.

These identifications are stated in Lawson–Michelsohn, Appendix D, equation (D.16), p. 396. Walpuski, Propositions 70.3 and 70.6, pp. 76–78, give the form decomposition and square-root construction explicitly.

28.2 The Dirac operator and its index

Write $\Omega^{0,k}(E_n)$ for the smooth E_n-valued $(0,k)$-forms. The *Dolbeault operator* $\bar\partial_A$ is the $(0,1)$ part of the covariant derivative. In a holomorphic frame it sends a coefficient f to $(\partial_{\bar z} f)\mathrm{d}\bar z$, as in Lecture 22. Its adjoint $\bar\partial_A^*$ is defined by the Hermitian L^2 inner products from the bundle metric and sphere area. The first slot of these inner products is antilinear. There are no $(0,2)$-forms on a complex curve.

Proposition 28.1 (The sphere Dirac operator · *sketched*)**.** *With the above chirality, the twisted Dirac operator is*

$$D = \sqrt{2}(\bar\partial_A + \bar\partial_A^*) = \begin{pmatrix} 0 & D^- \\ D^+ & 0 \end{pmatrix}, \qquad D^+ = \sqrt{2}\bar\partial_A, \qquad D^- = (D^+)^* = \sqrt{2}\bar\partial_A^*.$$

The block D^+ maps $\Omega^{0,0}(E_n)$ to $\Omega^{0,1}(E_n)$. The other block maps in the reverse direction.

Explanation. The cited spinor identification turns Clifford multiplication into exterior multiplication by an antiholomorphic covector and its adjoint contraction. Contracting the spin connection and monopole connection with it gives the displayed operator. This identification, including its $\sqrt{2}$ normalization, is Walpuski's Proposition 70.3. For a flat metric $\mathrm{d}x^2 + \mathrm{d}y^2$, the unit antiholomorphic coframe is $\mathrm{d}\bar z/\sqrt{2}$. Thus the positive block has coefficient $2\partial_{\bar z}$ and the negative block has coefficient $-2\partial_z$. Their product is $-4\partial_z\partial_{\bar z} = -\partial_x^2 - \partial_y^2$, the nonnegative scalar Laplacian of Lecture 15. To compare with the familiar flat Dirac equation, let σ_x and σ_y be the Pauli matrices. The corresponding self-adjoint spatial operator is $-\mathrm{i}(\sigma_x\partial_x + \sigma_y\partial_y)$. Multiplying its negative component by i gives the displayed chiral blocks. This verifies the local normalization and recalls the flat operator, not the global spinor construction. ∎

A *zero mode* is a smooth spinor annihilated by D. We use its self-adjoint realization on the closed sphere with the first-order Sobolev domain defined next. This is supplied by cited elliptic spectral theory (Lawson–Michelsohn, Theorem 5.8, pp. 196–197), with the domain identified by the elliptic estimate in Theorem 5.2. The first-order Sobolev domain consists of square-integrable sections whose first derivatives, understood by integration by parts, are also square-integrable in local charts. Its chiral block is *Fredholm*: between this domain and L^2 it has closed range and finite-dimensional kernel and cokernel. The cokernel is the target modulo the range, and is identified with $\ker D^-$ by the adjoint. These analytic facts follow from cited elliptic theory (Lawson–Michelsohn, Chapter III, Theorem 5.2, pp. 193–194). The *index* is the signed difference

$$\operatorname{ind} D^+ = \dim\ker D^+ - \dim\ker D^-.$$

Both dimensions are over $\mathbb{C}$. An index of zero alone does not say that either kernel vanishes.

Along a norm-continuous path of Fredholm chiral blocks from the fixed Sobolev domain to L^2, the index is constant by the cited Fredholm stability theorem (Lawson–Michelsohn, Proposition 7.1 and Corollary 7.4, pp. 201–202). Each kernel dimension can change, with equal changes in the two counts. The finite-dimensional model $T_t\colon \mathbb{C}^2 \to \mathbb{C}$, given by $T_t(a,b) = ta$, illustrates this: for $t \neq 0$ its kernel and adjoint kernel have dimensions 1 and 0. At $t = 0$ they have dimensions 2 and 1. Their difference stays one. This is a model for index stability, not a claim that these jumps occur within the special sphere family below. Perturbations that destroy the chiral block structure are outside this statement.

The square D^2 preserves chirality. Its positive-energy states pair across the two components: if $D^-D^+u = \epsilon u$ with $\epsilon > 0$, then $D^+u/\sqrt{\epsilon}$ is a normalized state of D^+D^- with the same eigenvalue. Its norm is one because $\|D^+u\|^2 = \langle u, D^-D^+u\rangle = \epsilon$ for $\|u\| = 1$. The reverse map is $D^-/\sqrt{\epsilon}$. The figure shows why a difference can remain only at zero.

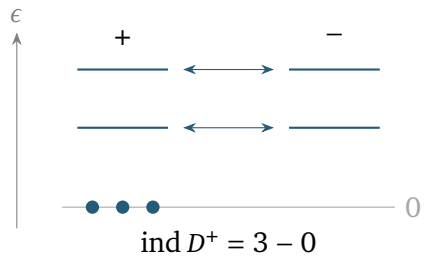


Figure 28.2: Positive squared-Dirac eigenvalues pair between chiralities. The schematic zero eigenspace has three positive modes and no negative mode, so its index is three.

28.3 Polynomial sections and monopole zero modes

For a holomorphic line E, define $H^0(E) = \ker \bar\partial_E$, the space of its global holomorphic sections. Define its Dolbeault cohomology by

$$H^1(E) = \Omega^{0,1}(E)/\bar\partial_E \Omega^{0,0}(E).$$

Every $(0,1)$-form is Dolbeault-closed in complex dimension one, explaining this quotient. The cited Dolbeault Hodge theorem identifies each class with a unique harmonic representative. In this dimension that representative is annihilated by $\bar\partial_E^*$. Consequently the cited spinor and Hodge inputs identify $\ker D^+ \cong H^0(E_n)$ and $\ker D^- \cong H^1(E_n)$ (Walpuski, Proposition 70.3, p. 77).

We now count H^0 directly, including the global condition at infinity. Let $m \in \mathbb{Z}$ and recall the frames $e_S = e_N z^m$ for $O(m)$. A section with coefficients f_N and f_S satisfies

$$f_S(w) = w^m f_N(1/w).$$

Both coefficients must be holomorphic at their own chart origin. The figure displays the monomial test for this condition.

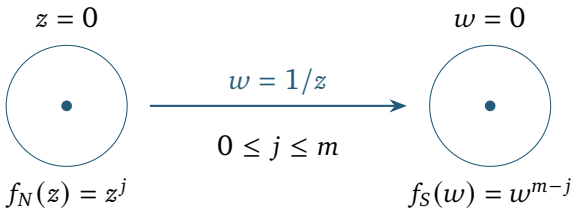


Figure 28.3: A northern monomial z^j becomes w^{m-j} in the southern frame. Holomorphicity at both poles requires $0 \le j \le m$.

Proposition 28.2 (Sections of a degree-m line · *proved given elementary Laurent uniqueness*)**.** *For $m \ge 0$, the holomorphic sections of $\mathcal{O}(m)$ have northern coefficients equal to the polynomials of degree at most m. Their dimension is $m+1$. For $m < 0$ there is no nonzero holomorphic section.*

Proof. The northern coefficient is an entire function, with Taylor expansion $f_N(z) = \sum_{j\ge 0} a_j z^j$ convergent for every $z \in \mathbb{C}$. The coefficient relation gives the Laurent expansion $f_S(w) = \sum_{j\ge 0} a_j w^{m-j}$ on a punctured neighbourhood of $w = 0$. Recall the elementary uniqueness of a convergent Laurent expansion: its coefficients are fixed by circle integrals, so a holomorphic extension has no negative powers. Thus $a_j = 0$ whenever $j > m$. For $m < 0$ this kills every coefficient. For $m \ge 0$ it leaves exactly $1, z, \dots, z^m$. Conversely each of these monomials gives a nonnegative southern power and hence a global section. Their distinct Taylor coefficients prove linear independence, so the dimension is $m+1$. ∎

To count the other chirality we use *Serre duality*, a cited theorem for compact complex curves, whose sphere case is $H^1(E) \cong H^0(K_{S^2} \otimes E^*)^*$. Here E^* is the dual line and the outer star denotes the dual vector space. See Walpuski, Remark 70.5, p. 77, or Gunning, *Function Theory on Compact Riemann Surfaces*, Theorem 1.17, p. 28. For $E = \mathcal{O}(m)$ this gives $\dim H^1(\mathcal{O}(m)) = \dim H^0(\mathcal{O}(-m-2))$. The polynomial result therefore determines both counts.

Theorem 28.3 (Monopole zero modes · *sketched*)**.** *On the round sphere with the monopole connection on $L_n = \mathcal{O}(n)$, for every $n \in \mathbb{Z}$,*

$$\dim\ker D^+ = \max(n,0), \qquad \dim\ker D^- = \max(-n,0), \qquad \operatorname{ind} D^+ = n.$$

The total number of zero modes is $|n|$. For $n > 0$ they all have positive chirality. For $n < 0$ they all have negative chirality. For $n = 0$ both kernels vanish for this sphere operator.

Proof. The positive component is $E_n = \mathcal{O}(n-1)$, so the polynomial count gives $\max(n,0)$. Serre duality identifies the negative count with $\dim H^0(\mathcal{O}(-n-1)) = \max(-n,0)$. Subtracting gives n for every integer n. Adding gives $|n|$. The global kernel identifications are the cited analytic steps in this derivation. ∎

Example 28.4 (Three zero modes in a monopole field)**.** Take the sphere monopole with $n = 3$ and the nonnegative Pauli operator $H = D^2$. An overall positive physical energy scale would not change its zero modes. Find its zero-energy states and their chirality. For a smooth spinor Ψ, the identity $\langle \Psi, D^2\Psi\rangle = \|D\Psi\|^2$ shows that $\ker H = \ker D$. The theorem gives three positive modes and no negative mode. Here the positive line has $m = n - 1 = 2$.

For a general $m \geq 0$, use Lecture 22's unit frame for $O(m)$. The local positive zero-mode equation and its monomial solutions are

$$\left(\partial_{\bar{z}} + \frac{mz}{2(1+|z|^2)}\right)\psi = 0, \qquad \psi_j(z) = (1+|z|^2)^{-m/2} z^j, \quad 0 \leq j \leq m.$$

Differentiating the weight in ψ_j gives $-mz\psi_j/[2(1+|z|^2)]$, which cancels the connection term. These are unit-frame coefficients of sections of E_n. They are not coefficients relative to $\mathrm{d}z^{1/2}$ or to a coordinate density. The spin connection has already entered through the degree shift from n to m. For $m = 2$ the northern coefficients are

$$\frac{1}{1+|z|^2}, \qquad \frac{z}{1+|z|^2}, \qquad \frac{z^2}{1+|z|^2}.$$

Their southern holomorphic coefficients are w^2, w and 1, respectively, so each section is holomorphic at the south pole.

For completeness, their squared norms use the round area $\mathrm{d}A = 4\mathrm{d}x\mathrm{d}y/(1+|z|^2)^2$, with $z = x + \mathrm{i}y$. The angular integral makes distinct monomials orthogonal. With $r = |z|$, the three radial norms are

$$8\pi \int_0^\infty \frac{r^{2j+1}}{(1+r^2)^4}\mathrm{d}r = \begin{cases} 4\pi/3, & j = 0, 2, \\ 2\pi/3, & j = 1. \end{cases}$$

Substitution $t = r^2$ reduces these to the integrals of $4\pi t^j/(1+t)^4$. Next put $u = (1+t)^{-1}$. The three integrals divided by 4π become $\int_0^1 u^2\mathrm{d}u = 1/3$, $\int_0^1 u(1-u)\mathrm{d}u = 1/6$ and $\int_0^1 (1-u)^2\mathrm{d}u = 1/3$, giving the displayed values. Dividing each coefficient by the square root of its stated squared norm gives an orthonormal basis. All three are zero-energy states of positive chirality.

The *lowest Landau level* is the lowest scalar magnetic-kinetic eigenspace. Haldane's spinless sphere model has $n+1$ lowest-level states for a nonnegative monopole magnitude n. A consulted derivation is Greiter, equations (2) and (20)–(21). His magnitude is $2s_0$, and the lowest level has angular momentum s_0, hence $2s_0 + 1$ states. His electron charge is negative. Here this comparison uses the flux magnitude, not a borrowed signed Chern convention. Its holomorphic description uses $O(n)$, rather than the spinor line $O(n-1)$. The corresponding polynomial count is $n+1$, giving four states when $n = 3$. The spin connection accounts for this one-state difference. This comparison concerns degeneracy. The scalar lowest Landau energy is not being set equal to the Dirac zero energy.

28.4 The surface index theorem

The polynomial count determines the left side of an index theorem for this sphere connection. The right side is the flux integer already computed in Lecture 22.

Theorem 28.5 (Atiyah–Singer on a surface · *cited*)**.** *Let M be a closed oriented Riemannian spin surface and L a Hermitian line bundle with unitary connection of curvature F_{conn}. Choose positive spinors in the square root of the canonical line for the chosen spin structure. For the corresponding chiral twisted Dirac operator,*

$$\operatorname{ind} D^+ = \int_M c_1(L) = \frac{\mathrm{i}}{2\pi}\int_M F_{\mathrm{conn}}.$$

This is the surface specialization of the twisted Dirac index theorem (Nakahara, equations (12.85)–(12.86)). The curvature here is anti-Hermitian, as in Lecture 22. For the sphere monopole, $F_{\mathrm{conn}} = -\mathrm{i}n \sin\theta \, \mathrm{d}\theta \wedge \mathrm{d}\varphi/2$. Its curvature integral gives $\mathrm{i}(-2\pi\mathrm{i}n)/(2\pi) = n$, agreeing with the kernel count. The theorem applies beyond this symmetric monopole connection. It fixes the difference of the kernels under smooth changes of the metric and connection on the same graded bundle. It does not generally fix either kernel dimension separately. An arbitrary prescribed pair of nonnegative dimensions need not be realizable by such a family.

28.5 The two-dimensional chiral anomaly

Remark 28.6 (Axial charge and index · *cited*)**.** For this application reinterpret two dimensions as Euclidean spacetime. The spatial-first coordinates are (x, τ) with orientation $\mathrm{d}x \wedge \mathrm{d}\tau$ and $z = x + \mathrm{i}\tau$. The Wick rotation is $t = -\mathrm{i}\tau$, so the local holomorphic argument becomes $z = x - t$. A fixed value of $x - t$ moves toward increasing x. Positive Euclidean chirality therefore continues to right movers. This local convention matches the Euclidean grading to the Lorentzian one.

For one massless Dirac fermion let j_R and j_L be the right and left number currents. Define the axial current by $j_5 = j_R - j_L$. Let Q_5 be its spatially integrated charge. With the vector current conserved, the cited integrated anomaly relation is

$$\Delta Q_5 = 2 \operatorname{ind} D^+.$$

It concerns the index of the corresponding Euclidean gauge background and assumes no axial charge escapes through the spatial boundary during the transition. It is not inferred by treating the spatial monopole sphere as Lorentzian spacetime. A background of index +1 gives a change +2 in this axial convention. The factor two comes from the Dirac fermion's two chiral contributions to the axial imbalance. Bilal prints the current relation in four dimensions: (3.30) in §3.5, p. 14, has divergence $-a$, and (3.41) in §3.7, p. 16, has integrated anomaly function -2 times the index, provided the Euclidean and Minkowski chiralities are matched. The regulated chirality-trace argument (3.39)–(3.40) is independent of dimension. Nonzero modes cancel in pairs. The zero modes contribute their signed difference. The fermion and antifermion measures supply the factor two. Repeating that argument with the two-dimensional measure gives the stated relation. The two minus signs give its positive factor. Bilal extends the index computation to arbitrary even dimension in (3.44)–(3.47). This is the cited quantum-field-theory input for the two-dimensional remark, rather than a claim that his printed (3.41) is a two-dimensional formula.

Self-check. For $n = 0$, determine the sphere index and both zero-mode counts. Explain what follows from the index alone.

Answer. The sphere theorem gives index zero and both kernel dimensions zero. The index alone says only that the two dimensions agree. Their vanishing uses the sphere-specific polynomial and Serre-duality count.

Summary. The positive spinor line on the outward sphere is $\mathcal{O}(-1)$. Twisting by the monopole line $\mathcal{O}(n)$ gives $\mathcal{O}(n-1)$. The polynomial proof and cited spinor, Hodge and Serre identifications give chiral counts $\max(n, 0)$ and $\max(-n, 0)$. Their difference is n, agreeing with the cited surface index theorem and the flux Chern number. For positive n, the spinor

count is n and the spinless lowest Landau-level count is $n+1$. With the stated spacetime, chirality and boundary conventions, the cited axial anomaly gives $\Delta Q_5 = 2\,\mathrm{ind}\,D^+$. The course's topological integers thus constrain both classical boundary sectors and quantum zero modes.

Reading. Lawson–Michelsohn, Chapter III, §5, pp. 192–197, and §7, pp. 201–202; Chapter IV, §11; Appendix D, p. 396. Walpuski, *Differential Geometry IV* (2022), Propositions 70.3 and 70.6 and Remark 70.5, pp. 76–78. Gunning, *Function Theory on Compact Riemann Surfaces*, Theorem 1.17, p. 28. Nakahara, §12.6. Bilal, §3.5–3.7, pp. 13–17. Haldane, *Phys. Rev. Lett.* 51 (1983), 605–608; Greiter, *Phys. Rev. B* 83 (2011), 115129, §II.